



INDIA'S HYDROCARBON OUTLOOK

(A Report on Exploration & Production Activities)

2024-2025



Directorate General of Hydrocarbons
(Ministry of Petroleum and Natural Gas, Government of India)



हाइड्रोकार्बन महानिदेशालय
पेट्रोलियम एवं प्राकृतिक गैस मंत्रालय | भारत सरकार
Directorate General of Hydrocarbons
Ministry of Petroleum and Natural Gas | Government of India

भारत का हाइड्रोकार्बन परिदृश्य
अन्वेषण व उत्पादन गतिविधियों पर एक रिपोर्ट 2024-25

India's Hydrocarbon Outlook
A Report on Exploration & Production Activities 2024-25

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Creation

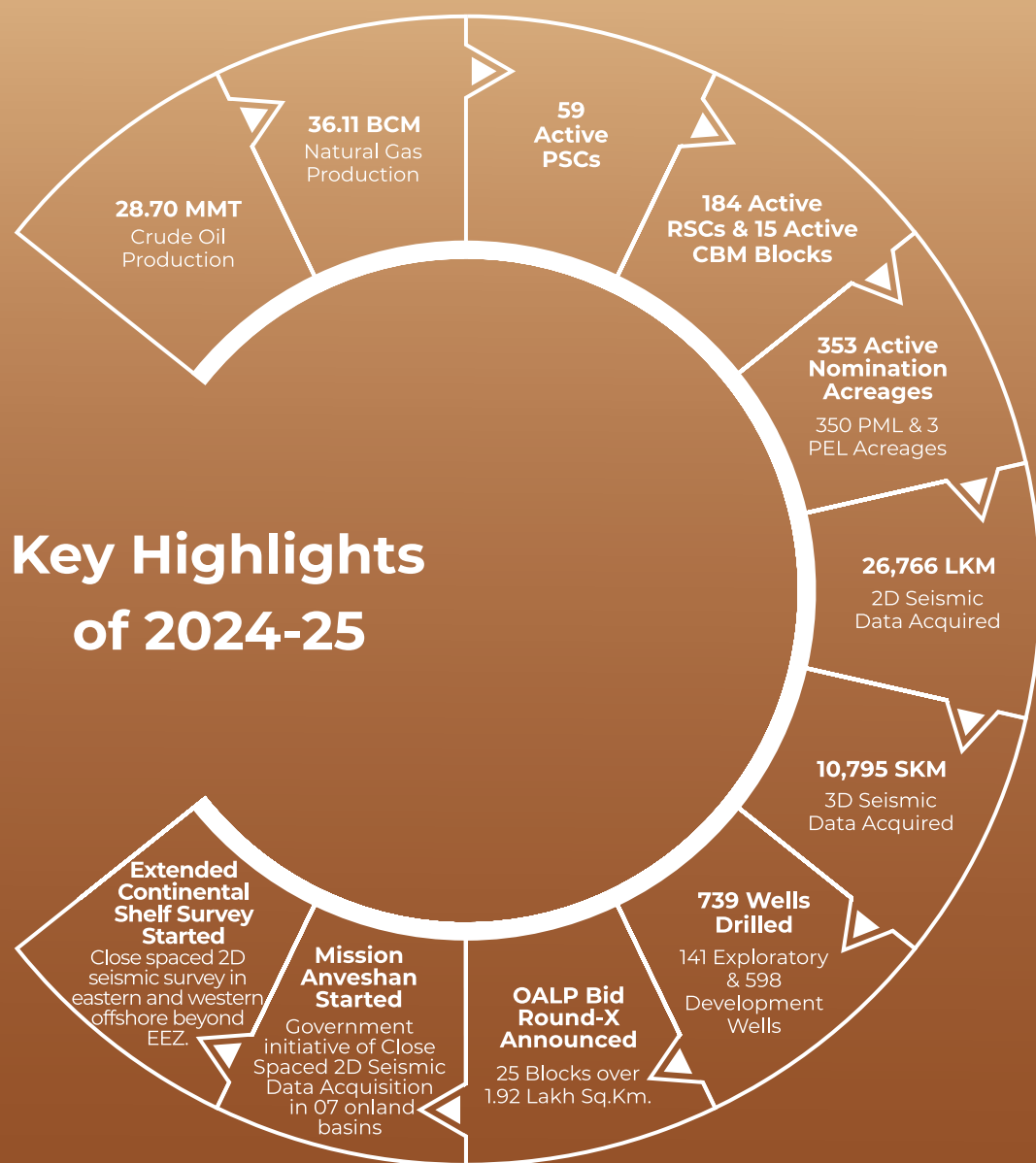
Directorate General of Hydrocarbons was formed through a Government of India resolution dated 08.04.1993 under the administrative control of Ministry of Petroleum & Natural Gas.



Objective

To promote sound management of the oil and natural gas resources while having a balanced regard for environment, safety, technological and economic aspects of the petroleum activity.





MMT- Million Metric Tonnes
BCM-Billion Cubic Meter

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हरदीप एस पुरी
HARDEEP S PURI



मंत्री
पेट्रोलियम एवं प्राकृतिक गैस
भारत सरकार
Minister
Petroleum and Natural Gas
Government of India

Foreword

Under the visionary leadership of Hon'ble Prime Minister Shri Narendra Modi, India's energy sector is undergoing a historic transformation. We are moving decisively towards building a secure, self-reliant and sustainable energy future through reforms, investment and innovation.

India's upstream oil & gas sector, once considered complex and risk-laden, is now witnessing renewed momentum. Our four-pillar revival strategy focuses on: (i) expanding quality geological data; (ii) ensuring a transparent and investor-friendly policy regime; (iii) incentivising advanced technology adoption; and (iv) improving access to risk capital.

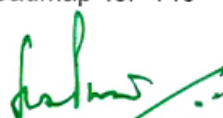
These efforts are showing tangible results. ONGC drilled 578 wells in FY 2024–25 which is the highest in last thirty-five years. Together with Oil India Ltd, 27 hydrocarbon discoveries have been made in recent years, adding reserves of over 75 million metric tonnes of oil equivalent and 2,700 MMSCM of gas.

The successful completion of OALP Round IX and launch of Round X, (covering 1.92 lakh sq km—the largest acreage offering in India's history) and the receipt of 60 bids for 28 blocks in Round IX alone, reflect growing investor confidence and the effectiveness of our progressive, flexible policies.

The Oilfields (Regulation and Development) Amendment Act, 2025 is a landmark reform, enabling hybrid leases, faster monetisation, and integrated energy projects, supporting offshore renewables and hydrocarbons alike. The National Seismic Programme, Mission Anveshan, and Airborne Gravity Gradiometry (AGG) Surveys have unlocked exploration potential in frontier and deepwater basins.

We have doubled our exploration acreage from 8% in 2021 to nearly 16% at present, with an aim to expand the acreage to 1 million sq km by 2030. India is now firmly positioned as a global E&P destination, backed by strong fundamentals and data-driven policy reforms.

This 32nd edition of the Indian Hydrocarbon Outlook documents the sector's structural transformation. As we move ahead, the convergence of hydrocarbons and renewables will define our inclusive, secure, and future-ready energy roadmap for 140 crore Indians.



(Hardeep S Puri)

New Delhi

27 June 2025

सुरेश गोपी
SURESH GOPI
സുരേഷ് ഗോപി



पेट्रोलियम एवं प्राकृतिक गैस
राज्य मंत्री
भारत सरकार
Minister of State
Petroleum and Natural Gas
Government of India

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June 18, 2025

FOREWORD

India's growing energy demand is both a symbol of our progress and a signal of our responsibility. As millions rise to new aspirations—through urbanization, industrialization, and rural transformation—our energy systems must keep pace with resilience, equity, and foresight.

At the heart of this evolution is our domestic Exploration and Production (E&P) sector. Through visionary reforms and unwavering commitment, India is unlocking the full potential of its sedimentary basins—from frontier zones to mature fields. These are not just technical milestones; they are milestones of national capacity, sovereignty, and opportunity.

With over one million square kilometres of sedimentary basins now open for exploration, a modernised policy framework in place, and strong investor interest across basins—India is laying the foundation for a more secure, self-reliant, and competitive upstream ecosystem.

A series of reforms have brought us here. The Hydrocarbon Exploration and Licensing Policy (HELP) has introduced flexibility, transparency, and ease of entry. The rationalisation of 'No-Go' zones has unlocked access to frontier areas including deepwater and ultra-deepwater plays. The strong response to Open Acreage Licensing Policy (OALP), Discovered Small Fields (DSF), and Coal Bed Methane (CBM) rounds reflects renewed investor confidence—both from established and first-time participants.

The enactment of the Oilfields (Regulation and Development) Amendment Act, 2025 further strengthens this momentum. By creating a unified lease structure, supporting infrastructure sharing, and simplifying renewal processes, the Act addresses long-standing concerns and aligns India's upstream regime with global standards.

India is also fostering global partnerships to augment domestic production while advancing decarbonization and clean energy initiatives. These developments will be instrumental over the coming decade in achieving our national goals of affordable, secure, and sustainable energy.

The Directorate General of Hydrocarbons (DGH) continues to play a vital enabling role—anchoring policy reforms, stewarding contracts, and supporting investment. The 32nd edition of the Indian Hydrocarbon Outlook captures the key trends, interventions, and next frontiers in India's upstream journey.

We have made meaningful progress—but the journey ahead is larger still. With sustained effort, collaboration, and execution, India's upstream sector can emerge as a pillar of national energy security and a platform for economic growth.



(Suresh Gopi)

पंकज जैन
सचिव
Pankaj Jain
Secretary



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India's upstream oil and gas sector has witnessed remarkable progress in recent years, driven by far reaching reforms – transition from Production Sharing Contracts to Revenue Sharing Contracts, reduction of 'No-Go Areas, enactment of Oilfields (Regulation and Development) Amendment (ORDA) Act 2025 and a renewed focus on data-driven strategies.

The recently enacted ORDA Act in particular reflects a landmark regulatory shift – introducing lease renewal provisions, infrastructure sharing, dispute resolution mechanisms, and decarbonization standards. By replacing mining-linked constraints with a unified lease framework, the Act modernizes India's upstream governance, aligning it with global best practices while ensuring legal clarity, operational flexibility, and fiscal predictability.

In parallel, the draft Petroleum and Natural Gas (PNG) Rules, 2025, reflect a comprehensive procedural overhaul. With 88 provisions across new chapters, the rules enhance regulatory precision and enable a greater degree of coherence to India's upstream framework.

Institutional reforms have been complemented by EoDB policy initiatives. The transformation of the National Data Repository into a cloud-based platform, Hydrocarbon Resource Assessment Studies and drilling of Stratigraphic Wells - are all pivotal steps enhancing supporting data-driven decision-making.

Building on this momentum, fresh bid rounds under OALP, DSF and CBM regimes are unlocking India's hydrocarbon potential with investor-friendly concessions . These rounds go beyond acreage—they reflect continuity of vision, deepening investor participation, and improving prospects for domestic production. The encouraging response from both new and global players underlines the growing credibility of our upstream ecosystem.

India's upstream sector today stands at a crossroad—enabled by reforms, guided by data, and powered by partnerships. The Directorate General of Hydrocarbons (DGH) continues to play a vital role as a policy advisor and facilitator of operational excellence.

The 32nd edition of the India Hydrocarbon Outlook 2024–25 offers useful insights into sectoral performance, opportunities, and the path ahead. I commend the DGH team for this timely and comprehensive publication and am confident it will serve as a key reference for stakeholders committed to building a robust and future-ready energy ecosystem for India.


[Pankaj Jain]



Foreword by the DG, DGH

As the world's third-largest energy consumer, India's growing industrial base, urban expansion, and improving living standards continue to drive the demand for reliable, affordable, and cleaner energy. Oil and gas remains a cornerstone of India's energy security strategy. Recognizing this, the Government has laid out a comprehensive roadmap to unlock domestic hydrocarbon potential.

One of the most transformative developments has been the opening of over 1 million sq. km of India's Exclusive Economic Zone (EEZ) for exploration-enabled by the removal of over 99% of earlier 'No-Go' areas. Backed by a government-funded ₹ 3,200 crore drilling of stratigraphic well programme and Mission Anveshan (encompassing 20,275 LKM of 20 seismic surveys in seven onland basins), India is systematically building a stronger geoscientific foundation across seven onland basins.

The National Data Repository (NOR 2.0) is now transitioning to a cloud-based platform for seamless global access. A dedicated international data center in Houston, along with platforms like Urja Pragati and Upstream India, is enhancing investor transparency, coordination, and outreach.

The Oilfields (Regulation and Development) Amendment Act, 2025 introduces a unified lease regime for all hydrocarbons, significantly improving procedural clarity, fiscal predictability, and investor confidence. The Act also strengthens compliance and introduces adjudicative mechanisms to resolve disputes fairly and efficiently.

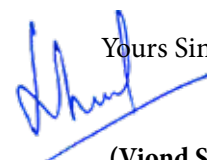
Progress under our key bid rounds continues to build. Under the Open Acreage Licensing Policy (OALP), over 172 blocks spanning 3.78 lakh sq. km have been awarded, with cumulative investment commitments exceeding US\$ 4.36 billion. Fourteen discoveries have been made under OALP-one already producing in Gujarat. OALP Round X is poised to offer another 1.92 lakh sq. km.

The Discovered Small Fields (DSF) policy has so far awarded 87 contract areas over four rounds with five contract areas currently in production. The 2025 launch of DSF-IV marks the newest opportunity, presenting 55 discoveries spread across nine contract areas estimated to hold approximately ~260 MMTOE of in-place resources. India's Coal Bed Methane (CBM) sector is gaining fresh traction-with US\$ 2.7 billion invested and output from 6 producing blocks reaching 2.2 MMSCMD. A new Special CBM Bid Round 2025 is launched which offers three blocks across two basins.

Our newly established Hydrocarbon Efficiency & New Energy (HENE) department is leading pilot initiatives on carbon capture, hydrogen integration, and emissions monitoring, helping position India's E&P ecosystem for a lower-carbon future.

This 32nd edition of the Indian Hydrocarbon Outlook captures the evolution of India's upstream sector-its reforms, achievements, and emerging frontiers. DGH remains committed to supporting a vibrant, data-driven, and globally competitive E&P ecosystem that fuels India's growth and strengthens national energy security.

I trust this publication will serve as a valuable reference for investors, policymakers, researchers, and all stakeholders shaping India's energy future.


Yours Sincerely,
(Viond Seshan)
Director General, DGH



1

Evolution of Oil and Gas Exploration in India

1.1 Seeds of Exploration

The Genesis of Oil in India (1828–1889)

India's oil journey began in Assam, where early observations of oil seepages by British geologists led to Asia's first mechanically drilled well at Makum in 1867. Though shallow, this success marked India's formal entry into petroleum exploration.

1.2 Emergence of Commercial Exploration (1889–1937)

The Assam Railways and Trading Company struck oil at Digboi in 1889, leading to the creation of the Assam Oil Company (AOC) in 1899. The Digboi refinery, operational since 1901, remains the world's oldest active refinery. By the early 20th century, the Burma Oil Company had acquired major upstream interests in the region, while Tata Engineering drilled India's early gas wells in Gujarat. A geophysical survey in 1937 at Naharkatiya marked the advent of seismic exploration.

1.3 National Awakening and Strategic Realignment (1947–1955)

Following independence, India prioritized domestic energy security. The 1948 Industrial Policy emphasized the role of hydrocarbons in industrial growth. The Geological Survey of India (GSI) resumed surveys, leading to the discoveries of Naharkatiya (1953) and Moran (1956) by AOC. The Government initiated partnerships with foreign experts, including support from the USSR, to build domestic E&P capabilities under the Second Five-Year Plan.



1.4 Institutional Foundations: ONGC and Oil India – The Twin Pillars (1955–1969)

To systematize petroleum exploration, the **Oil and Natural Gas Directorate** was set up in 1955 and later reconstituted as the **Oil and Natural Gas Commission (ONGC)** in 1956. It became a statutory body in 1959. ONGC quickly recorded breakthroughs in Cambay, Ankleshwar, Kalol, and other prolific areas.

In parallel, **Oil India Limited (OIL)** was incorporated in 1959 as a joint venture between the Government of India and Burma Oil Company to develop the Naharkatiya and Moran fields. By 1961, it evolved into a 50:50 partnership and was later nationalized as a fully government-owned enterprise. Together, **ONGC and OIL became the two flagship National Oil Companies (NOCs)**, driving India's upstream exploration through coordinated efforts across onshore and offshore terrains.

1.5 Game Changers: Offshore Discoveries and the Rise of Mumbai Offshore (1970s–1980s)

India's offshore era began with ONGC's seismic work in the Gulf of Cambay in 1962. The landmark discovery of **Bombay High (now Mumbai Offshore)** in 1974 transformed India's oil landscape. This field alone contributed to a sharp reduction in crude imports and ushered in an era of self-reliance. Subsequent discoveries included **Bassein** (major gas field), **Heera**, and **Neelam** on the west coast, and **PY-3** and **Ravva** on the east coast.

OIL, traditionally focused in Assam, also expanded to Orissa, Rajasthan, and the Andaman offshore in the late 1970s and 1980s—playing a strategic role in building India's geographic exploration footprint.

1.6 Transition Phase: The Pre-NELP Regime (1980–1995)

Despite major state-led discoveries, India's technological limitations and capital constraints prompted a shift toward public-private collaboration. The **Pre-NELP regime** introduced in 1980 marked India's first step toward liberalization of the upstream sector.

Under this regime:

- 28 blocks were awarded through successive

bidding rounds between 1980 and 1995.

- ONGC and OIL retained rights to participate post-discovery.
- Two **Development Rounds (1992–1993)** offered small and medium-sized discovered fields to private players, with Production Sharing Contracts (PSCs) signed for 29 fields.
- A standout success of the Pre-NELP era was the RJ-ON-90/1 block in Rajasthan, originally awarded to a consortium of Shell India Production Development (SIPD) and ONGC. Over time, Cairn Energy acquired Shell's participating interest and went on to discover the Mangala, Bhagyam, and Aishwariya fields—among the largest onshore oil finds in India. These discoveries transformed the Barmer basin into a major petroleum province and made a lasting contribution to India's domestic oil production.

Though early results were mixed, Pre-NELP established a precedent for competitive participation, leading to infrastructure upgrades, joint ventures, and the development of marginal fields.

1.7 Transition to Competitive Markets (1990s)

By the mid-1990s, the Government of India intensified efforts to attract private and foreign investments. Five bid rounds offered drawing participation from Shell, Enron, Aramco, and emerging domestic players like **Hindustan Oil Exploration Company (HOEC)**. ONGC also explored **Coal Bed Methane (CBM)** and **Enhanced Oil Recovery (EOR)** strategies, particularly in Gujarat.

These reforms laid the foundation for a fully liberalized framework under the New Exploration Licensing Policy (NELP).

1.8 Strengthening Oversight: Formation and Governance of DGH (1993)

As private participation in exploration grew, the Government recognized the need for an independent technical and regulatory body. Following recommendations from the Dasgupta and Kaul Committees, the **Directorate General of Hydrocarbons (DGH)** was established on **April 8, 1993**, under the **Ministry of Petroleum and Natural Gas (MoPNG)**.



1.8.1 Mandate and Objectives:

- Promote optimal and environmentally responsible development of India's hydrocarbon resources.

DGH-Objective

To promote sound management of the Indian Petroleum and Natural Gas resources having a balanced regard for the environment, safety, technological and economic aspects of the petroleum activity.

- Serve as the government's technical arm for upstream policy and regulatory oversight.
- Ensure reservoir health, data management, and transparency in operations.

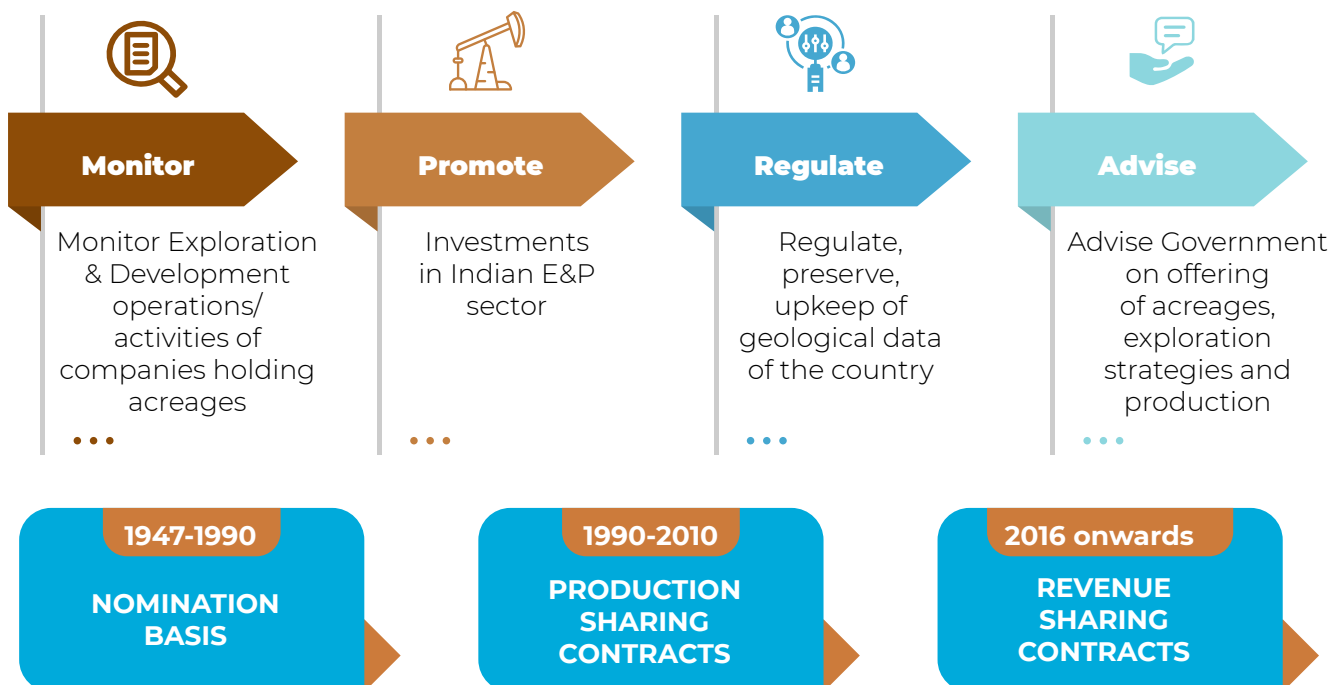
1.8.2 Key Functions:

- Appraising exploration and development plans.
- Evaluating reserves and exploitation strategies.
- Managing geoscientific data, licensing rounds, and operator compliance.

1.8.3 Governance Structure:

- Advisory Council: Comprising eminent industry experts, chaired independently with the Director General as Member Secretary.
- Administrative Council (since 2001): Chaired by the Secretary (MoP&NG), with members from MoPNG, OIDB, and DGH, responsible for budget, staffing, and performance oversight.

DGH has since become the cornerstone institution guiding India's upstream regulatory architecture.



1.9 Opening Up the Sector: New Exploration Licensing Policy (NELP) (1997-1999)

Launched in 1997 and implemented in 1999, **NELP** replaced nomination-based acreage allocation with open, competitive bidding. Both NOCs and private/foreign companies competed equally for blocks under a Production Sharing Contract (PSC) model.



Across nine bid rounds:

- 254 blocks were awarded.
- Key breakthroughs included commercial oil in **Rajasthan** and deepwater gas in **Krishna-Godavari**.

One of the most defining milestones under NELP was the discovery of the KG-DWN-98/3 block (commonly known as KG-D6) in the Krishna-Godavari Basin by Reliance Industries Limited. This deepwater block yielded major gas discoveries in the D1 and D3 fields and marked India's first significant success in ultra-deepwater exploration. The field commenced production in 2009 and, at its peak, accounted for nearly 40% of the country's domestic gas production, helping reduce import dependence and validating the deepwater potential of Indian offshore acreage.

NELP catalyzed a wave of investment, improved basin knowledge, and expanded India's offshore frontiers. In parallel with the launch of NELP, the Government of India also approved the Coal Bed Methane (CBM) Policy during the 1997–1999 period to harness unconventional gas resources. This policy created a dedicated legal and fiscal framework to promote CBM exploration and production from coal-bearing basins, separate from the existing petroleum legislation. Over the years, this initiative has matured into a commercially viable segment of India's gas portfolio. As of 2024–25, CBM production stands at approximately 2.2 MMSCMD, making a meaningful contribution to domestic natural gas supply and energy diversification.

1.10 Revenue Sharing and Renewed Investment (Post-2015)

To further streamline exploration operations and align with global best practices, the Government transitioned to **Revenue Sharing Contracts (RSCs)** – a simpler, more transparent fiscal model.

Two transformative policy frameworks were launched:

• **Discovered Small Field (DSF) Policy (2015):**

The Discovered Small Field (DSF) Policy was launched to monetize small and marginal oil and gas fields relinquished by National Oil Companies (NOCs), which were earlier considered commercially unviable under conventional regimes. Recognizing the need to unlock these stranded assets, the policy introduced a simplified and investor-friendly framework that included marketing and pricing freedom, no signature bonus requirements, and reduced royalty rates. Crucially, DSF rounds did not mandate prior E&P experience as a qualifying criterion, significantly lowering entry barriers and enabling broader participation.

This inclusive approach led to a surge in interest from new entrants, including first-time domestic operators, emerging energy firms, and non-traditional investors. Awarded under a transparent revenue-sharing contract model, DSF rounds attracted fresh capital, diversified the operator base, and brought new energy to fields that had long remained undeveloped. The policy has since proven effective in reviving stranded resources, boosting domestic production, and stimulating competition in India's upstream sector. • **Hydrocarbon Exploration and Licensing Policy (HELP, 2016):**

Replacing the PSC framework, HELP introduced:

- o A unified license for all hydrocarbons.
- o Open Acreage Licensing Programme (OALP) for continuous block selection.
- o Full pricing and marketing freedom.

HELP and DSF together have enhanced investor confidence and reinvigorated India's upstream momentum.



India's Oil & Gas Production Sources

Nomination Blocks/
Fields under ONGC
and OIL

The discovered fields
and producing Pre-
NELP(Exploration), Pre-NELP
(Discovered), NELP blocks
under the Production Sharing
Contract (PSC) regime

The contract areas
(CAs) Awarded under
DSF, RSC (HELP), CBM
Regime (Revenue sharing
contracts)

1.11 Sustained Government Initiatives in Exploration

In recent years, the Government of India has undertaken a series of transformative initiatives to boost hydrocarbon exploration across both conventional and frontier basins. Strategic campaigns such as the National Seismic Programme (NSP) have enabled extensive 2D seismic data acquisition in underexplored basins, laying the ground work for more informed bidding and exploration. Complementing this, Mission Anveshan was launched as an in-fill seismic campaign to sharpen subsurface imaging and fill critical data gaps. A major game-changer has been the opening up of previously restricted No-Go offshore areas, unlocking nearly 99% of India's Exclusive Economic Zone (EEZ) for E&P activities. To support data-driven exploration, the National Data Repository (NDR) has been a key enabler by providing centralized access to India's geoscientific data. The upcoming launch of NDR 2.0, with improved interface and analytical capabilities, is set to further empower operators and investors. Together, these efforts reflect a coherent national strategy to unlock the full potential of India's sedimentary basins and attract sustained investment in exploration.

1.12 Regulatory Reform for a Stable Future: The ORDA Amendment, 2025

The Oilfields (Regulation and Development)

Amendment Act, 2025 marks a pivotal reform in India's upstream legislative framework, aimed at fostering transparency, stability, and investor confidence. Key provisions include the transition to a uniform petroleum lease model, establishment of a centralized regulatory authority, and introduction of a structured dispute adjudication mechanism. The Act also decriminalizes certain procedural lapses by replacing them with monetary penalties, thereby reducing compliance burden and improving the ease of doing business. With a strong emphasis on environmental protection, safety, and the adoption of globally recognized petroleum industry practices, the amended Act brings much-needed legal stability and operational predictability. It is poised to serve as a cornerstone for India's next phase of exploration and production growth.

1.13 Looking Ahead: Unlocking India's Hidden Reserves

With vast sedimentary basin potential, India is strategically reassessing resource estimates across deepwater and frontier regions. DGH, supported by global partnerships, is driving data acquisition and structural mapping. Enabled by transparent licensing, digital platforms, and a future-ready regulatory model, India's upstream sector is poised for a quantum leap in hydrocarbon discovery and production.



Table 1.1:

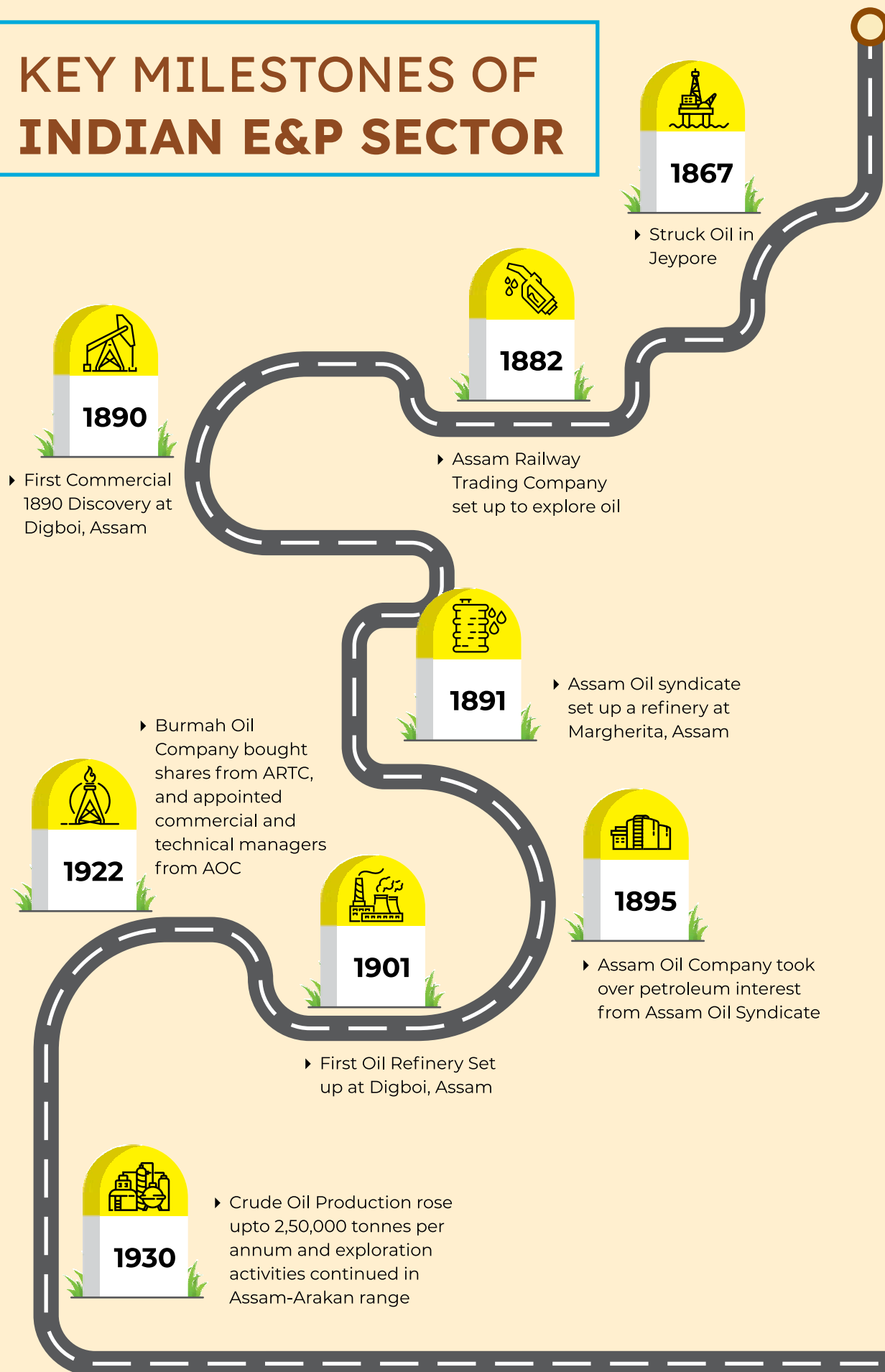
Exhibit – 1 Summarized Chronology of India's E&P Regimes

Regime	Timeframe	Key Highlights
Nomination	Pre-1980	Acreages awarded to ONGC and OIL. Exploration mostly in onland and shallow offshore areas. Dominated by public sector undertakings.
Pre-NELP Exploration Blocks	1980–1995	28 blocks awarded to private parties. ONGC/OIL had rights to participate post-discovery.
Pre-NELP Discovered Fields	1992–1993	29 fields offered to private operators under PSCs (with ONGC/OIL holding PI). Focused on small and medium-size discoveries.
NELP (I–IX)	1999–2012	254 blocks awarded across 9 rounds. Encouraged private and foreign participation on par with NOCs. Unlocked major discoveries in Rajasthan and KG basin.
DSF (I–III) & SDSF - 2024	2015–2025	85 contract areas awarded across 3 rounds. Designed for monetizing relinquished or marginal fields with simple terms and revenue sharing mechanism.
HELP (OALP I–IX)	2018–2025	172 blocks awarded under Open Acreage Licensing Programme. Operates under unified licensing and revenue sharing. Simplifies bidding and reduces regulatory burden.



KEY MILESTONES OF INDIAN E&P SECTOR

PRE-2000





1997-1999

- ▶ GoI formulated CBM Policy
- ▶ GoI approved NELP
- ▶ NELP-I launched offering 48 blocks



1987-1993

- ▶ Ravva Oil Field discovered in KG Offshore
- ▶ DGH Established
- ▶ ONGC is re-organised as a corporation



1981-1986

- ▶ Oil India Ltd became wholly owned GoI Enterprise
- ▶ Discovery of Gandhar Oil field, Gujarat
- ▶ Private participation in discovered fields allowed.
- ▶ Neelam Oil Field, Mumbai



1974-1977

- ▶ Bombay High discovered
- ▶ Discoveries of Bassein Gas Field, Mumbai Offshore, Kharsang oil field, Arunachal Pradesh
- ▶ Discovery of Heera oil field, Mumbai Offshore

- ▶ Discovery of Naharkatia Oil Field, Assam



1959-1961

- ▶ ONGC became statutory body
- ▶ Discovery of Ankleshwar oil field, Gujarat
- ▶ Oil India Pvt Ltd incorporated as 50% JV between GoI and BOC



1956

- ▶ Oil and Natural Gas Commission established
- ▶ Discovery of oil in Moran, Assam



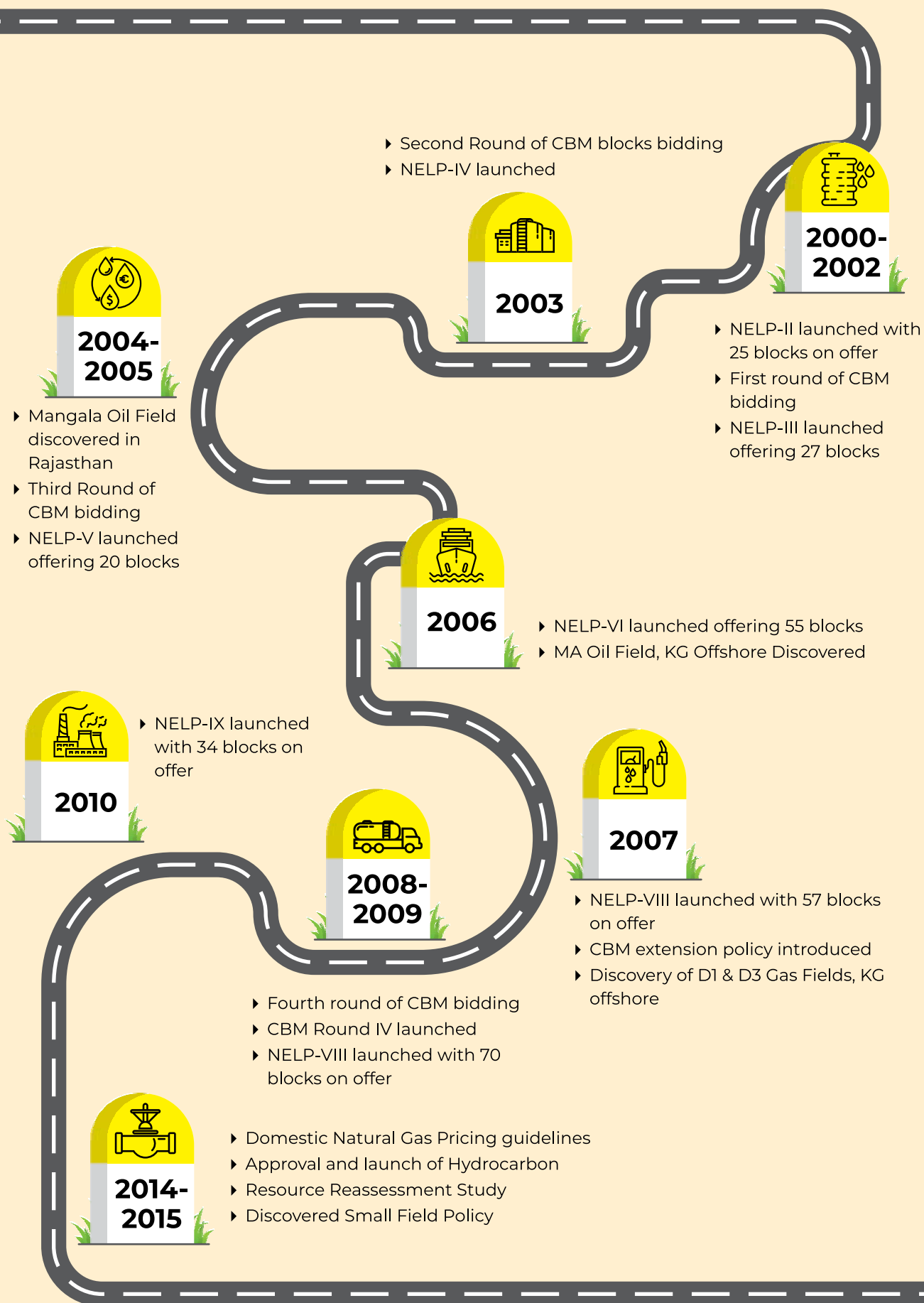
1953



1948

- ▶ Industrial Policy Resolution passed
- ▶ Oil fields Regulation and Development) Act, 1948 came into existence

POST - 2000



2024-2025

- ▶ OALP Bid Round X Announced
- ▶ ORDA Act 2025
- ▶ Special DSF Bid Round 2024 Launch

2023-2024

- ▶ First Oil Production from Challenging Deep Water Block KG-DWN-98/2 (Cluster-II) in KG-Basin
- ▶ Launch of OALP Bid Round - IX
- ▶ Mission anveshan
- ▶ Extended continental shelf survey

2022-2023

- ▶ Launch of OALP Bid Round - VIII
- ▶ Launch of SCBM-2022

2020-2021

- ▶ Launch of OALP Bid Round VI & VII
- ▶ Launch of DSF-III Bid Round
- ▶ Launch of SCBM-2021

- ▶ Launch of NDR
- ▶ Launch of HELP through OALP
- ▶ Early monetization of CBM
- ▶ Extension policy for Pre-NELP Contracts

2019-2020

- ▶ Launch of OALP Bid Round II, III, IV & V
- ▶ Self-Certification of PSC processes
- ▶ Natural Gas Marketing Reforms, 2020

2018

- ▶ Launch of OALP Bid Round-I, DSF Bid Round-II
- ▶ Enhanced Recovery Policy

2017

2016

- ▶ Launch of OALP Bid Round-I
- ▶ Enhanced Recovery Policy



2

India's Sedimentary Basins: Subsurface Insights

India's sedimentary basins have proven petroleum systems. The country is one of the 171 petroleum provinces independently mapped by U.S. Geological Survey (USGS) in 2012. The basin area occupies 50% of country's onland area and 73% of country's offshore within Exclusive Economic Zone (EEZ).

With 26 sedimentary basins mapped and assessed, India has discovered petroleum of 12.0 billion tonnes of oil equivalent and undiscovered risked potential of 12.9 billion tonnes of oil equivalent. While this is an estimate of conventional petroleum occurrence, unconventional petroleum exists in form of coal seam gas, shale plays, tight plays, basement fractures and gas hydrate.

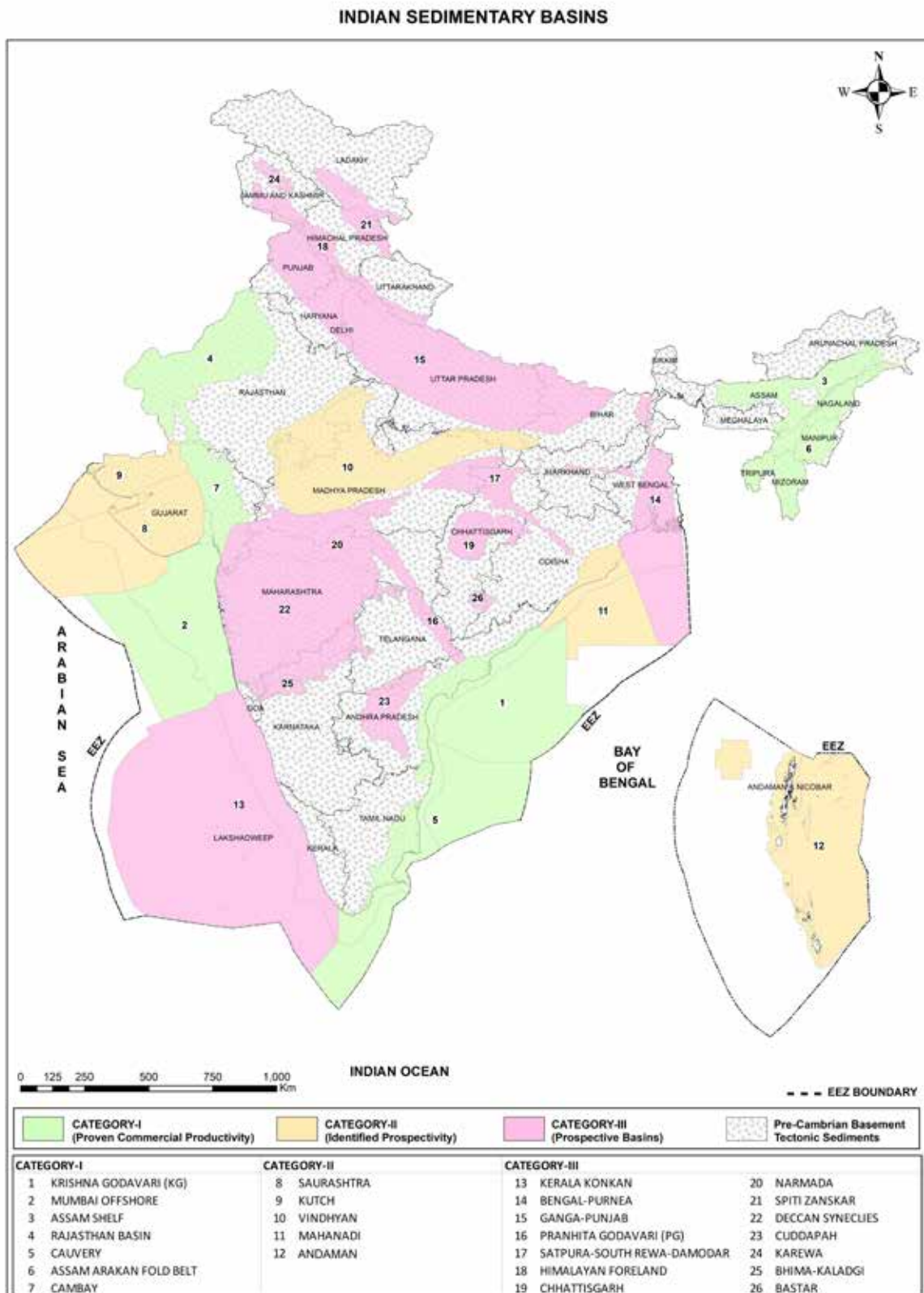
Sedimentary basins are divided into three categories mapping conventional resources and following the rationale of Petroleum Resources Management System (PRMS).

With the objective for complete appraisal of India's sedimentary basins, Government of India, over the past decade has been implementing various data acquisition campaigns through extensive seismic surveys both onland and offshore. As a result, 79.5% basin area are now appraised for some meaningful subsurface analysis. For more insights of hydrocarbon play systems, four stratigraphic wells are planned for drilling and further, all forms of hydrocarbon resources are now drawn up for re-assessment through expert consultancy. A country-level plan is chalked out to bring out hydrocarbon potential and their maturity in a holistic manner for a realistic assessment of fossil-fuels, to optimally align with the energy transition initiatives.



With a futuristic view, the current version portrays a succinct treatise of India's sedimentary basins in terms of basic geology, brief exploration history, present status and data availability. The sub-chapters are segregated into three clusters

aligned with major geological provinces (East Coast, West Coast and Other Onland basins) focusing on the upcoming Hydrocarbon Resource Assessment Study (HRAS 2025).



Source: DGH Internal

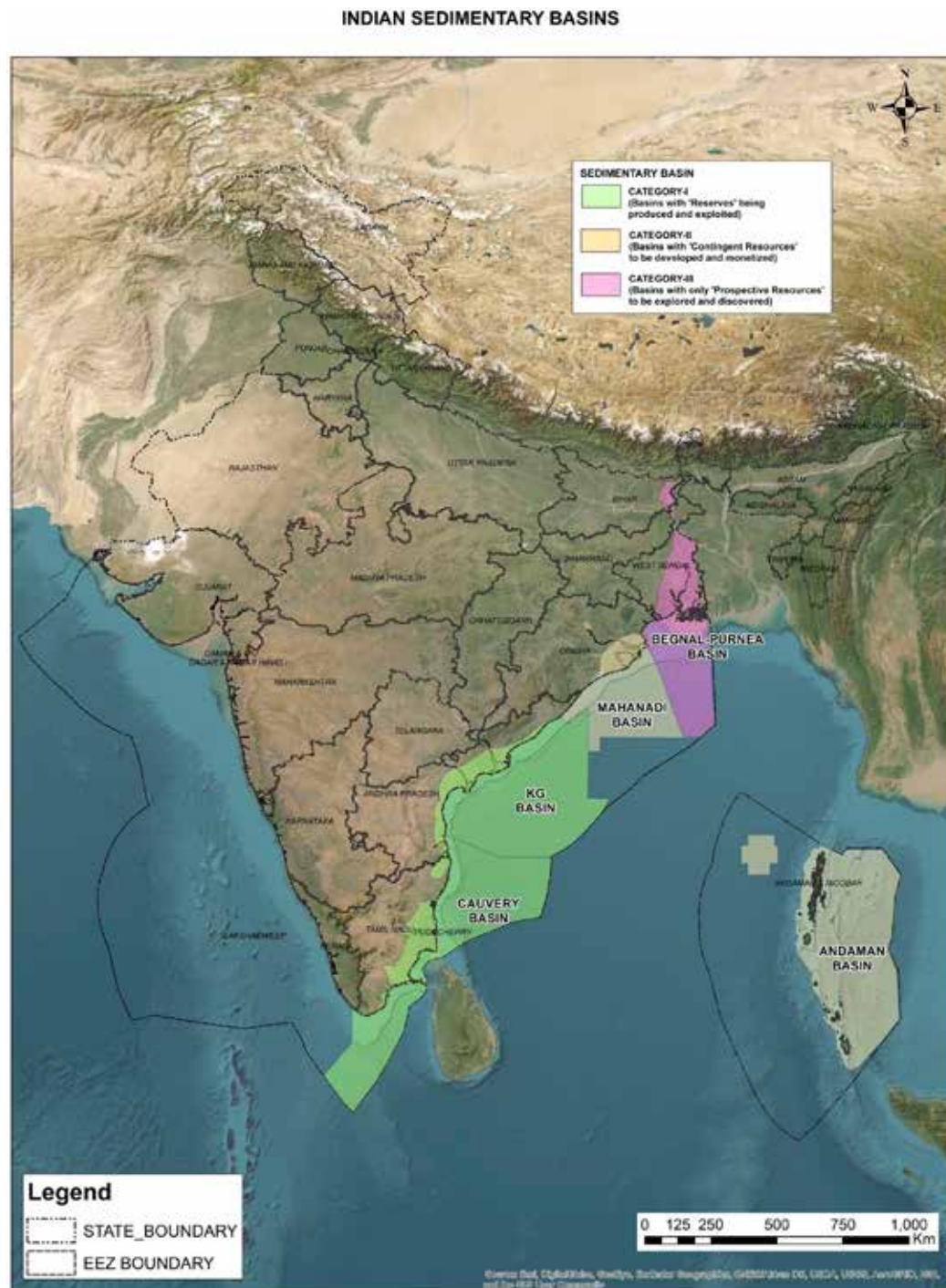


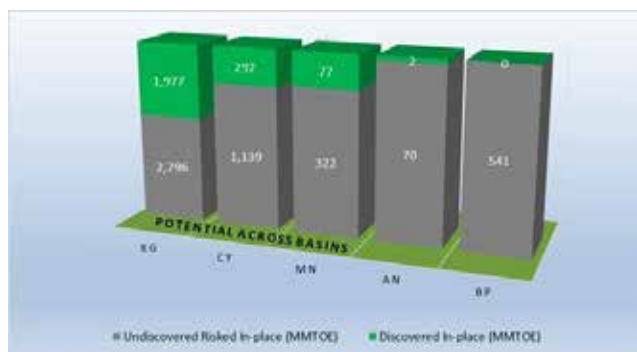
2.1 India's East Coast Basins

Out of 26 sedimentary basins, there are five basins located off India's east coast. Geologically, four basins namely (i) KG, (ii) Cauvery, (iii) Mahanadi and (iv) Bengal-Purnea belong to east coast passive continental margin under rift tectonics. The fifth basin, Andaman has tectonics associated to Island Arc system. Under the upcoming HRAS 2025, the resource assessment

of conventional plays, shales and gas hydrate will be conducted in these five basins as a part of Phase I scheduled tasks.

The KG basin tops the list in terms of both discovered and undiscovered hydrocarbon in-place. This is distantly followed by Cauvery and Mahanadi basins. Bengal-Purnea and Andaman are two emerging basins with increased exploration activities.





Source: DGH Internal



Source: DGH Internal

University of Houston, Texas, U.S.A. is engaged under GoI programme in research related studies for Andaman, Mahanadi and Bengal offshore areas.

GoI has also engaged NOCs, to coordinate with international experts from consulting agencies and IOCs to finalize and drill 4 stratigraphic wells, one each in four frontier offshore basins, including Andaman, Mahanadi, Saurashtra and Bengal.

M/s TGS, Perth, Australia carried out an integrated seismic study of India's east coast margin using proprietary tool, termed as 2D^{cubed}, whereby a 3D seismic migration volume is generated from a

set of 2D and 3D seismic surveys. The study is expected to provide a regional framework for planning future 2D/3D seismic surveys. The NDR data were used under NDR Data Policy on value-added study at a cost to company. Subsequently, TGS expanded the east coast study to develop another proprietary tool, christened as Facies Map Browser, that will provide extensive access to well-based stratigraphic interpretation and sequence-constrained facies maps.

In onland part of Cauvery basin, Airborne Gravity-Gradiometry (AGG) survey was conducted in areas where seismic surveys could not be undertaken due to operational challenges.

2.1.1 KRISHNA-GODAVARI (KG) BASIN

Krishna-Godavari (KG) is a Category I basin with commercial discovered in-place.

The onland part of the basin overlaps with the state(s) of Andhra Pradesh.

Krishna-Godavari (KG) is located along the east coast of Indian peninsula, lying between Mahanadi basin in the north and Cauvery basin towards south. The basin is most prospective and largely proven with maximum resources

(38% of total offshore), and is known for country's largest deepwater gas field. Both deepwater and shallow water are extensively appraised with large-scale datasets, which is an opportunity for intensive exploration of stratigraphic traps majorly the channelized deposits.

The KG basin consists of sediments with thickness of more than 7,000m, ranging in age from the Early Permian to Recent. The basin's onland part is mostly covered by the alluviums deposited by the two major river systems, viz.,

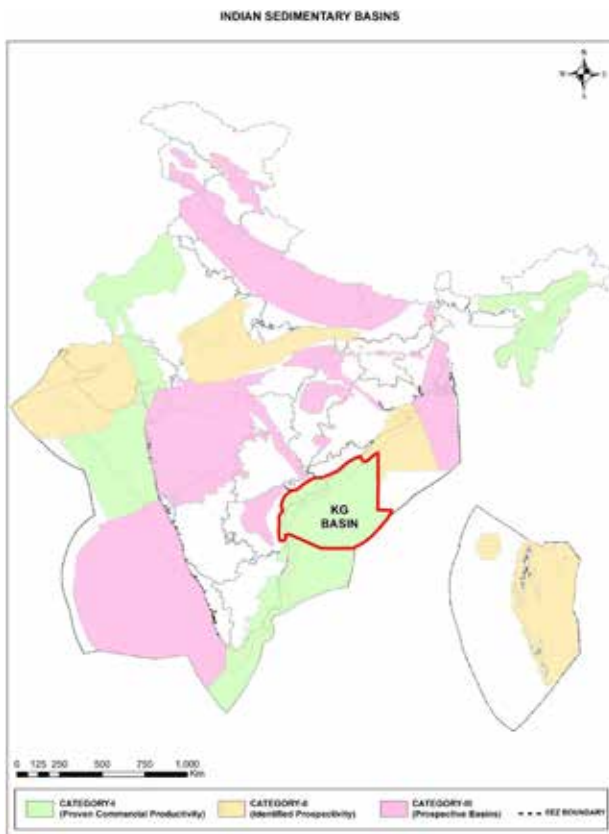


Godavari and Krishna and several stratigraphic sequences including Lower Gondwana that are outcropped near the basin margin.

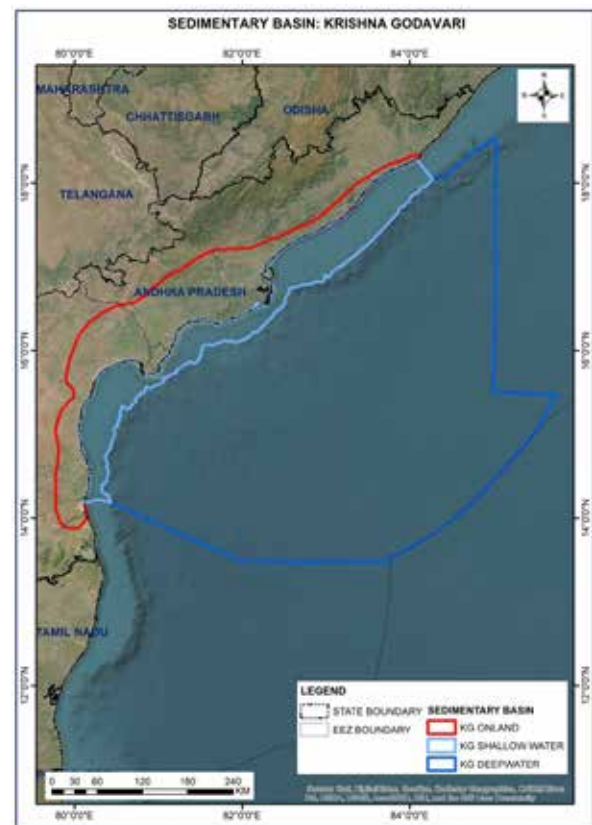
The reservoirs are primarily sandstones with isolated occurrences of limestone and unconventional reservoirs like fractured basalts. Effective source rocks are identified to be the

shales of Permian to Eocene. Biogenic reservoirs and gas hydrate deposits have been established in the basin.

Facies propagation in deepwater calls for mapping with high resolution seismic and well data. Syn-rift and Eocene play can be targets for exploration.



Source: DGH Internal



Source: DGH Internal

• Hydrocarbon prospectivity and Basin maturity:

In the basin, 10 thermogenic plays are identified within Basement, Permian, Mesozoic, Paleocene, Eocene, Miocene, Pliocene. Besides, 2 biogenic plays have been established in Mio-Pliocene sequences. Petroleum system was modelled in 3D basis adequate datasets.

Krishna-Godavari (KG) basin has a total hydrocarbon in-place of 4,772 MMTOE. This includes discovered in-place of 1,976 MMTOE and undiscovered risked in-place of 2,796 MMTOE. The discovered in-place is under commercial production. The basin has 58.6% of total in-place, potential to be explored and discovered.

• Overview of producing fields:

Commercial production in onland area commenced in 1965. Major producing fields with cumulative production in parenthesis are namely Pasarlupudi (12.5 BCM), Kesavadasapalem (5 BCM), Kesanapalli West (2.7 MMT), Mandapeta-Mandapeta(S) [4.4 BCM], Gopavaram (1.7 MMT). The area has 53 fields with 260 producing wells. The maximum flow rates achieved by any field on record are Oil 1,535 BOPD and Gas 1 MMSCMD.

Commercial production in shallow water area commenced in 1994. Major producing fields with cumulative production in parenthesis are namely Ravva (42 MMT, 13 BCM). The area has 6 fields with 39 producing wells. The maximum flow rates achieved by any field on record are Oil 9,491 BOPD.

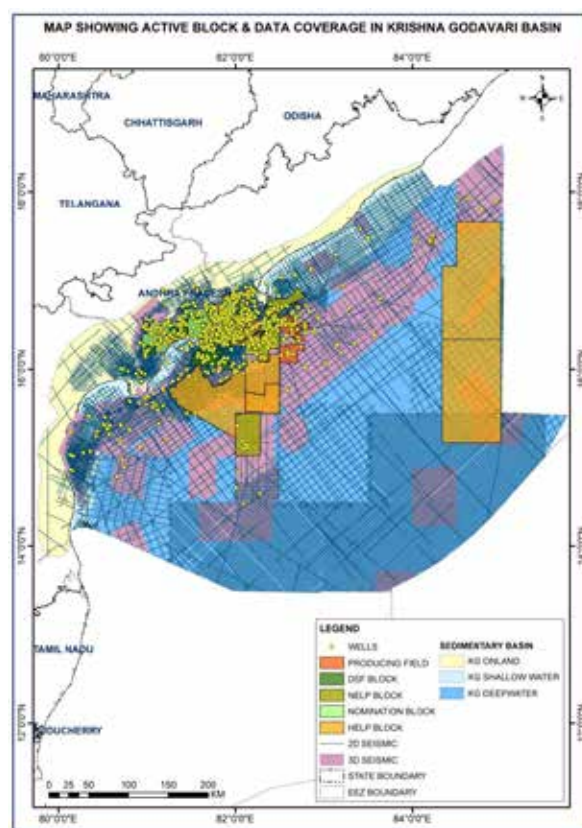


Commercial production in deepwater area commenced in 2009. Major producing fields with cumulative production in parenthesis are namely KG-DWN-98/3 (5.7MMT, 115 BCM), KG-DWN-98/2 (1.1 MMT, 2 BCM). The area has 3 fields with 58 producing wells. The maximum flow rates achieved by any field on record are Oil 20,875 BOPD and Gas 28 MMSCMD.

Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 167,140 LKM 2D seismic, 85,859 SKM 3D seismic and 1,450 conventional wells (1,023 exploratory). Multiple flavour of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.

The basin has 59 gas hydrate wells that were drilled under National Gas Hydrate Programme (NGHP).



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area (skm)	Appraised (skm)	Active acreage [Nos./Area (skm)]
Onland	31,456	31,456 (100%)	Nomination: 37/4,289 Fields: 1/55 DSF: 10/804
Shallow water	25,649	25,649 (100%)	Nomination: 7/593 Fields: 2/349 NELP: 1/495 OALP: 1/177 DSF: 8/1,379
Deepwater	172,895	172,895 (100%)	Fields: 1/1,174 NELP: 1/2,398 OALP: 5/26,049

The basin occupies an area of 230,000 sq km. It has been appraised to the extent of 230,000 sq km (100%). The active acreage nos. across regime(s) are 74 with area 37,764 sq. km (16% of basin area).

Bidding opportunities:

Onland:	1 DSF (Round IV) contract area with 301 sq km area [7 discoveries namely Vanadurru South, Bantumilli South-1, Chintalapalli-1, Medapadu-1, Bantumilli-2, Thanelanka-1, Yedurulanka-1]
Shallow water:	1 OALP (Round X) block with 2,968 sq km area; 2 DSF (Round IV) contract areas with 1,125 sq km area [11 discoveries namely [G-1-NE-1(AA), G-1-N-2, G-1-NE-2(AB), D-24(KGIII6-A1), GS-15-15, GS-15-22, GS-15-14] and [A3-2, H-2, D-36, KGS092NASRI-1]]
Deepwater:	3 OALP (Round X) blocks with 32,057 sq km area; 2 DSF (Round IV) contract areas with 922 sq km area [10 discoveries namely [D-04(KG-D6-F1), D-05(KG-D6-SH2), D-06(KG-D6-D1), D-07(KG-D6-K1), D-08(KG-D6-K2), D-42(KG-D6-L1), D-18(KG-D6-H1), D-23(KG-D6-P-1A)] and [CD-10-1, CD-11-1]]



2.1.2 CAUVERY BASIN

Cauvery is a Category I basin with commercial discovered in-place.

The onland part of the basin overlaps with the state(s) of Andhra Pradesh, Kerala, Puducherry, Tamil Nadu.

The basin is situated southerly along the east coast of Indian peninsula. Basin has significant production from Mesozoic and Basement. Exploration opportunity exists in the undrilled areas of ultra-deepwater, located more than 1,500m water depth in the northeast and southern part towards Gulf of Mannar.

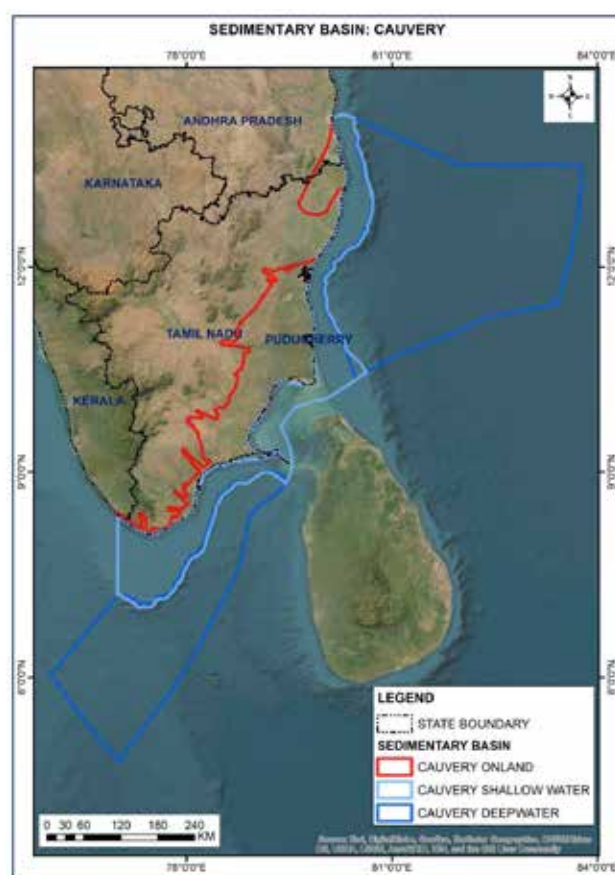
Commercial hydrocarbon occurrences are encountered in wide stratigraphic sequences with the oldest sediments of Jurassic including the fractured Basement rocks.

The basin hosts thick pile of sediments of the order of 8,000m, ranging from Late Jurassic. The onland part is mostly covered by the alluvium deposited by the major river system of Cauvery and several stratigraphic sequences including Lower Gondwana have cropped out along the basin margin areas.

Mapping of deeper plays, including Basement and facies propagation of Pre-Albian sequences are the focus area of exploration.



Source: DGH Internal



Source: DGH Internal

- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 9 plays are known within Basement, Jurassic, Cretaceous and Paleocene-to-Miocene. Petroleum system was modelled in 3D basis adequate datasets.

Cauvery basin has a total hydrocarbon in-place of 1,432 MMTOE. This includes discovered in-place of 294 MMTOE and undiscovered risked in-place of 1,138 MMTOE.

The discovered in-place is under commercial production. The basin has 79.5% of total in-place, potential to be explored and discovered.



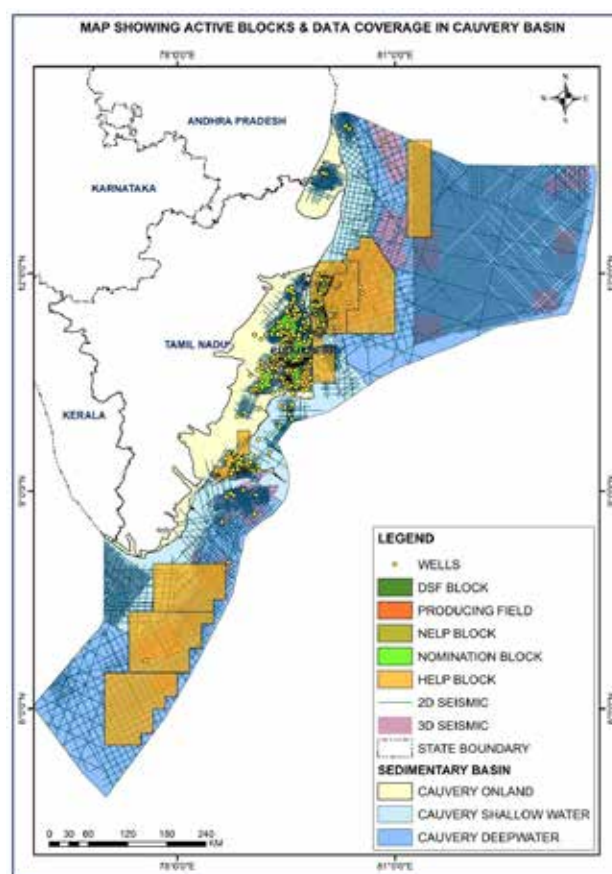
Overview of producing fields:

Commercial production in onland area commenced in 1986. Major producing fields with cumulative production in parenthesis are namely Narimanam (3.6 MMT, 1 BCM), Kanjirangudi (5.5 BCM), Periyapattinam (4.4 BCM), Kuthalam (3.7 BCM), Kamalapuram (2 MMT, 0.6 BCM). The area has 26 fields with 243 producing wells. The maximum flow rates achieved by any field on record are Oil 1,320 BOPD and Gas 2 MMSCMD.

Commercial production in shallow water area commenced in 1997. Major producing fields with cumulative production in parenthesis are namely PY-3 (3 MMT). The area has 2 fields with 2 producing wells. The maximum flow rates achieved by any field on record are Oil 4 BOPD.

Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 111,592 LKM 2D seismic, 51,003 SKM 3D seismic and 872 conventional wells (641 exploratory). Multiple flavour of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area (skm)	Appraised(skm)	Active acreage [Nos./Area (skm)]
Onland	37,825	37,825 (100%)	Nomination: 27/4,028 Fields: 2/267 OALP: 2/1,878
Shallow water	43,723	43,723 (100%)	Fields: 2/157 NELP: 1/858 OALP: 2/3,905
Deepwater	158,452	158,452 (100%)	OALP: 6/44,658

The basin occupies an area of 240,000 sq km. It has been appraised to the extent of 240,000 sq km (100%). The active acreage nos. across regime(s) are 42 with area 55,750 sq. km (23% of basin area)

Bidding opportunities:

Deepwater: 1 OALP (Round X) block with 9,991 sq km area



2.1.3 MAHANADI BASIN

Mahanadi is a Category II basin with sub-commercial discovered in-place, potential to be commercially produced.

The area geographically overlaps with the state(s) of Odisha.

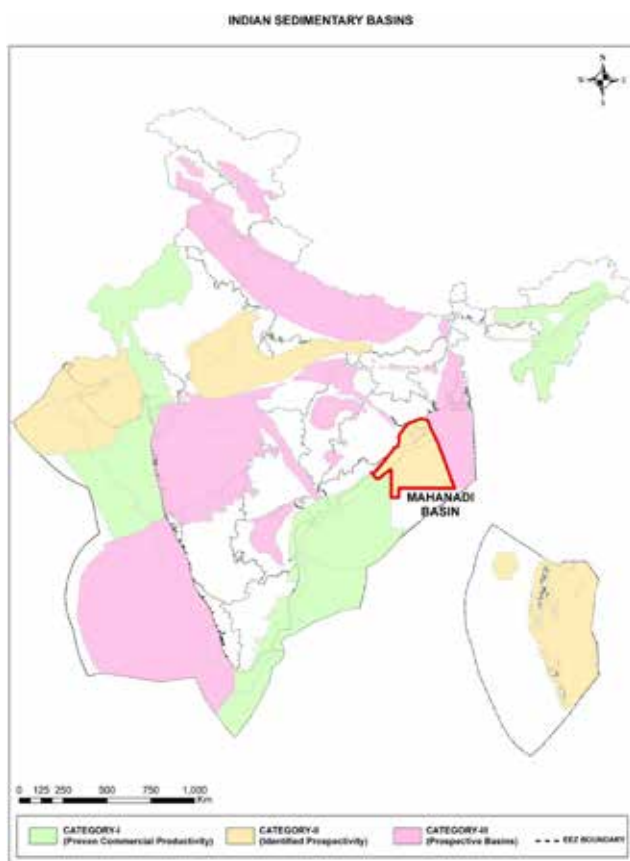
The basin is situated along the East Coast of Indian peninsula. Mahanadi basin is significantly appraised with strong analogy with the easterly located Bengal offshore that has numerous small-to-medium discoveries. Deepwater has significant discovered Miocene play in form of discrete and stacked reservoirs. Enhanced exploration opportunities exists for the prospective Pliocene plays of biogenic origin.

Hydrocarbon occurrences are spread over the Tertiary stratigraphic sequences ranging from the oldest sediments of Paleogene to the youngest sediments of Miocene-Pliocene.

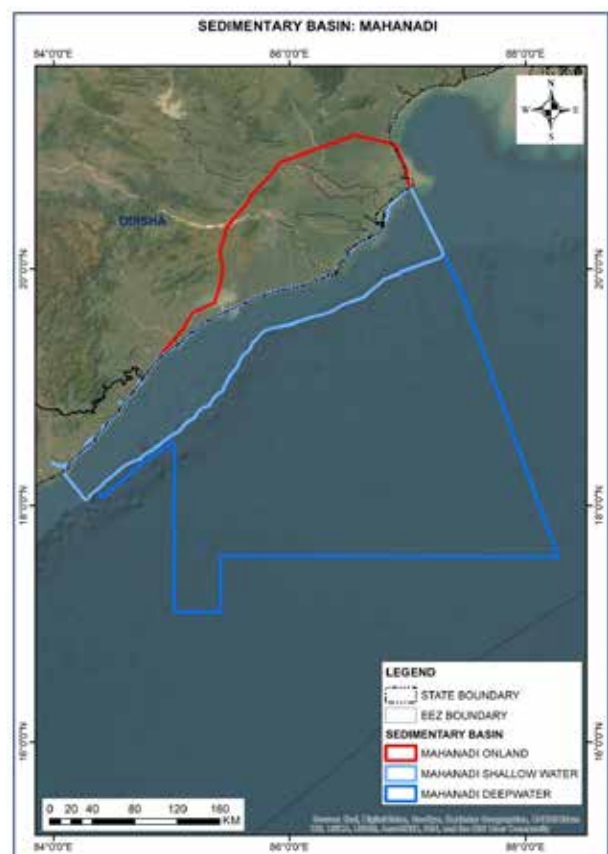
Several biogenic gas fields have been discovered with strati-structural entrapment conditions. Though hydrocarbon accumulations often indicate charging from in-situ shallower biogenic source sequences, there are instances where thermogenic accumulations have been established.

The basin hosts a thick pile of sediments, mainly ranging from Early Cretaceous to Recent. The basin's onland part is mostly covered by the Mesozoic sediments which is underlying the Miocene sediments, deposited by the major river system of Mahanadi. Stratigraphic sequences including Gondwanas are reported in some small rift grabens.

Petroleum system model results show prospective trend towards northeast part of offshore adjoining Bengal offshore. High resolution seismic acquisition for delineating deeper plays and supporting pore-pressure study in high pressure area are the key focus.



Source: DGH Internal



Source: DGH Internal



- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 10 plays exist within Cretaceous-Tertiary. Petroleum system was modelled in 3D basis adequate dataset

Onland sub-basin has a total hydrocarbon in-place of 22 MMTOE and this is entirely undiscovered risked in-place. The basin has 100% of total in-place, potential to be explored and discovered.

Offshore sub-basin has a total hydrocarbon in-place of 375 MMTOE. This includes discovered in-place of 77 MMTOE and undiscovered risked in-place of 298 MMTOE. The discovered in-place is subject to commercial development.

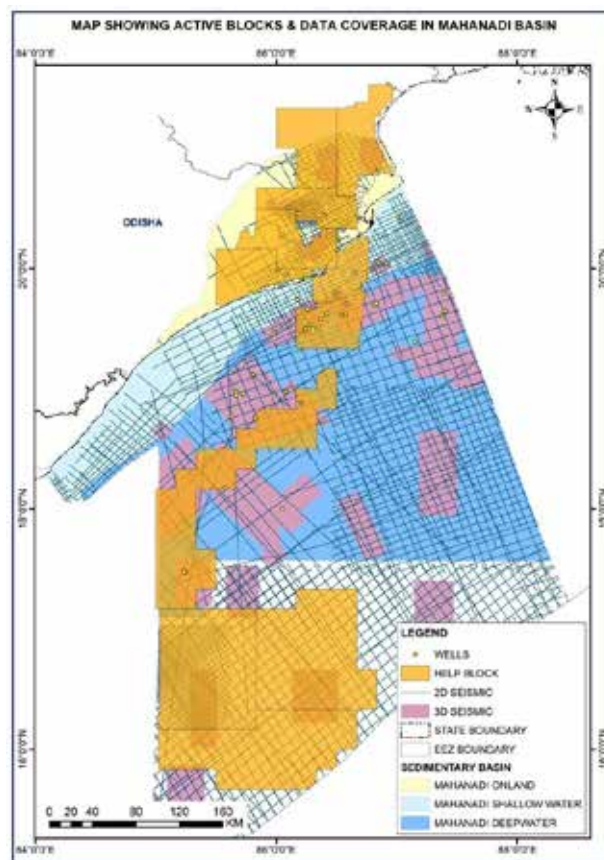
The basin has 79.5% of total in-place, potential to be explored and discovered.

- Basin datasets and studies:**

As of 31.03.2025, NDR has archived primary datasets of 37,609 LKM 2D seismic, 37,713 SKM 3D seismic and 52 conventional wells (45 exploratory). Multiple flavour of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.

The basin has 6 gas hydrate wells that were drilled under National Gas Hydrate Programme (NGHP).

The basin has been identified for drilling of one stratigraphic well under the Gol-approved plan. The basin is presently studied by University of Houston, Texas, USA.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area (skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
Onland	15,500	15,500 (100%)	OALP: 5/14,109
Shallow water	14,211	14,211 (100%)	OALP: 1/1,825
Deepwater	69,789	69,789 (100%)	OALP: 5/40,933

The basin occupies an area of 99,500 sq km. It has been appraised to the extent of 99,500 sq km (100%). The active acreage nos. across regime(s) are 11 with area 56,866 sq. km (57% of basin area)

Bidding opportunities:

Deepwater: 3 OALP (Round X) blocks with 23,242 sq km area

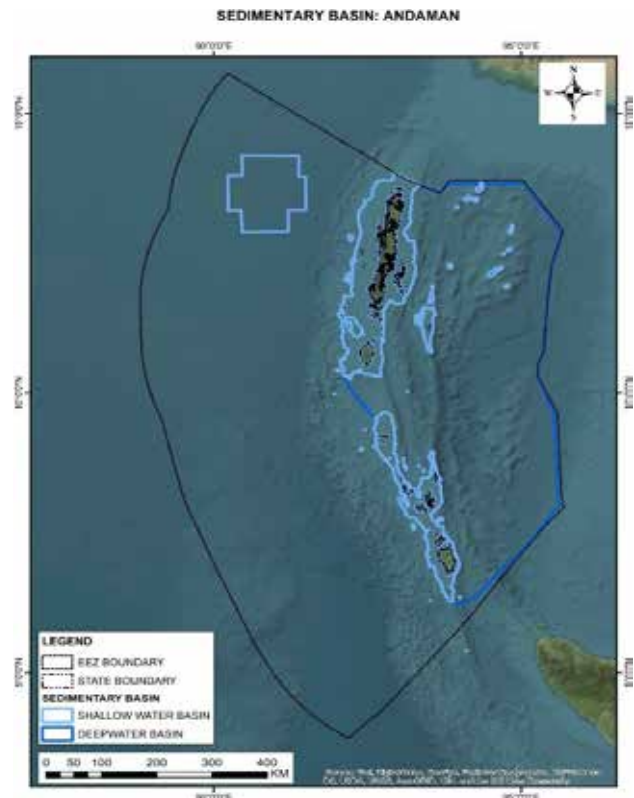


2.1.4 ANDAMAN BASIN

Andaman is a Category II basin with sub-commercial discovered in-place, potential to be commercially produced.



Source: DGH Internal



Source: DGH Internal

Andaman is an offshore basin, characterized primarily by siliciclastic shallow to deepwater sediments. Fore-arc has a significant gas discovery in Miocene, analogous to producing reservoirs of Myanmar and Indonesia gas fields.

Gas hydrate has also been established in Fore-arc. Back-arc area has sediments with prominent entrapment features on seismic.

Hydrocarbon occurrences are reported in Mid Miocene Play, geographically present in East Andaman part. Potential future gas discoveries are considered mainly within structural entrapment conditions. The hydrocarbon accumulation often indicates charging from deeper source sequences.

Geochemical data and structural complexity are key areas of focus. Southeastern area of East Andaman is considered prospective due to source rock maturity. Basin area is under revision to 248,908 sq km, subject to a detailed review during the next hydrocarbon resource assessment study.

• Hydrocarbon prospectivity and Basin maturity:

In the basin, 6 plays are identified within Eocene-to-Middle Miocene in East Andaman and Cretaceous-Tertiary in West Andaman. Petroleum system was modelled in 3D where datasets were adequate and Aerial Yield method was used where data were limited. Though biogenic system was envisaged, it was not modelled.

Offshore East sub-basin has a total hydrocarbon in-place of 55 MMTOE. This includes discovered in-place of 2 MMTOE and undiscovered risked in-place of 53 MMTOE. The discovered in-place is subject to commercial development. The basin has 96.4% of total in-place, potential to be explored and discovered.

Offshore West sub-basin has a total hydrocarbon in-place of 16 MMTOE and this is entirely undiscovered risked in-place. The basin has 100% of total in-place, potential to be explored and discovered.

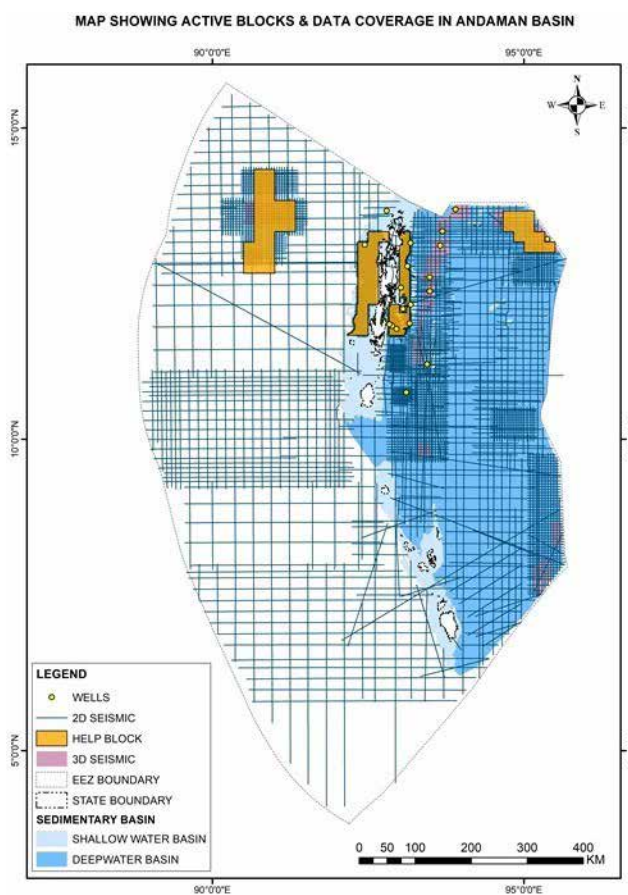


Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 142,029 LKM 2D seismic, 18,896 SKM 3D seismic and 22 conventional wells (all exploratory). Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.

The basin has 12 gas hydrate wells that were drilled under National Gas Hydrate Programme (NGHP).

The basin has been identified for drilling of one stratigraphic well under the GoI-approved plan. The basin is currently studied by University of Houston, Texas, USA.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area (skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
Shallow water	41,064	9,675 (24%)	OALP: 2/9,600
Deepwater West	16,862	16,862 (100%)	No active acreage
Deepwater East	190,982	190,982 (100%)	OALP: 2/13,645

The basin occupies an area of 248,908 sq km. It has been appraised to the extent of 217,519 sq km (87%). The active acreage nos. across regime(s) are 4 with area 23,245 sq. km (9% of basin area)

Bidding opportunities:

Deepwater East: 4 OALP (Round X) blocks with 47,058 sq km area



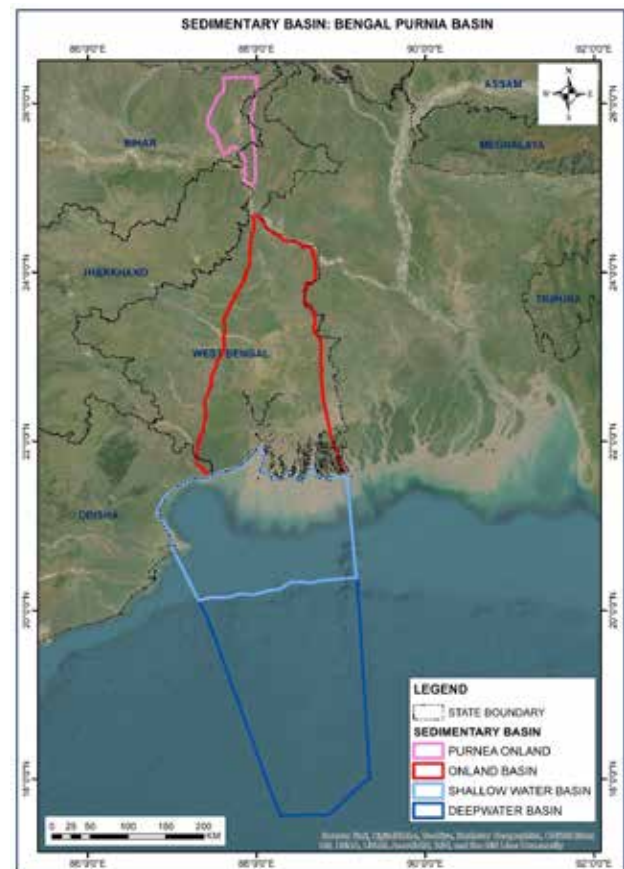
2.1.5 BENGAL-PURNEA BASIN

Bengal-Purnea is a Category III basin with largely undiscovered in-place, that calls for intense exploration for a potential discovery.

The onland area geographically overlaps with the state(s) of Odisha, West Bengal. Purnea area geographically overlaps with the state(s) of Bihar, West Bengal.



Source: DGH Internal



Source: DGH Internal

Bengal-Purnea is situated along the East Coast of Indian peninsula and it has three sub-basins namely Bengal onland, Bengal offshore and Purnea onland. The Bengal onland has established both oil and gas discoveries.

The offshore part of basin has two-third of potential lying in Middle Miocene play and six gas discoveries are notified. Occurrence of channelized deposits associated to subtle structures in the east-central area can be significant exploration targets.

In the basin, some biogenic gas shows are reported from the Tertiary stratigraphic interval ranging from the oldest sediments of Paleogene to the youngest sediments of Miocene-Pliocene complex, which are geographically distributed majorly over offshore areas.

This basin hosts a thick pile of sediments of 10,000 m and more, ranging from Permo-Triassic to Recent. The basin's onland part is mostly covered by the Mesozoic sediments, overlain by the thick Tertiary sediments comprising of Paleocene to Recent sediments, mainly deposited by the Ganga-Brahmaputra river system.

The hydrocarbon accumulations often indicate charging from in-situ shallower biogenic source sequences. The establishment of thermogenic plays (Mio-Pliocene) in recent discoveries in Bengal onland (Asokenagar, 2019) point to potential deep-seated source rocks.

Hinge zone of Bengal basin remains the exploration priority and the reservoir characterization studies of channel facies are the thrust area.



Hydrocarbon prospectivity and Basin maturity:

In the basin, 8 plays exist within Gondwana-Cretaceous-Tertiary in the Onland part. 5 plays are known within Eocene- Pliocene in Offshore part. 3 plays are identified within Gondwana (Karharbari-Barakar-Raniganj) of Purnea sub-basin. Petroleum system was modelled in 3D basis adequate datasets.

Onland area has a total hydrocarbon in-place of 307 MMTOE and this is largely undiscovered risked in-place. Offshore area has a total hydrocarbon in-place of 220 MMTOE and this is entirely undiscovered risked in-place. Purnea sub-basin has a total hydrocarbon in-place of 15 MMTOE and this is entirely undiscovered risked in-place.

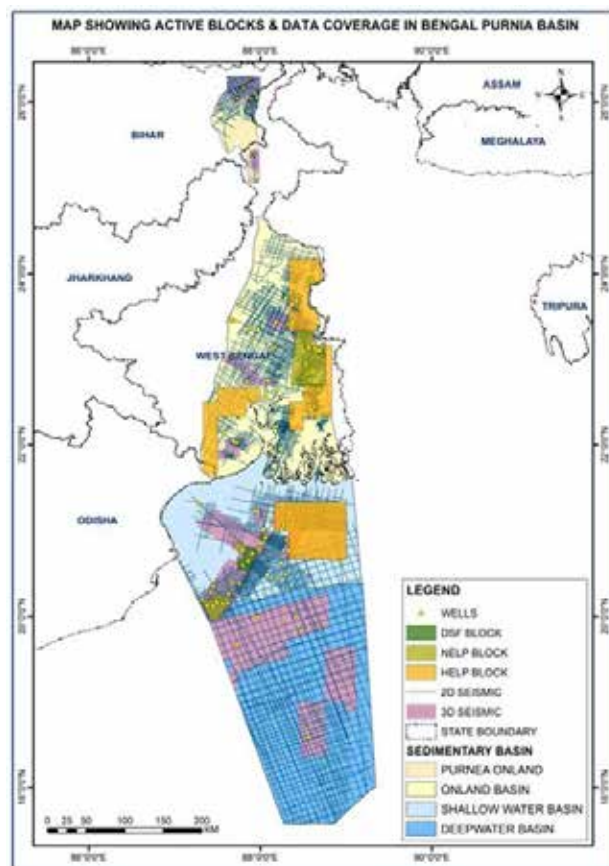
Overview of producing fields:

Commercial production in Bengal area commenced in 2020. The oil production started from Asokenagar discovery subsequent to implementation of early development scheme in the block WB-ONN-2005/4 (Asokenagar-1). Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 66,453 LKM 2D seismic, 30,380 SKM 3D seismic and 111 conventional wells (103 exploratory). Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.

The basin has 4 coal-seam gas wells (all exploratory).

The basin has been identified for drilling of one stratigraphic well under the Gol-approved plan. University of Western Australia, Perth had carried out the stratigraphic and geomorphic analysis of deepwater deposits and the basin is currently being studied by University of Houston, Texas, USA.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area (skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
Purnea	4,901	3,929 (80%)	CBM: 1/991
Bengal	37,513	34,427 (92%)	NELP: 1/2,365 OALP: 3/8,510
Shallow water	33,465	27,441 (82%)	NELP: 1/833 OALP: 1/5,755 DSF: 1/541
Deepwater	46,035	46,035 (100%)	No active acreage

The basin occupies an area of 121,914 sq km. It has been appraised to the extent of 111,832 sq km (92%). The active acreage nos. across regime(s) are 8 with area 18,995 sq. km (16% of basin area)

Bidding opportunities:

Shallow water: 1 OALP (Round X) block with 9,827 sq km area Deepwater: 1 OALP (Round X) block with 12,316 sq km area

Shallow water: 1 OALP (Round X) block with 9,827 sq km area

Deepwater: 1 OALP (Round X) block with 12,316 sq km area

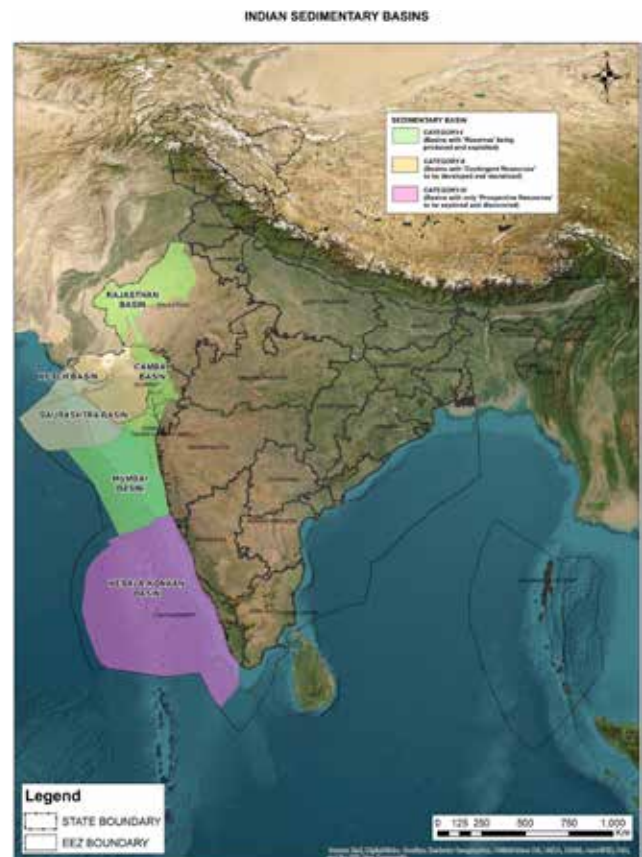


2.2 India's West Coast Basins

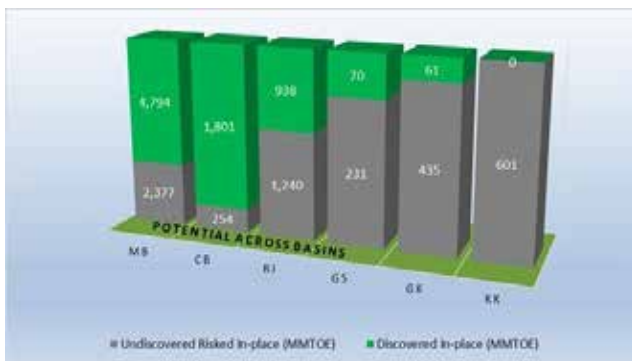
There are six basins located off India's west coast. Geologically, these basins namely (i) Kerala-Konkan, (ii) Mumbai, (iii) Saurashtra, (iv) Kutch, (v) Cambay and (vi) Rajasthan belong to west coast passive continental margin under rift tectonics. Under the upcoming HRAS 2025, the resource assessment of conventional plays and shales will be conducted in these six basins as a part of Phase II scheduled tasks.

The Mumbai basin tops the list in terms of both discovered and undiscovered hydrocarbon in-place. This is closely followed by Cambay and Rajasthan basins.

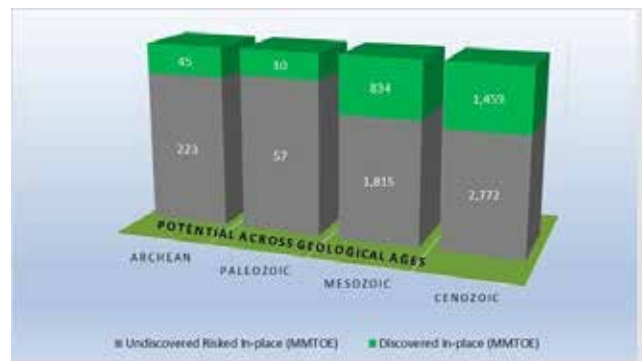
Saurashtra and Kutch are two emerging basins with increased exploration activities.



Source: DGH Internal



Source: DGH Internal



Source: DGH Internal

GoI has also engaged NOCs, to coordinate with international experts from consulting agency and IOC to finalize and drill 4 stratigraphic wells, one each in four frontier offshore basins, including Saurashtra.

M/s TGS, Perth, Australia started an integrated seismic study of India's west coast margin using

2D^{cubed} proprietary tool, whereby a 3D seismic migration volume is generated from a set of 2D and 3D seismic lines. The study is expected to provide a regional framework for planning future 2D/3D seismic survey. The NDR data were used under NDR data policy on value-added study at a cost to company. Previously, TGS conducted such study in India's east coast margin.

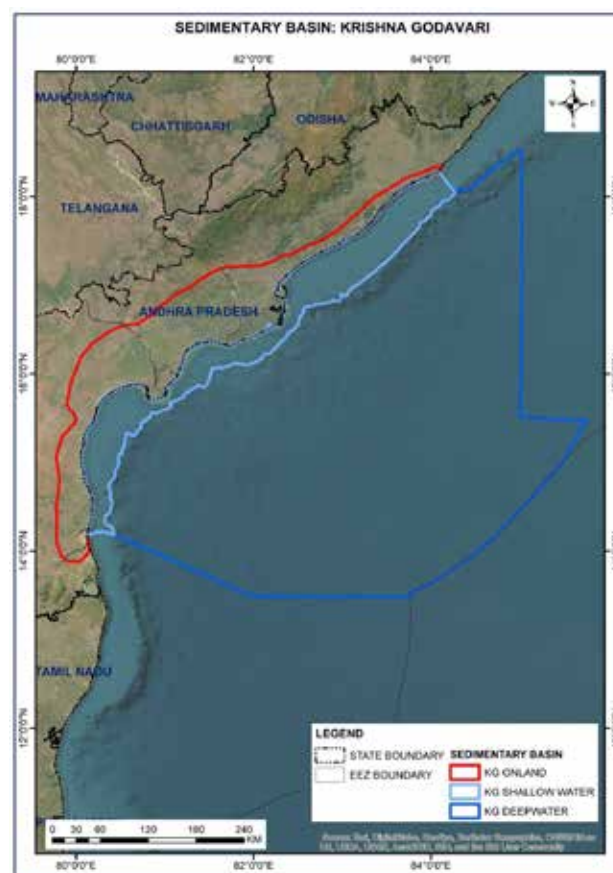


2.2.1 MUMBAI OFFSHORE BASIN

Mumbai Offshore is a Category I basin with commercial discovered in-place.



Source: DGH Internal



Source: DGH Internal

Mumbai basin is exclusively offshore and it is located along the West Coast of Indian peninsula. The basin is the most prolific petroliferous provinces of India, contributing nearly half of oil and gas production of the country annually.

The deepwater part of the basin is under-explored and opportunities lie to chase the basinward extension of prospective plays (Paleocene and Eocene). Similarly, Mesozoic plays lying below the Deccan basalts are still elusive in the basin, hence sub-basalt exploration for Mesozoic play remains an opportunity. The northerly located Saurashtra basin and southerly Kerala-Konkan basin have Mesozoic sequences identified or discovered.

Commercial hydrocarbon occurrences are spread over the Tertiary stratigraphic interval ranging from the oldest sediments of Paleogene to the youngest sediments of Plio-Pleistocene. The hydrocarbons are mostly thermogenic, however the basin has witnessed discovery of biogenic gas within shallow plays.

Deepwater areas have a very few wells drilled. Deeper sequences, including Paleocene-to-Eocene are the future targets of exploration, particularly in Tapti-Daman area.

- **Hydrocarbon prospectivity and Basin maturity:**

In the basin, 7 thermogenic plays are identified within Basement and Tertiary. Besides, 3 biogenic plays are established within Mid-to-Late Miocene and Plio-Pleistocene sequences. Petroleum system was modelled in 3D basis adequate datasets

Mumbai Offshore basin has a total hydrocarbon in-place of 7,171 MMTOE. This includes discovered in-place of 4,795 MMTOE and undiscovered risked in-place of 2,376 MMTOE.

The discovered in-place is under commercial production. The basin has 33.1% of total in-place, potential to be explored and discovered.

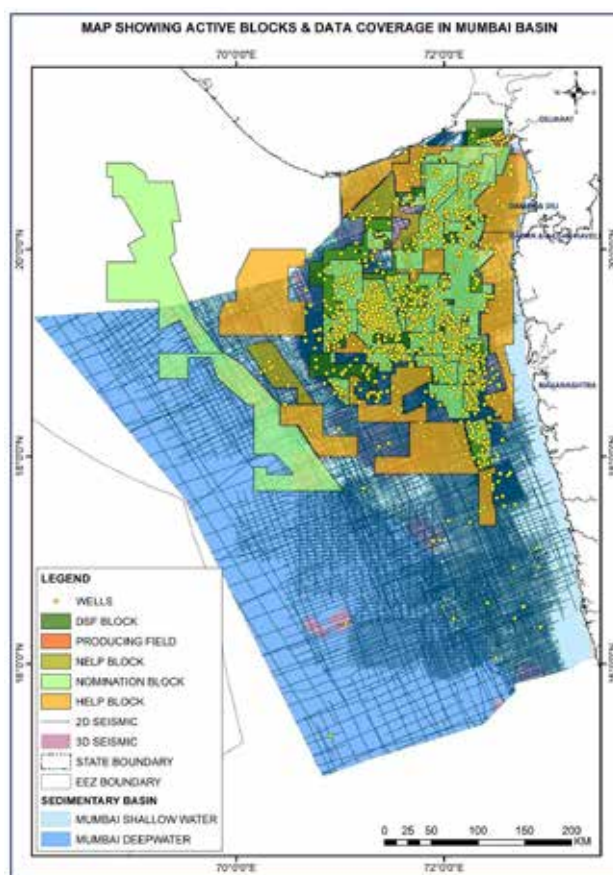


• Overview of producing fields:

Commercial production in shallow water area commenced in 1976. Major producing fields with cumulative production in parenthesis are namely Mumbai High (534 MMT, 226 BCM), Heera & South Heera (80 MMT, 23 BCM), Bassein & SB-11 (39 MMT, 290 BCM), Neelam (39 MMT, 16 BCM). The area has 26 fields with 1,747 producing wells. The maximum flow rates achieved by any field on record are Oil 127,994 BOPD and Gas 10 MMSCMD.

• Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 229,348 LKM 2D seismic, 115,269 SKM 3D seismic and 4,807 conventional wells (1,144 exploratory). Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area (skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
Shallow water	118,389	118,389 (100%)	Nomination: 29/25,905 NELP: 2/1,961 OALP: 11/27,915 DSF: 18/6,959
Deepwater	93,611	93,611 (100%)	Nomination: 2/16,708

The basin occupies an area of 212,000 sq km. It has been appraised to the extent of 212,000 sq km (100%). The active acreage nos. across regime(s) are 62 with area 79,449 sq. km (37% of basin area)

Bidding opportunities:

Shallow water: 2 OALP (Round X) blocks with 18,970 sq km area; 1 DSF (Round IV) contract area with 864 sq km area [8 discoveries namely C-23-9, C-39-14, BH-68, B-172-9, B-183-1, B-51-1, B-174-1, WO-5-11]



2.2.2 RAJASTHAN BASIN

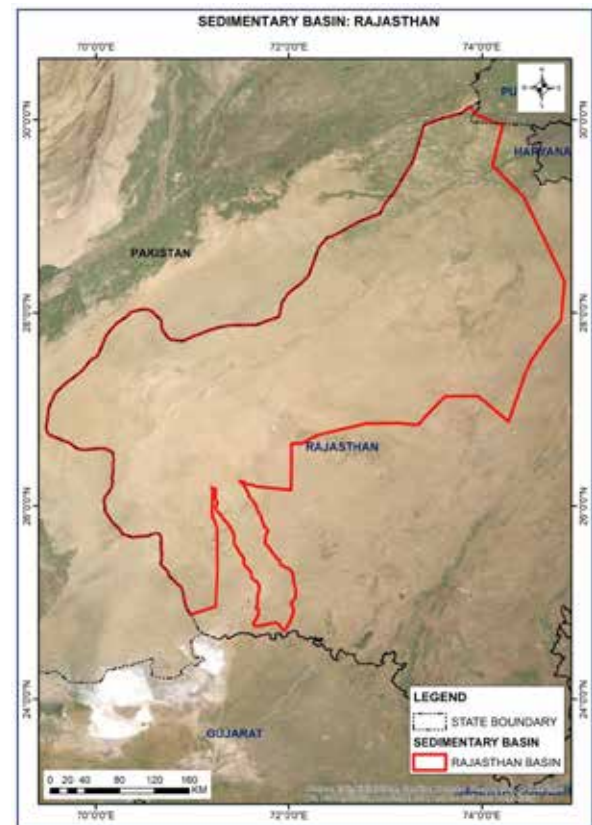
Rajasthan is a Category I basin with commercial discovered in-place.

Barmer and Jaisalmer sub-basins geographically overlaps with the state(s) of Rajasthan.

Bikaner-Nagaur sub-basin geographically overlaps with the state(s) of Punjab, Rajasthan.



Source: DGH Internal



Source: DGH Internal

Barmer sub-basin is situated in the western region of India and the most prominent among the three sub-basins of Rajasthan basin. The sub-basin accounts for one-fourth of oil production of the country.

The Jaisalmer sub-basin is mainly a Mesozoic rift basin set up during Permo-Triassic period.

The Bikaner-Nagaur is a shallow Proterozoic sub-basin with a Pre-Cambrian basement comprised of the Malani Igneous Suite and Delhi Metamorphites. The sub-basin is a part of Punjab platform of Indus basin, having a general dip towards west and sedimentary units thickening towards northwest.

Commercial hydrocarbon occurrences are found in different stratigraphic intervals ranging from the oldest sediments of Cretaceous (Ghaggar-Hakra) to the youngest sediments of Paleocene

and Eocene (Barmer Hill, Fatehgarh, Thumbli). There are also oil and gas deposits known from Deccan Trap and Basement.

Fatehgarh and Barmer Hill of Paleocene are major producing sequences of the subbasin.

Permo-Triassic & Jurassic plays in Jaisalmer and Cretaceous (Ghaggar-Hakra) plays in Barmer are potential exploration targets.

- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 7 plays are identified within Basement, Cretaceous, Volcanics, Paleocene-Eocene in Barmer sub-basin. 9 plays are known within Jurassic-Cretaceous, Paleocene-Eocene in Jaisalmer sub-basin. 3 plays are located within Pre-Cambrian in Bikaner-Nagaur sub-basin. Petroleum system was modelled in 3D basis adequate datasets.



Barmer sub-basin has a total hydrocarbon in-place of 1,680 MMTOE. This includes discovered in-place of 828 MMTOE and undiscovered risked in-place of 852 MMTOE. The discovered in-place is under commercial production. The sub-basin has 50.7% of total in-place, potential to be explored and discovered.

Jaisalmer sub-basin has a total hydrocarbon in-place of 307 MMTOE. This includes discovered in-place of 94 MMTOE and undiscovered risked in-place of 213 MMTOE. The discovered in-place is under commercial production. The sub-basin has 69.4% of total in-place, potential to be explored and discovered.

Bikaner-Nagaur sub-basin has a total hydrocarbon in-place of 190 MMTOE. This includes discovered in-place of 15 MMTOE and undiscovered risked in-place of 175 MMTOE. The discovered in-place is under commercial production. The sub-basin has 92.1% of total in-place, potential to be explored and discovered.

• Overview of producing fields:

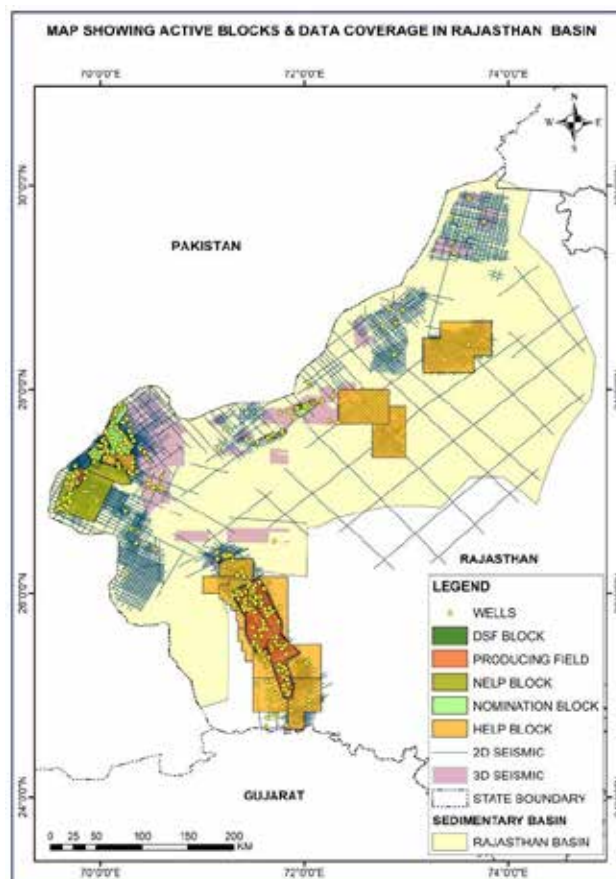
Commercial production in onland area commenced in 1996. Major producing fields with cumulative production in parenthesis are namely RJ-ON-90/1 (102 MMT, 16 BCM), Dandewala (4.6 BCM), RJ-ON/6 (3.2 BCM). The area has 6 fields with 770 producing wells. The maximum flow rates achieved by any field on record are Oil 70,666 BOPD and Gas 4 MMSCMD.

• Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 42,156 LKM 2D seismic, 17,774 SKM

3D seismic and 1,533 conventional wells (313 exploratory). Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.

The basin has 17 coal-seam gas wells (all exploratory).



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area (skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
Onland	126,000	86,940 (69%)	Nomination: 6/1,287 Fields: 2/3,294 NELP: 2/2,538 OALP: 12/13,835 DSF: 3/167

The basin occupies an area of 126,000 sq km. It has been appraised to the extent of 86,940 sq km (69%). The active acreage nos. across regime(s) are 25 with area 21,121 sq. km (17% of basin area)

Bidding opportunities:

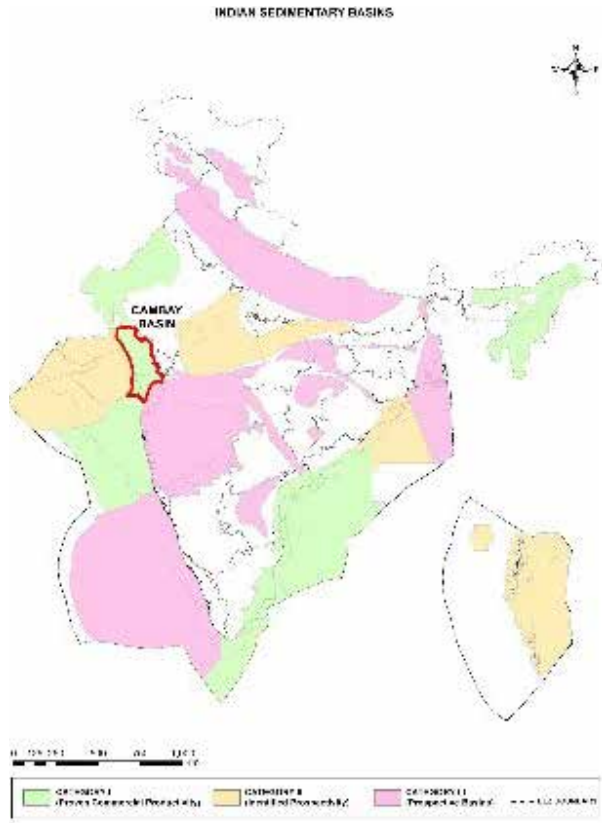
Onland: 1 OALP (Round X) block with 5,953 sq km area



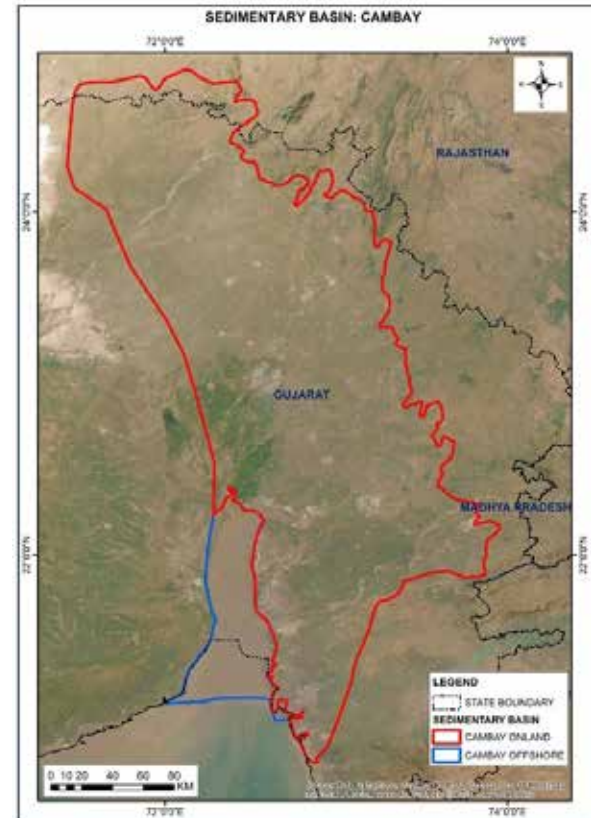
2.2.3 CAMBAY BASIN

Cambay is a Category I basin with commercial discovered in-place.

The area geographically overlaps with the state(s) of Gujarat, Rajasthan.



Source: DGH Internal



Source: DGH Internal

The basin is characterized primarily by siliciclastic rift-fill sediments, situated in the western region of Indian peninsula. The basin is endowed with most matured petroleum provinces of India with major producing fields and 88% of total hydrocarbons already discovered.

The Cambay basin, though extensively exploited, is still under active exploration for deep-seated plays. Commercial hydrocarbon occurrences are spread over different stratigraphic intervals ranging from the oldest sediments, including Deccan Trap volcanics of Cretaceous-Paleocene to the youngest sediments of Miocene. Several oil and gas fields have been discovered, out of which two-third of the discoveries has been made in Middle Eocene.

Cambay shale of Paleocene is the major source rock charging Middle Eocene reservoirs of Kalol (North and Central) and Hazad (Southern) deltaic reservoirs with Tarapur (Oligocene) acting as the regional cap rock. During later exploration,

Miocene plays (Babaguru) are proved to be potential reservoirs of oil and gas in southern onland part of the basin and the offshore area that extends into the Gulf of Cambay.

Paleogene plays (Akhiljuni, Gandhar) and Mesozoic remain the key focus area of the basin.

- **Hydrocarbon prospectivity and Basin maturity:**

In the basin, 7 plays exist within Basement, Paleocene, Eocene and Miocene. Petroleum system was modelled in 3D basis adequate datasets.

Onland sub-basin has a total hydrocarbon in-place of 2,054 MMTOE. This includes discovered in-place of 1,801 MMTOE and undiscovered risked in-place of 253 MMTOE. The discovered in-place is under commercial production. The basin has 12.3% of total in-place, potential to be explored and discovered.



• Overview of producing fields:

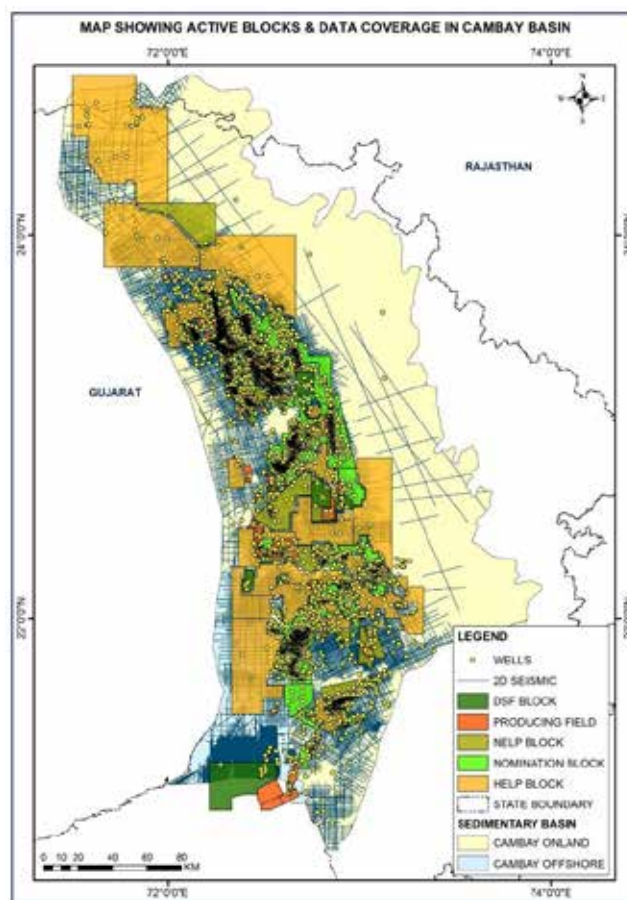
Commercial production in onland area commenced in 1958. Major producing fields with cumulative production in parenthesis are namely Ankleshwar (66.9 MMT, 11.6 BCM), Gandhar (42.5 MMT, 40.9 BCM), North Kadi (36 MMT), Santhal (25.9 MMT), Sobhasan (20.2 MMT), Kalol (18.4 MMT, 12.3 BCM). The area has 113 fields with 5,079 producing wells. The maximum flow rates achieved by any field on record are Oil 11,854 BOPD and Gas 1 MMSCMD.

Commercial production in shallow water area commenced in 2002. Major producing fields with cumulative production in parenthesis are namely CB-OS/2 (6.9 MMT, 7.6 BCM). The area has 2 fields with 29 producing wells. The maximum flow rates achieved by any field on record are Oil 3,490 BOPD.

• Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 86,542 LKM 2D seismic, 34,691 SKM 3D seismic and 9,722 conventional wells (3,082 exploratory). Multiple flavour of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.

The basin has 1 coal-seam gas well (exploratory).



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area (skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
Onland	48,882	44,374 (91%)	Nomination: 154/5,806 Fields: 45/818 NELP: 10/2,006 OALP: 23/11,669 DSF: 11/875

The basin occupies an area of 53,500 sq km. It has been appraised to the extent of 48,992 sq km (92%). The active acreage nos. across regime(s) are 255 with area 26,664 sq. km (50% of basin area)

Bidding opportunities:

Onland: 1 OALP (Round X) block with 126 sq km area; 1 DSF (Round IV) contract area with 110 sq km area [4 discoveries namely Pasunia-1, Pasunia-2, Ognaj-1, Uber-2]; 1 CBM (SCBM) block with 189 sq km area



2.2.4 SAURASHTRA BASIN

Saurashtra is a Category II basin with sub-commercial discovered in-place, potential to be commercially produced. The area geographically overlaps with the state(s) of Daman and Diu, Gujarat.



Source: DGH Internal

The basin is geologically contiguous with Kutch basin and separated by the Saurashtra Arch in the offshore part, which trends north-northwest-south-southeast direction. The basin has presence of prospective plays of Middle Jurassic to Early Cretaceous with significant potential of sub-basalt Mesozoic plays. There has been a gas discovery in Early Cretaceous reservoirs at 4,500m depth, below 2,500m thick weathered basalts. The northerly located proven Tertiary Plays of Kutch basin is extended into this basin.

Major part of Saurashtra mainland is covered with Deccan Trap basalts with thin veneer of Miocene and younger sediments, exposed along Porbandar coast. The Cenozoic sediments have been encountered in the wells drilled in the offshore Saurashtra basin. In offshore, the basin lies north of Mumbai Offshore Basin, while the onland part borders with Cambay Basin on its eastern flanks. The deeper offshore Saurashtra borders with the Indus fan.



Source: DGH Internal

The basin was built with Pre-Cambrian basement overlain by Mesozoic sediments followed by Deccan traps. A thin cover of Neogene and Quaternary sediments occurs at the top. The drainage of Saurashtra shows a radial pattern from the central highland.

The source rocks maturity of Cretaceous is poor and it remains immature in onland.

Sub-basalt imaging, variable basalt thickness and sub-optimal onland seismic data are some areas of focus for exploring Cretaceous plays.

- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 16 plays are located within Middle Jurassic-to-Early Cretaceous in Onland and Middle Jurassic-to-Trip-to Middle Miocene in Offshore. Petroleum system was modelled in 3D where datasets were adequate and Aerial Yield method was used where data were limited.

Onland sub-basin has a total hydrocarbon in-place of 33 MMTOE and this is entirely undiscovered risked in-place. The basin has 100% of total in-place, potential to be explored and discovered.

Offshore sub-basin has a total hydrocarbon in-place of 267 MMTOE. This includes discovered in-place of 70 MMTOE and undiscovered risked in-place of 197 MMTOE. The discovered in-place is subject to commercial development.

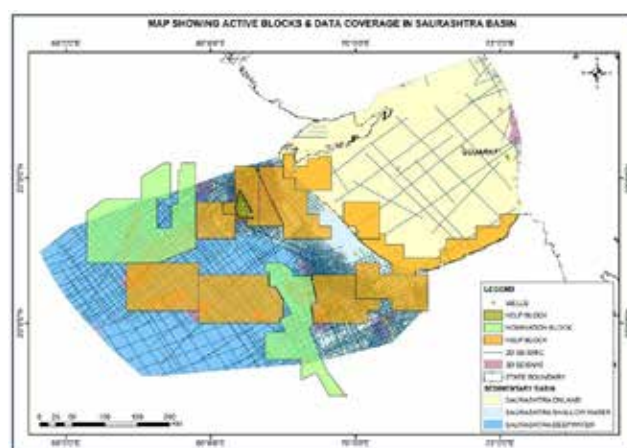


The basin has 73.8% of total in-place, potential to be explored and discovered.

- Basin datasets and studies:**

As of 31.03.2025, NDR has archived primary datasets of 55,094 LKM 2D seismic, 20,186 SKM 3D seismic and 39 conventional wells (all exploratory). Multiple flavour of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.

The basin has been identified for drilling of one stratigraphic well under the Gol-approved plan.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area(skm)	Appraised(skm)	Active acreage [Nos./Area(skm)]
Onland	75,076	44,295 (59%)	OALP: 4/11,194
Shallow water	42,617	42,617 (100%)	NELP: 1/552 OALP: 6/21,171
Deepwater	76,421	76,421 (100%)	OALP: 3/18,888

The basin occupies an area of 194,114 sq km. It has been appraised to the extent of 163,333 sq km (84%). The active acreage nos. across regime(s) are 14 with area 51,805 sq. km (27% of basin area)

Bidding opportunities:

Shallow water: 2 OALP (Round X) blocks with 9,627 sq km area

Deepwater: 1 OALP (Round X) block with 9,060 sq km area



2.2.5 KUTCH BASIN

Kutch is a Category II basin with sub-commercial discovered in-place, potential to be commercially produced. The area geographically overlaps with the state(s) of Gujarat, Rajasthan.



Source: DGH Internal

The basin is geologically contiguous with Saurashtra basin, separated by the Saurashtra Arch in the offshore part. The Mesozoic sediments ranging in age from Middle Jurassic to Cretaceous are exposed in the mainland Kutch and encountered in the wells drilled in offshore. The Cenozoic strata are also exposed in the southwestern part of Kutch mainland and encountered in the offshore wells.

The basin has significant opportunity to consolidate discovered resources of Tertiary Plays. Being analogous to Saurashtra Basin, opportunity also lies for exploring Mesozoic Plays.

Stratigraphically, the northern limit of Kutch outcrops is continued beyond Indo-Pakistan border. In the offshore of Kutch and further south, the basin is contiguous with Saurashtra offshore basin.

Middle Miocene and Early Eocene are the main plays and additional hydrocarbons are



Source: DGH Internal

established in Paleocene, Late Cretaceous, and Early Cretaceous in the basin. All the hydrocarbon accumulations are related to mild inversion tectonics involving Mesozoic and Tertiary sections.

Improved sub-basalt imaging is a major push towards assessment of Mesozoic sequences. Basin area is under revision to 69,986 sq km, subject to a detailed review during the next hydrocarbon resource assessment study.

- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 17 plays are identified within Middle Jurassic-to-Cretaceous in Onland and Middle Jurassic-to-Trip-to- Middle Miocene in Offshore. Petroleum system was modelled in 3D where datasets were adequate and Aerial Yield method was used where data were limited.

Onland sub-basin has a total hydrocarbon in-place of 39 MMTOE and this is entirely undiscovered risked in-place. The basin has 100% of total in-place, potential to be explored and discovered.

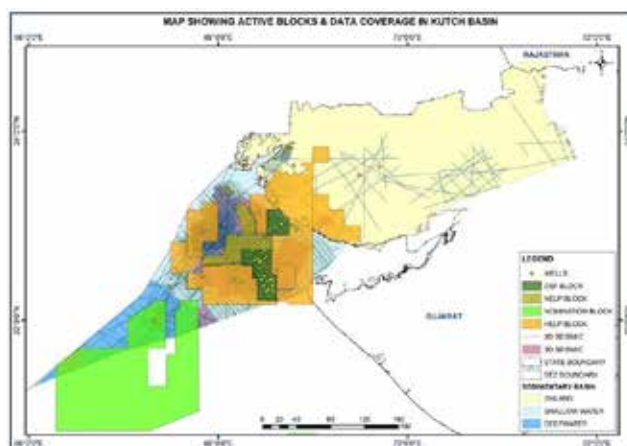
Offshore sub-basin has a total hydrocarbon in-place of 457 MMTOE. This includes discovered in-place of 60 MMTOE and undiscovered risked in-place of 397 MMTOE. The discovered in-place is subject to commercial development.

The basin has 86.9% of total in-place, potential to be explored and discovered.



Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 38,404 LKM 2D seismic, 12,988 SKM 3D seismic and 79 conventional wells (all exploratory). Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area(skm)	Appraised(skm)	Active acreage [Nos./Area(skm)]
Onland	42,186	23,513 (56%)	OALP: 3/4,697
Shallow water	20,500	20,500 (100%)	NELP: 2/1,879 OALP: 5/9,435 DSF: 2/2,092
Deepwater	7,300	7,300 (100%)	Nomination: 1/16,580

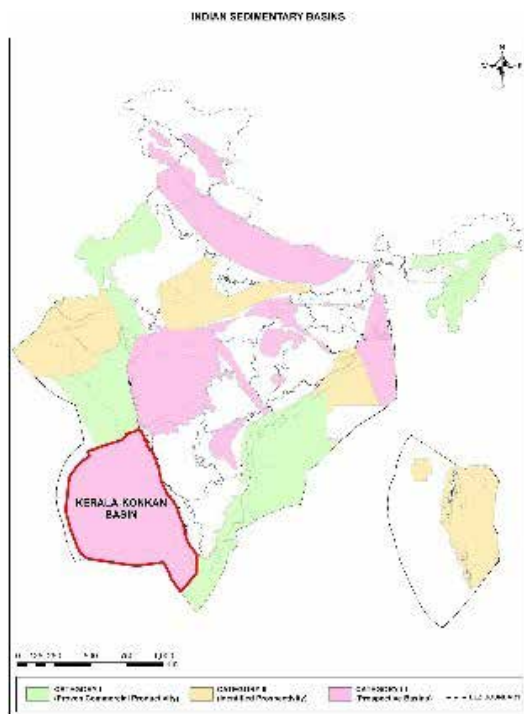
The basin occupies an area of 69,986 sq km. It has been appraised to the extent of 51,313 sq km (73%). The active acreage nos. across regime(s) are 13 with area 34,684 sq. km (50% of basin area)

Bidding opportunities:

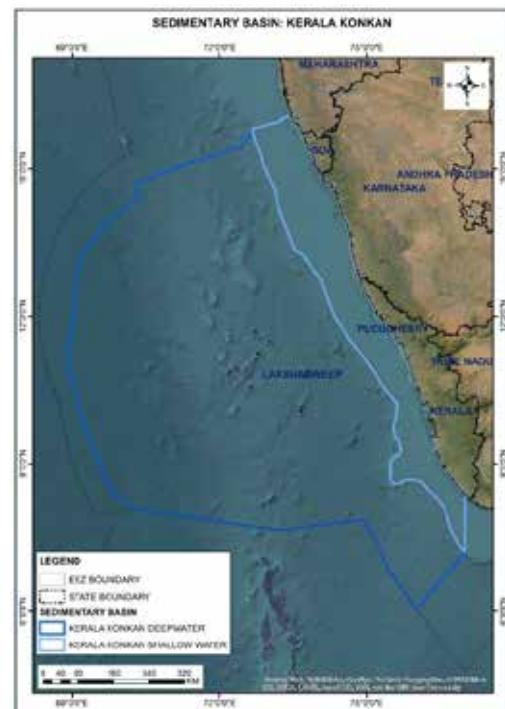
Shallow water: 1 DSF (Round IV) contract area with 2,102 sq km area [7 discoveries namely GKS091NFA-1, GKS091NDA-1, GKS101NAA-1, GKS101NCA-1, GSS041NAA-1, GSS041NAA-2, GK-1]



2.2.6 KERALA-KONKAN (KK) BASIN



Source: DGH Internal



Source: DGH Internal

Kerala-Konkan (KK) is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for a potential discovery.

Kerala-Konkan Basin is characterized by carbonate-siliciclastic shallow to deepwater sediments. The basin covers mainly the offshore provinces with isolated fringes of Miocene and younger sediments in coastal onland area.

Deepest play is the Early Cretaceous and shallowest is the Oligo-Miocene.

Resources are entirely undiscovered and half of total resources envisaged in Late Cretaceous play below basalts.

Poor sub-basalt imaging can be addressed by long-offset seismic survey, magneto-telluric, controlled source electromagnetic surveys. Shelfal areas around Cochin Low in Kerala sub-basin is the key focus area for exploration.

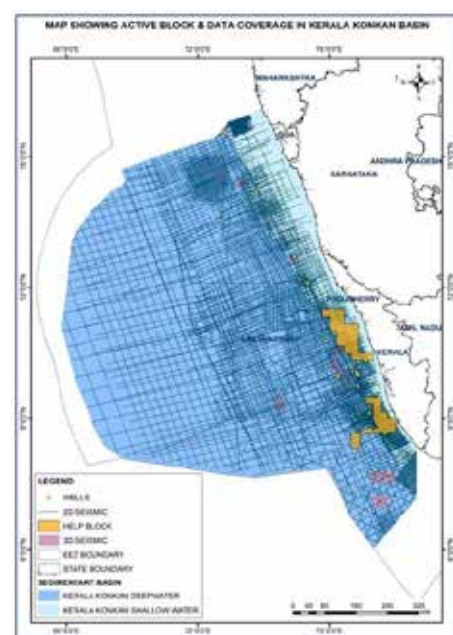
- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 5 plays are located within Early Cretaceous-to-Miocene. Petroleum system was modelled in 3D basis adequate datasets.

Kerala-Konkan (KK) basin has a total hydrocarbon in-place of 602 MMTOE and this is entirely undiscovered risked in- place.

- Basin datasets and studies:**

As of 31.03.2025, NDR has archived primary datasets of 154,002 LKM 2D seismic, 13,717 SKM 3D seismic and 20 conventional wells (all exploratory). Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Source: DGH Internal



Basin appraisal and regime-wise active acreage:

Location	Basin area(skm)	Appraised(skm)	Active acreage [Nos./Area(skm)]
Shallow water	90,380	90,380 (100%)	OALP: 2/10,237
Deepwater			
Shallow water	489,620	489,620 (100%)	OALP: 1/1,113

The basin occupies an area of 580,000 sq km. It has been appraised to the extent of 580,000 sq km (100%). The active acreage nos. across regime(s) are 3 with area 11,350 sq. km (2% of basin area)

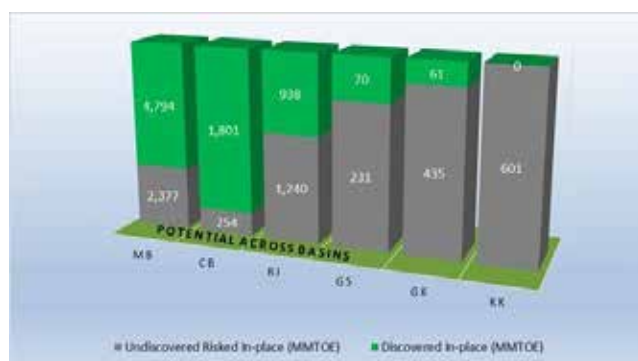
2.3 India's other Onland Basins

There are fifteen basins located onland that are geologically not associated to rift tectonics. The basins which are associated to Himalayan orogeny forming fold- belt, foreland and shelf regions include (i) Assam-Arakan, (ii) Assam Shelf, (iii) Ganga-Punjab, (iv) Spiti-Zaskar, (v) Karewa, (vi) Himalayan Foreland. Other basins namely (vii) Vindhyan, (viii) Deccan Syneclise, (ix) Bhima-Kaladgi, (x) Chhattisgarh, (xi) Bastar, (xii) Cuddapah are mostly Proterozoic sag basins. Three basins namely (xiii) Pranhita-Godavari, (xiv) Narmada, (xv) Satpura-South Rewa-Damodar are associated to rift tectonics. Under the upcoming HRAS 2025, the resource assessment of conventional plays, shales and coal-seam gas will be conducted in these fifteen basins as a part of Phase III scheduled tasks.

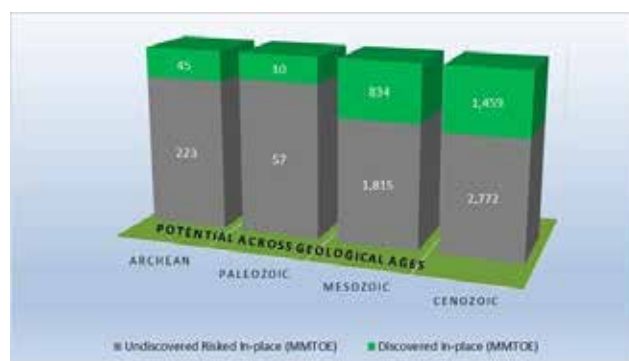
The Assam Shelf basin tops the list in terms of discovered and total hydrocarbon in-place including undiscovered. This is followed by Assam-Arakan Fold- belt. Remaining basins were until NSP campaign bereft of any seismic data and could not be studied for hydrocarbon estimates. Thus, aerial yield methods were applied for speculative estimates.



Source: DGH Internal



Source: DGH Internal



Source: DGH Internal

For some basins like Bastar, Assam-Arakan Fold-belt, Airborne Gravity-Gradiometry (AGG) survey was conducted where seismic surveys could not

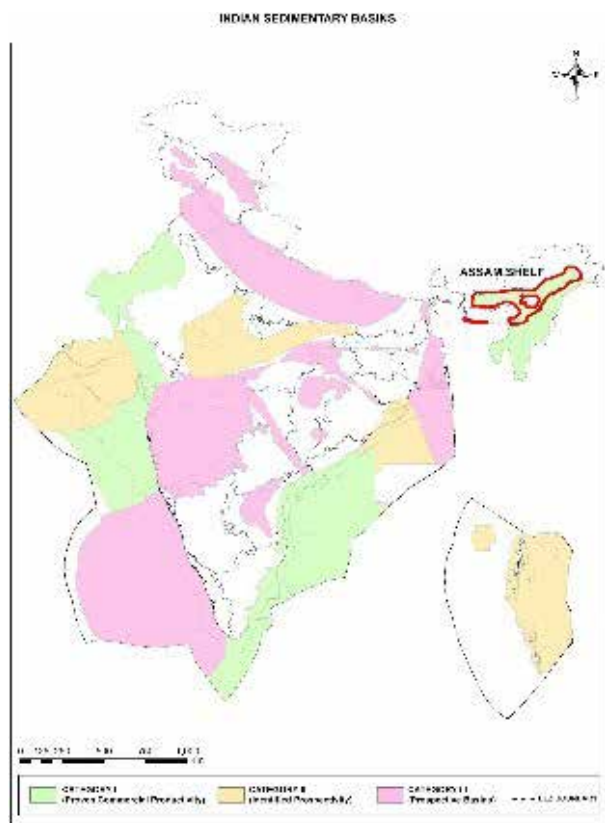
be undertaken due to operational and logistic challenges.



2.3.1 ASSAM SHELF BASIN

Assam Shelf is a Category I basin with commercial discovered in-place.

The basin overlaps with the state(s) of Arunachal Pradesh, Assam, Meghalaya, Nagaland.



Source: DGH Internal

The Assam Shelf Basin is situated in the north-east region of India.

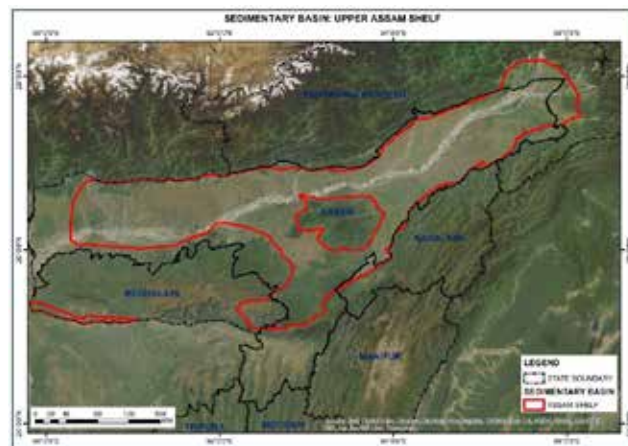
Commercial hydrocarbon occurrences are spread over different stratigraphic intervals ranging from the oldest sediments of Paleocene (Tura) to the youngest sediments of Pliocene (Girujan), apart from fractured plays of Basement.

Mature source rocks lie below the Naga thrust belt, which is pivotal to future exploration focus. Shallow play near Naga-Schuppen belt remains focus area of exploration.

2.3.2 Hydrocarbon prospectivity and Basin maturity:

In the basin, 8 plays are identified within Basement and Tertiary. Petroleum system was modelled in 3D basis adequate datasets.

Assam Shelf basin has a total hydrocarbon in-



Source: DGH Internal

place of 2,700 MMTOE. This includes discovered in-place of 1,826 MMTOE and undiscovered risked in-place of 874 MMTOE.

The discovered in-place is under commercial production. The basin has 32.4% of total in-place, potential to be explored and discovered.

2.3.3 Overview of producing fields:

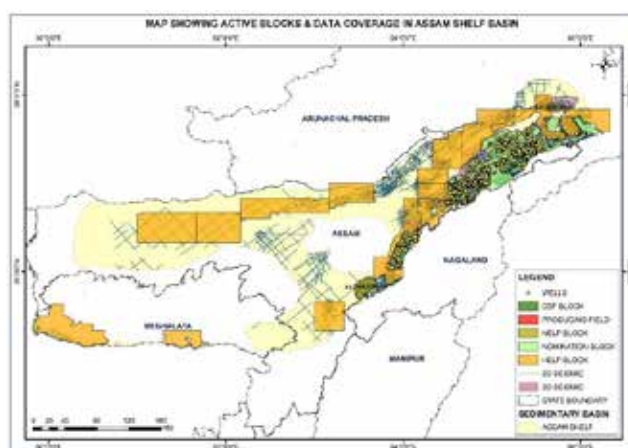
Commercial production in onland area commenced in 1889. Major producing fields with cumulative production in parenthesis are namely Khoraghat (1.4 MMT, 734 BCM), Borholla (2.8 MMT, 391 BCM), , Greater Naharkatiya (56 MMT, 33 BCM), Lakwa-Lakhmani-Kuargaon (40.5 MMT, 6.8 BCM), Greater Hapjan (25.5 MMT, 8.2 BCM), Moran (19.2 MMT, 6.6 BCM). The area has 47 fields with 1,593 producing wells. The maximum flow rates achieved by any field on record are Oil 20,863 BOPD and Gas 3 MMSCMD.

2.3.4 Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 34,926 LKM 2D seismic, 12,470 SKM 3D seismic and 3,494 conventional wells (1,228 exploratory). Multiple flavour of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.

The basin has 17 coal-seam gas wells (2 exploratory).





Source: DGH Internal

Basin appraisal and regime-wise active acreage:

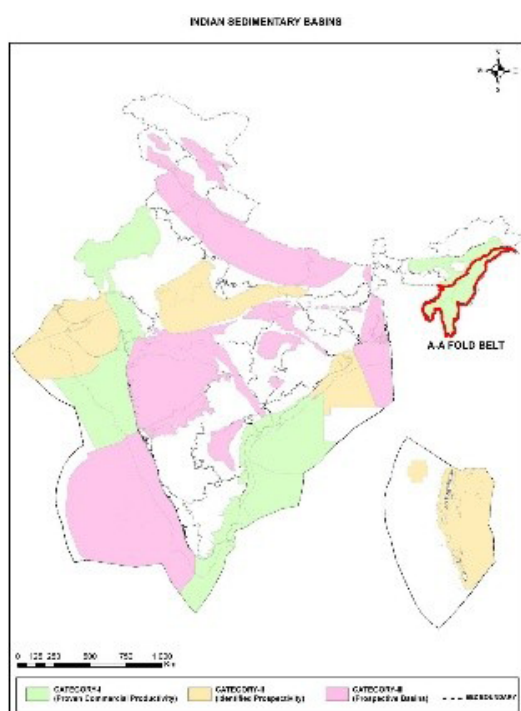
Location	Basin area(skm)	Appraised(skm)	Active acreage [Nos./Area(skm)]
Onland	56,000	51,520 (92%)	Nomination: 58/5,625 Fields: 1/56 NELP: 2/567 OALP: 22/21,430 DSF: 6/163

The basin occupies an area of 56,000 sq km. It has been appraised to the extent of 51,520 sq km (92%). The active acreage nos. across regime(s) are 89 with area 27,841 sq. km (50% of basin area).

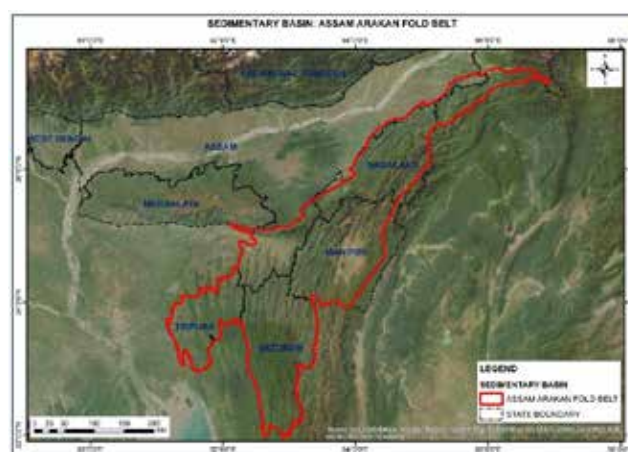
2.3.5 ASSAM-ARAKAN FOLD BELT (AARB) BASIN

Assam-Arakan Fold Belt (AARB) is a Category I basin with commercial discovered in-place.

The basin overlaps with the state(s) of Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Tripura.



Source: DGH Internal



Source: DGH Internal

The basin is characterized primarily by siliciclastic deposits of fluvial to shelf margin setup, situated in the northeast region of India.

Commercial hydrocarbon occurrences are mainly encountered within Miocene and



Pliocene formation except some small discoveries in Oligocene and Eocene sequences. Several gas pools have been discovered within the structural closure while hydrocarbon accumulations often indicate charging from deeper sequences.

The basin hosts a thick pile of tertiary sediments, ranging from Paleocene to Recent, which overlies the Archean basement consisting of granitic and metamorphic rocks.

Mio-Pliocene petroleum system is discovered and producing. Deepest Oligocene play (Rengi) is yet to be thoroughly explored.

The fact that structural traps primarily had formed since Miocene, sequential restoration and migration of thrusts are the focus area.

- **Hydrocarbon prospectivity and Basin maturity:**

In the basin, 7 plays are identified within Eocene-to-Pliocene. Petroleum system was modelled in 3D where datasets were adequate and Aerial Yield method was used where data were limited.

Assam-Arakan Fold Belt (AAFB) basin has a total hydrocarbon in-place of 1,176 MMTOE. This includes discovered in-place of 176 MMTOE and undiscovered risked in-place of 1,000 MMTOE.

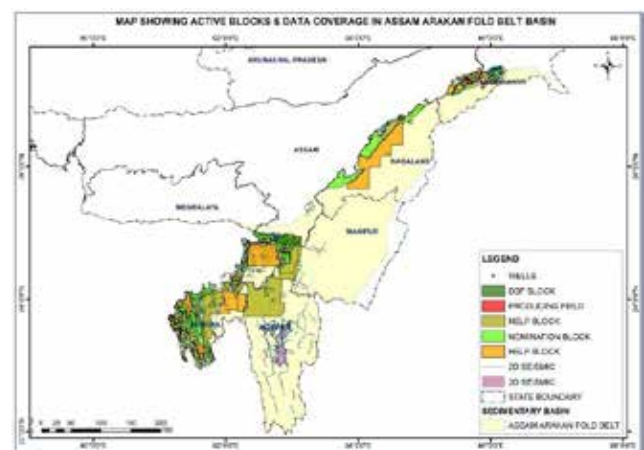
The discovered in-place is under commercial production. The basin has 85% of total in-place, potential to be explored and discovered.

- **Overview of producing fields:**

Commercial production in onland area commenced in 1976. Major producing fields with cumulative production in parenthesis are namely Agartala Dome (9.5 BCM), Konaban (7.7 BCM), Manikyanagar-Rokhia (4.2 BCM), Baramura (2.5 BCM), Sundulbari (1.6 BCM). The area has 11 fields with 375 producing wells. The maximum flow rates achieved by any field on record are Oil 15 BOPD and Gas 1 MMSCMD.

- **Basin datasets and studies:**

As of 31.03.2025, NDR has archived primary datasets of 18,287 LKM 2D seismic, 4,637 SKM 3D seismic and 497 conventional wells (362 exploratory). Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area(skm)	Appraised(skm)	Active acreage [Nos./Area(skm)]
Onland	80,825	74,359 (92%)	Nomination: 37/7,353 Fields: 2/87 NELP: 7/7,104 OALP: 14/6,569 DSF: 3/638

The basin occupies an area of 80,825 sq km. It has been appraised to the extent of 74,359 sq km (92%). The active acreage nos. across regime(s) are 63 with area 21,751 sq. km (27% of basin area).

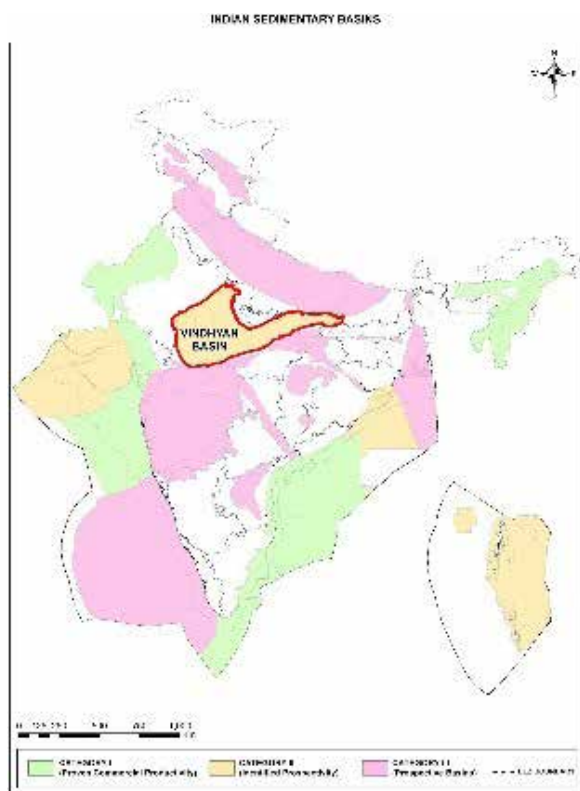
Bidding opportunities:

Onland: 1 DSF (Round IV) contract area with 645 sq km area [8 discoveries namely Laxmijan-1A, Bihubar-1, Barsilla-1, Charaideo-1, Tukbai-2, Patharia-2, Patharia-5, Kheram-2]



2.3.6 VINDHYAN BASIN

Vindhyan is a Category II basin with sub-commercial discovered in-place, potential to be commercially produced. The basin overlaps with the state(s) of Bihar, Madhya Pradesh, Rajasthan, Uttar Pradesh.



Source: DGH Internal

Vindhyan Basin occupies the numero uno position amongst all the Proterozoic petroliferous basins of India due to its large areal extent and significant discoveries in recent years.

The exploratory efforts for hydrocarbon in the basin commenced in 1970's. It was only after the initial gas finds at wells Jabera-1, Nohta-2 and Damoh-4, the hydrocarbon prospectivity perception of the basin took an attention and subsequently led to commercial discovery at Hatta-2 in 2018.

In the basin, three major stratigraphic horizons, viz. Rohtas, Mohana and Jardepahar were identified.

Reservoir layers are tight with low porosity and permeability, requiring advanced extraction technology.

- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 3 plays are known within Pre-Cambrian-to-Paleozoic (Jardepahar-Mohana-



Source: DGH Internal

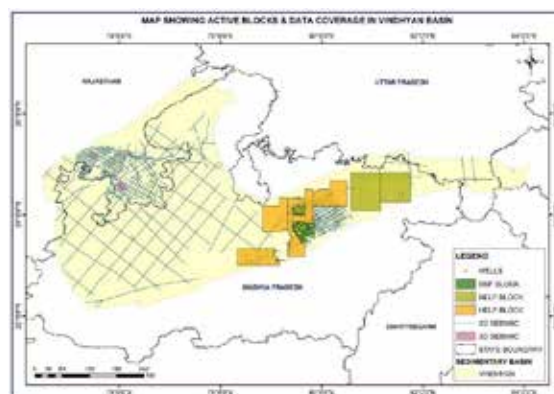
Rohtas sequences). Aerial Yield method was used for each play following intra-basin calibration. Datasets used were 6 Vertical Seismic Profiles (VSP), 59,771 stations of Aeromagnetic survey, 6,001 stations of Gravity-Magnetic (GM) data and 5 profiles of Deep Seismic Sounding (DSS).

Vindhyan basin has a total hydrocarbon in-place of 685 MMTOE. This includes discovered in-place of 1 MMTOE and undiscovered risked in-place of 684 MMTOE.

The discovered in-place is subject to commercial development. The basin has 99.9% of total in-place, potential to be explored and discovered.

- Basin datasets and studies:**

As of 31.03.2025, NDR has archived primary datasets of 26,469 LKM 2D seismic, 1,381 SKM 3D seismic and 33 conventional wells (32 exploratory). Multiple flavour of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Source: DGH Internal



Basin appraisal and regime-wise active acreage:

Location	Basin area(skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
Onland	202,888	71,011 (35%)	NELP: 3/9,155 OALP: 5/13,260 DSF: 1/1,472

The basin occupies an area of 202,888 sq km. It has been appraised to the extent of 71,011 sq km (35%). The active acreage nos. across regime(s) are 9 with area 23,886 sq. km (12% of basin area).

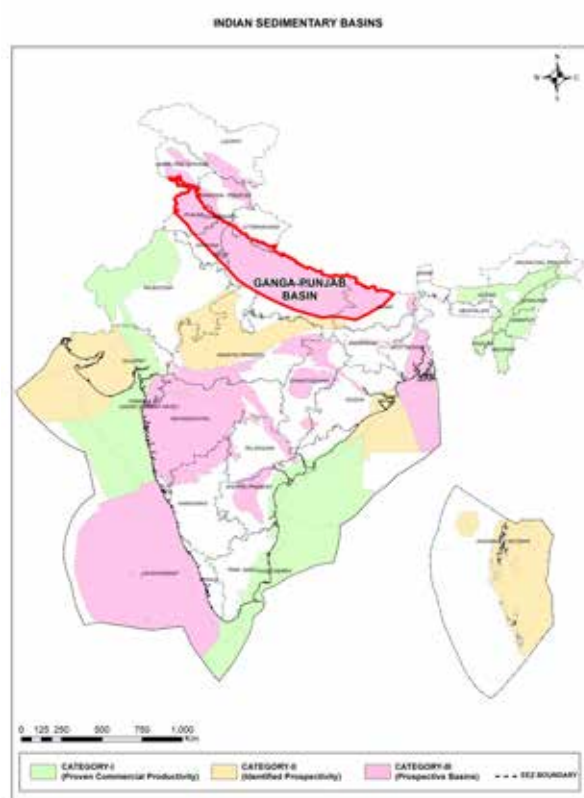
Bidding opportunities:

Onland: 1 OALP (Round X) block with 4,275 sq km area

2.3.7 GANGA-PUNJAB BASIN

Ganga-Punjab is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for a potential discovery.

The basin overlaps with the state(s) of Bihar, Chandigarh, Delhi, Haryana, Himachal Pradesh, Jammu and Kashmir, Madhya Pradesh, Punjab, Uttar Pradesh, Uttarakhand.



Source: DGH Internal

The Indo-Gangetic plain, which is the extensive alluvial plain of Ganga, Indus and Brahmaputra rivers and their tributaries, separates the Himalayan range from peninsular India. Ganga valley and Punjab plain are the integral part Indo-Gangetic plain, representing the river Ganga and Indus catchment area respectively.



Source: DGH Internal

Ganga-Punjab basin is covered by thick alluvium, which conceals the pre-collisional (Indo-Eurasian plates) Proterozoic and post-collisional Tertiary records. Post-collision flexuring of the Indian lithosphere and thrust-fold loading produce the Himalayan peripheral foreland in Early Miocene. The basin expanded during Middle Miocene and attained its present configuration in Late Quaternary.

The exploration in the Ganga-Punjab basin started in late 1950. M/s ONGC started exploration in the year 1978-79 in the Gandak depression of the Ganga sub-basin. M/s Cairn and M/s OIL carried out seismic surveys in the Kashipur block of Sarda depression.

In Ganga sub-basin two speculative plays have been identified i.e. Neo-Proterozoic unconventional play and Middle Miocene



conventional play. Whereas in Punjab sub-basin only one hypothetical play i.e. Middle Miocene Play is envisaged as only Tertiary sediments, resting directly over metamorphic basements. Ganga sub-basin Neo-Proterozoic play is unconventional tight gas play like Vindhyan play. Middle Miocene play (Lower Siwalik) has been considered as conventional play.

- **Hydrocarbon prospectivity and Basin maturity:**

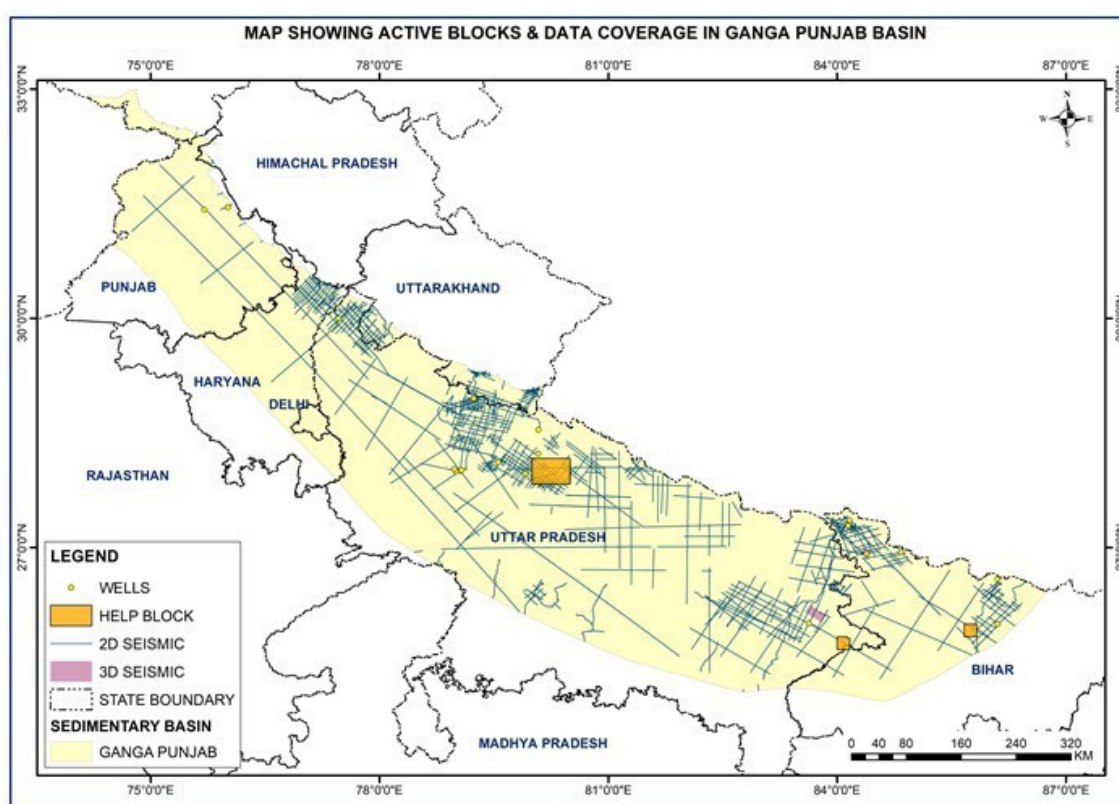
In the basin, 2 plays are known within Pre-Cambrian and Mid Miocene. Aerial Yield method was used on analogy of Vindhyan basin (Pre-

Cambrian sequences) and Duarmara gas field of Assam state (Tertiary sequences).

Ganga-Punjab basin has a total hydrocarbon in-place of 126 MMTOE and this is entirely undiscovered risked in- place.

- **Basin datasets and studies:**

As of 31.03.2025, NDR has archived primary datasets of 25,023 LKM 2D seismic, 420 SKM 3D seismic and 19 conventional wells (all exploratory). Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area(skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
Onland	304,000	104,465 (34%)	OALP: 3/2,428

The basin occupies an area of 304,000 sq km. It has been appraised to the extent of 104,465 sq km (34%). The active acreage nos. across regime(s) are 3 with area 2,428 sq. km (<1% of basin area).

Bidding opportunities:

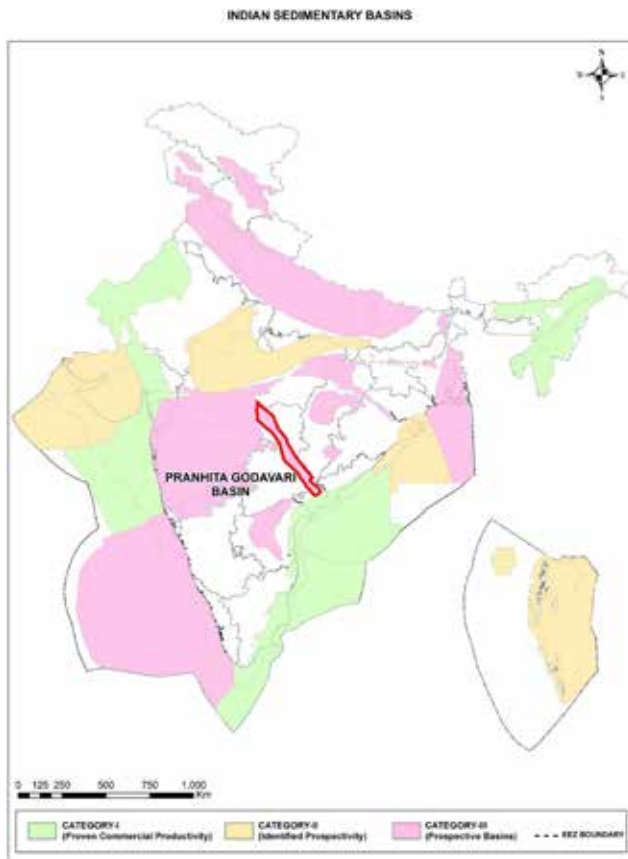
Onland: 1 OALP (Round X) block with 5,241 sq km area



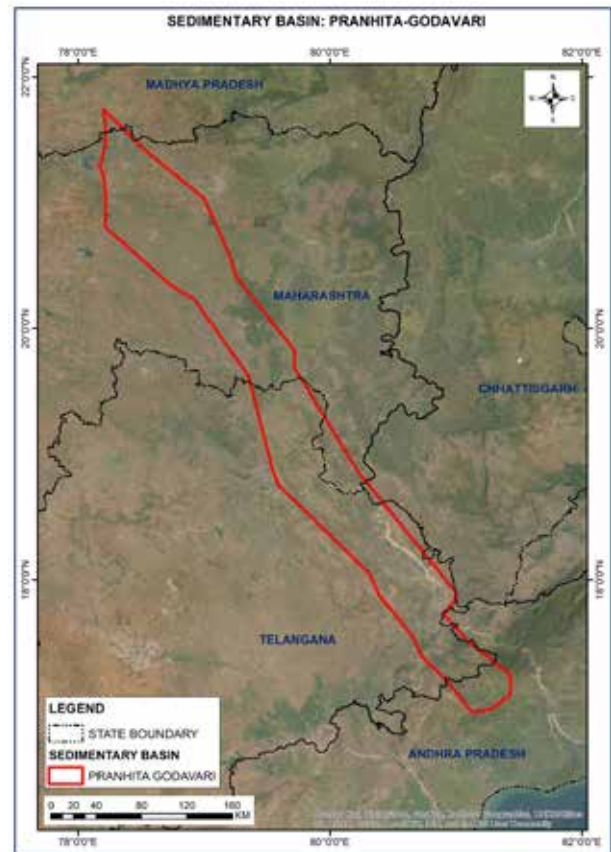
2.3.8 PRANHITA-GODAVARI (PG) BASIN

Pranhita-Godavari (PG) is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for potential discovery.

The basin overlaps with the state(s) of Andhra Pradesh, Chhattisgarh, Madhya Pradesh, Maharashtra, Telangana.



Source: DGH Internal



Source: DGH Internal

The basin is a NW-SE trending intracratonic rift, located over Peninsular India and bounded by metamorphic rocks of Bastar to its northeast and Dharwar cratons to its southwest.

Potential plays are envisaged in Talchir, Barakar, Lower, Middle and Upper Kamthi Formations.

Geological data indicates presence of about 4 to 5 Km of sediments. The oldest sediments known from surface exposures are the Lower Proterozoic overlain by Gondwana sediments.

- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 4 plays exist within Lower Permian-to-Lower Triassic. Aerial Yield method was used on analogy with KG Basin. The study incorporated geological maps, GM data and geochemical data from source rock samples.

Pranhita-Godavari (PG) basin has a total hydrocarbon in-place of 94 MMTOE and this is entirely undiscovered risked in-place.

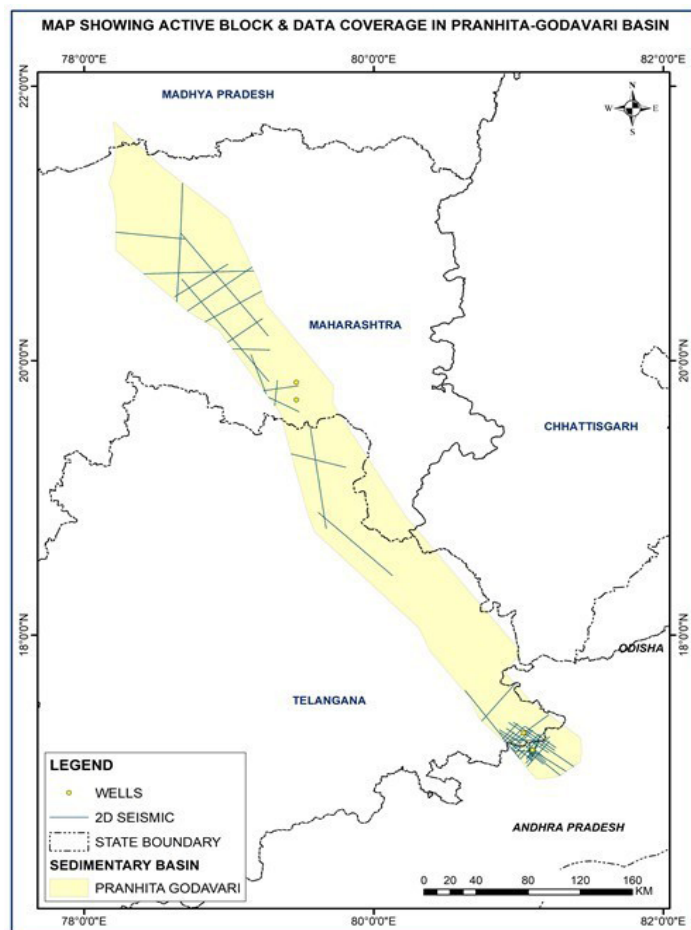


Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 1,488 LKM 2D seismic and 2 conventional wells (all exploratory). Multiple flavour of reprocessed seismic datasets are also

archived whenever the same are provided by E&P Operators.

The basin has 2 coal-seam gas wells (all exploratory).



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area(skms)	Appraised (skms)	Active acreage [Nos./Area(skms)]
Onland	30,000	10,500 (35%)	No active acreage

The basin occupies an area of 30,000 sq km. It has been appraised to the extent of 10,500 sq km (35%).

2.3.9 SATPURA-SOUTH REWA-DAMODAR BASIN

Satpura-South Rewa-Damodar is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for a potential discovery. The basin, however is commercially producing unconventional resource like coal seam gas.

Satpura area geographically overlaps with the state(s) of Madhya Pradesh. South Rewa area geographically overlaps with the state(s) of Chhattisgarh, Madhya Pradesh, Odisha, Uttar Pradesh. Damodar area geographically overlaps with the state(s) of Jharkhand, West Bengal.





Source: DGH Internal

The Gondwana rocks of Satpura-South Rewa-Damodar Basin are exposed along a linear belt and known for commercial production of coal seam gas.

The basin has limited geoscientific data, specific to exploration of conventional hydrocarbons. The elements of conventional petroleum system in the basin are largely speculative.

The basin has Gondwana sediments - both Lower Permian plays and the Upper Mesozoic.

The basin has hypothetical Gondwana petroleum system, having a close analogy with the Cooper Basin of Australia. There are three plays identified. viz. Early Permian, Mid. to Late Permian and Triassic to Cretaceous across all three sub-basins viz. Satpura, South Rewa and Damodar.

Instances of igneous intrusions call for an integrated approach during mapping hydrocarbon resources.



Source: DGH Internal

• Hydrocarbon prospectivity and Basin maturity:

In the basin, Satpura and South Rewa have Early Permian-to-Cretaceous plays while Damodar has Early Permian-to-Triassic. In 3 sub-basins of Satpura, South Rewa and Damodar, Aerial Yield method was used on analogy of Cooper Basin (Australia).

Satpura sub-basin has a total hydrocarbon in-place of 19 MMTOE and this is entirely undiscovered risked in-place. South Rewa sub-basin has a total hydrocarbon in-place of 39 MMTOE and this is entirely undiscovered risked in-place.

Damodar sub-basin has a total hydrocarbon in-place of 3 MMTOE and this is entirely undiscovered risked in-place.

• Overview of producing fields:

Commercial production in Damodar area commenced in 2007. Major producing fields with cumulative production in parenthesis are namely Raniganj-East (2.9 BCM), Raniganj-South (1.8 BCM). The area has 4 fields with 702 producing wells.

Commercial production in South Rewa area commenced in 2011. Major producing fields with cumulative production in parenthesis are namely Sohagpur-West (2.3 BCM). The area has 2 fields with 391 producing wells. The maximum flow rates achieved by any field on record are Gas 1 MMSCMD.

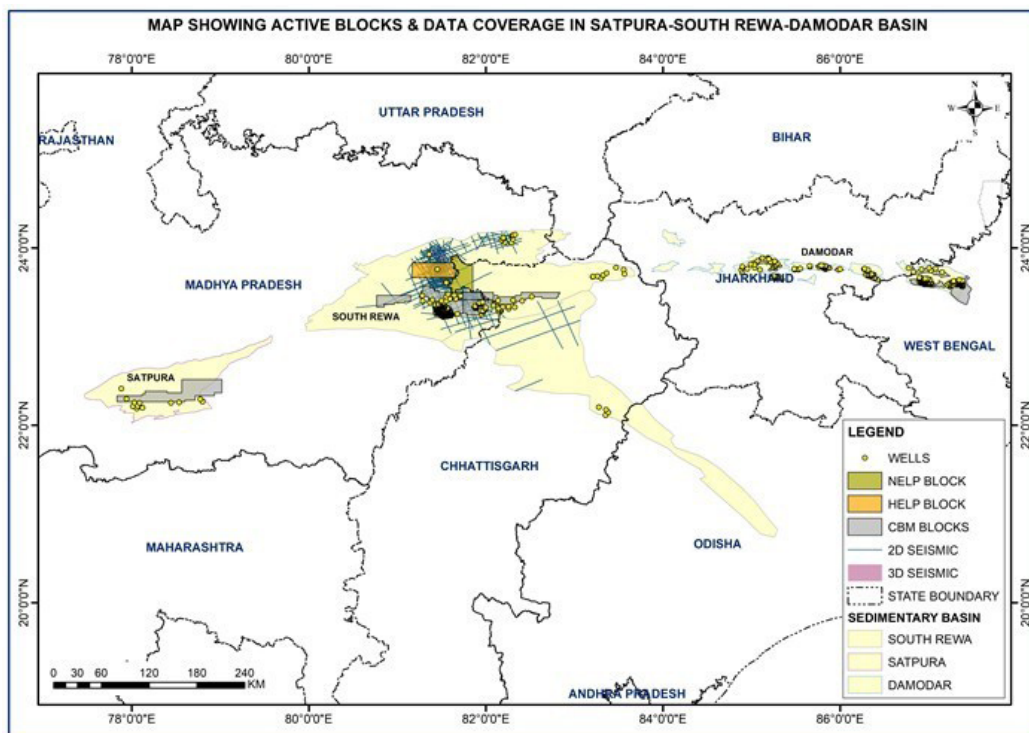


Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 6,541 LKM 2D seismic, 554 SKM 3D seismic and 2 conventional wells (all exploratory). Multiple flavour of reprocessed seismic datasets

are also archived whenever the same are provided by E&P Operators.

The basin has 1158 coal-seam gas wells (61 exploratory).



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area(skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
South Rewa	44,667	18,476 (41%)	NELP: 1/790 OALP: 1/906 CBM: 7/2,958
Satpura	7,890	2,632 (33%)	CBM: 1/1,771
Damodar	4,623	4,623 (100%)	CBM: 6/1,453

The basin occupies an area of 57,180 sq km. It has been appraised to the extent of 25,731 sq km (45%). The active acreage nos. across regime(s) are 16 with area 7,878 sq. km (14% of basin area).

Bidding opportunities:

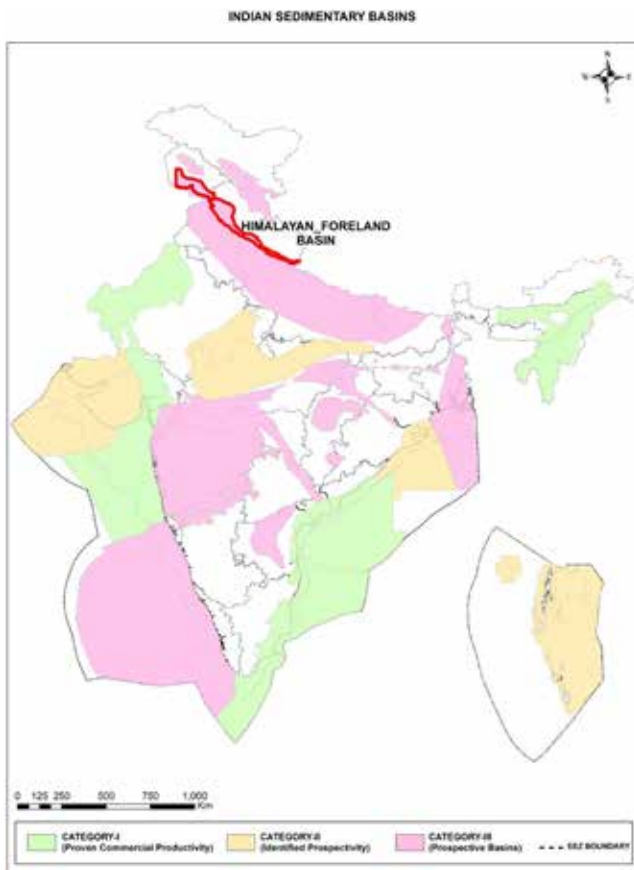
South Rewa: 2 CBM (SCBM) blocks with 331 sq km area

2.3.10 HIMALAYAN FORELAND BASIN

Himalayan Foreland is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for a potential discovery.

The basin overlaps with the state(s) of Chandigarh, Haryana, Himachal Pradesh, Jammu and Kashmir, Punjab, Uttar Pradesh, Uttarakhand.





Source: DGH Internal

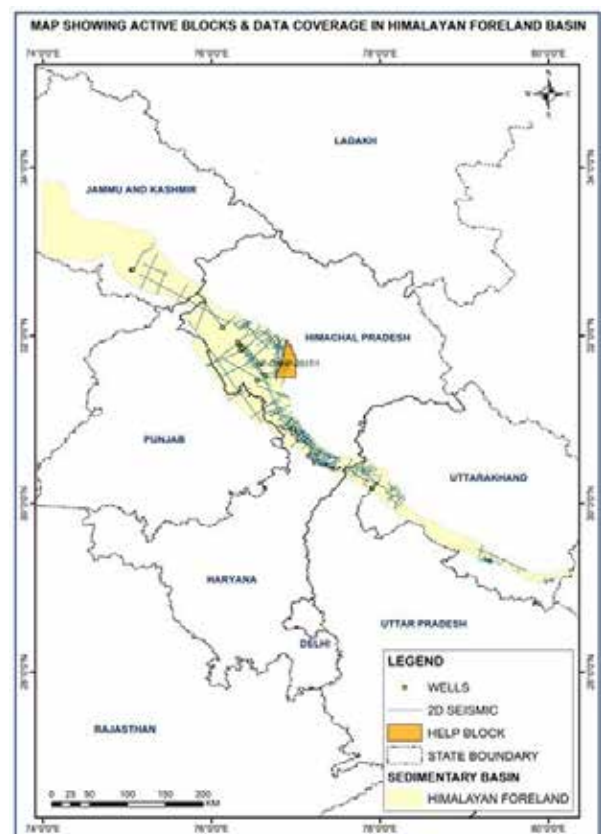


Source: DGH Internal

Himalayan foreland basin is an east-west trending elongate basin that extends from Pakistan in the west to India in the east. The basin came into existence at the close of Mesozoic Era following the first India-Asia impingement, resulting in the deposition of the initial foredeep sediments in front of the rising Himalaya.

Hydrocarbon exploration in Himalayan Foothills of erstwhile-named Punjab Basin was initiated in 1956 by M/s ONGC. Under geophysical campaign, aeromagnetic surveys were conducted in 1957, which was followed by seismic and gravity-magnetic (GM) surveys in 1957-58. Major part of the basin is covered by GM surveys. Such surveys bring out clearly that northwest-southeast Himalayan trend is the dominant lineament in the basement. However, there is a strong perception of northeast-southwest Aravalli trend in the area lying between Kalka and Dehradun.

Poonch-Ravi sector, Doon Sector and Rishikesh-Tanakpur sector call for conceptual analysis in view of data gaps.



Source: DGH Internal



- **Hydrocarbon prospectivity and Basin maturity:**

In the basin, 4 plays are identified within Pre-Cambrian, Tertiary and Pleistocene. 10,000m of sedimentary thickness was envisaged in the basin. Aerial Yield method was used on analogy of Kohat Potwar Basin (Pakistan).

Himalayan Foreland basin has a total hydrocarbon in-place of 46 MMTOE and this is entirely undiscovered risked in- place.

- **Basin datasets and studies:**

As of 31.03.2025, NDR has archived primary datasets of 5,338 LKM 2D seismic and 21 conventional wells (all exploratory). Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.

Basin appraisal and regime-wise active acreage:

Location	Basin area(skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
Onland	30,110	19,270 (64%)	OALP: 1/666

The basin occupies an area of 30,110 sq km. It has been appraised to the extent of 19,270 sq km (64%). The active acreage nos. across regime(s) are 1 with area 666 sq. km (<1% of basin area).

Bidding opportunities:

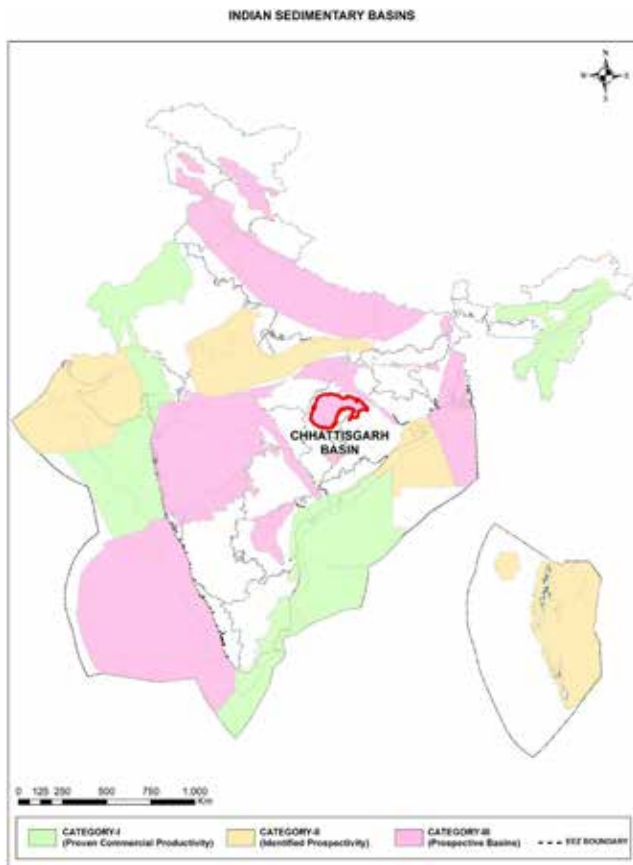
Onland: 1 OALP (Round X) block with 991 sq km area



2.3.11 CHHATTISGARH BASIN

Chhattisgarh is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for a potential discovery.

The basin overlaps with the state(s) of Chhattisgarh, Odisha.



Source: DGH Internal

The basin is located over the Bastar craton of Peninsular India.

The elements of petroleum system in the basin are largely speculative.

Entire Proterozoic sequence of the basin has been considered as a single Proterozoic Play and sub-divided into four assessment units based on geological rationales.

Study of sub-trappean sequences is the focus area.

- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 1 play is known in Pre-Cambrian. 2,500m of sedimentary thickness was envisaged



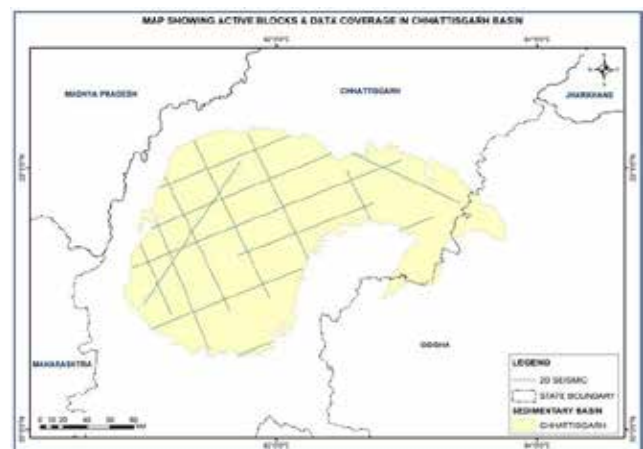
Source: DGH Internal

in the basin. Aerial Yield approach has been used on analogy of Vindhyan Basin.

Chhattisgarh basin has a total hydrocarbon in-place of 25 MMTOE and this is entirely undiscovered risked in-place.

- Basin datasets and studies:**

As of 31.03.2025, NDR has archived primary datasets of 1,999 LKM 2D seismic. Multiple flavour of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Source: DGH Internal



Basin appraisal and regime-wise active acreage:

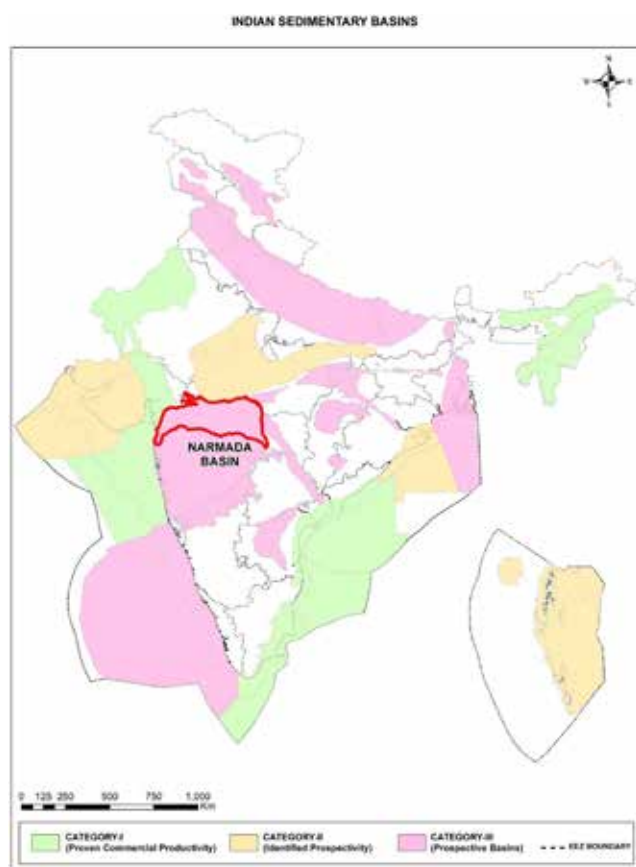
Location	Basin area(skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
Onland	32,000	32,000 (100%)	No active acreage

The basin occupies an area of 32,000 sq km. It has been appraised to the extent of 32,000 sq km (100%).

2.3.12 NARMADA BASIN

Narmada is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for a potential discovery.

The basin overlaps with the state(s) of Dadra and Nagar Haveli, Daman and Diu, Gujarat, Madhya Pradesh, Maharashtra, Telangana.



Source: DGH Internal

The basin is located in the northwestern part of Indian Peninsula and it extends up to the eastern limit of Cambay Basin while eastward it merges with the western boundary of Satpura.

The elements of petroleum system in the basin are largely speculative.

The basin is fully covered by Deccan Trap volcanics. However, Pre-trappean Gondwana



Source: DGH Internal

sediments are speculated over the basement rocks in the basin.

The basin has Pre-Trappean Gondwana play, which is divided into four assessment units based on available geological rationales.

Basin is largely covered by Deccan trap basalts. Sub-trappean sediments are the focus area of exploration.

- Hydrocarbon prospectivity and Basin maturity:**

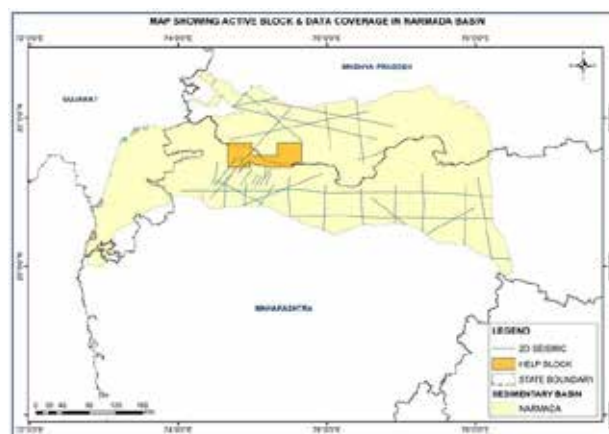
In the basin, 1 play is identified within sub-trappean Gondwana. 400-1,500m of sedimentary thickness was envisaged the basin. Aerial Yield method was used on analogy of Cooper Basin, Australia.

Narmada basin has a total hydrocarbon in-place of 9 MMTOE and this is entirely undiscovered risked in-place.



• Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 2,094 LKM 2D seismic, 37 SKM 3D seismic. Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

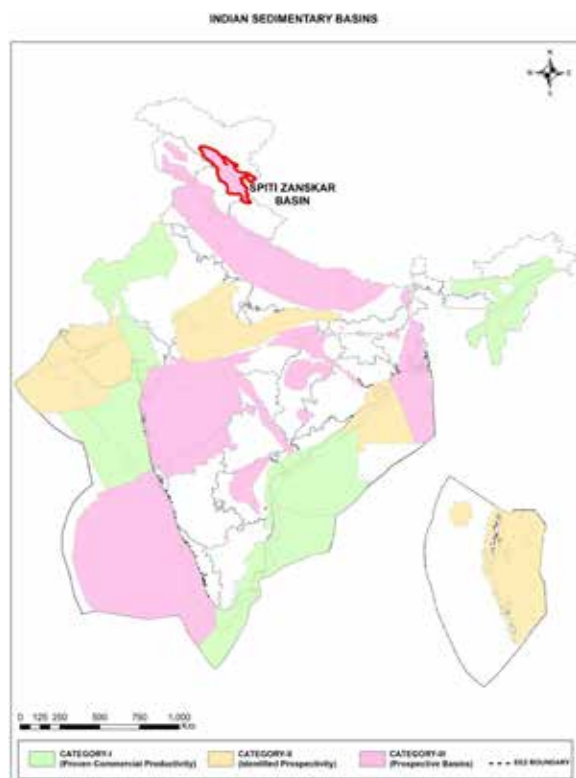
Location	Basin area(skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
Onland	95,215	35,230 (37%)	OALP: 1/2,999

The basin occupies an area of 95,215 sq km. It has been appraised to the extent of 35,230 sq km (37%). The active acreage nos. across regime(s) are 1 with area 2,999 sq. km (<1% of basin area).

2.3.13 SPITI-ZANKSKAR BASIN

Spiti-Zankskar is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for a potential discovery.

The basin overlaps with the state(s) of Himachal Pradesh, Jammu and Kashmir, Uttarakhand.



Source: DGH Internal



Source: DGH Internal



The basin is situated within north-northwestern Himalayan Mountains along northern margin of India. The Basin exposes about 11,000 metre of Paleozoic, Mesozoic and a part of Lower Tertiary sediments.

The elements of petroleum system in the basin are speculative.

The basin comprises of sedimentary sequence of Precambrian to Eocene age.

The Paleozoic and Mesozoic Plays of the Potwar Basin of Pakistan are considered as close analogue. These plays have four assessment units identified based on geological rationales.

The basin has inherent complex structure, which calls for detail study for identifying probable entrapment.

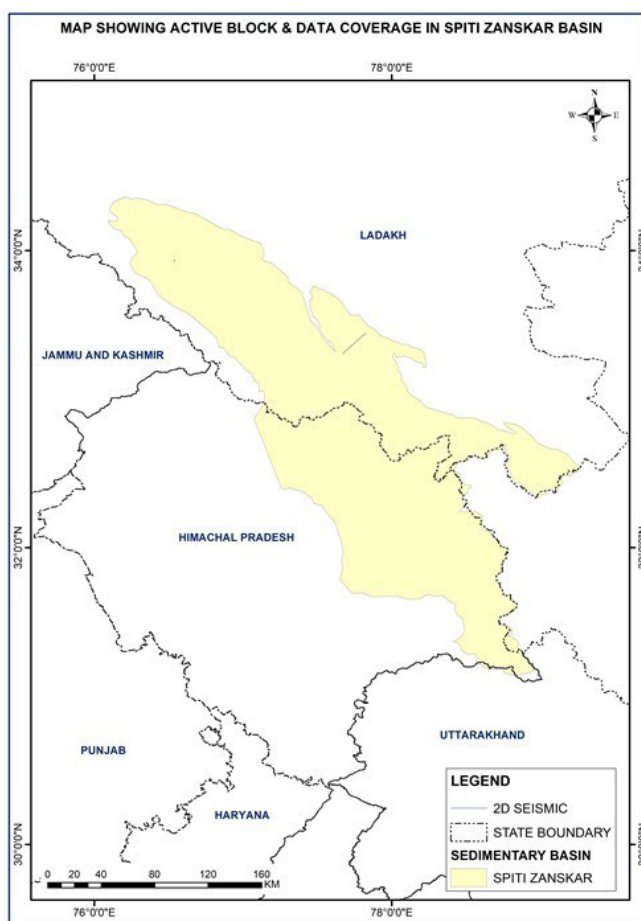
• Hydrocarbon prospectivity and Basin maturity:

In the basin, 2 plays are present within Paleozoic and Mesozoic. 11,000m of sedimentary sequence was envisaged in the basin. Aerial Yield method was used on analogy of Kohat Potwar Basin (Pakistan).

Spiti-Zaskar basin has a total hydrocarbon in-place of 10 MMTOE and this is entirely undiscovered risked in-place.

• Basin datasets and studies:

As of 31.03.2025, NDR has archived primary datasets of 42 LKM 2D seismic. Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area(skm)	Appraised (skm)	Active acreage [Nos./Area(skm)]
Onland	32,000	32,000 (100%)	No active acreage

The basin occupies an area of 32,000 sq km. It has been appraised to the extent of 8,640 sq km (27%).



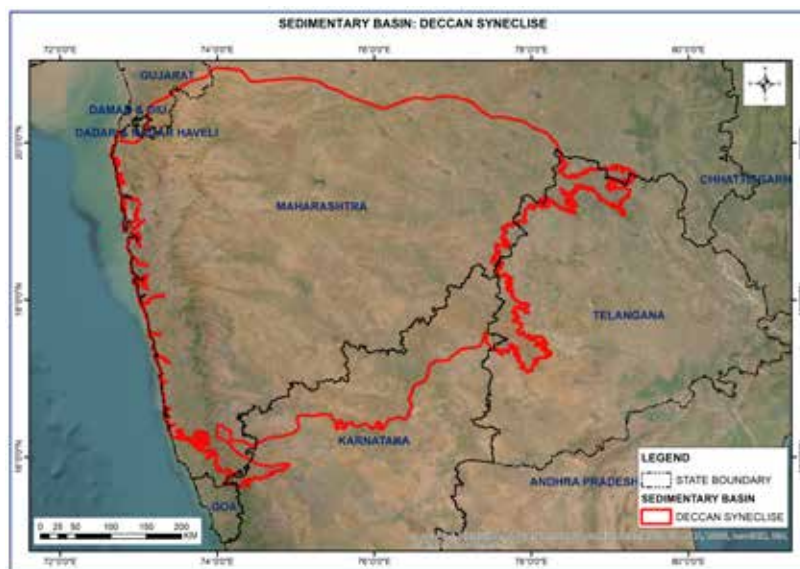
2.3.14 DECCAN SYNECLISE BASIN

Deccan Syneclise is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for a potential discovery.

The basin overlaps with the state(s) of Dadra and Nagar Haveli, Goa, Gujarat, Karnataka, Maharashtra, Telangana.



Source: DGH Internal



Source: DGH Internal

Deccan Syneclise basin is located in the western and south-central part of India. The large part of the basin is underlain by Deccan basalts.

The elements of petroleum system in the basin are speculative.

Based on gravity and DSS data about 1.5-2.0 Km thick sediments are speculated to be present underneath Deccan Trap basalts.

The inferred geological history of this area favours presence of Pre-Trappean Gondwana and Proterozoic sequences in the area. The basin has been divided into three assessment units on the basis of tectonic framework. Two units in north-east being proximal to Satpura Basin may represent Pre-Trappean Gondwana sequence whereas the third unit represents the Proterozoic sequence.

Sub-basalt imaging to study sub-trappean sediments is the focus area.

- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 2 plays exist within Precambrian and Late Paleozoic (Gondwana). 1,500m sedimentary thickness was envisaged in the basin. Aerial Yield method was used on analogy of Vindhyan Basin

(Pre-Cambrian) and Cooper Basin, Australia (Gondwana).

Deccan Syneclise basin has a total hydrocarbon in-place of 12 MMTOE and this is entirely undiscovered risked in- place.

- Basin datasets and studies:**



Source: DGH Internal

As of 31.03.2025, NDR has archived primary datasets of 11,076 LKM 2D seismic. Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Basin appraisal and regime-wise active acreage:

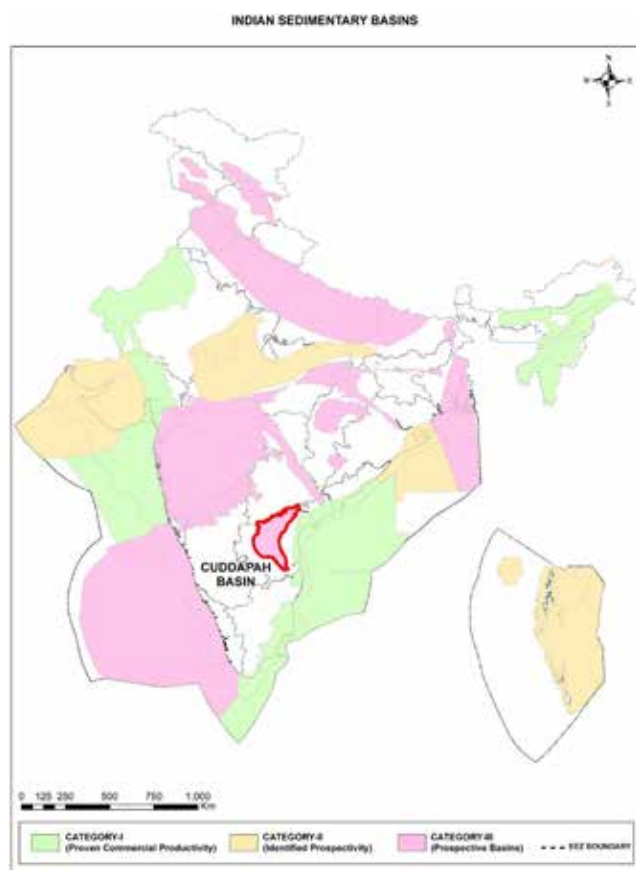
Location	Basin area (skm)	Appraised (skm)	Active acreage [Nos./Area (skm)]
Onland	237,500	121,125 (51%)	No active acreage

The basin occupies an area of 237,500 sq km. It has been appraised to the extent of 121,125 sq km (51%).

2.3.15 CUDDAPAH BASIN

Cuddapah is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for a potential discovery.

The basin overlaps with the state(s) of Andhra Pradesh, Telangana.



Source: DGH Internal



Source: DGH Internal

Cuddapah is a crescent-shaped basin, located in the eastern part of Dharwar craton. The elements of petroleum system in the basin are speculative.

The Palaeo-to-Meso-Proterozoic Basuhari-Rohtas play of Vindhyan Basin is the close analogue. Entire Proterozoic sequence of the

basin is considered a single Proterozoic Play, subdivided into five assessment units based on geological rationales.

- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 1 play is known within Pre-Cambrian. 6,000-10,000m sedimentary thickness was

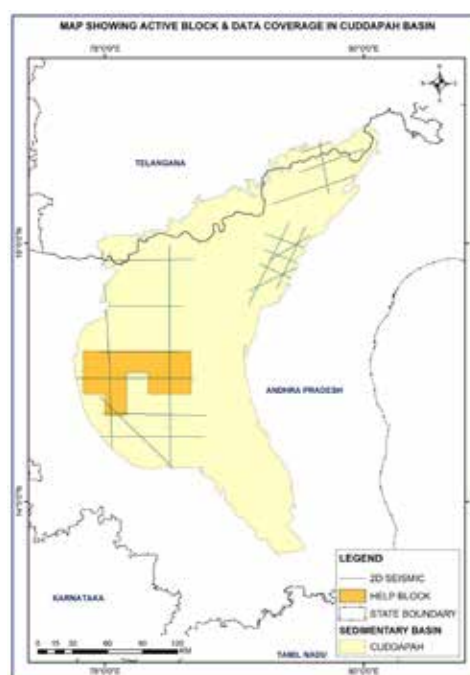


envisaged in the basin. Aerial Yield method was used on analogy of Vindhyan Basin.

Cuddapah basin has a total hydrocarbon in-place of 5 MMTOE and this is entirely undiscovered risked in-place.

- Basin datasets and studies:**

As of 31.03.2025, NDR has archived primary datasets of 1,674 LKM 2D seismic. Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area (skm)	Appraised (skm)	Active acreage [Nos./Area (skm)]
Onland	40,100	11,760 (29%)	OALP: 1/3,306

The basin occupies an area of 40,100 sq km. It has been appraised to the extent of 11,760 sq km (29%). The active acreage nos. across regime(s) are 1 with area 3,306 sq. km (<1% of basin area).

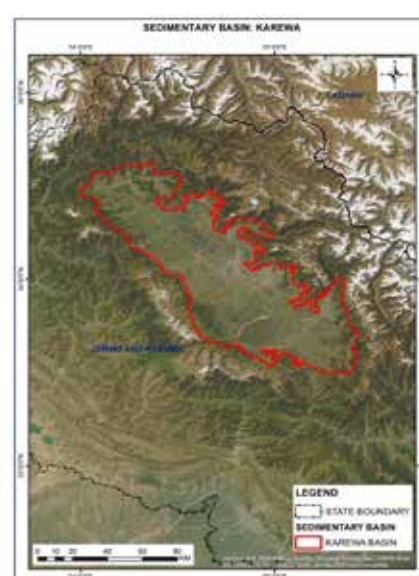
2.3.16 KAREWA BASIN

Karewa is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for a potential discovery.

The basin overlaps with the state(s) of Jammu and Kashmir.



Source: DGH Internal



Source: DGH Internal



Karewa is an intermontane basin, located in Kashmir valley of northwestern Himalayas. It is an oval-shaped basin between Pir-Panjal Range in the southwest and Zaskar Range to its northeast.

The elements of petroleum system in the basin are largely speculative.

The identified three plays viz. Paleozoic Play, Triassic Limestone Play and Plio-Pleistocene Play has six assessment units each.

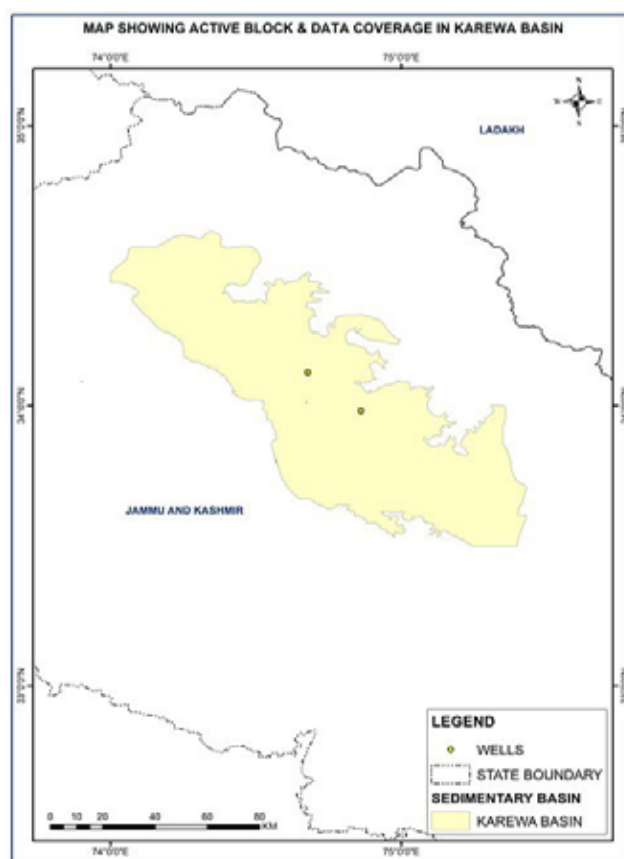
- **Hydrocarbon prospectivity and Basin maturity:**

In the basin, 3 plays are present within Paleozoic, Mesozoic and Plio-Pleistocene. Aerial Yield approach is used on analogy of Kohat Potwar Basin (Pakistan). 14,000m sedimentary thickness has been envisaged in the basin.

Karewa basin has a total hydrocarbon in-place of 6 MMTOE and this is entirely undiscovered risked in-place.

- **Basin datasets and studies:**

As of 31.03.2025, NDR has archived primary datasets of 2 conventional wells (all exploratory).



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area (skm)	Appraised (skm)	Active acreage [Nos./Area (skm)]
Onland	6,671	2,402 (36%)	No active acreage

The basin occupies an area of 6,671 sq km. It has been appraised to the extent of 2,402 sq km (36 %).

Bidding opportunities:

Onland: 1 OALP (Round X) block with 284 sq km area

2.3.17 BHIMA-KALADGI BASIN

Bhima-Kaladgi is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for a potential discovery.

Bhima area geographically overlaps with the state(s) of Karnataka, Telangana. Kaladgi area geographically overlaps with the state(s) of Karnataka, Maharashtra.





Source: DGH Internal

Bhima-Kaladgi are two Proterozoic sub-basins situated over Dharwar craton. The elements of petroleum system in the basin are largely speculative.

Bhima sub-basin is characterized by Neo-Proterozoic sedimentary sequence whereas Kaladgi sub-basin has thick Pale to Neo-Proterozoic sediments. Entire Proterozoic sequences in Bhima-Kaladgi Basin has been considered as single play, sub-divided into three assessment units each on the basis of their respective tectonic set-up.

Paleo-to-Meso-Proterozoic Basuhari-Rohtas play of Vindhyan Basin has been used in the analogue study. Sub-basalt imaging to study sub-trappean sediments is the focus area.

- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 1 play is identified within Pre-Cambrian. Aerial Yield method was used



Source: DGH Internal

on analogy of Vindhyan Basin. Sedimentary thickness of 270m (Bhima) and 3,000-4,500m (Kaladgi) were envisaged.

Bhima sub-basin has a total hydrocarbon in-place of 0.6 MMTOE and this is entirely undiscovered risked in-place.

Kaladgi sub-basin has a total hydrocarbon in-place of 6 MMTOE and this is entirely undiscovered risked in-place.

- Basin datasets and studies:**

As of 31.03.2025, NDR has archived primary datasets of 1,087 LKM 2D seismic. Multiple flavours of reprocessed seismic datasets are also archived whenever the same are provided by E&P Operators.



Source: DGH Internal

Basin appraisal and regime-wise active acreage:

Location	Basin area (skm)	Appraised (skm)	Active acreage [Nos./Area (skm)]
Kaladgi	8,820	2,616 (30%)	No active acreage
Bhima	5,280	5,280 (100%)	No active acreage

The basin occupies an area of 14,101 sq km. It has been appraised to the extent of 7,896 sq km (56%)



2.3.18 BASTAR BASIN

Bastar is a Category III basin with entirely undiscovered in-place, that calls for intense exploration for a potential discovery.

The basin overlaps with the state(s) of Chhattisgarh, Odisha.



Source: DGH Internal



Source: DGH Internal

Bastar Basin, one amongst the several Proterozoic sedimentary basins is located over Peninsular India. The elements of petroleum system in the basin are largely speculative.

Entire Proterozoic sequence of the basin has been considered as a single play, sub-divided by a fault into two assessment units.

Paleo-to-Meso-Proterozoic Basuhari-Rohtas play of Vindhyan Basin has been used in the analogue study. Assessment of thickness of sedimentary sequence is the focus area.

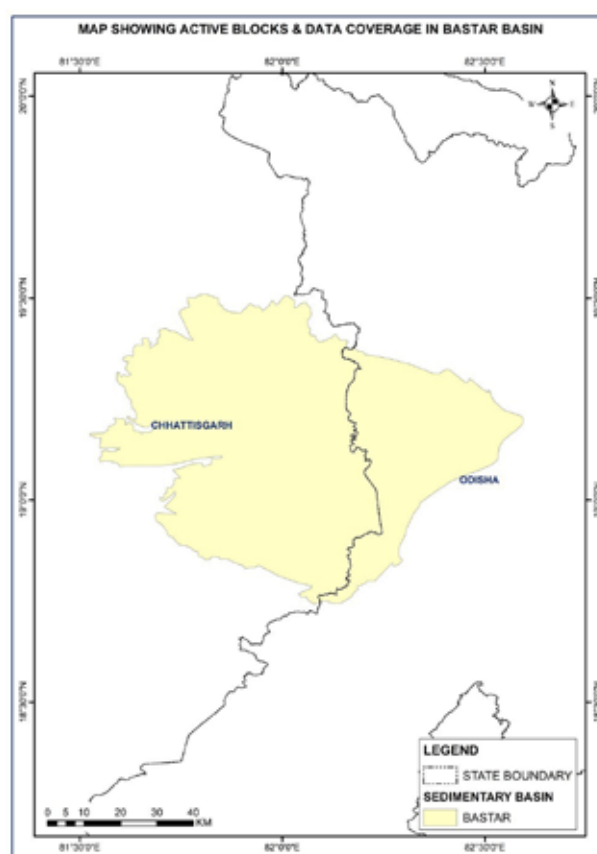
- Hydrocarbon prospectivity and Basin maturity:**

In the basin, 1 play is known within Pre-Cambrian. Aerial Yield method was used on analogy of Vindhyan Basin.

Bastar basin has a total hydrocarbon in-place of 1 MMTOE and this is entirely undiscovered risked in-place.

- Basin datasets and studies:**

As of 31.03.2025, there is no seismic/well data available at NDR.



Source: DGH Internal



Basin appraisal and regime-wise active acreage:

Location	Basin area (skm)	Appraised (skm)	Active acreage [Nos./Area (skm)]
Onland	5,360	5,360 (100%)	No active acreage

The basin occupies an area of 5,360 sq km. It has been appraised to the extent of 5,360 sq km (100%).

As on 31.03.2025, in terms of single-flavour primary datasets, NDR has archived 1,266,413 line km 2D seismic data, 457,937 sq km 3D seismic data, 22,777 wells including 8,180 exploratory wells. These data exclude all value-added reprocessed variants and those acquired but not yet submitted to NDR.

Among two ongoing 2D seismic acquisition campaigns, Mission Anveshan(MA) has been planned across seven onland basins (Rajasthan, Ganga-Punjab, Cuddapah, Chhattisgarh, KG, Deccan Syncline and Saurashtra) to acquire closed-grid data around the leads identified from 2D seismic data acquired under National Seismic Programme (2016-2022). As on 31.03.2025, 2,954 line km was surveyed under MA against the planned quantum of 20,275 line km. In another 2D seismic survey, christened as Extended Continental Shelf (ECS) Survey has the objective to establish India's claim for extended continental shelf boundary in East and Western offshore beyond present day EEZ. Out of total 30,000 line km planned, as on 31.03.2025, 22,876 line km data were acquired. The western offshore acquisition has been completed and east to be completed. The campaign is supervised by DGH under Ministry of Petroleum and Natural Gas in close coordination with Ministry of Earth

Sciences, Ministry of Defence and Indian Navy. Both MA and ECS are undertaken by NOCs (ONGC and OIL).

Out of 3.36 million sq km basin area, 2.62 million sq km (79.5%) has been appraised and 0.51 sq km area(15%) is active under different operating regimes.

As a part of Govt-sponsored initiatives, country-level assessment of conventional and unconventional hydrocarbons has been formulated with proposed participation of NOCs and major private E&P Operators. Such study will be carried out by expert consulting agency to be engaged through competitive bidding. The study will create National Hydrocarbon Atlas (NHA) and help integrate with discovered resources on National PRMS Chart. All project deliverables will be software-agnostic and expert-audited with futuristic update with new data.

The Chapter herein presented is intended to give an overview of India's sedimentary basins with latest facts segregated at basin-level and will serve as a quick reference to hydrocarbon prospectivity. Additionally, it may be used as a brief dossier for Indian Sedimentary Basins and Hydrocarbon potential.





3

Exploration and Production Activities

The Indian government's commitment to streamlining business regulations in the Exploration and Production (E&P) sector is evident in the recent wave of policy reforms. This focus on improving the Ease of Doing Business has led to a significant rise in E&P activity. This surge is expected to play a crucial role in achieving the government's ambitious goal of reducing dependence on imported hydrocarbons.

Recognizing that increased domestic oil and gas production is key to reducing import reliance, the government has prioritized boosting E&P activities. This includes offering more exploration acreage and discovered fields for development.

The substantial E&P efforts undertaken in 2014-25 have set the stage for sustained growth in the Indian E&P sector for years to come. This chapter provides a comprehensive overview of E&P activities during the 2024-25 fiscal year, including details on hydrocarbon discoveries and statistics on exploration, development, and production across the country.

3.1 Seismic Activities

In FY 2024-25, majority of the data acquisition was carried out under Nomination and RSC regime (in OALP Blocks). The data acquisition efforts during FY 2024-25 under various regimes have been highlighted in Table 3.1. The data indicates that 936.37 GLKM of 2D seismic data was acquired. For 3D seismic, a total of 10795.3 SKM was collected, with 2573.69 SKM attributed to nominated agencies (ONGC and OIL) and a substantial 8221.61 SKM from the Contract Regime (PSC+RSC), predominantly from offshore operations. This highlights significant



exploration activity across both 2D and 3D methodologies.

Details of seismic data acquisition activities in Nomination, PSC & RSC regime during FY 2024-25 is presented in Table 3.1.

Table 3.1 Seismic Data Acquired during FY 2024-25

Sl. No.	Subject	Parameter	ONGC (Nomination)	OIL (Nomination)	Contract Regime (PSC+RSC)	Mission Anveshan & ECS	Grand Total
1	2D Seismic data acquired	Onland (GLKM)	191	71	674	25,830	26,766
		Offshore (GLKM)	0	0	0	0	0
Total 2D Seismic Acquired			191	71	674	25,830	26,766
2	3D Seismic data acquired	Onland (SKM)	519	482	1,202	0	2,204
		Offshore (SKM)	1572	0	7,020	0	8,592
Total 3D Seismic Acquired			2,091	482	8,222	0	10,795

3.2 Drilling Activities

During FY 2024-25, a total of 141 exploratory wells were drilled while a total of 598 development wells were drilled by NOCs and Pvt./JVs. Majority of the wells were drilled by ONGC in its onland

nomination areas while in offshore development wells were drilled by ONGC only. Country's drilling activities are summarized in Table 3.2.

Table 3.2 Drilling Activities during FY 2024-25

Sl. No	Subject	Parameter	ONGC (Nomination)	OIL (Nomination)	Contract Regime (PSC+RSC+CBM)	Total
1	Exploratory wells	Onland	52	15	35	102
		Offshore	19	0	20	39
Total Exploratory wells			71	15	55	141
2	Development wells	Onland	371	32	103	506
		Offshore	92	0	0	92
Total Development wells			463	32	103	598

The regime-wise drilling activities are presented in Table 3.3

Table 3.3 Regime wise drilling activities during FY 2024-25

Regime-wise Drilling during FY 2024-25			
Sl. No.	Regime	Development	Exploratory
1	NOMINATION	495	86
2	PSC	42	5
3	DSF	4	0
4	OALP	0	48
5	CBM	57	2
TOTAL		598	141



Year-wise trend of drilling activities are as depicted in Figure 3.1 & 3.2:

Figure 3.1: Year Wise Exploratory Wells Drilling Trend

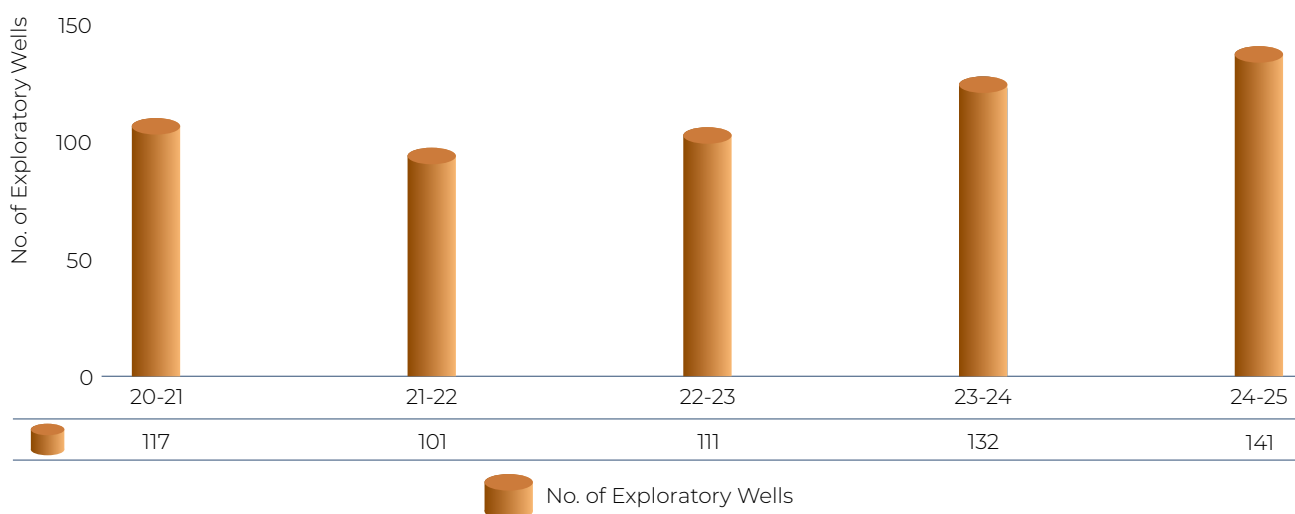
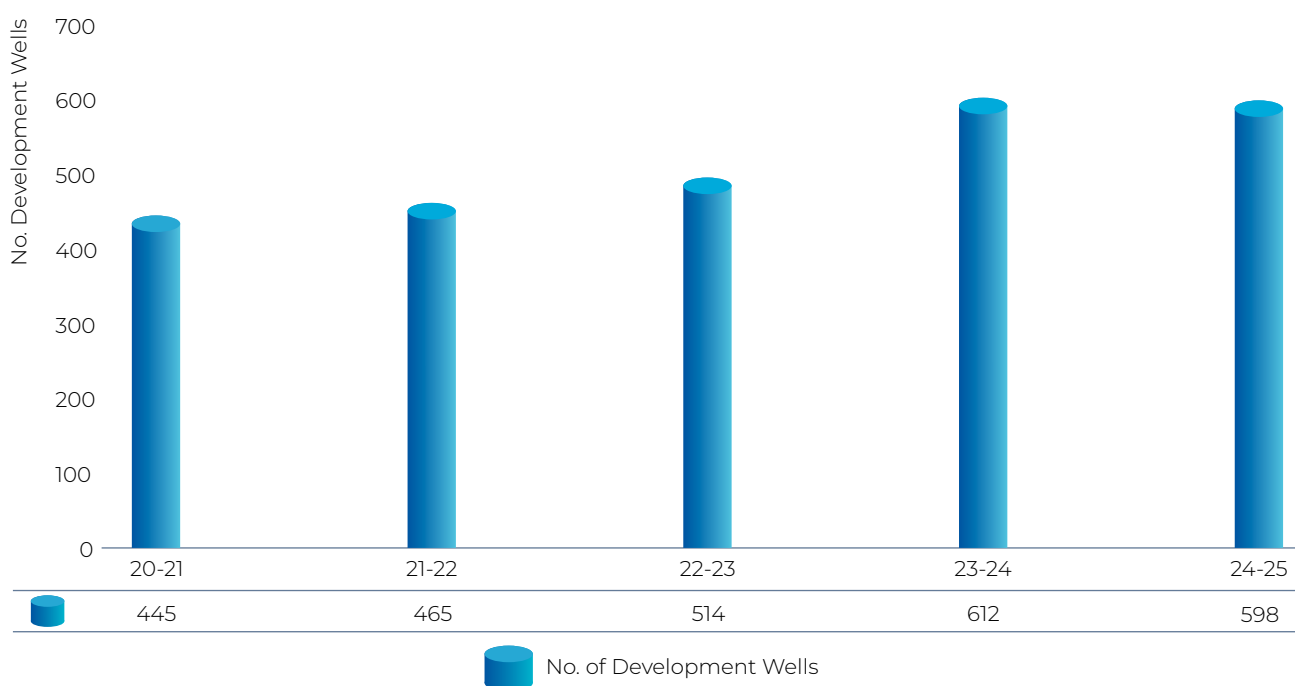


Figure 3.2: Year Wise Development Wells Drilling Trend



3.3 Oil and Gas Production

In FY 2024-25, the country achieved 91.8% of its crude oil production target by producing 28.70 MMT against crude production target of 31.27 MMT, while the domestic gas production stands at 36.11 BCM against the target of 39.67 BCM.

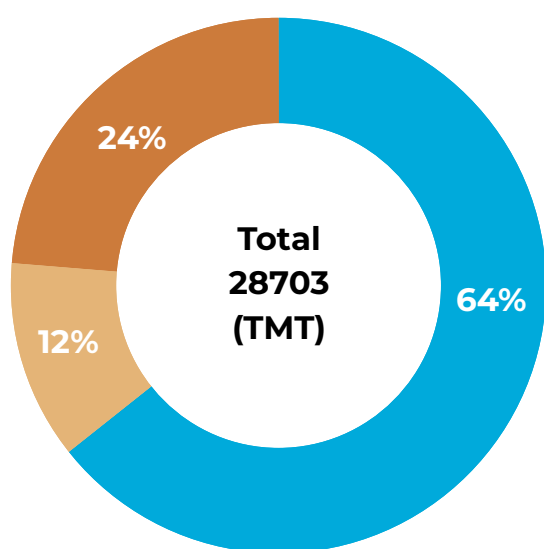
The year witnessed a decrease in crude oil production by 2.2% and in gas production by 0.9% as compared to the previous year i.e FY 2023-24.

The production performance during the year can be explained by the following tables and figures

Table 3.4: Oil and Gas production in the country – 2024-25

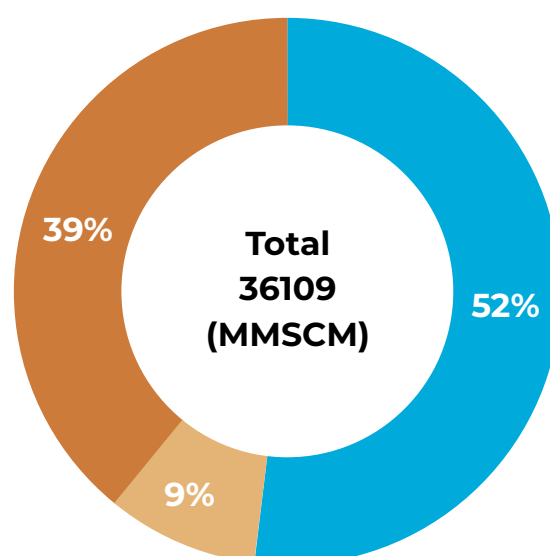
Operator (Regimes)	OIL TMT					Gas (MMSCM)				
	2023-24	2024-25		% Ach w.r.t		2023-24	2024-25		% Ach w.r.t	
	Actual	Target	Actual	Target	23-24	Actual	Target	Actual	Target	23-24
ONGC (N)	19210	19659	18466	93.9%	96.1%	19316	19998	18795	94.0%	97.3%
OIL (N)	3344	3958	3444	87.0%	103.0%	3090	3929	3171	80.7%	102.6%
PSC + RSC+CBM	6803	7655	6793	88.7%	99.9%	14033	15746	14143	89.8%	100.8%
Country	29357	31272	28703	91.8%	97.8%	36439	39673	36109	91.0%	99.1%

Figure 3.3: Crude oil production of country in FY 2024-25, TMT



● ONGC (Nomination) ● OIL (Nomination) ● PSC + RSC

Figure 3.4: Natural Gas Production of Country in FY 2024-25, MMSCM



● ONGC (Nomination) ● OIL (Nomination) ● PSC, RSC and CBM



Table 3.5: Basin and Operator-wise production of country in FY 2024-25

Sl. No.	Operator	Basin	Oil (MMT)	Gas (BCM)	O+OEG (MMT)
National Oil Companies (NOCs) – Nomination					
1	OIL Nomination	Assam-Arakan Shelf	3.411	2.972	6.384
2		Rajasthan	0.033	0.199	0.232
Total (OIL Nomination)			3.444	3.171	6.615
1	ONGC Nomination	Assam-Arakan Fold Belt	0.002	1.224	1.23
2		Assam-Arakan Shelf	1.049	0.339	1.39
3		Cambay	4.548	0.810	5.36
4		Cauvery	0.200	0.984	1.18
5		Krishna Godavari	0.233	0.975	1.208
6		Mumbai	12.434	14.463	26.90
Total (ONGC Nomination)			18.466	18.795	37.26
Total NOCs			21.910	21.966	43.876
NOCs/PVT/JV Companies (PSC)					
1	CEIL	Cambay	0.206	0.253	0.459
2		Assam-Arakan Shelf	0.0002	0.027	0.027
3		Krishna Godavari	0.469	0.075	0.545
4		Rajasthan	3.394	1.544	4.938
5	ESSAR	Cambay	0.000	0.000	0.000
6	FOCUS	Rajasthan	0.002	0.043	0.045
7	GEOENPRO	Assam-Arakan Shelf	0.019	0.005	0.023
8	GNRL	Cambay	0.004	0.007	0.010
9	GSPC	Cambay	0.028	0.006	0.033
10	HOEC	Assam-Arakan Shelf	0.014	0.184	0.197
11		Cambay	0.004	0.005	0.009
12		Cauvery	0.000	0.001	0.001
13		Mumbai	0.036	0.054	0.090
14	JTI	Cambay	0.028	0.008	0.035
15	KIRI	Cambay	0.000	0.002	0.002
16	MERCATOR	Cambay	0.000	0.000	0.000
17	NIPPON	Cambay	0.00	0.00	0.00
18	OILEX	Cambay	0.00	0.00	0.00



Sl. No.	Operator	Basin	Oil (MMT)	Gas (BCM)	O+OEG (MMT)
19	OILMAX	Assam-Arakan Shelf	0.01	0.07	0.08
20		Cambay	0.00	0.00	0.00
21	ONGC	Cambay	0.06	0.07	0.13
22		Cauvery	0.06	0.03	0.09
23		Krishna Godavari	1.06	0.75	1.81
24		Bengal	0.00	0.00	0.00
25	PAN INDIA	Cambay	0.0004	0.00	0.00
26	RIL	Krishna Godavari	0.95	10.22	11.17
27	SELAN	Cambay	0.04607	0.01664	0.06271
28	SACF	Cambay	0.00002	0.00000	0.00002
29	SHELL	Mumbai	0.00	0.00	0.00
30	SUNPETRO	Cambay	0.39	0.02	0.41
Total NOCs/PVT/JV COMPANIES (PSC)			6.793	13.387	20.180
COAL BED METHANE (CBM)					
1	ESSAR	Raniganj East		0.336	0.336
2	GEECL	Raniganj South		0.079	0.079
3	ONGC	Bokaro		0.047	0.047
4		Jharia		0.0002	0.0002
5	RIL	Sohagpur East		0.002	0.002
6		Sohagpur West		0.292	0.292
Total CBM				0.756	0.756
India Grand Total			28.703	36.109	64.812

3.3.1 Key Basins: The Geological Powerhouses

India's hydrocarbon production is anchored in several key basins, each with unique geological characteristics and contributions:

- ▶ **Krishna-Godavari Basin:** A Category I basin, considered the most prospective with maximum resources. It is home to the country's largest deepwater gas fields, contributing 11.775 BCM of gas (32.61% of total) in 2024-25. It has also produced 2.7 MMT of crude oil in 2024-25.
- ▶ **Mumbai Offshore Basin:** Historically significant, accounting for 43.45% of India's oil output with 12.47 MMT oil, also contributing 14.517 BCM of gas (40.20%) in 2024-25.
- ▶ **Rajasthan Basin:** A key onshore basin, contributing approximately 11.95% to oil production (3.43 MMT) and 1.786 BCM of gas (4.95%) in 2024-25.
- ▶ **Cambay Basin:** Emerging as the second highest contributor to oil production at 18.5% (5.3 MMT), with 1.205 BCM of gas (3.34%) in 2024-25.
- ▶ **Assam and Tripura Basins:** Historically significant, contributing 3.593 BCM from Assam-Arakan Shelf (9.95%) and 1.224 BCM from Assam-Arakan Fold Belt (3.39%) in 2024-25, along with 4.5 MMT of oil (15.7%)



3.3.2 The basin wise crude oil and natural gas production details are as follows:

Figure 3.5: Country's Basin-Wise Oil Production in 2024-25 (MMT)

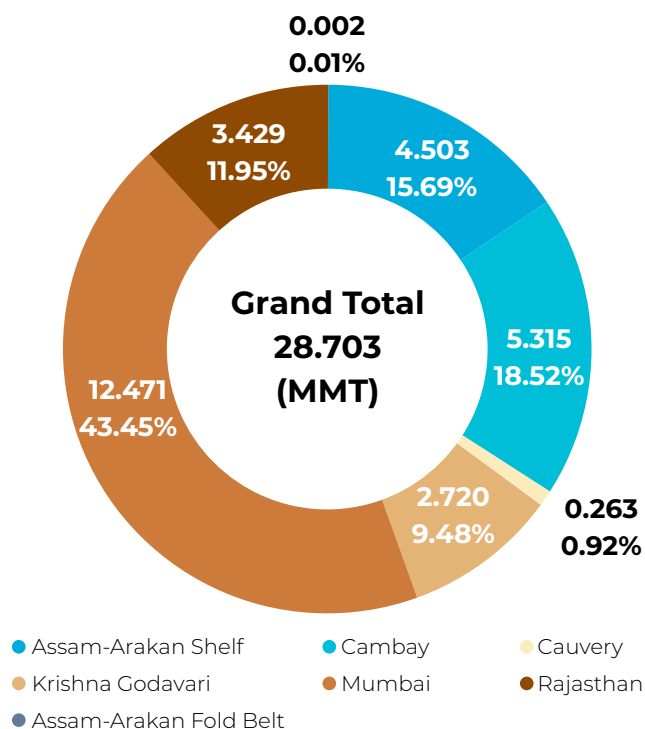


Figure 3.6: Country's Basin-Wise Gas Production in 2024-25 (BCM)

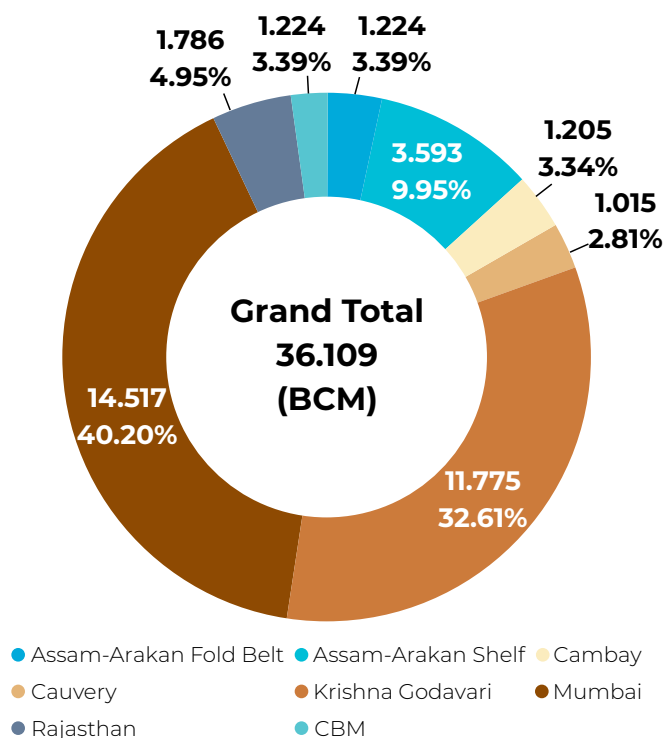


Figure 3.7: Country's Location-Wise Oil Production in 2024-25 (MMT)

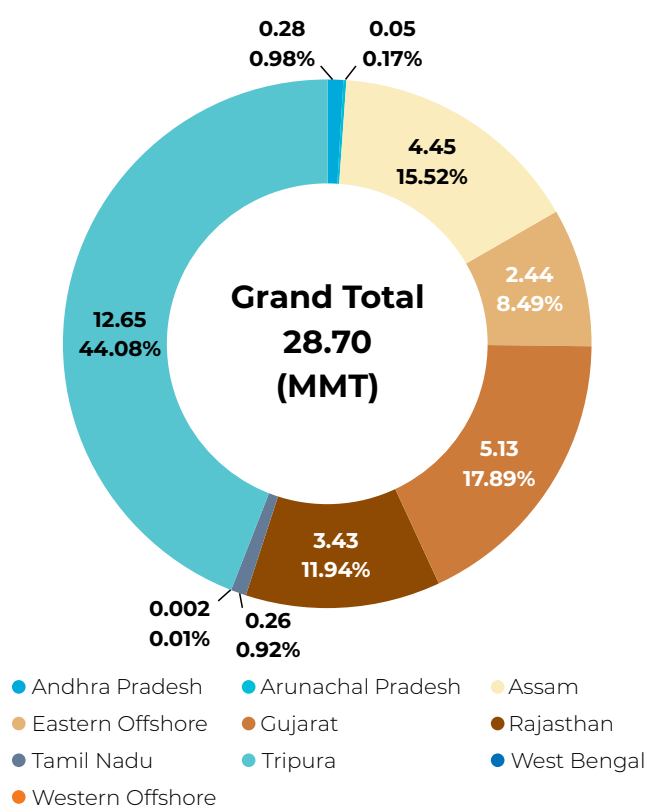
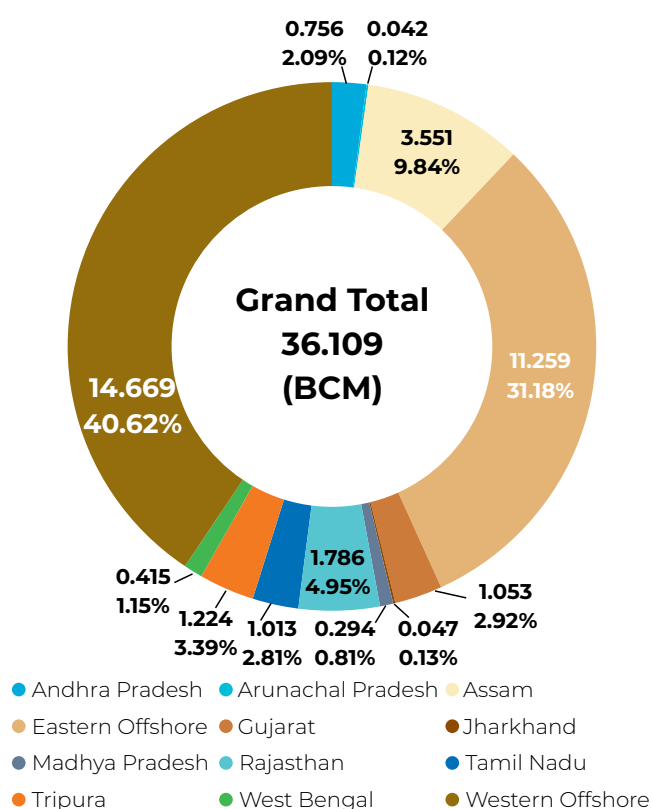


Figure 3.8: Country's Location-Wise Gas Production in 2024-25 (BCM)



3.3.3 Production Trends: Navigating Challenges and Opportunities

Crude oil production has generally declined from 36.96 MMT in 2015-16 to 28.70 MMT in 2024-25, attributed to depletion of reserves and limited contribution from new discoveries. In contrast, natural gas production has shown modest growth, rising from 32.25 BCM in 2015-16

to 36.11 BCM in 2024-25. During this period, the share of PSC production rose from 24.3% to 37%, signalling a shift towards private companies and joint venture

Notably, after a decline until 2020-21, gas production recovered, driven largely by increased contribution from NELP blocks, particularly from the deepwater blocks in Krishna Godavari basin.

Table 3.6: Contribution by ONGC, OIL and NOCs/Pvt./JVs in country's Oil and Gas production in last 10 years

Year	Oil Production (MMT)				Gas Production (BCM)				
	ONGC (Nomination)	OIL (Nomination)	PSC	Total Oil (MMT)	ONGC (Nomination)	OIL (Nomination)	PSC	CBM	Total Gas (BCM)
2014-15	22.26	3.41	11.78	37.45	22.02	2.72	8.68	0.23	33.65
2015-16	22.37	3.23	11.36	36.96	21.18	2.84	7.84	0.39	32.25
2016-17	22.21	3.26	10.53	36.00	22.09	2.94	6.31	0.56	31.90
2017-18	22.25	3.38	10.06	35.68	23.43	2.88	5.60	0.73	32.65
2018-19	21.04	3.29	9.87	34.20	24.67	2.73	4.77	0.71	32.88
2019-20	20.63	3.11	8.44	32.17	23.75	2.67	4.12	0.66	31.19
2020-21	20.18	2.94	7.37	30.49	21.87	2.48	3.68	0.64	28.67
2021-22	19.45	2.99	7.25	29.69	20.63	2.89	9.82	0.68	34.02
2022-23	19.49	3.16	6.53	29.18	19.97	3.04	10.77	0.67	34.45
2023-24	19.21	3.34	6.80	29.36	19.32	3.09	13.38	0.65	36.44
2024-25	18.466	3.444	6.793	28.703	18.795	3.171	13.378	0.756	36.109





4

Fuelling India's E&P Activities with DATA

4.1 Historical Perspective

Oil and Gas Industry is one of the most data-rich industries and upstream data, that is, Exploration & Production (E&P) data is a significant part of it. Historical and new data have always been crucial to the success of any E&P project. Throughout E&P lifecycle, huge amount of data is continuously generated during the planning, exploration, field development, and management phases. The industry has recognized the importance of leveraging data for making key business decisions from the early days of exploration.

India has a rich history of E&P activity dating back to the 19th century. Methodical E&P activity began after the first commercially successful well was discovered in Digboi, Assam, in 1889 through the first oil discovery in independent India was made by Assam Oil Company Limited (AOC) in Nahorkatiya in 1953, followed by Moran in 1956, both in upper Assam. AOC was later incorporated into Oil India Private Limited in 1959, which became Oil India Limited (OIL) as a PSU (in 1981) to increase the pace of exploration and production in Assam. In 1959, the Government set up Oil and Natural Gas Commission (ONGC), which intensified and spread exploration to various parts of the country. ONGC systematically started geophysical surveys in other sedimentary basins considered prospective, resulting in the discovery of the Ankleshwar oil field in Gujarat in 1960.

Offshore exploration in India was initiated with the experimental seismic survey in the Gulf of Cambay in 1962 and later in the western offshore. Detailed seismic surveys in the western offshore



resulted in the discovery of a giant structure in Bombay Offshore in 1972-73, later termed as Bombay High Field. ONGC and OIL generated considerable amounts of E&P data through geoscientific surveys and drilling, making them the earliest dominant players in the Indian E&P industry. However, private sector E&P companies also contributed to data generation under Pre-NELP (1980-95) and pre-NELP field rounds (1992-93) offers and discovered field rounds since the 1980s.

The real boost in exploration efforts and appraisal of Indian basins came during the NELP regime, particularly in offshore areas, resulting in a quantum jump in data volume from the preceding era. Both NOC and private operators contributed to this using the latest technology of the time. Multi-client surveys, particularly, projects INDIASPACE 1 (2006) and INDIASPACE 2 (2009) were conducted under Non-Exclusive Multi-Client (NEMC) Policy which for the first time mapped west coast and east coast sedimentary basins of India on a regional/basinal scale. Lately, new data is being acquired by the Operators in the blocks awarded in Open Acreage Licensing Policy (OALP) and Discovered Small Field (DSF) rounds under Hydrocarbon Exploration and Licensing Policy (HELP) launched in 2016. With a major thrust on the seismic acquisition on Indian sedimentary basins, huge seismic data have been acquired under the Government funded schemes like onland National Seismic Programme (NSP)-2016-21, Andaman offshore Seismic Program (2020), Exclusive Economic Zone (EEZ) Surveys (2023) with seismic 2D lines upto EEZ boundary in east and west coasts. In logistically difficult basin areas AGG Surveys were carried out in 11 states including all the

Nort-Eastern states. The survey was completed in March 2024. With the continuous support from GOI for hydrocarbon exploration, seismic survey named as "Mission Anveshan" is in progress in the onshore areas and "Extended Continental Shelf (ECS)" in offshore to augment and infill the existing coverage of seismic data. Service providers have also been reprocessing vintage 2D/3D and CSEM data under the "value-addition" clause of NDR data policy to generate new insights from vintage data using their latest proprietary technologies.

4.2 E&P data – A National Asset

E&P data is considered a valuable national asset, and its ownership lies with the government. The Petroleum and Natural Gas (Amendment) rules of 2006 mandate that all E&P operators in India provide to the Central Government or its designated agency with any data obtained as a result of petroleum operations in the country, free of cost. The E&P data policy issued by MoPNG in 2017 outlines the framework for data assimilation, disclosure, sharing, accessibility, and dissemination through the National Data Repository (NDR). The policy designates DGH as the agency for data submission.

Technological advancements in data processing, interpretation, and visualization provide opportunities to gain new insights into even age-old E&P data. The exponential growth in data volume and diversity of data types enables the processing of diverse data sets to identify relational patterns, which can improve understanding of the subsurface, design exploration campaigns, and optimize oil and gas production. Effective data management is crucial to develop a robust national E&P database



and accelerating exploration and production activities in the country. Therefore, safeguarding E&P data is a matter of national significance.

4.3 National Data Repository

The National Data Repository (NDR) was established by the Government of India in 2014 under the Directorate General of Hydrocarbons (DGH) to assimilate, preserve, and disseminate a vast amount of data acquired by public and private companies, operators, agencies, and contractors during E&P activities over several decades. The data was previously scattered across different work centres of ONGC, OIL, and DGH, or held by the respective operating companies.

The main objective of the NDR is to organize and regulate the data for future exploration and development activities, as well as for research and development by educational institutions and government bodies. The NDR serves as a national-level establishment capable of managing a massive volume of data and providing access to it for public use.

To implement the project, the government awarded a turnkey contract for the "Build, Populate and Operate" model for the NDR to M/s Halliburton Offshore Services in March 2014. After the initial set up and data loading, the NDR was officially launched for public use on 28th June 2017. The establishment of the NDR has facilitated the efficient management and utilization of E&P data and has helped to promote exploration and development activities in India's hydrocarbon sector.

The NDR is a key enabler of Open Acreage Licensing Policy (OALP), Discovered Small Fields (DSF) exploration for unconventional resources like CBM rounds by facilitating maps and providing reliable data to the different E&P companies & investors. In HELP regime, E&P operators have the freedom to carve out blocks of their interest and size, accordingly all types of data available to the carved out areas are provided by the NDR under the new policy regime.

In addition to its use in the oil and gas sector, the NDR's data are being utilized for research and development by educational/ academic institutions and government bodies. Overall, the NDR is a valuable resource for the Indian energy industry, offering a centralized repository of E&P data for reliable decision-making and future development.

4.4 Objectives of NDR

The broad objectives of NDR are as follows:

- ▶ To assimilate, validate, archive, preserve, reproduce, and disseminate data relating to prospecting, exploration, development, and production of hydrocarbons.
- ▶ To create and maintain a modern state of the art platform for reporting, managing, mapping and visualization using the latest technologies.
- ▶ To ensure that industry, academia, research bodies and other users have ready access to the E&P data in NDR for promoting E&P research
- ▶ To ensure transparent access to quality E&P data and promote conditions for innovation and investment in exploration and production to discover and drive the development of new discoveries

4.5 Secondary Data Centre

DGH established its Secondary Data Centre (SDC) in high-tech Software Technology Parks of India (STPI), a Tier-III data centre at Bhubaneswar. SDC is located in a different seismic zone and geography than Primary Data Centre (PDC), NOIDA.

Data replication and synchronisation is maintained by 1 GBPS dedicated link between PDC and SDC. SDC works as a Business Continuity/Disaster Recovery Centre of Primary Data Centre (PDC) of National Data Repository of DGH, Noida. The Secondary Data Centre (SDC) is fully operational since 4th August 2018. SDC will be able to resume essential business operations in case of a disaster at PDC.

Apart from being data repository and recovery centre, SDC also provides quality industrial and educational trainings to the students of various academic institutions and organizations.

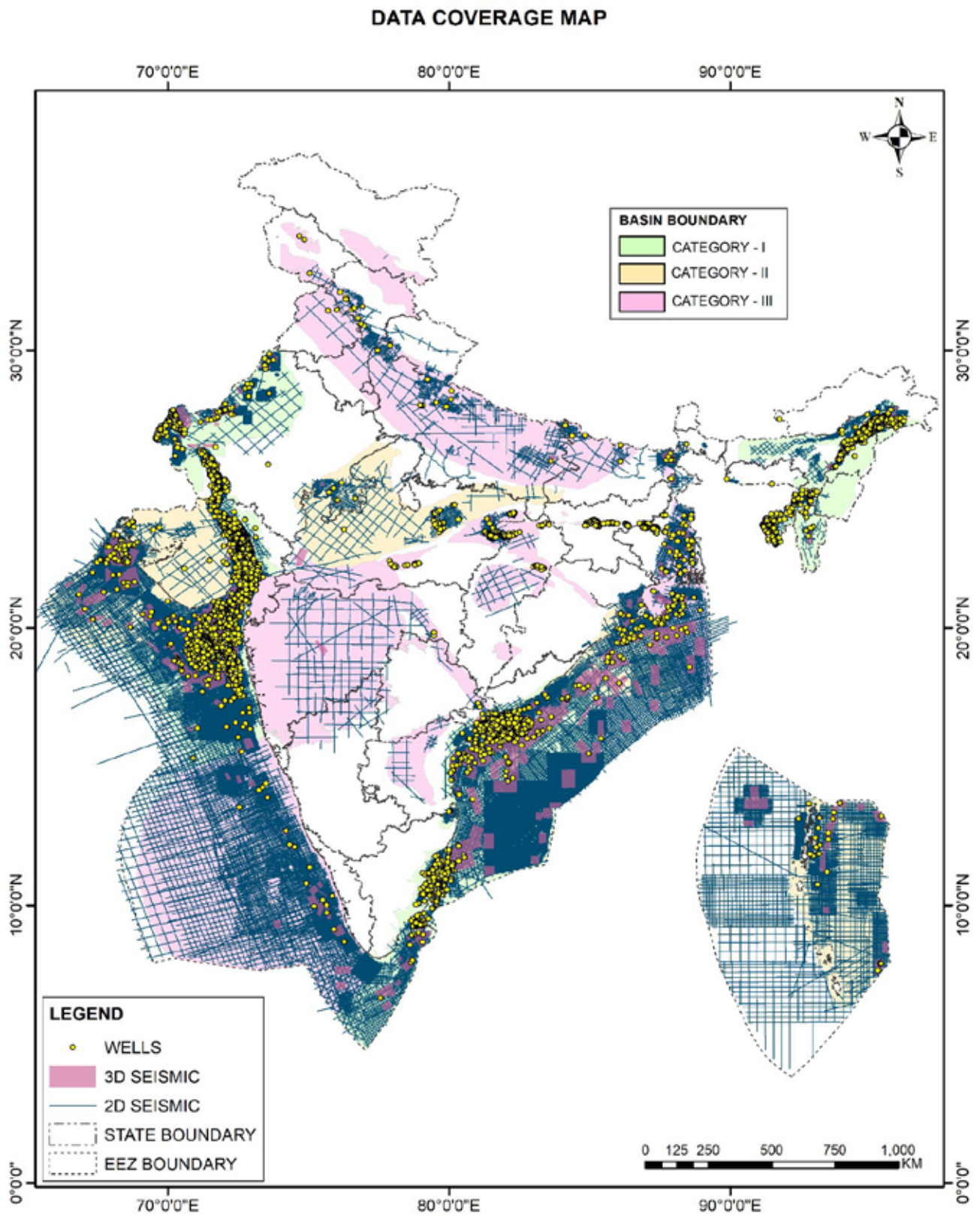
4.5.1 Data available in NDR

Table 4.1: NDR: Data Availability

Cumulative Data available in NDR as of 31st March, 2025	
2D Seismic data	4.01 Million LKM
3D Seismic data	1.25 Million SKM
Well data	24,434 No. of Wells
Well reports	52,941 No. of reports



Figure 4.1: Map of India showing coverage of 2D/3D Seismic and well data in NDR



During FY 2024-2025 the incremental data added to NDR database is tabulated below.

Table 4.2:

Incremental data added during 1st April 2024 to 31st March 2025	
2D Seismic data	316,804 LKM
3D Seismic data	110,077 SKM
Well data	818 Nos.
Well reports	3237 Nos.

Indian sedimentary basins have been categorized in Category-I, II & III types depending upon their hydrocarbon production/ exploration status. The basin wise data availability at NDR are tabulated below (as on 31st March 2025).

Basin Category	Seismic 2D (LKM)	Seismic 3D (SKM)	Wells
 CATEGORY - I	689,990.86	321,702.66	22,777
 CATEGORY - II	299,605.75	91,163.36	225
 CATEGORY - III	276,816.57	45,071.45	177

Besides these, NDR database is also having the well data and reports of unconventional resources like CBM (1172 wells), NGHP (66 wells) and Shale Gas wells (35 wells). Considering the huge data volume availability at NDR, large number of national & International companies like ONGC, OIL, TGS, ExxonMobil, BP Exploration, RIL, Vedanta Ltd. etc. purchase data from NDR for understanding of the Indian sedimentary basins.

4.6 Role of NDR in OALP/DSF bid rounds

NDR is playing a pivotal role in the ongoing OALP & DSF rounds under regime. It provides a gateway for operators/investors to carve out/ select blocks/acreages of choice based on existing data, concessions, and no-go areas for putting up expression of interest/bids. NDR constantly updates its portal with the latest data, E&P related information, active license areas, restricted areas and other cultural features. NDR is providing easy access of quality data to the investors thus helping them make robust evaluation of blocks/acreage for investment decision. NDR is not only preparing data packages for blocks offered under every round but also facilitating interested E&P investors with

visualization of G&G projects in the physical data room where they can understand and interpret available data within offered blocks in order to focus their interest according to their business models. In addition, NDR is also open to its users/ academia/Startups for online browsing as well as evaluation/viewing of E&P data of any basinal area in its physical data room.

4.7 Key Achievements of Year 2024-25

• Promoting exploration interests through Data visualisation room

In year 2024-25, NDR's physical data rooms (PDR) were booked by 11 entities which includes E&P operators, service providers, academic institutions and Govt organizations. Seismic data Interpretation projects were created in leading Industry Software application for Blocks under OALP-X bid round and DSF-IV bid round which are on offer by DGH. Apart from bid rounds, several customised G&G interpretation projects of 2D, and 3D seismic data along with well data were prepared and made ready for visualization to different E&P companies. Specific G&G projects are also made incorporating 2D and 3D seismic data available in East coast sedimentary Basins of India, viz. Bengal, Mahanadi, KG and Andaman basin for showcasing data to various global E&P operators.

Data packages of all the OALP-IX & X blocks were prepared for bid rounds so that prospective investors could order data specific to their areas of interest.

• Data Visualization Centre at India Energy week (IEW) 2025

Indian Energy week (IEW) was held in New Delhi during 11th- 14th February 2025. Based on the concept of physical data visualisation room of NDR at Noida, a data visualisation centre of was set up at DGH stall of IEW at Delhi, India. To facilitate this data centre, specific interpretation projects, were created in Decision Space Geoscience (DSG) software with seismic, well data, and selected 3D seismic volumes, providing stakeholders a clear understanding of the availability and quality of data in their areas of interest specially in the OALP-X blocks. This Data Centre allowed participants to visualize and assess the data of blocks on offer through hands-on experience in a workstation environment. This initiative attracted significant attention from companies, investors, delegates, and academia during the conference.



The tailor-made projects offered insights into subsurface basin analysis through seismic and well data, motivated stakeholders to visit the physical data centre in Noida or directly acquire specific data from NDR. This, in turn, enabled them to perform detailed analyses and comprehensive due diligence for bidding rounds of OALP, DSF and CBM.

The Data Centre was with a large high-resolution graphics screen showcasing the Indian sedimentary basins potential & NDR data availability along with its upgradation plan of the same. The Data Visualisation Centre at IEW was a significant success, offering stakeholders valuable insights and facilitating their decision-making processes regarding India's subsurface energy resources.

Figure 4.2: DGH Stall with Interactive Workstations showcasing OALP -X Blocks



4.8 Data Dissemination in Year 2024-2025

Large volume of 2D/3D Seismic and well data and associated reports have been shared with stakeholders/operators since the inception of NDR. In FY2024-25, a total of **663,363 LKM 2D seismic, 161,341 SKM of 3D seismic and 1579 no of well data** have been disseminated to as many as 20 E&P companies and 10 research/ academic institutes. Through this, DGH has been able to generate revenue of Rs. 16.52 Cr in FY 2024-25 from data sale on promotional/nominal rates under the NDR Policy.

4.8.1 Collaboration with University of Houston, Texas, USA:

Directorate General of Hydrocarbon (DGH) entered a Memorandum of Understanding (MoU) with the University of Houston (UH) 7th February 2023 for five years. The key Objectives were:

- 1) To establish the UH-DGH Data Centre which will host geoscience data of Indian sedimentary basins/blocks at Houston to reach out to major energy companies for facilitating them to participate in bidding process.
- 2) To collaborate with UH to take up subsurface/ geoscientific study of select Indian offshore basins for developing regional geological framework and identifying most prospective plays for exploration thereby enhance perspectivity perception and overall value addition from exploration point of view.

Under the MoU, DGH agreed to provide subsurface data for hosting at the UH-DGH data centre at Houston as well as for any collaborative technical study as mentioned above at (2). In turn, UH agreed to provide infrastructure/space at University of Houston Campus and engage their experts to undertake geoscientific study to enhance and showcase play-level prospectivity at basinal scale of select offshore basins, notably, Mahanadi-Bengal and Andaman.

Basin scale interpretation studies for the three basins viz. Bengal offshore, Mahanadi offshore and Andaman area are being carried out by the University of Houston team with special emphasize on the OALP -X blocks falling in these basins. Detail analysis of those blocks has been carried out in the University of Houston team. The result of those analysis brings out several

structural and stratigraphic leads in those blocks. These leads/prospects were also shown in workstation in India Energy week conference (IEW) 2025, which attracted many E&P investors/ participants of the conference.

4.8.2 Data rejuvenation, reprocessing and reorganization through Value-addition model of NDR:

Under the provision of Article 3.7 of NDR Data Policy 2017, NDR shared well data of West coast to M/S TGS to develop the 3D cube from the 2D data. The project has been completed during FY 2024-2025.

The West Coast of India is a challenging area covered by an array of vintage 2D seismic data, ranging from 1970's to 2010's. The processing applied to each dataset is also highly variable ranging from post stack time migration to pre-stack depth migration. The existing 3D coverage is quite sparse and to enable regional screening of the area, TGS (under the value-added provision of NDR policy), initiated a project using its 2Dcubed technology which would enable informed decisions on the requirement of future 3D seismic acquisitions.

2Dcubed is a proprietary software of M/S TGS for guiding interpolation of 2D seismic data along 3D geological horizons, generated from 2D dip fields and layer models which allows distances of kilometres to be bridged, something that would otherwise not be possible with conventional interpolation techniques.

(As per NDR policy DGH will have limited visualization rights for 5 years and no selling right of the data during that period).

4.9 Industry exposure training at SDC, Bhubaneswar

Since its inception SDC, Bhubaneswar is organising internship programme for the students across various institutes in the country. Through these internship programs students are taught about different theoretical aspects of oil and gas exploration and production, practical aspects of seismic data acquisition & interpretation, attribute analysis, mapping, overview of electrical logs and its interpretation technique, well correlation, estimation of reservoir parameters and hydrocarbon in-place calculation. The students were also exposed to the industry standard software like Petrel, Geolog etc. during hands on practice sessions. SDC



also arranged a visit to on-land drilling wellsite which provided trainees a unique opportunity to understand field operations.

The internship program at SDC Bhubaneswar commenced in FY 2021–22, and as of Mar 2025, a total of six such programs have been successfully conducted. Altogether, 76 students from 12

different universities and institutes across India have been trained, accumulating over 800 cumulative working hours.

Overwhelmingly positive responses were received the participants who joined from different parts of India for the industry exposure trainings.

Figure 4.3: Statistics of Internship Programme at SDC-NDR, DGH

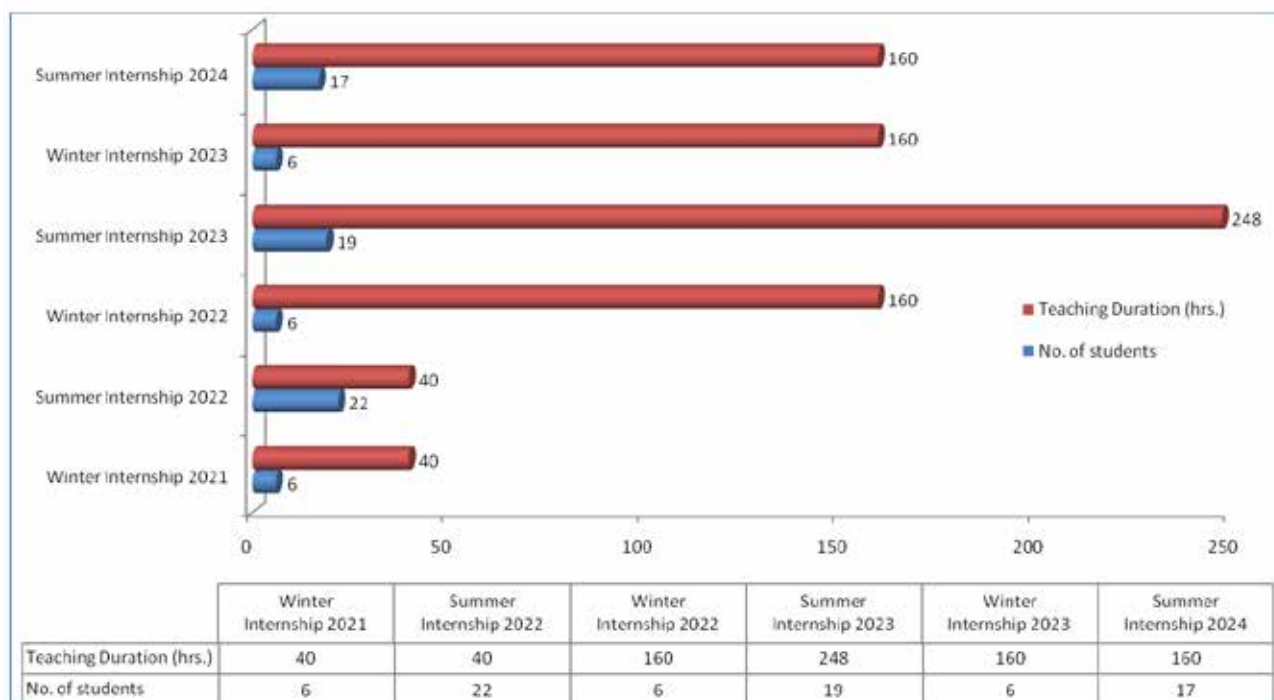


Figure 4.4: Glimpses of internship program organised at SDC, Bhubaneswar





4.9.1 In-house subsurface studies:

First time initiatives have been taken to form a **core report repository** at NDR. **During FY 2024–25, a total of 1,785 wells** were studied with the objective of compiling their core reports. Among these, core data was found to have been acquired for 879 wells. Of these 879 wells, core-related data, including petrophysical reports for 111 wells, are now available at the NDR.

4.9.1 Promoting E&P research in Indian academic institute/university:

As per NDR Policy, students of recognised universities and research institutes are eligible to obtain data up to 50 Sq. km of area free of charge for E&P related research & development purpose. In this financial year, students of IIT Kharagpur, IIT Bombay, IIT ISM, and Kurukshetra University received data free of charge for research work.

4.10 National Data Repository 2.0

The NDR has been operational since June 2017, so far successfully meeting the demands of all of its stakeholders such as investors, operators, service-providers, R&D institutes, academia, and government institutions. However, with the rapid transformation of digital technology and the emergence of cloud technology, along with advancements in E&P data management, DGH

has decided to upgrade present NDR to Next Generation cloud based National data repository. Next generation National Data Repository 2.0 (NDR 2.0) aims to substantially improve the data accessibility (downloading), data reporting, submission (uploading) and enhance storage (for all E&P data types including seismic field data) and create a cloud-enabled database platform. To fulfil this objective DGH has engaged project management consultant, Digital India Corporation (DIC), a company under Ministry of Electronics and Information Technology.

4.10.1 Salient features of NDR 2.0:

- i. Cloud based Primary Data Centre (PDC) and Disaster Recovery (DR) solution on Meity empanelled IT Cloud.
- ii. Enhanced disk storage based sub-system to facilitate efficient raw and raw/Pre-Stack data management and delivery.
- iii. New look GIS-based portal with dashboard having extended search capabilities and enhanced user experience on data preview, improved data searching, data visualization and quick-look analysis with custom settings.
- iv. Secured Self-service data loading with automated data QC/ validation and



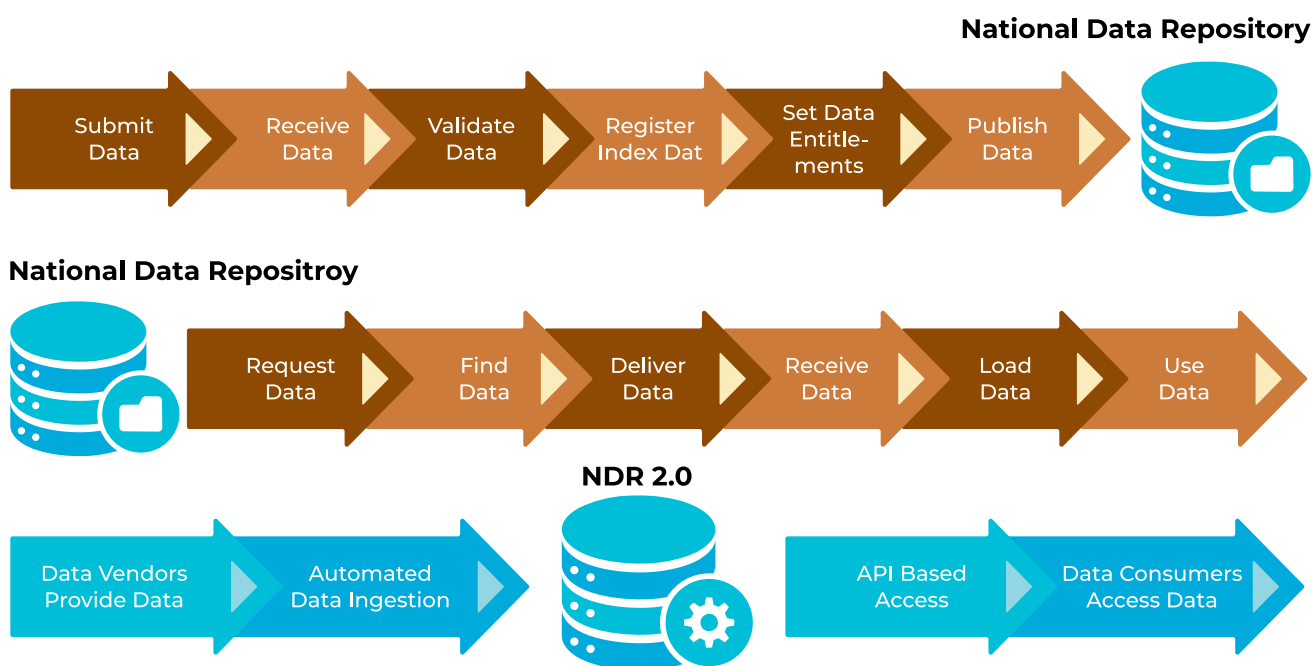
downloading for entitled users, will reduce time for data submission and delivery through contactless process.

- v. Online data ordering, payment through web based online portal and online delivery of stack seismic data, well data and reports up to certain volume.

- vi. 24x7 Virtual Data Room - Online/anywhere/ anytime data visualization and interpretation facility.

After implementation of NDR 2.0 the data submitted by the operator as well as data dissemination to the interested companies will be much simpler with integration of AI/ML technology (Figure.4).

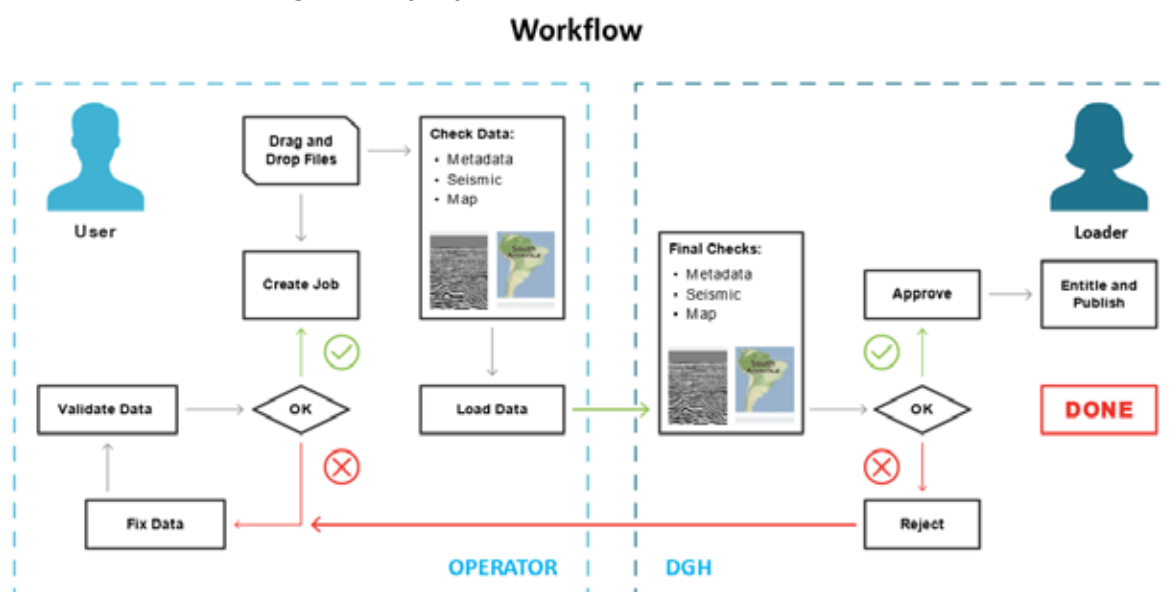
Figure 4.5: Comparison of Data Assimilation & Dissemination in Present NDR1.0 & NDR 2.0



Manual intervention will be minimal except some key points and the total process will be much faster. The Figure 5. Below shows the

envisaged process for the data transfer and dissemination as envisaged for NDR2.0.

Figure 4.6: Workflow diagram of proposed NDR 2.0



4.10.2 SEISMIC SURVEY AND DATA ACQUISITION:

Mission Anveshan:

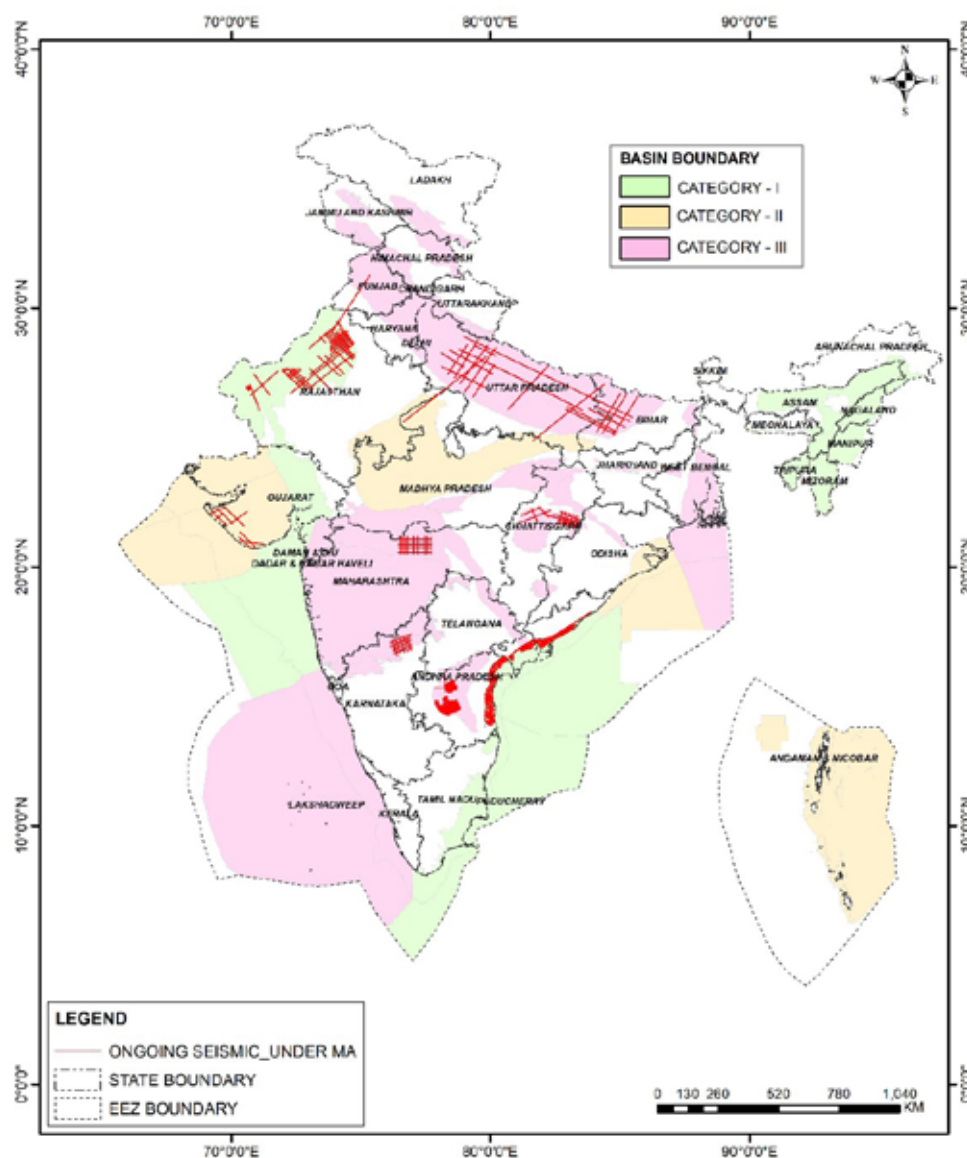
Mission Anveshan has been launched in continuity with the National Seismic Program (NSP) aiming at the hydrocarbon resources in India's onshore basins. Mission Anveshan is a scheme of 2D Seismic Data Acquisition, Processing, and Interpretation (API) of **20,275 LKM** spread over seven onland Indian sedimentary basins, namely **Rajasthan, Ganga-Punjab, Cuddapah, Chhattisgarh, Krishna-Godavari, Deccan Syneclise, and Saurashtra**, with an outlay of INR 792 crores.

The objective of Mission Anveshan is a

comprehensive appraisal of the identified sedimentary basins based on leads generated during the interpretation of National Seismic Program (NSP) data. Under Mission Anveshan, Oil India Limited (OIL) started seismic data acquisition in Rajasthan and Ganga-Punjab basin on 29th October 2024. In Cuddapah, Krishna-Godavari and Deccan Syneclise Basin seismic data acquisition started in January - March 2025 period by ONGC Ltd. The seismic data acquisition in other basins will be taken subsequently.

Till **31st March 2025**, a total of approximately **2,954 LKM** (ONGC: 350.920 LKM and OIL: 2,602.920 LKM) data have been acquired. The project is expected to be completed during FY 2025-26.

Figure 4.7: Seismic profiles under Mission Anveshan Survey







5

India's Hydrocarbon Reserves & Resources

India's energy demand is projected to grow significantly in the coming decades, potentially at a rate higher than the global average. With crude oil import dependence exceeding 88% in 2024–25, the need to bolster domestic energy production is both urgent and strategic.

To address this, India must target a quantum jump in domestic crude oil output by:

1. Adding incremental production through expansion of improved and enhanced oil recovery (IOR/EOR) projects to newer areas/fields.
2. Adding incremental water injection volume to target voidage replacement ratio of 1.2 in mature fields through conversion of oil wells to injectors as well as drilling of new injection wells if required,
3. Optimizing artificial lift performance through improved design, mode change and uptime as necessary.
4. Optimizing well spacing through drilling of more infill/replacement wells (producers and injectors) in area/fields as necessary.
5. Reviving non-flowing or marginal wells in a time-bound manner to increase near-term production.
6. Use of new technology in selected wells/fields with a potential to increase production/injection rate per well such as drilling of a few large RAC (Reservoir Area Contact) wells.
7. Grant of PML extension to operators should be based on the concrete efforts by the operator to achieve committed production



Additionally, contingent resources-those that are currently non-commercial or technically constrained-should be prioritized. Unlocking these resources will require:

- ▶ Swift resolution of contingencies such as infrastructural, technological, or regulatory barriers.
- ▶ Adoption of alternate development strategies, tailored to field-specific conditions, to expedite commercial production.

India also possesses significant undiscovered or yet-to-find (YTF) in-place volumes, estimated at approximately 13 billion tonnes of oil equivalent. These prospective resources represent a substantial opportunity and call for an aggressive, targeted exploration campaign, focusing on:

- ▶ Established basins with proven hydrocarbon systems.
- ▶ Underexplored and frontier areas with potential for new discoveries.
- ▶ Innovative exploration techniques and advanced geoscientific studies to de-risk investments and identify new plays.
- ▶ Basin wise analysis of success rate (exploratory expenditure vs addition of reserves in last 25 years/since inception) for optimal allocation of resources to more prolific basins and other frontier areas which are largely unexplored.
- ▶ Collaboration with International E&P companies with proven track record to explore difficult and challenging frontier areas through pragmatic approach by clubbing of mature oil fields with low recovery and other incentives as sweetener.

5.1 Implementation of PRMS for Standardized Hydrocarbon Resource Reporting in India

To ensure consistency, transparency, and international alignment in hydrocarbon resource estimation, the Directorate General of Hydrocarbons (DGH) has initiated a periodic update of the country-level database on oil and gas reserves and resources. This update is conducted in accordance with the **Petroleum Resources Management System (PRMS)** standard, and applies across all operating regimes-both Nomination and Contract.

Effective from April 1, 2021, all Indian Exploration & Production (E&P) operators were mandated to adopt and adhere to the PRMS framework for reporting their company-level resource and reserve portfolios. PRMS is a globally recognized standard developed to ensure the consistent definition, classification, and estimation of hydrocarbon resources. The latest revision of PRMS, completed by the Society of Petroleum Engineers (SPE) Oil and Gas Reserves Committee (OGRC), was approved in June 2018 by the SPE Board and co-sponsored by six leading industry organizations:

- ▶ World Petroleum Council (WPC)
- ▶ American Association of Petroleum Geologists (AAPG)
- ▶ Society of Petroleum Evaluation Engineers (SPEE)
- ▶ Society of Exploration Geophysicists (SEG)
- ▶ European Association of Geoscientists and Engineers (EAGE)
- ▶ Society of Petrophysicists and Well Log Analysts (SPWLA)

Under this mandate, all E&P operators are required to undertake estimation, classification, categorization, auditing, and reporting of hydrocarbon reserves and resources in accordance with PRMS standards. Reports, along with supporting technical data, must be submitted annually to the DGH as of



Furthermore, all major technical submissions to the DGH-such as Declarations of Commerciality (DOCs), Field Development Plans (FDPs), and Revised FDPs-must be accompanied by a statement of resources and reserves, reflecting the project's maturity and the probability of commercial success, in line with PRMS classifications.

In this framework, DGH maintains discovered Petroleum Initially-In-Place (PIIP) data under the following categories:

- ▶ Production (Cumulative)
- ▶ Commercial PIIP, with corresponding **Reserves**
- ▶ Sub-commercial PIIP, with corresponding Contingent **Resources**

This structured and transparent approach ensures better resource management, regulatory oversight, and investor confidence, contributing significantly to India's long-term energy security.

Figure 5.1: Sub classes based on project maturity

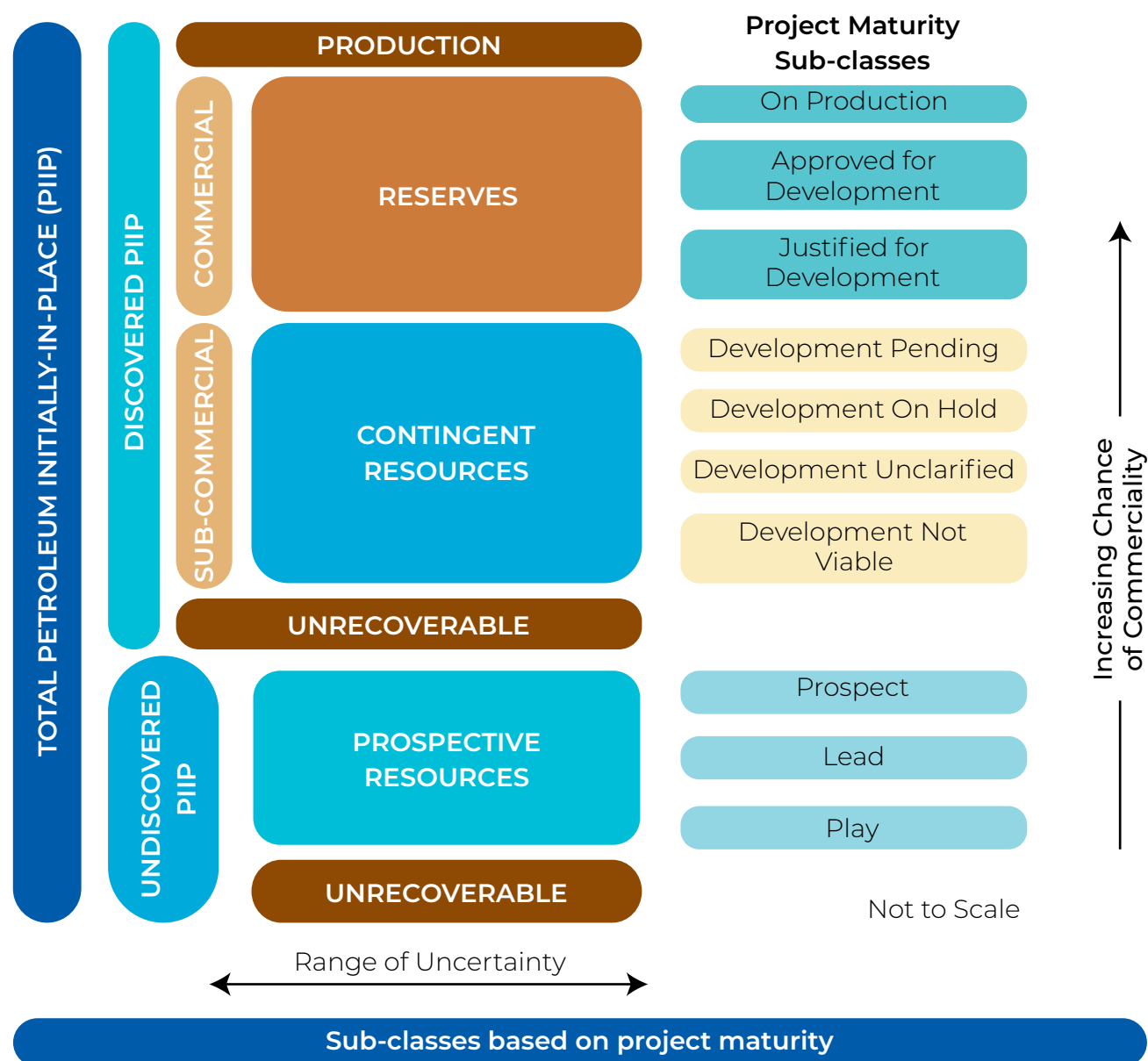
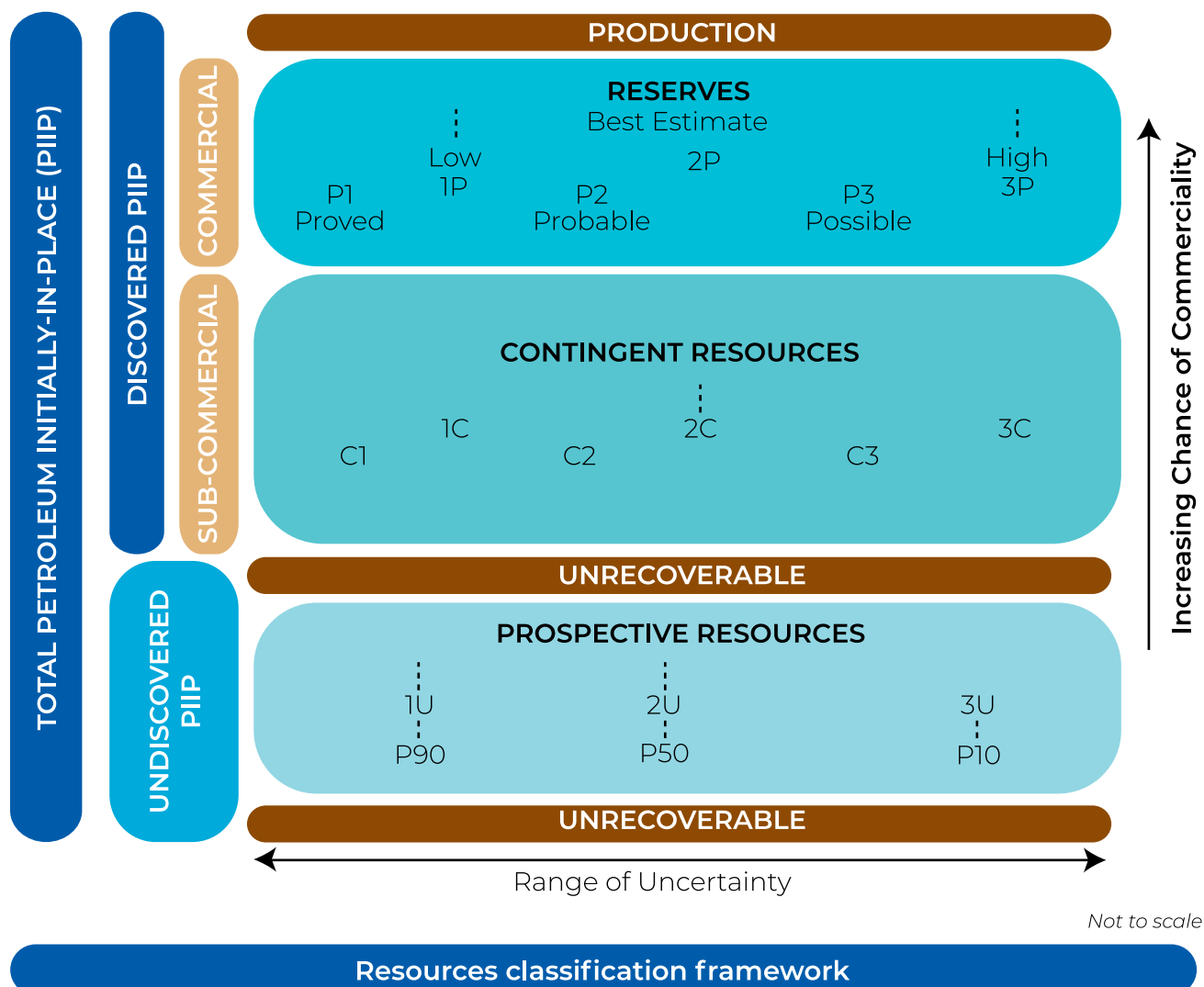


Figure 5.2: Resources classification framework



5.2 India's Hydrocarbon Reserves and Resources – Facts & Figures

5.2.1 Overview

As of 1st April 2024, Indian Exploration and Production (E&P) operators have reported an estimated 671 Million Metric Tonnes (MMT) of recoverable oil and 1,094 Billion Cubic Metres (BCM) of recoverable gas. These figures represent the sum of 2P reserves (Proved + Probable) and 2C contingent resources at the national level, in alignment with PRMS standards. As of 1st April 2024, the total 2P reserves at the national level are estimated at **434 Million Metric Tonnes (MMT)** of oil and **643 Billion Cubic Metres (BCM)** of natural gas, while the total 2C contingent resources stand at **237 MMT** of oil and **451 BCM**

of gas. India's upstream sector operates under five distinct regulatory regimes:

- ▶ Nomination
- ▶ Production Sharing Contract (PSC)
- ▶ Open Acreage Licensing Policy (OALP)
- ▶ Discovered Small Fields (DSF)
- ▶ Coal Bed Methane (CBM)

These regimes support both conventional and unconventional hydrocarbon resource development and production.

Fields under the Nomination regime, operated by Oil and Natural Gas Corporation Ltd. (ONGC) and Oil India Limited (OIL), contribute significantly to



India's resource base:

- ▶ Account for 70% of total recoverable oil and 61% of total recoverable gas (2P + 2C).
- ▶ Hold 83% of the total 2P oil reserves (434 MMT) and 58% of the 2P gas reserves (643 BCM).
- ▶ Comprise 47% of 2C oil resources (237 MMT) and 64% of 2C gas resources (451 BCM).

Nomination acreage also contributes substantially to assets under other regimes, including PSC.

The CBM regime, a major contributor to unconventional gas development in India, holds a 9% share of the country's 2P gas reserves.

5.2.2 Reserve Reporting Framework

- ▶ For Nomination fields, 2P reserves are reported based on the economic life of the field.
- ▶ In contrast, for PSC, DSF, and OALP regimes, reserves are reported up to the end of the contractual term (PSC/RSC duration).

5.2.3 Development Potential

Despite a long production history, fields under the Nomination regime still exhibit significant potential:

- ▶ The Estimated Ultimate Recovery (EUR) for oil and condensate is only 27% of the in-place volume.
- ▶ The recovery factor stands at 22% as of 1st April 2024.

This presents a strong case for National Oil Companies (NOCs) to revisit and revitalize major oil fields. Strategic actions may include:

- ▶ Aggressive Infill drilling campaigns for injectors and producers to reduce well spacing
- ▶ Optimization of water injection systems through continuous monitoring and surprise visit of injection wells and plants to ensure that injection water quality has not been compromised and injection on time has been close to 1.0

The Directorate General of Hydrocarbons (DGH) expresses its full support and readiness to collaborate in this mission to maximize

production from Nomination fields while achieving VRR of 1.2.

5.2.4 Prioritizing Contingent Resources (2C)

India holds a substantial volume of 2C contingent resources, which must be prioritized for conversion to reserves. To achieve this, the following steps are recommended:

- ▶ Address technical and regulatory contingencies hindering commercial development.
- ▶ Focus on new and redevelopment projects with defined implementation timelines.
- ▶ Commercially implement Enhanced Oil Recovery (EOR) methods at full field scale, especially those already proven effective at the pilot stage.

5.2.5 Role of DGH

As the nodal agency for overseeing petroleum reserves and resource management in India, DGH maintains continuous engagement with E&P operators to:

- ▶ Facilitate the adoption of international best practices in reservoir management.
- ▶ Encourage data-driven and performance-oriented field strategies to enhance recovery.

Note: In above context, EUR (Estimated Ultimate Recovery) refers to the commercially recoverable Petroleum Initially-In-Place (PIIP), i.e., the sum of cumulative production and remaining reserves.



Table 5.1: India's Hydrocarbon Reserves status as of 01.04.2024

Regime	2P In-Place		2P EUR		Accretion during 2023-2024		Reserves (2P)		Contingent Resources (2C)	
	Oil+ Cond. (MMT)	Gas (BCM)	Oil+ Cond. (MMT)	Gas (BCM)	Oil+ Cond. (MMT)	Gas (BCM)	Oil+ Cond. (MMT)	Gas (BCM)	Oil+ Cond. (MMT)	Gas (BCM)
Nomination (ONGC & OIL)	6209.4	2761.8	1697.6	1421.0	9.3	10.7	361.9	374.8	110.2	289.4
PSC	956.1	770.8	229.9	346.9	3.8	19.8	67.1	187.4	121.1	106.7
DSF	63.8	91.6	5.4	24.5	1.2	6.3	5.3	24.1	5.0	19.1
OALP	6.2	34.1	0.0	0.0	0.0	0.0	0.0	0.0	0.9	20.1
CBM	0.0	342.7	0.0	63.7	0.0	-1.2	0.0	57.2	0.0	15.4
Grand Total	7235.4	4001.1	1933.0	1856.1	14.2	35.6	434.3	643.4	237.1	450.8

Note: As per data received from NOCs and Private/JV Companies; (-ve) accretion denotes revision in reserves/resources after adoption of PRMS standard. PRMS data as on 01.04.2025 is under preparation.

Figure 5.3: Regime-wise Petroleum In-place, EUR, 2P Reserves, 2C Resources (Oil + Condensate, in MMT) as of 01.04.2024

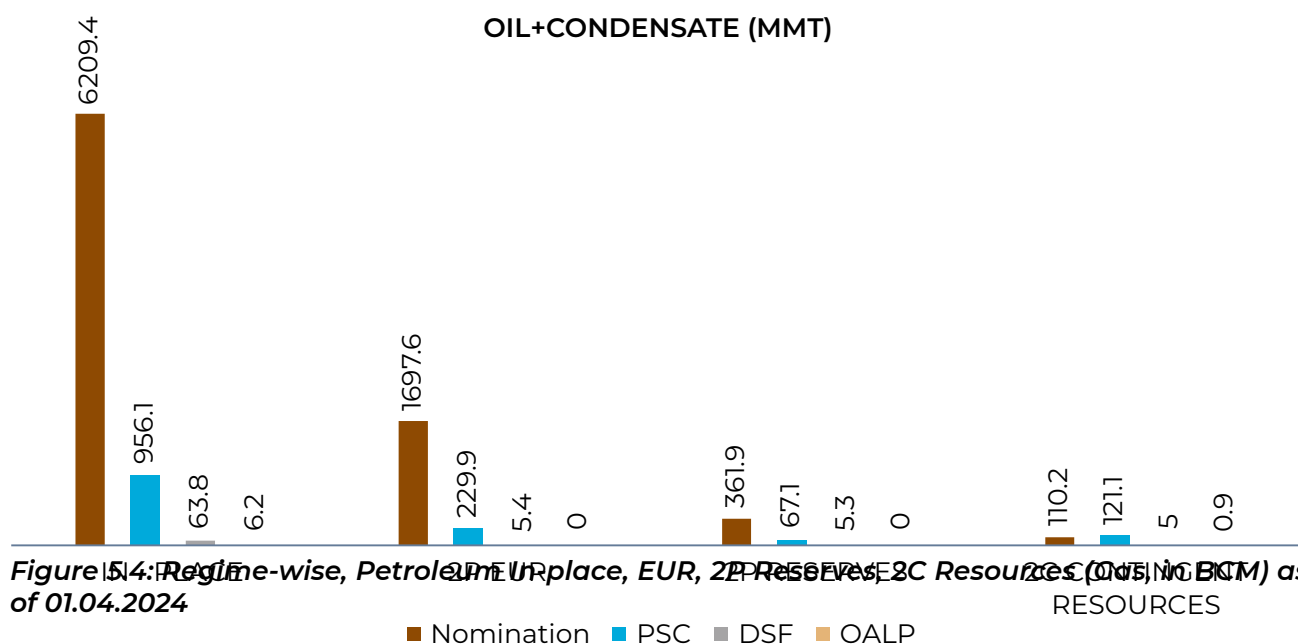


Figure 5.4: Regime-wise, Petroleum In-place, EUR, 2P Reserves, 2C Resources (Oil + Condensate, in BCM) as of 01.04.2024



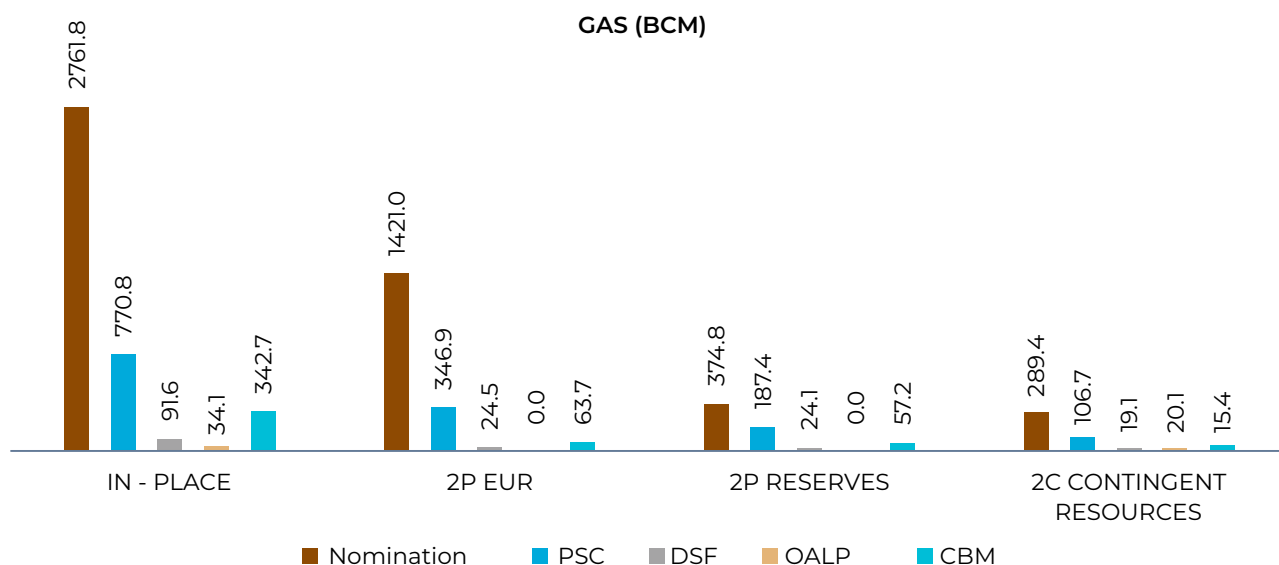


Table 5.2: Oil and Condensate (MMT) In-place-Reserves-Resources across Location/State as of 01.04.2024

Location/State	In-Place	2P EUR	2P Reserves	2C Contingent Resources
ONLAND	3,617.8	911.8	231.6	185.9
ARUNACHAL PRADESH	44.8	5.2	2.3	0.6
ASSAM	1,346.5	375.8	104.1	41.3
GUJARAT	1,504.3	380.5	92.6	26.3
MADHYA PRADESH	0.0	0.0	0.0	0.0
RAJASTHAN	532.5	122.0	23.4	108.1
WEST BENGAL	1.0	0.02	0.02	0.1
MIZORAM	0.0	0.00	0.0	0.0
NAGALAND	12.5	1.0	0.0	2.4
TRIPURA	0.0	0.1	0.1	0.0
TAMILNADU	102.9	17.1	5.8	2.7
ANDHRA PRADESH	73.2	9.9	3.3	4.4
JHARKHAND	0.0	0.0	0.0	0.0
OFFSHORE	3,617.7	1,021.2	202.7	51.2
EASTERN OFFSHORE	269.3	82.8	32.2	8.4
WESTERN OFFSHORE	3348.3	938.4	170.5	42.8
Grand Total	7,235.4	1,933.0	434.3	237.1

Figure 5.5: Location-wise Oil and Condensate (MMT) In-place-Reserves-Resources across as of 01.04.2024



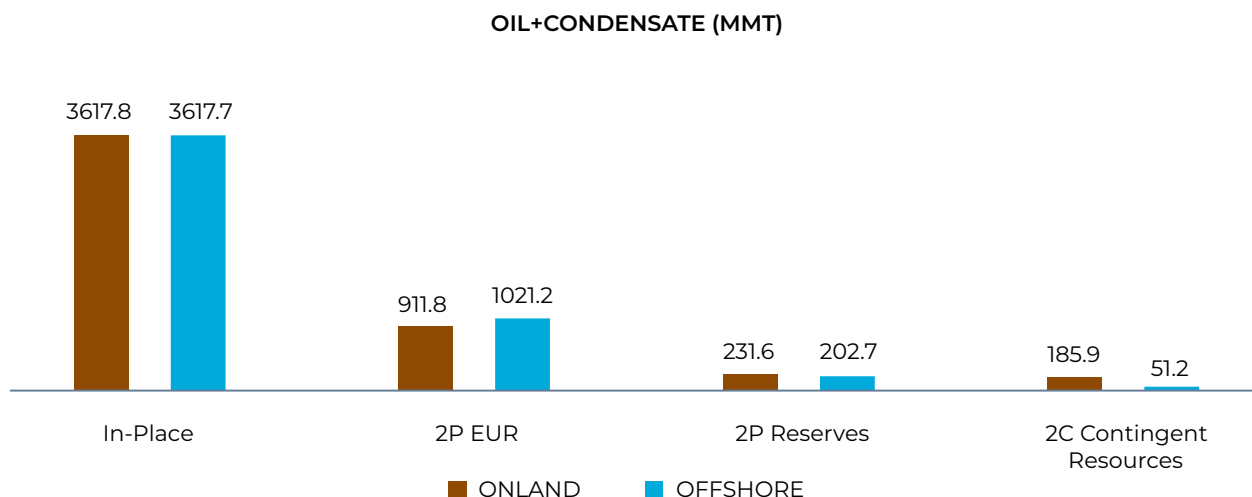


Figure 5.6: State-wise Onland Oil and Condensate (MMT) 2P Reserves + 2P Resources as of 01.04.2024

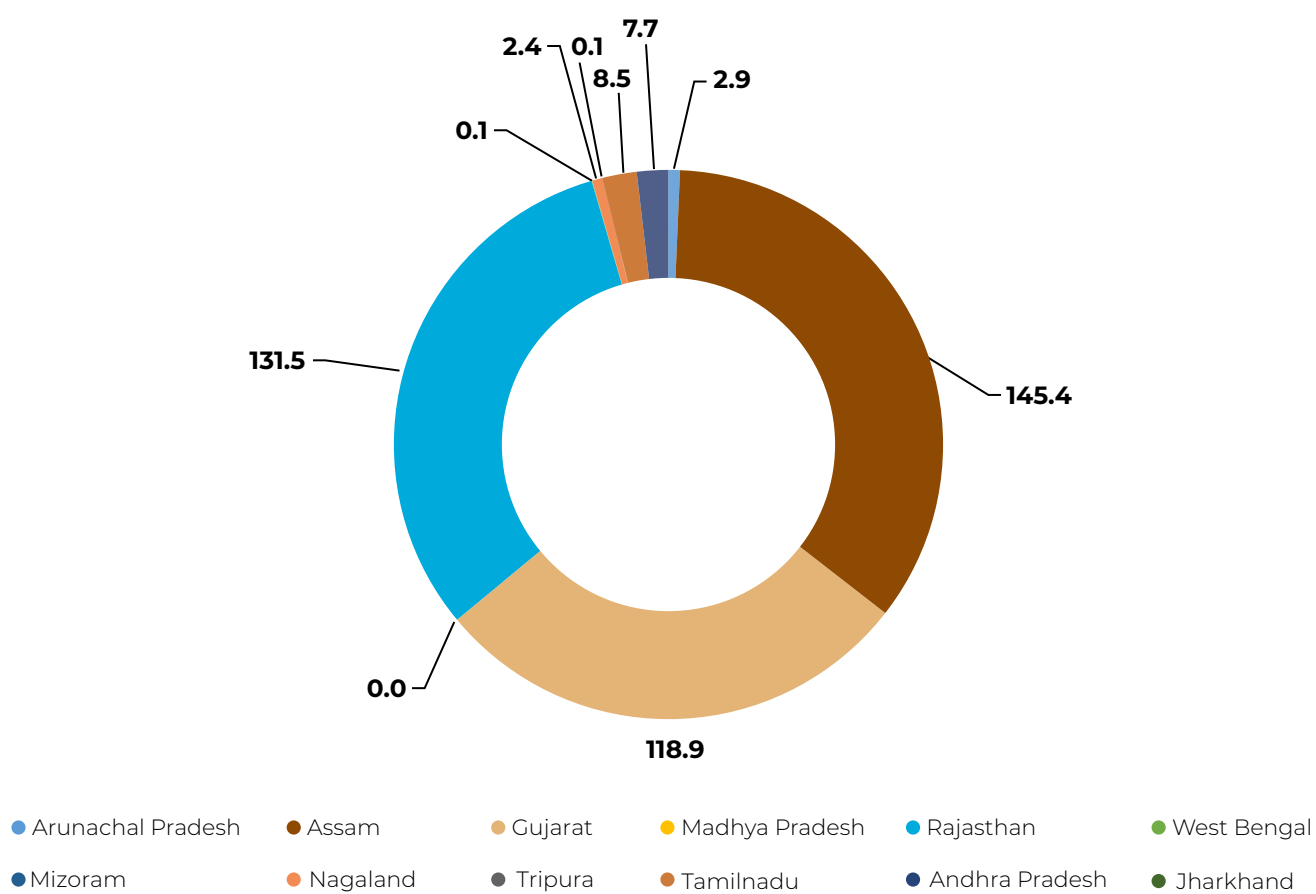


Table 5.3: Location-wise Gas (BCM) In-place-Reserves-Resources as of 01.04.2024

Location/State	2P In-Place	2P EUR	2P Reserves	2C Contingent Resources
ONLAND	1,778.7	700.9	277.8	210.7
ARUNACHAL PRADESH	18.0	4.2	0.00	7.3
ASSAM	535.9	277.3	73.9	90.6
GUJARAT	274.5	135.9	36.5	19.0
MADHYA PRADESH	107.1	25.8	23.7	0.6
RAJASTHAN	166.3	73.4	49.5	14.1
WEST BENGAL	181.5	34.0	29.6	14.8
MIZORAM	0.0	0.0	0.0	0.00
NAGALAND	1.0	0.2	0.0	0.1
TRIPURA	97.3	40.5	16.5	11.7
TAMILNADU	161.7	47.4	23.2	13.3
ANDHRA PRADESH	177.1	58.0	20.8	38.5
JHARKHAND	58.3	4.2	4.1	0.7
OFFSHORE	2,222.4	1,155.3	365.6	240.0
EASTERN OFFSHORE	567.9	261.8	136.0	127.3
WESTERN OFFSHORE	1,654.5	893.5	229.6	112.7
Grand Total	4,001.1	1,856.1	643.4	450.8

Note: As per data received from NOCs and Private/JV Companies

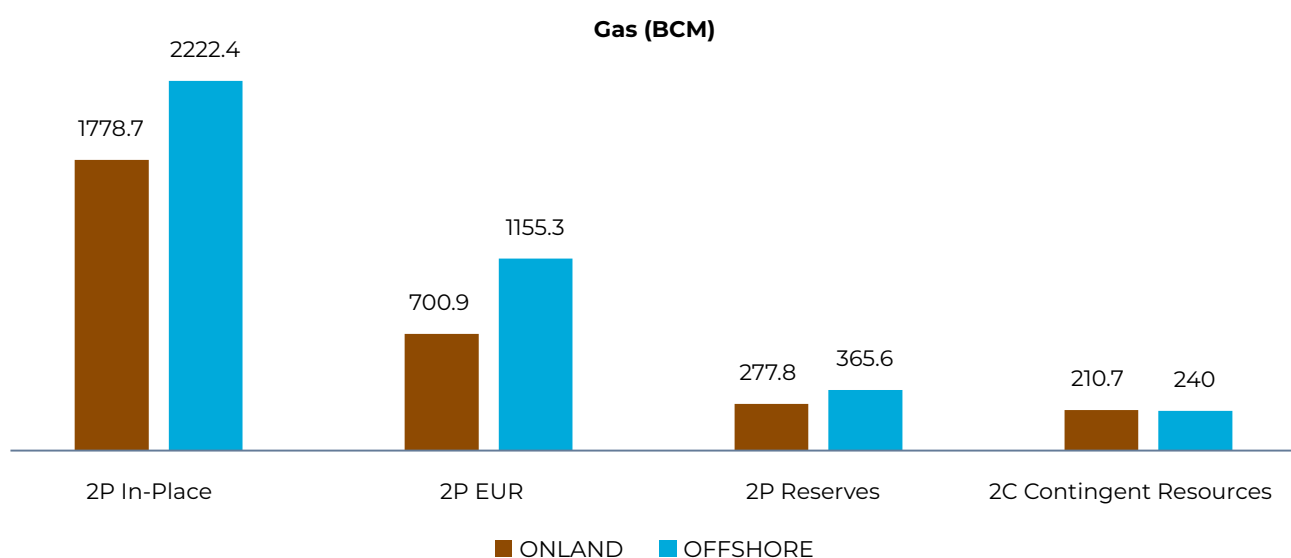
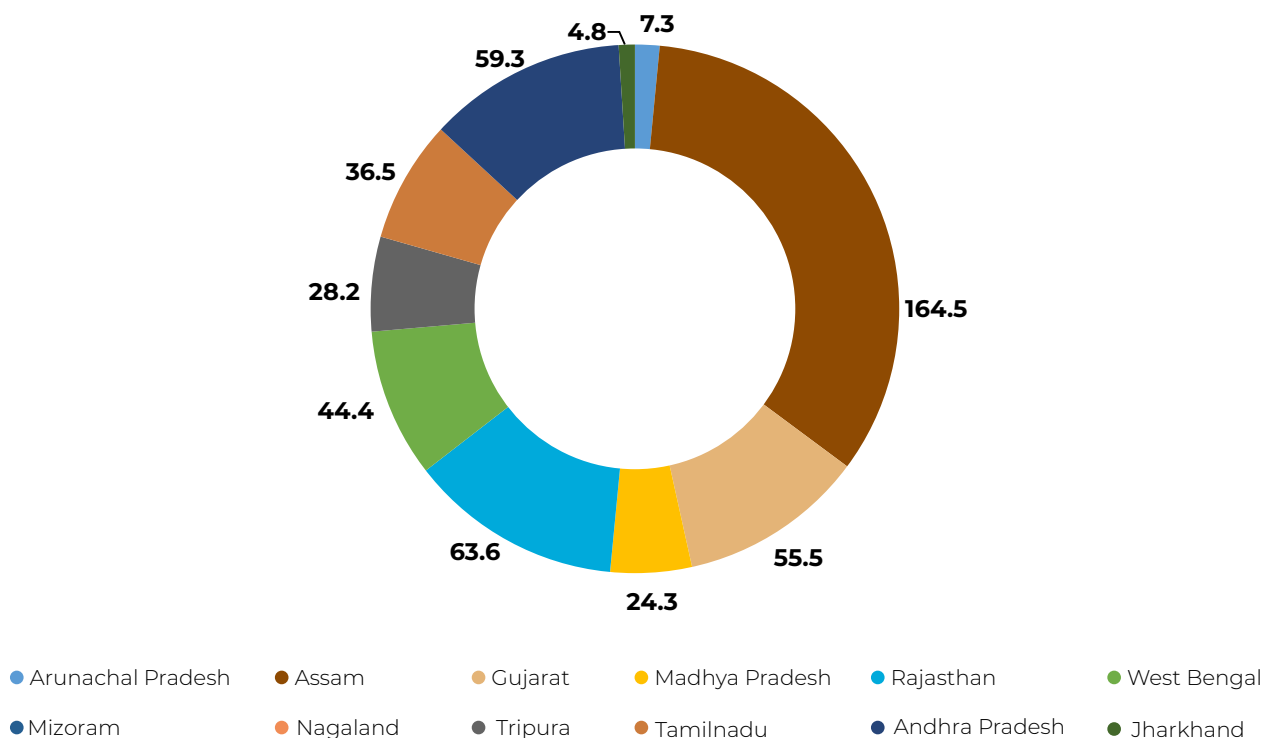
Figure 5.7: Location-wise Gas (BCM) In-place-Reserves-Resources across country as of 01.04.2024

Figure 5.8: State-wise Onland Gas (BCM) 2P Reserves + 2P Resources across country as of 01.04.2024



5.2.6 India's Reserve Replacement Ratio (RRR) & Reserves to Production ratio (R/P)

Reserves Replacement Ratio (RRR) for a particular year in following manner:

$$\text{RRR} = \frac{\text{Reserves accreted}}{\text{total production}} \text{ during a FY}$$

Change in EUR (Production+ reserves) denotes reserves accretion (+ve/-ve) during a financial year.

Country's RRR FY 2024-25	
Oil + Condensate	Gas
0.51	0.99

The Reserves-to-Production ratio (R/P) is an estimate of the number of years oil and gas reserves will continue to produce based on current production rates. It does not account for future reserve accretion; hence it tends to increase with the time.

$$\text{R/P} = \frac{\text{total reserves}}{\text{total production during a FY}} \text{ years}$$

Country's R/P FY 2023-24	
Oil + Condensate	Gas
16 years	18 years

5.2.7 Performance review of Nomination blocks

The Nomination regime continues to play a pivotal role in India's energy landscape,

contributing to over 80% of the nation's 2P oil reserves and more than 60% of 2P gas reserves. Recognizing its strategic importance, the Directorate General of Hydrocarbons (DGH) has embarked on a focused initiative to evaluate and



enhance the performance of these legacy assets.

As part of this initiative, a set of priority nomination blocks-operated by major national oil companies-has been shortlisted for detailed performance review. This comprehensive assessment will aim to:

- ▶ Identify underperforming assets and untapped potential
- ▶ Recommend strategies for production enhancement and recovery improvement
- ▶ Promote best-in-class reservoir management practices

The review process is scheduled to be undertaken in the upcoming financial year and will lay the groundwork for actionable interventions to maximize production and extend the economic life of India's key oil and gas fields.

5.2.8 Benchmarking of Indian Oil & Gas fields

In the context of mature fields with declining production, there is an increasing need to assess and improve field performance across India's upstream sector. As part of this effort, the Directorate General of Hydrocarbons (DGH) is initiating steps to establish cut-off values for recovery factors, which can serve as performance benchmarks for Indian oil and gas fields.

To advance this initiative, DGH has already commenced deliberations with leading technical service providers and domain experts. The objective is to develop standardized recovery benchmarks based on:

- ▶ Field maturity and reservoir characteristics

- ▶ Global best practices and comparable analogues
- ▶ Historical production trends and in-place resource estimates

These benchmarks will serve as critical reference points for evaluating underperforming fields and identifying areas where enhanced recovery techniques or development re-planning could be applied

5.2.9 Procurement of Reservoir Simulation Software

To strengthen its technical capabilities and support state-of-the-art evaluation of Field Development Plans (FDPs), the Directorate General of Hydrocarbons (DGH) has established a dedicated Reservoir Engineering (RE) workstation.

As part of this upgrade, DGH has procured the Computer Modelling Group (CMG) suite of reservoir simulation software, a globally recognized platform for advanced reservoir modelling and simulation. This acquisition is aimed at enhancing the technical rigor and efficiency in evaluating FDPs submitted by E&P operators, particularly for:

- ▶ Reservoir performance forecasting
- ▶ Enhanced oil recovery (EOR) modelling
- ▶ Complex well and fluid behavior simulations

The integration of CMG software into DGH's technical workflow marks a significant step toward modernizing resource evaluation practices and ensuring more informed and data-driven decision-making.







6

New Investment Opportunities in E&P Sector

6.1 Introduction

Since 1999, the New Exploration Licensing Policy (NELP) had governed oil and gas exploration and production. Its cornerstone, the Production Sharing Contracts (PSCs), Operated on a profit-sharing basis, necessitating extensive governmental micromanagement for cost control and revenue maximization. However, administering these contracts proved problematic, spawning disputes and arbitrations with the contractors. Issues encompassed cost recovery limits, procurement challenges, investment multiples calculation methodology, and a lack of operator incentives for cost containment, all detrimentally affecting profit petroleum. Further, separate policies and fiscal terms were formulated to administer various hydrocarbon types, including CBM, shale oil/gas, and conventional hydrocarbons, fostering inefficiencies in resource exploitation. Under NELP, exploration was confined to government-offered blocks, constraining access to potentially lucrative opportunities in other unexplored areas. To tackle these systemic and operational shortcomings, the government introduced a Revenue Sharing Contract Regime, heralding significant policy overhauls.

Two major policies were formulated to implement the Revenue Sharing Contract Mechanism are Discovered Small Field Policy & Open Acreage Licence Policy.

6.2 Discovered Small Field Policy (DSF)

The Discovered Small Field (DSF) Policy was introduced by the Government of India in 2015. The Cabinet Committee on Economic Affairs (CCEA) gave its nod to the policy on September 2, 2015, followed by formal notification on October



14, 2015. DSF Policy falls under the broader framework of the Hydrocarbon Exploration and Licensing Policy (HELP).

DSF Policy introduced a transparent, investor-friendly, and globally benchmarked mechanism to offer discovered but unmonetized oil and gas fields to domestic and international players with the primary objective to monetize these already discovered hydrocarbon fields by offering them under a more investor-friendly, low-risk, and transparent policy framework and reduce the import dependency.

The DSF Policy represents a significant step in India's energy strategy, aiming to leverage its domestic hydrocarbon resources more effectively, reduce import dependency, promote small and Private Operators, and foster investment and technology transfer in the energy sector. The DSF Bid Round-I, launched on May 25, 2016, marked the onset of a new era in the country's oil and gas sector.

Following the successful implementation of DSF Bid Round-I, the Cabinet Committee on Economic Affairs (CCEA) in its meeting dated February 7, 2018, approved extension of this Policy to include un-monetized discoveries/ small fields from Production Sharing Contract (PSC) regime along with un-monetized discoveries of Nomination regime. Government also decided to extend the DSF Policy for future bidding rounds. For this purpose, a committee comprising of Additional Secretary / Additional Secretary & Financial Advisor (MoPNG), Joint Secretary (Exploration, MoPNG) and Director General of Hydrocarbons will identify Fields / Discoveries to be put on offer in consultation with CMD (ONGC) and CMD (OIL) for future bidding round. The DSF Policy extension was notified vide Notification No. O-31018/149/2016-ONG-III dated April 5, 2018.

6.3 Salient Fiscal and Policy Features of DSF Policy

- ▶ **100% participation allowed from Foreign Companies/JV:** The DSF policy permits full ownership and participation by foreign entities and joint ventures, encouraging global investment and expertise in India's E&P sector.
- ▶ **No Oil Cess:** Crude oil produced under DSF contracts is exempt from oil cess, reducing the overall fiscal burden on contractors and enhancing project viability.
- ▶ **Moderate Royalty Rates:** The policy set moderate royalty rates, which are fees paid to the government for the right to extract hydrocarbons. This balanced approach helps to attract investors while ensuring government revenue.
- ▶ **Exemption from Custom Duty:** Operators are granted custom duty exemptions on imported equipment and materials, significantly reducing capital costs during exploration and development.
- ▶ **No Upfront Signature Bonus:** Unlike traditional bidding processes, these contracts do not require an upfront signature bonus, reducing the financial burden on operators during the initial stages of exploration and production.

DSF Timelines

#6

SDSF Launched

May 28, 2024

#5

DSF-III Launched

June 10, 2021

#4

DSF-II Launched

August 09, 2018

#3

DSF-I Launched

May 25, 2016

#2

Formal Notification

October 14, 2015

#1

CCEA Approval

September 2, 2015





- ▶ **The policy provides pricing and marketing:** Operators have pricing and marketing freedom for oil and gas, allowing them to determine the pricing of their products and explore various marketing strategies.
- ▶ **No Carried Interest by NOCs:** The policy does not require National Oil Companies (NOCs) to hold a carried interest in the projects, providing more autonomy to contractors.
- ▶ **Single PML License:** The Policy offers a single Petroleum Mining Lease (PML) license that covers both conventional and non-conventional hydrocarbons, simplifying administrative processes for operators.
- ▶ **Flexibility in Exploration Activities:** During the contract period, there are no restrictions on exploration activities, allowing operators to adapt their strategies based on evolving conditions and information.
- ▶ **No prior experience:** No prior technical experience is required for bidding, thus enabling wider participation, including start-ups and new entrants in the E&P domain.

6.4 OVERVIEW OF DSF BID ROUNDS

Up to FY 2024-25, the Government of India conducted three DSF bid rounds. Also, Special DSF Bid Round (SDSFR 2024) was launched on May 28, 2024 for which bidding process is over and contract would be awarded shortly. Further, a DSF Bid Round-IV is soon to be launched in early FY 2025-26. All bid rounds utilize a transparent online portal, offering bidders and investors strong support for a seamless experience.

Together, the earlier bid rounds including the SDSFR 2024 offered 105 Contract Areas with 204 hydrocarbon discoveries across 10 sedimentary basins (7 Category-I and 3 Category-II), covering around 17,593 sq. km.

Out of these, 85 Contract Areas have been awarded to 35 Indian and foreign companies, spanning 16,508 sq. km under DSF Bid Rounds-I, II and III. SDSR 2024 will be awarded in early FY 2025-26.

These awarded blocks include 175 discoveries, with an estimated 464 Million Metric Tonnes of oil equivalent (MMtoe) in-place reserves.

Table 6.1: Summary of DSF Contract Areas till FY 2024-25

Sl. No.	DESCRIPTION	DSF-I	DSF-II	DSF-III	SDSFR 2024	Total
1	Contract Areas Offered	46	25	32	2	105
2	Discoveries on Offer	67	59	75	3	204
3	Contract Areas Awarded	30	24	31	To be awarded in April 2025	85
4	Discoveries Awarded	43	58	74		175
5	Area Awarded (sq. km)	777	3,004	12,727		16,508
6	In-place Reserves Awarded (MMtoe)	45	189	230		464
7	New Entrants (Companies)					29



6.5 DISCOVERED SMALL FIELD BID ROUND-I (2016)

To reduce India's dependence on imported oil and gas, the Government of India launched the Discovered Small Field (DSF) Bid Round-I on May 25, 2016, under the vision of the Hon'ble Prime Minister.

In this round, 46 Contract Areas with 67 discoveries were offered across 9 sedimentary basins, covering ~1,551 sq. km with an estimated 86 Million Metric Tonnes (MMtoe) of in-place reserves.

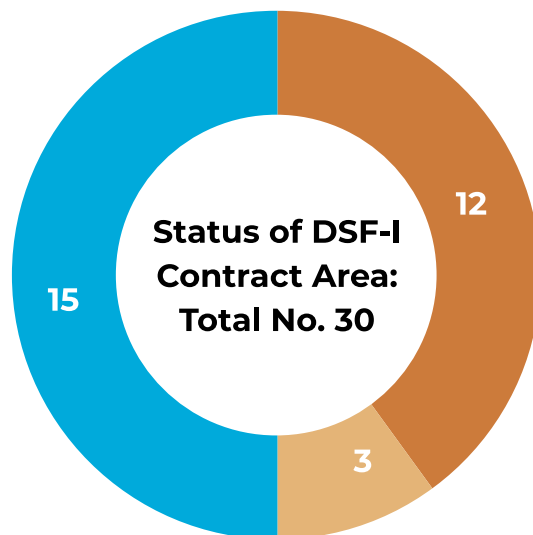
As a result, 30 Contract Areas were awarded - 23 onland and 7 shallow offshore, covering 777 sq. km with 44.66 MMtoe of reserves.

The Revenue Sharing Contracts for these blocks were signed on March 27, 2017, just 10 months after the launch.

6.5.1 Key Highlights of DSF Bid Round-I are as outlined below:

- ▶ A total of 134 bids were received for 34 contract areas

- ▶ Out of the 22 shortlisted companies, 15 were first-time entrants into the Indian E&P sector.
- ▶ Ultimately, 20 companies (either solo or in consortium) signed RSCs, including 13 companies new to the upstream industry in India.



● Active ● PML Awaited ● Relinquished/Terminated

Table 6.2: Status of Contract Areas awarded under DSF Bid Round-I

Sl. No.	Basin		Basin Category	Contract Areas awarded under DSF Bid Round-I			
				Active	PML Awaited	Terminated/elinquished	Total
1	Assam Shelf		Cat-I	2	0	5	7
2	Assam- Arakan fold belt	0		0	2	2	
3	Cambay	5		0	0	5	
4	Cauvery	0		2	0	2	
5	Krishna Godavari	3		1	1	5	
6	Mumbai Offshore	2		0	3	5	
7	Rajasthan	0		0	2	2	
8	Kutch Offshore		Cat-II	0	0	1	1
9	Vindhyan			0	0	1	1
Grand Total				12	3	15	30



6.6 DISCOVERED SMALL FIELD BID ROUND-II (2018)

Building upon the momentum of the DSF Bid Round-I, the Government of India launched DSF Bid Round-II on August 9, 2018, with renewed focus and enhanced investor-centric features. The round followed the formal extension of the DSF Policy of 2015 for future rounds in April 2018, which retained the fiscal incentives of the original policy while reducing the royalty rates in alignment with the Hydrocarbon Exploration and Licensing Policy (HELP).

In this round, the Government offered 25 Contract Areas encompassing 59 discoveries spread across seven sedimentary basins, covering an expansive 3,042 sq. km. These blocks held an estimated 190 Million Metric Tonnes (O+OEG) of in-place hydrocarbon reserves.

Following evaluation, 24 Contract Areas were awarded, including 15 onshore and 9 shallow offshore blocks with 58 discoveries, covering approximately 3,004 sq. km and holding 188.9 MMtoe of in-place reserves.

The Revenue Sharing Contracts (RSCs) for 23 Contract Areas were signed on March 7, 2019, with one additional RSC signed in January 2021.

6.6.1 Key Highlights of DSF Bid Round-II are as outlined below:

- ▶ A total of 145 bids were received for 24 contract areas.
- ▶ 14 companies were shortlisted for 23 contract areas with 8 being new to the E&P sector.

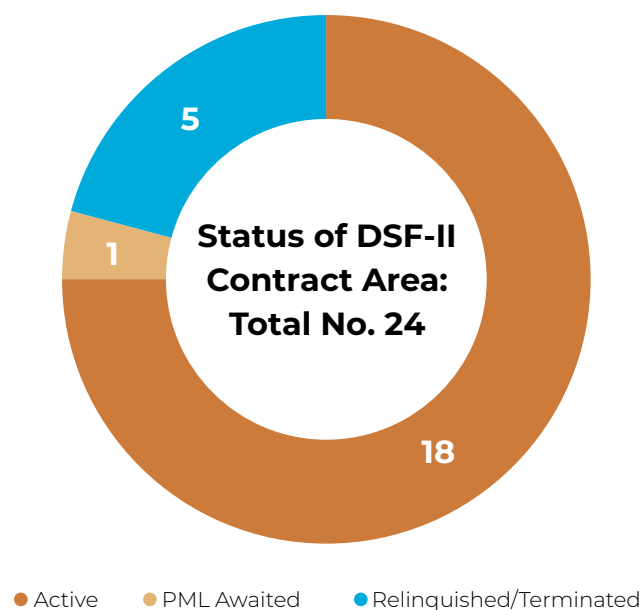


Table 6.3: Status of Contract Areas awarded under DSF Bid Round-II

Sl. No.	Basin	Basin Category	Contract Areas awarded under DSF Bid Round-II			
			Active	PML Awaited	Terminated/ Relinquished	Total
1	Assam Shelf	Cat-I	3	0	0	3
2	Assam- Arakan fold belt		1	0	2	3
3	Cambay		2	0	2	4
4	Krishna Godavari		4	1	1	6
5	Mumbai Offshore		6	0	0	6
6	Rajasthan	Cat-II	1	0	0	1
7	Bengal-Purnea		1	0	0	1
Grand Total			18	1	5	24



6.7 DISCOVERED SMALL FIELD BID ROUND-III (2021)

Building on the success of DSF Bid Rounds I and II, and recognizing the potential of untapped discoveries, the Government of India launched DSF Bid Round III on June 10, 2021.

In this round, Government offered 32 Contract Areas with 75 discoveries across nine sedimentary basins covering about 13,000 square kilometres of area and having 232 Million Metric Tonnes (O+OEG) of in-place reserves.

Following the evaluation, 31 Contract Areas were awarded, including 11 onland and 20 offshore blocks with 74 discoveries covering approximately 12,727 square kilometres of area and having 230 Million Metric Tonnes (O+OEG) of in-place reserves.

The Revenue Sharing Contracts (RSCs) for 30 Contract Areas were signed on September 09, 2022.

During DSF Bid Round-III, additional provisions and incentives were introduced to encourage operators in oil and gas exploration and production for early monetization. The major revised provisions are tabulated below:

6.7.1 Key Highlights of DSF Bid Round-II are as outlined below:

- ▶ The DSF-III round received an 107 bids for 32 Contract Areas, with 27 companies participating-23 of them private players.
- ▶ The round welcomed five new entrants, including two foreign companies.
- ▶ A standout feature of DSF-III is that nearly 90% of the hydrocarbon resources offered were located in offshore areas. This marks a strategic shift towards leveraging India's blue economy potential.
- ▶ To enhance investor appeal, DSF-III introduced more competitive terms, including the issuance of PML grant letters for offshore blocks at the time of contract signing, and incentives for early monetization of discoveries.
- ▶ Recognizing the value of data in reducing exploration risk, the round facilitated faster access to geological and geophysical data, enabling bidders to make informed decisions and streamline their project timelines.

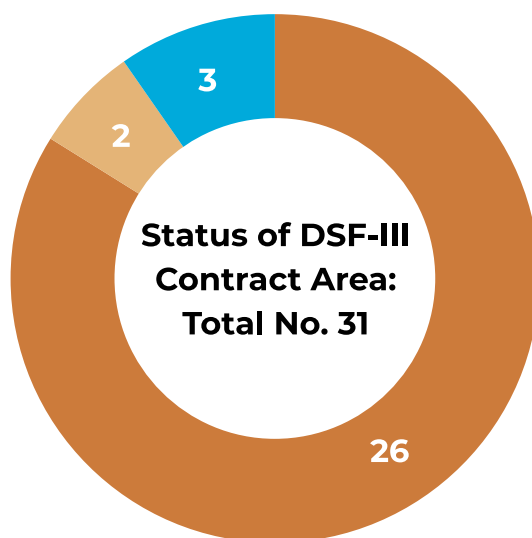
Table 6.4: DSF Previous vs Revised Provision

Feature	Previous Provision	Revised Provision
Biddable Parameter	Weightage: Biddable Work Program: 20% Govt. Revenue Share: 80% No HRP Ceiling	Weightage: Biddable Work Program: 50% Govt. Revenue Share: 50% HRP Upper Ceiling: 50%
Early Production	No Incentive	Incentivizing early commercial production during development period (Oil: Min of 5 US\$/bbl and Quoted LRP) (Gas: Min of 0.2 US\$/MMBtu and Quoted LRP)
Liquidated Damage (LD)	High LD in case of failing to meet Minimum Work Programme	75% reduction in LD amount
Exit Option	No Provision	Option to exit without LD Payment in case of delay in statutory clearances beyond 2 years



Table 6.5: Status of Contract Areas awarded under DSF Bid Round-III

Sl. No.	Basin	Basin Category	Contract Areas awarded under DSF Bid Round-III			
			Active	PML Awaited	Terminated/ Relinquished/ Contract Cancelled	Total
1	Assam Shelf	Cat-I	1	0	0	1
2	Assam- Arakan fold belt		1	0	1	2
3	Cambay		3	0	0	3
4	Cauvery		0	1	0	1
5	Krishna Godavari		9	0	1	10
6	Mumbai Offshore		8	0	1	9
7	Rajasthan		2	0	0	2
8	Kutch	Cat-II	2	0	0	2
9	Vindhyan		0	1	0	1
Grand Total			26	2	3	31



● Active ● PML Awaited ● Relinquished/Terminated/
Award Cancelled



6.7.2 Consolidated Status of DSF Bid Rounds

- The Contract Areas awarded under DSF regimes are as under:

Table 6.6: Consolidated status of DSF Bid Rounds

Sl. No.	STATE	CONTRACT AREA	DSF-I	DSF-II	DSF-III	Total
1	MUMBAI OFFSHORE	MB/OSDSF/B-37/2016				20
		MB/OSDSF/B127E/2016				
		MB/OSDSF/B15/2016	5			
		MB/OSDSF/B80/2016				
		MB/OSDSF/B9/2016				
		MB/OSDSF/CA/2018				
		MB/OSDSF/D18/2018				
		MB/OSDSF/D31/2018		6		
		MB/OSDSF/D33/2018				
		MB/OSDSF/NMT/2018				
		MB/OSDSF/SB15/2018				
		MB/OSDSF/B15/2021				
		MB/OSDSF/B154N/2021				
		MB/OSDSF/B174/2021				
		MB/OSDSF/B203/2021				
		MB/OSDSF/B37/2021			9	
		MB/OSDSF/B66/2021				
		MB/OSDSF/BH68/2021				
		MB/OSDSF/RATNA/2021				
		MB/OSDSF/WO5/2021				
2	ASSAM	AA/ONDSF/BARSILLA/2016				16
		AA/ONDSF/CHARAIDEO/2016				
		AA/ONDSF/DIPLING/2016				
		AA/ONDSF/DUARMARA/2016	8			
		AA/ONDSF/HILARA/2016				
		AA/ONDSF/JERAIPATHAR/2016				
		AA/ONDSF/LAXMIJAN/2016				
		AA/ONDSF/PATHARIA/2016				
		AA/ONDSF/TIPHUK/2018				
		AA/ONDSF/DISAIJAN/2018				
		AA/ONDSF/HAZARIGAON/2018		5		
		AA/ONDSF/MADHAKALI/2018				
		AA/ONDSF/UMATARA/2018				

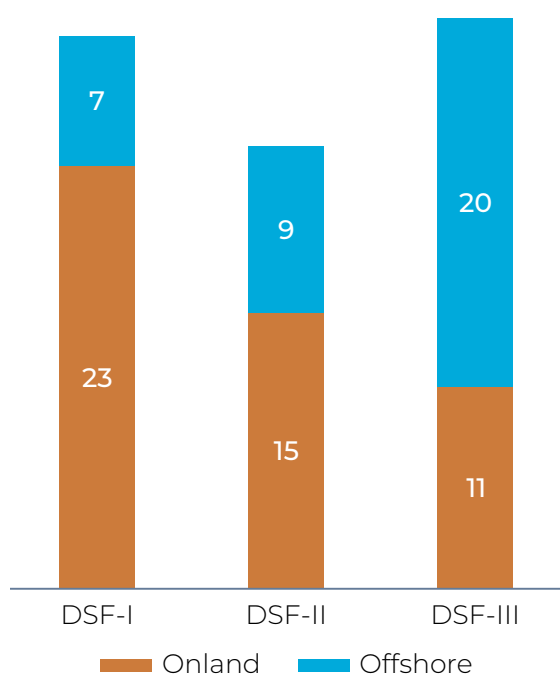


Sl. No.	STATE	CONTRACT AREA	DSF-I	DSF-II	DSF-III	Total
3	ANDHRA PRADESH	AA/ONDSF/PATHARIA/2021				
		AA/ONDSF/TIPHUK/2021			3	
		AA/ONDSF/TUKBAI/2021				
		KG/ONDSF/ACHANTA/2016				11
		KG/ONDSF/BHIMANAPALLI/2016				
		KG/ONDSF/KORAVAKA/2016	4			
		KG/ONDSF/ SANARUDRAVARAM/2016				
		KG/ONDSF/ GOKARNAPURAM/2018				
		KG/ONDSF/KAZA/2018		4		
		KG/ONDSF/PALAKOLLU/2018				
		KG/ONDSF/SURYARAOPETA/2018				
		KG/ONDSF/DANGERU/2021				
		KG/ONDSF/KAVITAM/2021			3	
		KG/ONDSF/ SANARUDRAVARAM/2021				
4	GUJARAT	CB/ONDSF/ELAO/2016				10
		CB/ONDSF/KAMBOI/2016				
		CB/ONDSF/KHAMBEL/2016	5			
		CB/ONDSF/SOUTH PATAN/2016				
		CB/ONDSF/WEST BECHRAJI/2016				
		CB/ONDSF/A1/2018				
		CB/ONDSF/D45/2018		4		
		CB/ONDSF/SANGANPUR/2018				
		CB/ONDSF/VADATAL/2018				
		CB/ONDSF/INDRORA/2021			1	
5	KG OFFSHORE	KG/OSDSF/GSKV1/2016	1			10
		KG/OSDSF/G4/2018				
		KG/OSDSF/GSKW/2018		2		
		KG/OSDSF/G4/2021				
		KG/OSDSF/GS21/2021				
		KG/OSDSF/CHANDRIKA/2021				
		KG/OSDSF/GS49/2021			7	
		KG/OSDSF/YS6/2021				
		KG/OSDSF/RAVVA/2021				
		KG/DWDSF/GD10/2021				

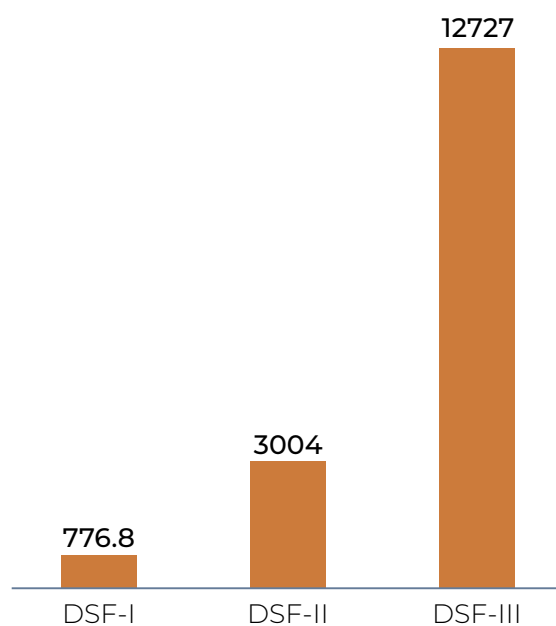


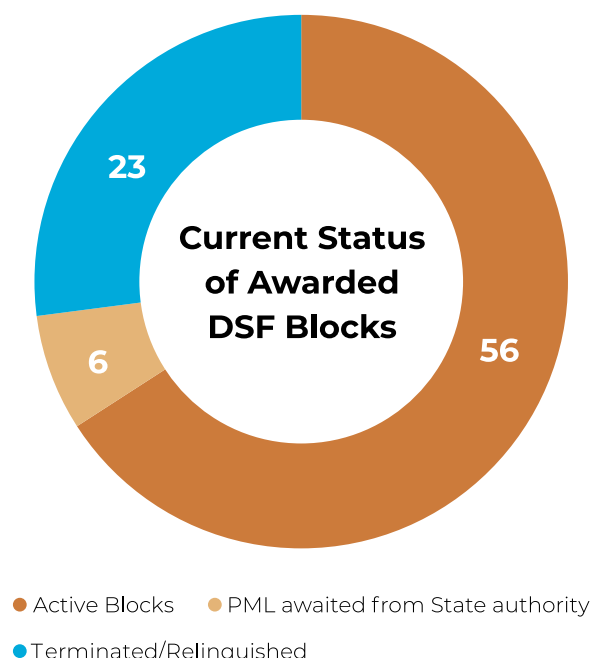
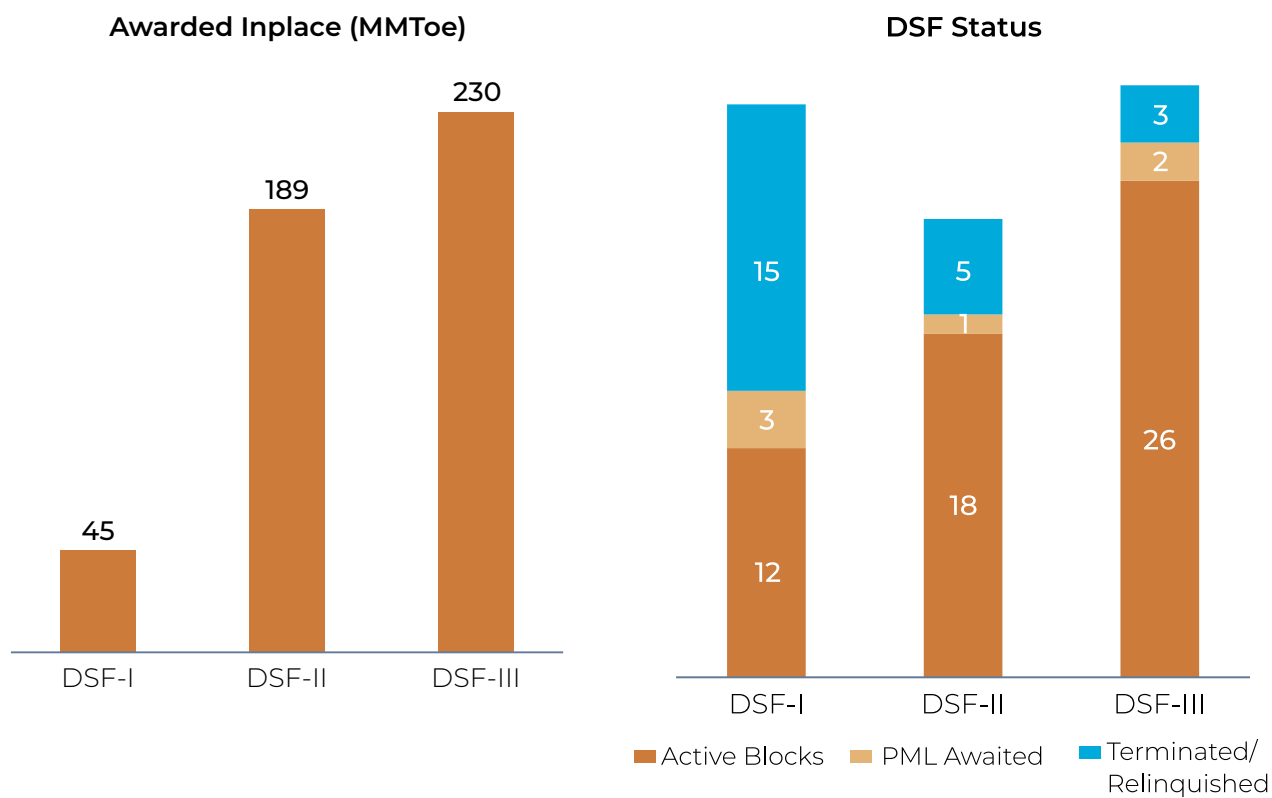
Sl. No.	STATE	CONTRACT AREA	DSF-I	DSF-II	DSF-III	Total
6	RAJASTHAN	RJ/ONDSF/BAKHRI TIBBA/2016	2			5
		RJ/ONDSF/SADEWALA/2016				
		RJ/ONDSF/CHINNEWALA/2018		1		
		RJ/ONDSF/BAKHRI TIBBA/2021			2	
		RJ/ONDSF/PUNAM/2021				
7	TAMIL NADU	CY/ONDSF/KARAIKAL/2016	2			3
		CY/ONDSF/NEDUVASAL/2016				
		CY/ONDSF/VADATHERU/2021			1	
8	KUTCH OFFSHORE	GK/OSDSF/KD/2016	1			3
		GK/OSDSF/GK1/2021			2	
		GK/OSDSF/GK28/2021				
9	MADHYA PRADESH	VN/ONDSF/NOHTA/2016	1			2
		VN/ONDSF/NOHTA/2021			1	
10	TRIPURA	AA/ONDSF/TULAMARA/2018		1		1
11	ARUNANCHAL PRADESH	AA/ONDSF/KHEREM/2016	1			1
12	CAMBAY OFFSHORE	CB/OSDSF/AMBE/2021			2	2
		CB/OSDSF/GULFA/2021				
13	MAHANADI OFFSHORE	NEC/OSDSF/D11/2018		1		1
Grand Total			30	24	31	85

85 Blocks Awarded



Area Awarded (Sq. km.)





- Status of DSF I, II & III Contract Areas is as under:

Table 6.7: Present status of DSF Contract Areas

S. No	Status	DSF-I	DSF-II	DSF-III	Total
1	Awarded Contract Area (CA)	30	24	31	85
2	No of PMLs awaited	3	1	2	6
3	Relinquished/ Terminated/ Contract Cancelled	15	5	3	23
4	Operational / Active	12	18	26	56

- State-wise status of Contract Areas awarded under DSF-I-II & III regime

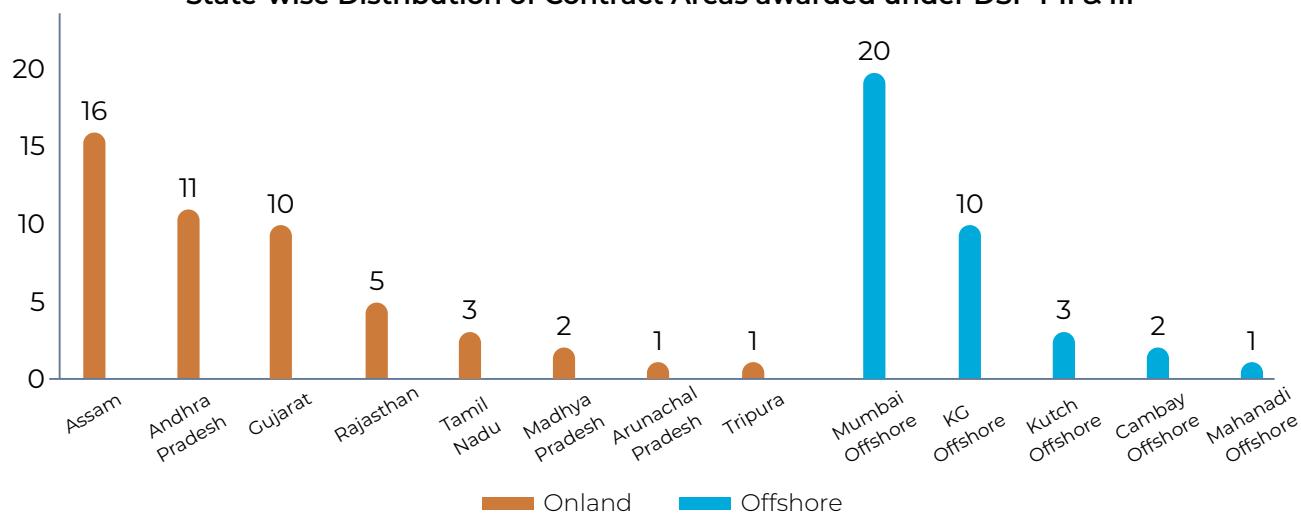
Table 6.8: State-wise Awarded Contract Areas in DSF Bid Rounds

DSF	States	Active	PML Awaited	Terminated/ Relinquished/ Award Cancelled	Total
DSF-I	Andhra Pradesh	2	1	1	4
	Arunachal Pradesh	0	0	1	1
	Assam	2	0	6	8
	Gujarat	5	0	0	5
	Madhya Pradesh	0	0	1	1
	Rajasthan	0	0	2	2
	Tamil Nadu	0	2	0	2
	KG Offshore	1	0	0	1
	Kutch Offshore	0	0	1	1
	Mumbai Offshore	2	0	3	5
DSF-I Total		12	3	15	30
DSF-II	Andhra Pradesh	3	1	0	4
	Assam	3	0	2	5
	Gujarat	2	0	2	4
	Rajasthan	1	0	0	1
	Tripura	1	0	0	1
	KG Offshore	1	0	1	2
	Mahanadi Offshore	1	0	0	1
	Mumbai Offshore	6	0	0	6
DSF-II Total		18	1	5	24



DSF	States	Active	PML Awaited	Terminated/ Relinquished/ Award Cancelled	Total
DSF-III	Andhra Pradesh	3	0	0	3
	Assam	2	0	1	3
	Gujarat	1	0	0	1
	Madhya Pradesh	0	1	0	1
	Rajasthan	2	0	0	2
	Tamil Nadu	0	1	0	1
	Cambay Offshore	2	0	0	2
	KG Offshore	6	0	1	7
	Kutch Offshore	2	0	0	2
	Mumbai Offshore	8	0	1	9
DSF-III Total		26	2	3	31

State-wise Distribution of Contract Areas awarded under DSF-I-II & III



Distribution of Contract Areas awarded under DSF-I-II & III



Table 6.9: State-wise status of PML granted to the Operators of DSF Blocks

State-wise status of PMLs granted to the Operators of DSF Blocks					
Sl. No.	States	No. of Live* CAs under DSF regime	No of PML's granted/ Active CAs	PML Awaited	Area of Granted PML
1	Andhra Pradesh	10	8	2	767.93
3	Assam	7	7	0	656.81
4	Gujarat	8	8	0	391.40
5	Madhya Pradesh	1	0	1	0
6	Rajasthan	3	3	0	167.00
7	Tamil Nadu	3	0	3	0
8	Tripura	1	1	0	47.23
9	Cambay Offshore	2	2	0	801.20
10	KG Offshore	8	8	0	1379.3
11	Kutch Offshore	2	2	0	2092.3
12	Mahanadi Offshore	1	1	0	541.44
13	Mumbai Offshore	16	16	0	6285.06
	Grand Total	62	56	6	13129.67

* Awarded Contract Areas excluding the Terminated/Relinquished Blocks.



6.8 Hydrocarbon Exploration and Licencing Policy (HELP)

The Government of India notified the Hydrocarbon Exploration and Licensing Policy (HELP) on 30th March 2016 to revamp the E&P sector by shifting from the Production Sharing Contract (PSC) regime to a more transparent Revenue Sharing Contract (RSC) model. A key feature of HELP was the introduction of the Open Acreage Licensing Policy (OALP), which allowed companies to select and propose exploration blocks throughout the year, without waiting for formal bidding rounds. This reform significantly improved the ease of doing business by reducing government intervention in operational matters and enabling investor-driven exploration. The first OALP Bid Round was launched in January 2018, marking the operational rollout of this progressive licensing framework.

HELP comes with attractive and liberal terms. Some of the Salient features of the Policy is given below: -

- ▶ One single license for exploring of all types of hydrocarbons both conventional and unconventional
- ▶ Revenue Sharing model instead of cost recovery
- ▶ Companies can choose and propose their own exploration blocks
- ▶ Expression of Interest (EoI) can be submitted any time during the year
- ▶ Freedom to sell oil and gas at market prices
- ▶ Lower royalty rates, especially in difficult and frontier areas
- ▶ No oil cess under HELP
- ▶ Exploration is allowed throughout the contract period
- ▶ Simple and transparent bidding process

To make Policy more attractive, Government further notified the policy reforms on 28th February 2019, where many of the processes and approvals were relaxed to promote "Ease of Doing Business". These reforms were focused to increase exploration activities, attract domestic and foreign investment in unexplored/unallocated areas of sedimentary basins. These reforms were made applicable from OALP Round IV onwards. These reforms provide many

other benefits in addition to mentioned above, some of the features are as follows:

- ▶ More weightage to Work Programme than the Revenue Share for Category-I Basin.
- ▶ Revenue Share from Category II & III type of basins were removed except in case of windfall gain.
- ▶ Concession in Royalty rates
- ▶ Flexibility to revise FDP
- ▶ Liberal freedom to Transfer/Exit the blocks
- ▶ No obligation to run Drill Stem Test
- ▶ Introduction of alternate dispute resolution mechanism and Empowered Coordination Committee (ECC) for expediting statutory clearances/approvals.

To enhance the participation, the amendments in the bid documents for bid round VIII has been carried out. The objective of such amendments is to address the issues of existing operators which they are facing in existing contractual framework and to make them compatible for the needs of global E&P players so that the bid documents can be made globally competitive. Moreover, these amendments bring in more ease of doing business in the upstream sector of India. Major of the modifications are listed below:

- ▶ Larger acreage areas upto 20,000 Sq.km for play based exploration
- ▶ Modification in modalities for carving out open acreage blocks with partial '1x1' unit.
- ▶ Modifications in Operatorship Experience for Deep/ Ultra-deep blocks bidding to maximize the participation.
- ▶ Removal of Dual Liquidated Damage for commencement of Development and Production Operations
- ▶ Enhanced Stabilization Period for incentivizing early monetization of Discoveries.
- ▶ 3 Year Retention Period for Sub-Commercial Discoveries
- ▶ Modified provisions for change in Consortium during EOI and bidding stage.
- ▶ Enhanced Scope of Force Majeure



To enhance the Bidding in Category- II & III Basins, Some Special incentives are also introduced apart from the above: -

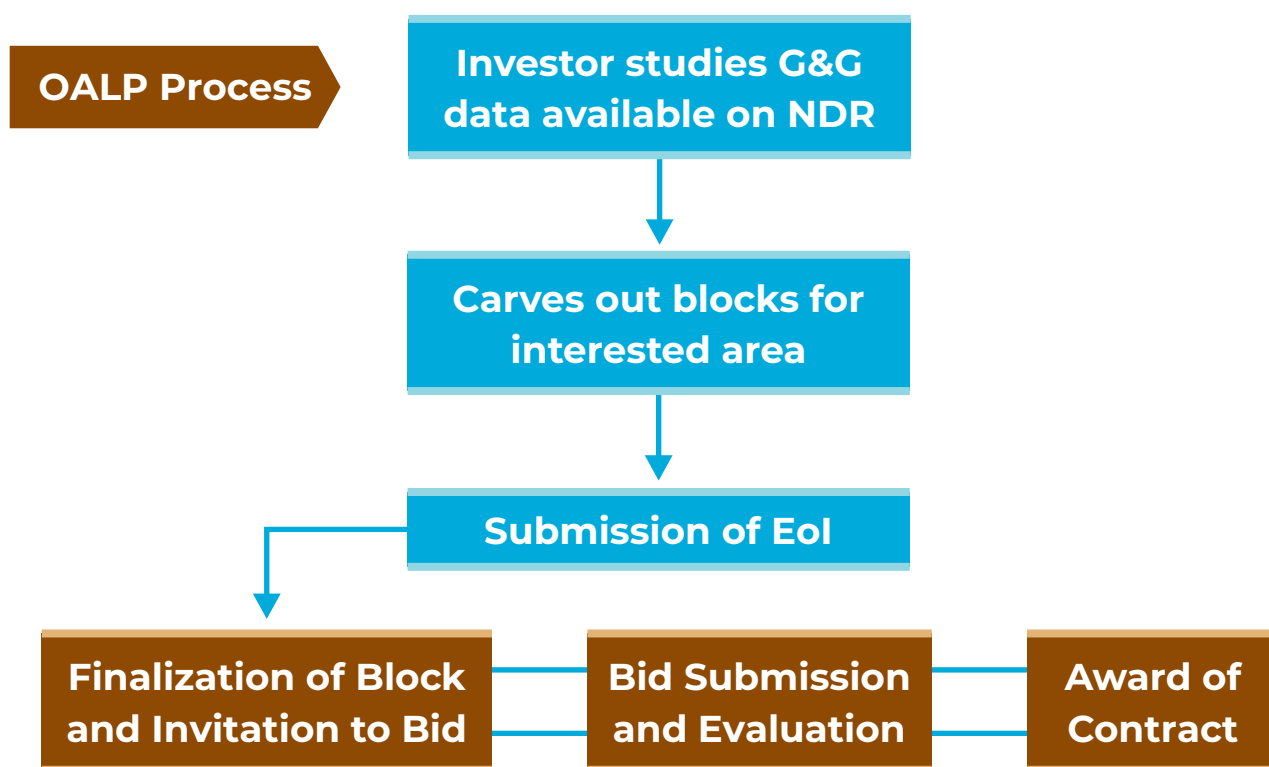
- ▶ To reduce the risk of exploration, Only Seismic Survey is biddable under work programme commitment in Cat-II and III.
- ▶ Extended Exploration Period for Category-II & III Basins.
- ▶ Swapping of CWP with other surveys and Exploration Wells allowed

6.8.1 Bidding Process under OALP Regimes:

OALP mechanism allow the investor to carve out the blocks of their choice by assessing E&P data available at National Data Repository (NDR) & by submitting an Expression of Interest (Eol). OALP

also encourages Round the year submission of Expression of Interest (EOI) in 3 windows (April 1 to July 31, August 1 to November 30 and December 1 to March 31) by the interested bidders. The govt. also launched state-of-art National Data Repository, a database comprising all the geo-scientific data of hydrocarbon resources in the country, in 2017 in a bid to enable potential investors to take informed decisions.

Investor are given freedom to carve out the Eols as per Modalities. The submitted Eols will be scrutinized in accordance with the provisions of Modalities. Finalized Eols will be considered as Blocks and put-up for bidding. Contractor is required to submit the bid as per publish NIO and modalities. Eol originator is eligible for originator incentive.



Eol Submission is a continuous process. A total of 15 Eols with an area of 59,382 sq.km have received in Eol Window – XIX, XX & XXI during the FY 2024-25. The details are given below: -

S.No.	Eol Window	Period	No. of Eol	Area, Sq. Km
1	Eol Window – XIX	01.04.2024-31.07.2024	8	23171.64
2	Eol Window – XX	01.08.2024-30.11.2024	2	5252.61
3	Eol Window – XXI	01.12.2024-31.03.2025	5	30957.75 (Under examination)



6.8.2 Summary of Participation

In Nine rounds of bid under OALP, 177 blocks were on offer and 172 exploration blocks covering an area of 3,78,653 sq. Km. were awarded to successful bidders. Five (5) un-awarded blocks were offered under OALP-Round-III and in all

of them CBM was focus; however, no bids were received for those CBM blocks. 312 bids were received for remaining 172 blocks that are spread over 19 Sedimentary Basins.

Summary of participation in OALP rounds is as under:

OALP Bid Round	Blocks on offer	Number of participants	No. of bids	Area on offer (sq.km.)	Area awarded (sq.km.)	No. of Blocks Awarded
OALP-I	55	9	110	59,283	59,283	55
OALP-II	14	8	33	29,233	29,233	14
OALP-III	23*	5	42	31,722	29,765	18
OALP-IV	7	2	8	18,510	18,510	7
OALP-V	11	3	12	19,789	19,789	11
OALP-VI	21	3	24	35,346	35,346	21
OALP-VII	8	4	10	15,766	15,766	8
OALP-VIII	10	6	13	34,365	34,365	10
OALP-IX	28	7	60	1,36,596	1,36,596	28
Total	177	47	312	3,80,610	3,78,653	172

* No bid received for five CBM Blocks.

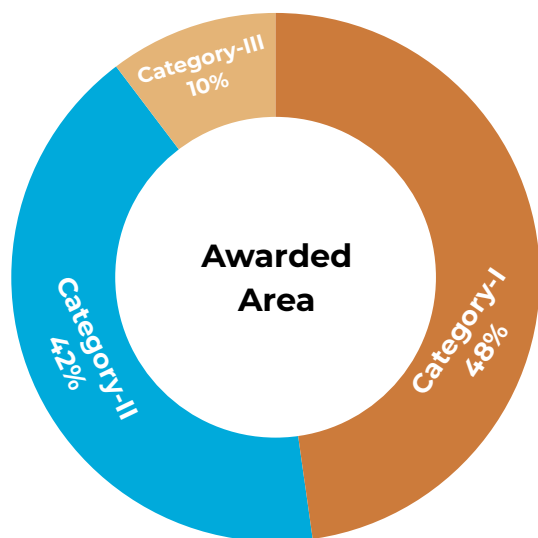
Overall awarded area of 3,78,653 sq. Km is split in to three categories of sedimentary basins as follows:

Category of Basin	Number of Blocks	Area (Sq. km.)	% of area
Category-I	116	1,80,829.56	47.76%
Category-II	41	1,58,772.37	41.93%
Category-III	15	39,050.94	10.31%
Grand Total	172	3,78,652.87	100.00%

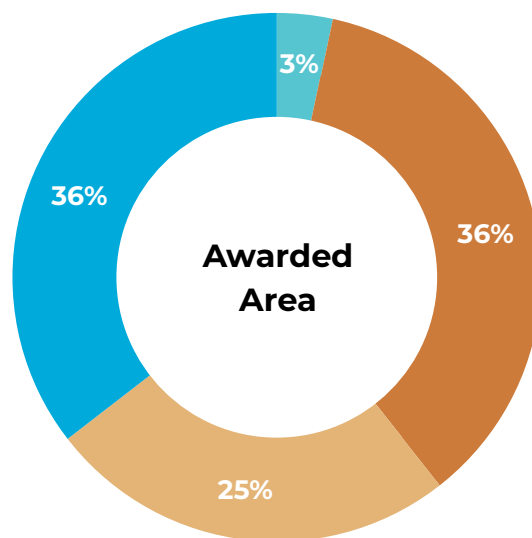
Table 1: Basin-wise area awarded under OALP rounds



Distribution of Acreage Awarded till OALP-IX, by Basin Category



Distribution of Acreage Awarded till OALP-IX, by location type



● Onland ● Shallow Water ● Ultra Deep Water ● Deep Water

Basin wise segregation of Awarded Blocks (OALP I to IX)

Basin Category	Sl. No.	Basin Name	No. of Blocks	Area Awarded
Category I	1	Krishna Godavari	11	34,053
	2	Mumbai	11	27,957
	3	Assam-Arakan Shelf	23	23,835
	4	Rajasthan	15	19,482
	5	Cauvery	12	52,094
	6	Assam-Arakan Fold Belt	14	6,569
	7	Cambay	30	16,840
Sub Total Category-I			116	1,80,829.56
Category II	8	Saurashtra	13	51,252
	9	Kutch	8	14,132
	10	Vindhyan	5	13,260
	11	Mahanadi	11	56,866
	12	Andaman-Nicobar	4	23,262
Sub Total Category-II			41	1,58,772
Category III	13	Kerala Konkan	3	11,350
	14	Bengal	5	17,396
	15	Ganga	3	2,428
	16	Pranhita Godavari		0
	17	Satpura-South Rewa-Damodar	1	906
	18	Himalayan-Foreland	1	666



Basin Category	Sl. No.	Basin Name	No. of Blocks	Area Awarded
	19	Chhattisgarh		0
	20	Narmada	1	2,999
	21	Spiti-Zaskar		0
	22	Deccan Syneclise		0
	23	Cuddapah	1	3,306
	24	Karewa		0
	25	Bhima-Kaladgi		0
	26	Bastar		0
Sub Total Category-III			15	39,050.94
Grand Total			172	3,78,652.87

6.8.3 Bidding Round Completed in FY 2024-25:

• OALP Bid Round-IX :

Government had launched OALP Bid Round-IX on 3rd January 2024 offering 28 Blocks with an area of around 1,36,596 sq.km for Exploration and Development through International Competitive Bidding. Out of 28 Blocks, 23 Blocks are carved out based on the EoIs received during different EoI Windows and five (5) blocks were carved out by DGH.

These 28 Blocks are spread over eight (8) Sedimentary Basins where 16 Blocks are cover under Category-I Basins and 12 Blocks are cover under Category-II Basins.

OALP Round-IX is mainly focused on Deep / Ultra Deep-water exploration where out of total Area offered, Ultra-Deep-Water blocks covers an area of 96,073 Sq. km in 11 Blocks which account of more than 70% of total area. Remaining Nine (9) Blocks with an area of 13875 Sq.km are in Onland and Eight (8) Blocks with an area of around 26648 Sq.Km. are in Shallow Water.

The bidding was closed for OALP bid round IX on 21st September 2024 and 60 bids were submitted and received. OALP-IX witnessed a remarkable increase in bidding activity, with the average number of bids per block rising to 2.4 from just 1.3 in the previous round. A total of Seven (7) bidders including three (3) public and four (4) private parties participated in the Bid Round.

Total seven (07) Companies- Three (03) PSUs viz. M/s ONGC Limited., M/s OIL and M/s IOCL and four (04) private sector companies viz. M/s Reliance Industries Limited (RIL), M/s British Petroleum Exploration (Alpha) Limited, M/s Sun Petrochemical Private Limited and M/s Vedanta Limited participated in the bidding process.

The submitted bids were evaluated and Blocks were awarded to H1 bidders. The details are tabulated below :-

Name of H1 Bidder	No. of Blocks Awarded
M/s OIL	6
M/s ONGC	11
M/s ONGC + M/s BPEAL + M/s RIL (Consortium)	1
M/s ONGC + M/s OIL (Consortium)	3
M/s Vedanta	7
	28

Total Committed Investment in this round is Expected to be 990 MMUSD.

Expected Committed Work Plan:

2D Seismic (LKM)	11742
3D Seismic (SKM)	19247
Wells	45
Committed Investment	990 MMUSD



6.8.4 Current Investment Opportunity:

• OALP Bid Round- X

OALP Bid Round-X was launched at the India Energy Week Summit 2025 on 11th Feb, 2025 in New Delhi. In this bid round, 25 blocks spread across 13 Sedimentary Basins covering an area of 1,91,986.21 sq. km are on offer for bidding to investors. OALP Bid Round-X is the largest bid round in terms of acreage under the HELP regime

This bid round primarily focuses on offshore exploration, featuring 19 offshore blocks spanning a total area of 1,75,115 sq. km. Additionally, 6 onland blocks cover an area of 16,871 sq. km.

Among the offshore blocks:

- ▶ **6 blocks** (41,391 sq. km) are located in **Shallow Offshore**,

- ▶ **12 blocks** (1,23,733 sq. km) are in **Ultra-Deep Water**, and

- ▶ **1 block** (9,991 sq. km) falls under **Deep Water**.

Moreover, notably, 16 blocks, accounting for 97,919.6 Sq.Km. (51%) fall within areas previously designated as "No-Go" zones.

Additionally, for OALP Round X, a new e-bidding portal has been introduced to enhance transparency and security, along with features designed to improve ease of doing business for investors.

A 70% of total area falls in Deep/ Ultra Deep water whereas 21.5% of area is in Shallow Water offshore. Remaining area is fall under Onland. The Bid Closing date for this Round is 31.07.2025.

The Blocks on offer are tabulated below:-

S. No.	Basin Name	Basin Category	Block Name	Block Area (Sq.km)	Well Target Depth (m)	Minimum Net worth (MMUSD)	Bid Bond (USD)
ONLAND BLOCKS, 6 Blocks (16871 Sq.km)							
1	RAJASTHAN	I	RJ-ONHP-2024/1	5953.44	1400	23.98	200000
2	CAMBAY		CB-ONHP-2024/1	126.44	2600	5	9000
3	VINDHYAN	II	VN-ONHP-2024/1	4275.13	Not Biddable	17.95	200000
4	KAREWA	III	KR-ONHP-2024/1	283.83	Not Biddable	5	20000
5	HIMALAYAN FORELAND		HF-ONHP-2024/1	990.86	Not Biddable	7.44	68800
6	GANGA-PUNJAB		GP-ONHP-2024/1	5241.1	Not Biddable	21.94	200000
SHALLOW WATER BLOCK, 6 Blocks (41392 Sq.km)							
7	MUMBAI OFFSHORE	I	MB-OSHP-2024/1	5838.03	2950	54.6	200000
8			MB-OSHP-2024/2	13131.72	2070	98	200000
9	KRISHNA-GODAVARI	II	KG-OSHP-2024/1	2967.83	2750	37.08	170600
10	SAURASHTRA		GS-OSHP-2024/1	3125.84	Not Biddable	37.8	192000
11			GS-OSHP-2024/2	6501.38	Not Biddable	58.66	200000



S. No.	Basin Name	Basin Category	Block Name	Block Area (Sq.km)	Well Target Depth (m)	Minimum Net worth (MMUSD)	Bid Bond (USD)
12	BENGAL-PURNEA	III	BP-OSHP-2024/1	9826.81	Not Biddable	79.34	200000
DEEP WATER BLOCK, 1 Blocks (9991 Sq.km)							
13	CAUVERY	I	CY-DWHP-2024/1	9990.96	2000	131.35	200000
ULTRA-DEEP WATER BLOCK, 12 Blocks (123733 Sq.km)							
14	KRISHNA-GODAVARI	I	KG-UDWHP-2024/1	12610.14	2950	202.17	200000
15			KG-UDWHP-2024/2	9511.65	1600	174.57	200000
16			KG-UDWHP-2024/3	9935.27	2600	177	200000
17	MAHANADI	II	MN-UDWHP-2024/1	5520.09	Not Biddable	138	200000
18			MN-UDWHP-2024/2	10553.23	Not Biddable	187.14	200000
19			MN-UDWHP-2024/3	7169.14	Not Biddable	153	200000
20	SAURASHTRA		GS-UDWHP-2024/1	9059.6	Not Biddable	171.99	200000
21	ANDAMAN-NICOBAR		AN-UDWHP-2024/1	12816.65	Not Biddable	201	200000
22			AN-UDWHP-2024/2	10027.9	Not Biddable	177	200000
23			AN-UDWHP-2024/3	8732.15	Not Biddable	165	200000
24		AN-UDWHP-2024/4	15481.03	Not Biddable	226.65	200000	
25	BENGAL-PURNEA	III	BP-UDWHP-2024/1	12315.99	Not Biddable	202.14	200000

Each offered Block under OALP-X is technically analyzed by International Experts for Hydrocarbon Potential and is freely available for Investors. A comprehensive Detailed Project Report (DPR) for each block proposed in the upcoming bidding rounds will be prepared and made available to investors. The DPR will include geographical mapping, existing and planned infrastructure, and an assessment of probable statutory and environmental clearance requirements. This will enhance transparency, reduce entry-level risk, and enable informed decision-making by potential bidders.

Initiatives for facilitating clearances/approvals and enhanced coordination:

- Empowered Coordination Committee (ECC):** This committee has been constituted under the chairmanship of Cabinet Secretary to discuss the defense, environment, license etc. related issues pertaining to the Hydrocarbon sector.
- Creation of 'Hydrocarbon Clearance Cell' (HCC):** A dedicated 'Hydrocarbon Clearance Cell' has been institutionalized within DGH. The cell liaises with Central/State



Government authorities for expeditious grant of clearances/approvals.

iii. Creation of North-East Coordination Committee (NECC): ECC under HCC of DGH created in September 2021 and 18 Coordinators are placed with various Departments of State Govts of Assam, Arunachal Pradesh and Tripura for ease of processing Clearance and Approval proposals under various environmental regulations as well as expediting the other issues related to Oil & Gas E&P activities requires administrative/ regulatory supports from the State Govt.

iv. UrjaPragati Portal (Upstream Response by Joint Action for proactive Governance and Timely Implementation): To facilitate E&P operators and monitor the approvals/clearances, URJA PRAGATI web-based interactive portal has been launched recently by DGH to prioritize and flag the long pending issues concerning the upstream hydrocarbon sector involving

various stakeholders such as Operators, Central Ministries and State Governments. The issues to be dealt through this portal covers clearances related to PEL (Petroleum Exploration License), PML (Petroleum Mining Lease), FC (Forest Clearance), WLC (Wildlife Clearance), CTE (Consent To Establish)/CTO (Consent To Operate), ESZ (Eco Sensitive Zone) and CRZ (Coastal Regulation Zone).

v. Upstream India Portal: The 'Upstream India' portal was launched to provide support to E&P operators. It offers services such as sharing of facilities and infrastructure available at DGH, creation of a Knowledge Sharing Platform, and organization of regular workshops, seminars, conferences, training sessions, and discussions on the latest government policies, technologies, and usage of inventories.

vi. Urja Suraksha Samanvay: GIS based monitoring and decision support system for swift information exchange.





7

Unconventional Petroleum Resources

7.1 Introduction:

Unconventional energy resources refer to sources of energy that are extracted and produced using methods that differ from traditional fossil fuel extraction. Unconventional resources exist in petroleum accumulations that are pervasive throughout a large area and are not significantly affected by hydrodynamic influences (also called “continuous-type deposit”). These resources are often more challenging to develop and require advanced technology and techniques. Key unconventional energy resources include:

7.1.1 Coal Bed Methane (CBM):

During the process of coalification or thermal maturation of coal beds, valuable quantities of hydrocarbon gas get accumulated, primarily in the adsorbed state. CBM is considered a cleaner energy source compared to coal, as its combustion produces less carbon dioxide and other pollutants. It is extracted by reducing the pressure within the coal seams through water pumping, which releases the adsorbed methane gas.

There are four types of terms used to describe the methane emanating from coal beds:

- a. **Coalbed Methane (CBM) or Coal Seam Gas (CSG)** - A generic term for the methane-rich gas naturally in coal seams typically comprising 80% to 90% methane with lower proportions of ethane, propane, nitrogen, and carbon dioxide. In international use, this term refers to the methane recovered from unmined coal seams using surface boreholes.



- b. **Coal Mine Methane (CMM)** - Methane gas captured at working mine by underground methane drainage techniques. Any gas captured underground, whether drained in advance or after mining, and any gas drained from the surface of well is including in this definition.
- c. **Ventilation air methane (VAM)** - Methane emitted from coal seams that enters the ventilation air and are exhausted from the ventilation shaft at a low concentration, typically in the range of 0.1% to 1.0% by volume.
- d. **Abandoned Mine Methane (AMM)** - The methane gas recovered from abandoned coal mines.

7.1.2 Shale Gas and Shale Oil:

Gas or oil within shales are the un-expelled part during petroleum generation process and found trapped within the pore space of shale as the source rock. Genetically, shales have no intrinsic permeability for self-flow to a well bore.

Like all other unconventional resources of continuous nature, shale plays too have less geological risks than conventional but more development risks.

Shale gas and oil have significantly increased global energy supplies and have transformed the energy landscape in countries like the USA. The commercial extraction of shale gas and oil involves the use of horizontal drilling followed by

hydraulic fracturing (fracking). These techniques open up more reservoirs contact and create sufficient fracture conductivity to release the trapped gas and oil.

7.1.3 Tight Gas:

Tight gas refers to natural gas trapped in low-permeability rocks other than shales, viz. sandstones, siltstones or carbonates. Like shale gas, tight gas extraction requires hydraulic fracturing to create flow conduits. Tight gas is genetically conventional deposits, however due to location in the deeper part of the basins, it loses intra granular porosity due to excessive over-burden pressure. For the same reasons, such deposits are often developed into high pressure and high temperature conditions making gas extraction more challenging.

7.1.4 Gas Hydrates:

Gas hydrates are crystalline ice-like structures that contain methane and other gases. They are found in deep ocean several hundred meter below seabed and also in permafrost regions. It is formed under moderately high pressure but low temperature conditions.

Extraction is still experimental, involving methods like depressurization, thermal stimulation, and inhibitor injection to dissociate the hydrate. Gas hydrates hold vast amount of natural gas as 1-unit hydrate volume expands into ~160-unit gas to surface and represent a potential future energy source.



7.1.5 Oil Sands (Tar Sands):

Oil shales consist of a mixture of shales, sands, water, clay, and bitumen, a heavy viscous form of crude oil. It is genetically immature source rocks formed due to failed process of petroleum generation.

Oil is extracted by surface mining and in-situ techniques, such as steam-assisted gravity drainage. Oil shales are a significant source of heavy oil, but their development raises environmental concerns due to greenhouse gas emissions and land disturbance.

7.2 Unconventional hydrocarbons for energy security Energy Security

7.2.1 Energy Security

India is transitioning towards a gas-based economy, aiming to increase the share of natural gas to 15% in its energy mix by 2030. In this context, the contribution of unconventional hydrocarbons is crucial to achieving this goal.

• Coal Bed Methane (CBM):

India has established CBM in-place with commercially producing deposits.

The development of these resources can substantially augment domestic gas supplies.

Table 7.1: State-wise distribution of CBM Resources in India

No.	STATE	Estimated CBM resources (in BCM)	Estimated CBM resources (in TCF)	GIIP (in TCF)
1	Jharkhand	722.08	25.5	1.88
2	Rajasthan	359.62	12.7	0
3	Gujarat	351.13	12.4	0
4	Odisha	243.52	8.6	0
5	Chhattisgarh	240.69	8.5	0
6	Madhya Pradesh	218.04	7.7	3.65
7	West Bengal	218.04	7.7	6.56
8	Tamil Nadu	104.77	3.7	0
9	Telangana & Andhra Pradesh	99.11	3.5	0
10	Maharashtra	33.98	1.2	0
11	Northeast	8.50	0.3	0
Total CBM Resource		2599.48	91.8	12.09

*Conversion factor: 1 cubic metre = 35.3147 cubic feet

The development of these resources can substantially augment domestic gas supplies. By tapping into its CBM reserves, India can ensure a more stable and secure supply of natural gas, which is critical for power generation, industrial use, and residential consumption.

Shale Gas and Shale Oil:

A preliminary resource assessment of Shale gas/oil was carried out by three different agencies.

1. In 2011, the United States Geological Survey (USGS) estimated the technically recoverable Shale gas/oil as 6.1 TCF for 3 basins: Cambay,

Krishna-Godavari (KG), and Cauvery. Later in 2014, it estimated a technically recoverable volume of 62 million barrels of shale oil in Cambay Basin alone.

2. In 2013, the National Oil Company, ONGC estimated Shale Gas resources of 187.5 TCF from 5 sedimentary basins: Cambay, KG, Cauvery, Ganga, and Assam.
3. CMPDI in July 2013 had estimated around 45.8 TCF in the Gondwana basin.

It has been envisaged that shale gas and oil exploration in basins/sub-basins



like Cambay, Krishna-Godavari (KG) and Damodar Valley can significantly help meet the growing energy demand.

• Gas Hydrates:

Gas hydrate deposits, located in the KG and Mahanadi offshore represent a significant untapped energy resource. Although commercial production is still in the research phase, gas hydrates could provide a substantial boost to India's future energy security.

7.2.2 Economic Growth

The development of unconventional hydrocarbons contributes to economic growth through job creation, infrastructure development, and revenue generation. CBM sector alone has generated significant employment in the states of Madhya Pradesh, Jharkhand and West Bengal.

As of 31.03.2025, USD 2.6 billion has been invested for exploration, development, and production activities in eight operational CBM blocks. CBM production contributed USD 148 million as Royalty to State Governments and USD 7 million to the Central Government.

7.3 Status of Unconventional Hydrocarbons

7.3.1 Coal bed Methane (CBM)

• Governing Policies:

1. In 1997, the Government of India formulated a policy for the development of Coal Bed Methane (CBM). A Memorandum of Understanding (MoU) was signed between the Ministry of Coal (MoC) and the Ministry of Petroleum and Natural Gas (MoPNG) to work cooperatively on this initiative. According to the policy, MoPNG was designated as the administrative ministry, with the Directorate General of Hydrocarbons (DGH) appointed as the nodal agency for CBM development in India.
2. In 2007, CBM Phases & Extensions Policy was framed to provide a transparent and consistent framework for granting extension in exploration phases, under CBM Contracts.
3. In 2015 (re-notified in 2018), the Government of India, granted permission to Coal India Limited (CIL) and its subsidiaries to explore and produce CBM from its areas

under Coal Mining Lease allotted to them, thereby, dispensing the requirement of having additional License from Ministry of Petroleum and Natural Gas. This was formulated to increase the area under CBM exploration and to enhance and accelerate the CBM production in the country from Coal mining areas.

4. In 2016, Unified Licensing Policy under Hydrocarbon Exploration and Licensing Policy (HELP) was introduced wherein all types of hydrocarbon resources, both conventional and unconventional were allowed to be explored and exploited. The Open Acreage Licensing Policy (OALP) to carry out exploration and production from areas which are either, free or relinquished and Discovered Small Field (DSF) Policy to exploit resources from already discovered fields are two such policies within the HELP regime.
5. In 2017, a policy framework for Early Monetization of CBM was introduced to develop alternate sources of natural gas including CBM and promote gas economy. This policy was formulated to provide marketing and pricing freedom for Coal Bed Methane (CBM) and streamline the operational issues in the existing blocks.
6. In 2018, the Government of India notified a Policy framework for Exploration and Exploitation of Unconventional hydrocarbons in existing acreages under existing Production Sharing Contracts (PSC), Coal Bed Methane (CBM) Contracts and Nomination fields.

• Award of CBM acreages

1. As per extant CBM policy 1997, The Ministry of Coal, through the Central Mine Planning and Design Institute (CMPDI), conducted studies to identify prospective CBM areas within the coal-bearing regions of the country. From 2001 to 2008, four CBM bidding rounds were held, resulting in the award of 30 CBM blocks. Additionally, two blocks were allocated on a nomination basis, and one block was awarded through the Foreign Investment Promotion Board (FIPB) route in the year 2001. During this period (2001-2008), a total area of 16,598 square kilometres was awarded for CBM exploration and production. These awarded blocks were located across various states including Andhra Pradesh, Assam,



Chhattisgarh, Gujarat, Jharkhand, Madhya Pradesh, Maharashtra, Odisha, Rajasthan, Tamil Nadu, and West Bengal.

- The total estimated CBM resource for the 33 awarded blocks is approximately 62 trillion cubic feet (TCF) or 1767 billion cubic meters (BCM), of which 10.5 TCF (296.9 BCM) has been established as Gas-in-Place (GIP).
- Further, in pursuit of the goal of energy

security and to increase the gas share in the country's energy mix GOI launched Special CBM bid rounds (SCBM) in 2021 & 2022 with attractive fiscal and administrative terms under the Open Acreage Licensing Policy (OALP) through International Competitive Bidding. During these two SCBM rounds 7 blocks of area 4579 sq.km were awarded. The prognosticated resources of these 7 awarded blocks are around 294 BCM.

Awarded Blocks

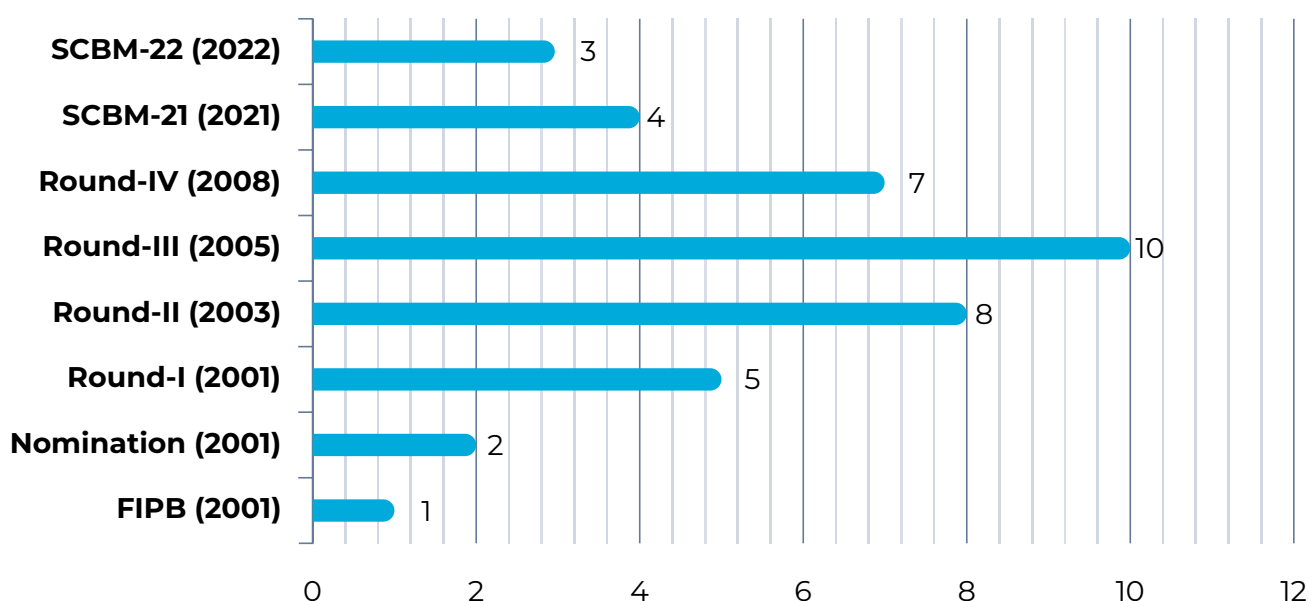


Figure 7.1 Awarded CBM Blocks

• Status of the awarded CBM blocks

- The first commercial production from the CBM blocks commenced in the year 2007 from the Raniganj South block operated by M/s. Great Eastern Energy Corp. Ltd. (GEECL). Thereafter three (3) more CBM Blocks, Raniganj (East) Operated by M/s. Essar Oil & Gas Exploration & Production Ltd (EOGEPL), Sohagpur (West) operated by M/s. Reliance Industries Ltd. (RIL) and Bokaro operated by M/s. Oil and Natural Gas Ltd. (ONGC) started its commercial production in 2016, 2017 & 2019 respectively. In addition to this incidental CBM gas is being produced during the testing of CBM wells in Jharia block operated by ONGC.
- At present, the active acreage for CBM exploration & production is around 7010 Sq km. Fifteen (15) CBM Blocks are under operation currently of which 7 are in exploration, 3 in development and 5 are in production stage. The cumulative CBM production up to FY 24-25 is 7.14 BCM.

Table 7.2: Snapshot of CBM resources in India

As on 31st March 2025	
Total CBM rounds completed	6
No. of CBM Blocks awarded in 6 rounds	40



As on 31st March 2025	
Area covered under 40 blocks	21,177 Sq. Km
Estimated CBM Resource in Country	2600 BCM (91.8 TCF)
CBM Resources (from 33 Blocks)	1767.06 BCM (62.4 TCF)
Established CBM Reserves (GIP) (8 CBM Blocks)	342 BCM (12.10 TCF)
Present Area for CBM Operations (15 Blocks)	7009 Sq. Km
Commercial Production commenced	Jul-07 1994
Total No. of Development Wells drilled	1198
Avg. Gas Production (FY 2024-25)	2.1 MMSCMD
No. of CBM Blocks in Development/Production Phase	8
No. of CBM Blocks in Exploration	7
No. of CBM Blocks Under Relinquishment	12
No. of CBM Blocks Relinquished	12
No. of blocks under Arbitration	1
Annual CBM Production in FY 2024-25	756.06 MMSCM
Cumulative Production up to FY 2024-25	7.14 BCM

Table 7.3: Awarded CBM Blocks

No.	Block	State	Present Area (SKM)	Contractor (PI%)	Contract signed on	Present Status
CBM BLOCKS OFFERED ON NOMINATION/FIPB ROUTE						
1	Raniganj (South)	West Bengal	210	GEECL (100)	31.05.2001	Production
2	Raniganj (North)	West Bengal	311.8	ONGC (74)-CIL (26)	06.02.2003	Development
3	Jharia	Jharkhand	67.1	ONGC (74)-CIL (26)	06.02.2003	Development/ Incidental production
CBM ROUND-I						
4	RG(East)-CBM-2001/I	West Bengal	500	EOGEPL (100)	26.07.2002	Production
5	SP(East)-CBM-2001/I	Madhya Pradesh	495	RIL (100)	26.07.2002	Development
6	SP(West)-CBM-2001/I	Madhya Pradesh	500	RIL (100)	26.07.2002	Production
7	BK-CBM-2001/I	Jharkhand	75	ONGC (80)-IOC (20)	26.07.2002	Production
8	NK-CBM-2001/I	Jharkhand	271.5	ONGC (55)-IOC (20)-PEPL (25)	26.07.2002	Development



No.	Block	State	Present Area (SKM)	Contractor (PI%)	Contract signed on	Present Status
CBM ROUND-II						
9	SH(N)-CBM-2003/II	Chhattisgarh	825	RIL (100)	06.02.2004	Relinquished
10	BS (1)-CBM-2003/II	Rajasthan	1045	RIL (100)	06.02.2004	Under relinquishment
11	BS (2)-CBM-2003/II	Rajasthan	1020	RIL (100)	06.02.2004	Under relinquishment
12	SK-CBM-2003/II	Jharkhand	70	ONGC (100)	06.02.2004	Under Relinquishment
13	NK(W)-CBM-2003/II	Jharkhand	267	ONGC (100)	06.02.2004	Relinquished
14	ST-CBM-2003/II	Madhya Pradesh	714	ONGC (100)	06.02.2004	Relinquished
15	WD-CBM-2003/II	Maharashtra	503	ONGC (100)	06.02.2004	Relinquished
16	BS (3)-CBM-2003/II	Gujrat	790	ONGC (70)-GSPC (30)	06.02.2004	Relinquished
CBM ROUND-III						
17	SP(N)-CBM-2005/III	Madhya Pradesh	609	R-Infra (55)-RNRL (45)	07.11.2006	Under Relinquishment
18	SR-CBM-2005/III	Madhya Pradesh	330	DIL (90)-Coal Gas (10)	07.11.2006	Under Relinquishment
19	RM-CBM-2005/III	Jharkhand	469	Dart Energy (35)-GAIL (35)-EIG (15)-TATA Power (15)	07.11.2006	Under Relinquishment
20	GV(N)-CBM-2005/III	Telangana	386	Coal Gas (10)-DIL (40)-Adinath (50)	07.11.2006	Relinquished
21	BB-CBM-2005/III	West Bengal	248	British Petroleum (100)	16.11.2006	Relinquished
22	MR-CBM-2005/III	Chhattisgarh	634	Dart Energy (35)-GAIL (35)-EIG (15)-TATA Power (15)	07.11.2006	Under relinquishment
23	TR-CBM-2005/III	Chhattisgarh	458	Dart Energy (35)-GAIL (35)-EIG (15)-TATA Power (15)	07.11.2006	Under relinquishment
24	BS (4)-CBM-2005/III	Rajasthan	1168	REL (45)-RNRL (45)-Geopetrol (10)	07.11.2006	Under Relinquishment
25	BS (5)-CBM-2005/III	Rajasthan	739	REL (45)-RNRL (45)-Geopetrol (10)	07.11.2006	Under Relinquishment

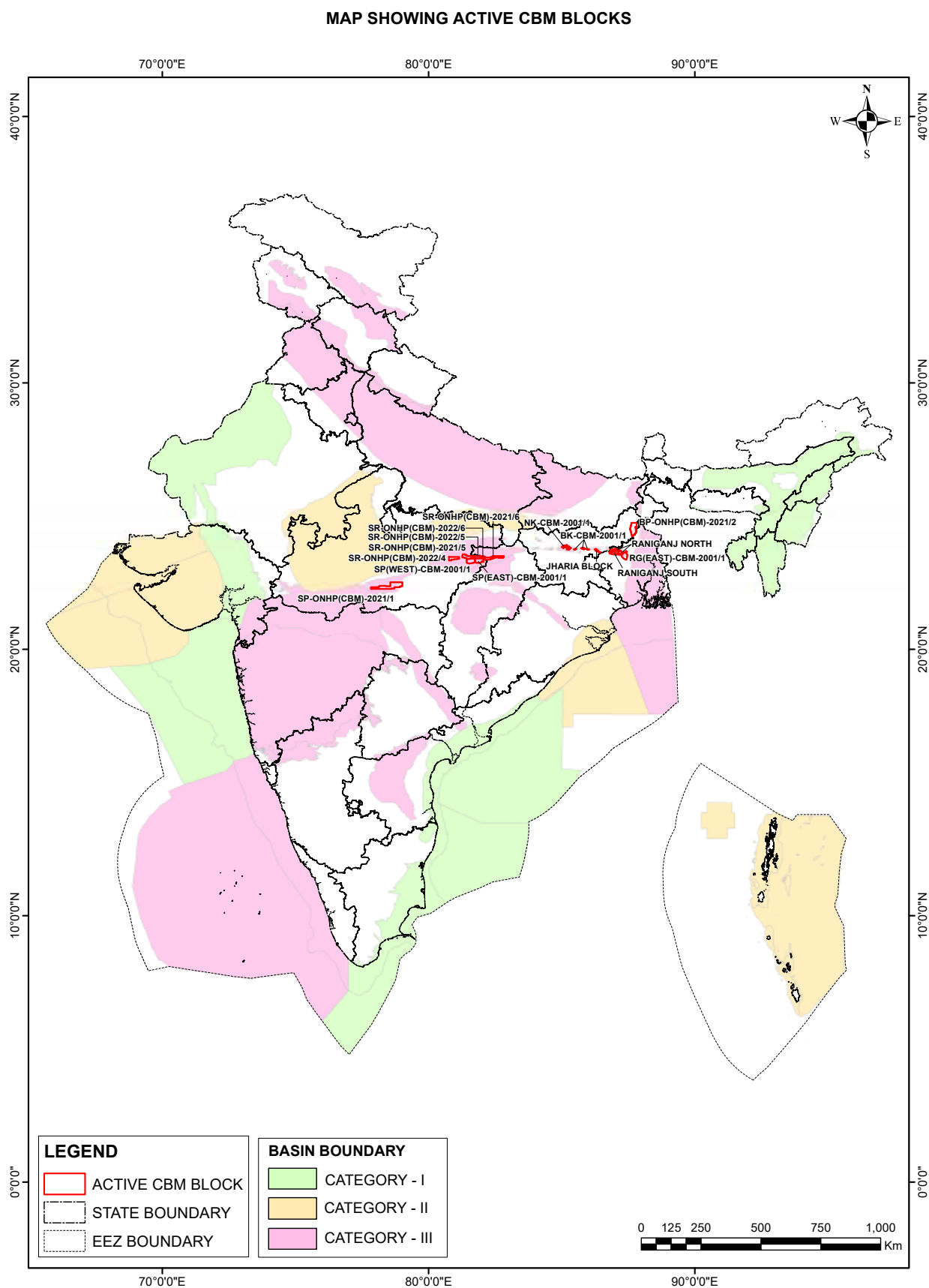


No.	Block	State	Present Area (SKM)	Contractor (PI%)	Contract signed on	Present Status
26	KG (E)-CBM-2005/III	Andhra Pradesh	735	REL (45) – RNRL (45) – Geopetrol (10)	07.11.2006	Relinquished
CBM ROUND-IV						
27	AS-CBM-2008/IV	Assam	113	Dart Energy (10)-OIL (90)	29.07.2010	Under Relinquishment
28	MG-CBM-2008/IV	Tamil Nadu	667	GEECL (100)	29.07.2010	Under Arbitration
29	RM(E)-CBM-2008/IV	Jharkhand	1128	EOGEPL (100)	29.07.2010	Under Relinquishment
30	TL-CBM-2008/IV	Odisha	557	EOGEPL (100)	29.07.2010	Relinquished
31	IB-CBM-2008/IV	Odisha	209	EOGEPL (100)	29.07.2010	Relinquished
32	SP(NE)-CBM-2008/IV	Madhya Pradesh & Chhattisgarh	339	EOGEPL (100)	29.07.2010	Under Relinquishment
33	ST-CBM-2008/IV	Madhya Pradesh	714	Dart Energy (80)-TATA Power (20)	29.07.2010	Relinquished
SPECIAL CBM BID ROUND-2021						
34	BP-ONHP(CBM)-2021/2	Jharkhand	991	ONGC	09.09.2022	PEL Awaited
35	SP-ONHP(CBM)-2021/1	Madhya Pradesh	1771.5	Invenire Petrodyne	09.09.2022	PEL Granted
36	SR-ONHP(CBM)-2021/5	Madhya Pradesh	515	ONGC	09.09.2022	PEL Granted
37	SR-ONHP(CBM)-2021/6	Chhattisgarh	584	Vedanta	09.09.2022	PEL Granted
SPECIAL CBM BID ROUND-2022						
38	SR-ONHP(CBM)-2022/4	Madhya Pradesh	418	Transcontinental Natural Resources Ltd	03.01.2024	PEL Granted
39	SR-ONHP(CBM)-2022/5	Madhya Pradesh	211	Transcontinental Natural Resources Ltd	03.01.2024	PEL Granted
40	SR-ONHP(CBM)-2022/6	Chhattisgarh	88	Oilmax Energy Private Ltd.	03.01.2024	PEL Granted

*PEL- Petroleum Exploration License, granted by respective State Governments



Figure 7.2: The geographical distribution of operational/active CBM blocks is shown in below map



• CBM Production Over the Years

At present out of fifteen (15) active/operational CBM Blocks, eight (8) CBM blocks are under production & development. The table below

presents historical data on production and development wells drilled across the eight blocks.

Table 7.4: Gas Initially in Place (GIIP), CBM Reserves, Production and Development wells

Sl. No.	Block Name	Put on Production	GIIP	GIIP	Rec. Reserves (BCM)	Rec. Reserves (TCF)	Cum. Production Up to FY 24-25	Cum. Production	Development Wells
			(BCM)	(TCF)			(BCM)	(BCF)	
1	RG(E)-CBM-2001/I	Jul-2016	106.84	3.77	22.12	0.78	5.88	207.56	458
2	SP(W)-CBM-2001/I	Mar-2017	55.52	1.96	13.06	0.46	2.32	81.90	354
3	RANIGANJ (SOUTH)	Jul-2007	74.19	2.62	11.88	0.42	1.82	64.25	156
4	BK-CBM-2001/I	Aug-2019	25.76	0.91	2.62	0.09	0.06	2.12	101
5	JHARIA*	Jan-2012	11.95	0.42	0.76	0.03	0.04	1.33	27
6	NK-CBM-2001/I	-	15.69	0.55	0.79	0.03	0.00	0.00	62
7	SP(E)-CBM-2001/I	-	47.86	1.69	10.09	0.36	0.01	0.33	37
8	RANIGANJ (NORTH)	-	4.88	0.17	0.00	0.00	0.00	0.00	3
Total			342.69	12.09	61.32	2.17	10.13	357.48	1198

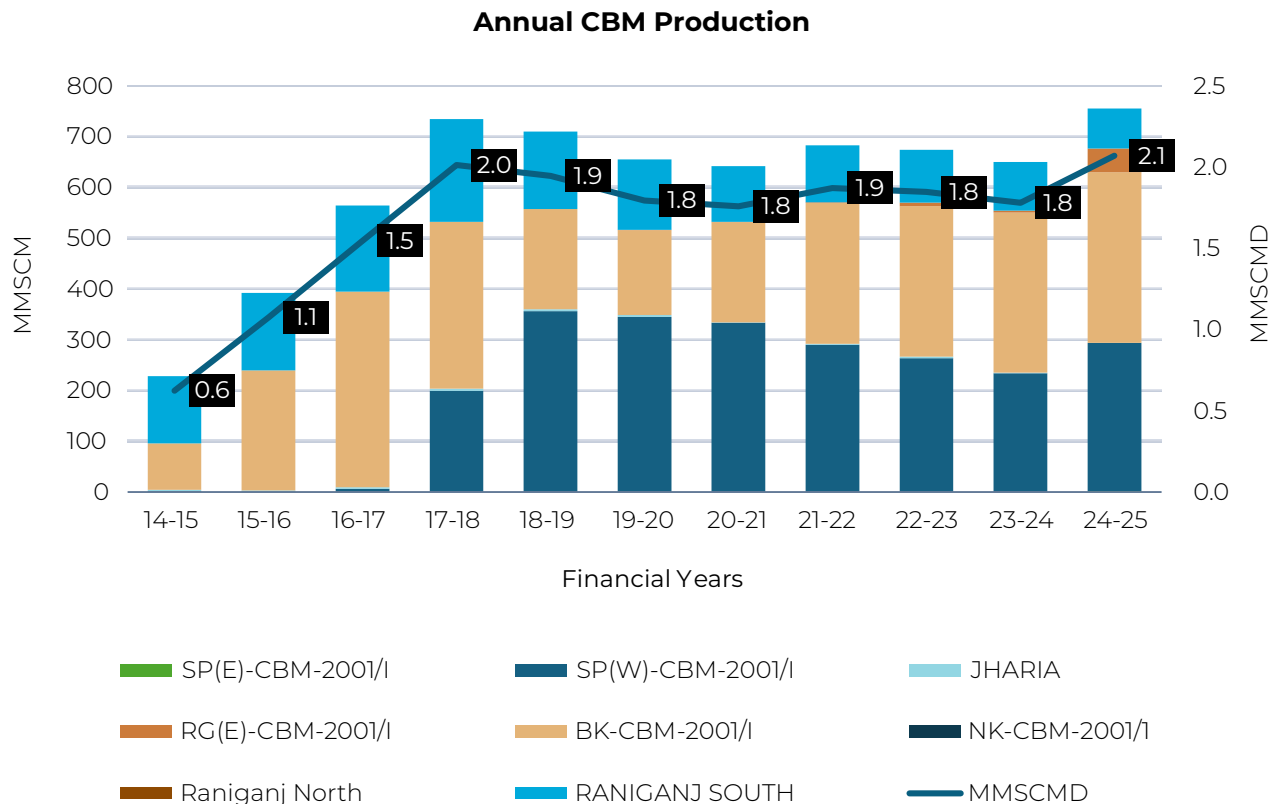
*Jharia is on test production since 2012.



Table 7.5 CBM Production (MMSCM)

Below tables presents the Year-on-year CBM production volumes for last 10 years

Sl No	Block Name	14-15	15-16	16-17	17-18	18-19	19-20	20-21	21-22	22-23	23-24	24-25
1	SP(W)-CBM-2001/I	1.5	1.4	6.4	199.8	356.7	344.9	333.7	290.1	263.9	233.8	292
2	RG(E)-CBM-2001/I	91.3	236.5	385.5	328.4	197.7	166.9	197.1	277	296.3	315.1	336
3	RANIGANJ SOUTH	132.4	152.9	169.6	202.6	152.6	138.9	109.9	112.3	104.0	95.8	79
4	BK-CBM-2001/I	0	0	0	0	0	0.2	0.6	1.6	7.0	3.9	47
5	JHARIA	2.5	2	3.1	4	3.6	4.4	1.1	2.4	3.1	1.5	0
6	SP(E)-CBM-2001/I	0.6	0	0.1	0	0	0	0	0	0.0	0.4	2
7	NK-CBM-2001/I	0	0	0	0	0	0	0	0	0.0	0.0	0
8	Raniganj North	0	0	0	0	0	0	0	0	0.0	0.0	0
Total		228.30	392.80	564.70	734.80	710.60	655.30	642.40	683.40	674.36	650.45	756.00
MMSCMD		0.6	1.1	1.5	2.0	1.9	1.8	1.8	1.9	1.8	1.8	2.1

**Figure 7.3 Annual CBM Production**



• Case Studies: Insights into significant projects

A. RG(East)-CBM-2001/1: The CBM Block RG(East)-CBM-2001/1, covering an area of approximately 500 Sq.km, was awarded to M/s Essar Oil Ltd in 2002. The block is in Production since July 2016. Some of key highlights of the CBM project in RG(East) are:

- ▶ Drilling of more than 450 development wells.
- ▶ Current production rate is 0.95 MMSCMD.
- ▶ Implementation of technological advancements as:
 - a. Advanced Chemical Stimulation: Coal fines accumulation near the wellbore reduces water and gas output and leads to productivity loss and well failure issues like rotor locking. To address this, surfactant flushing has been implemented to clear near-wellbore damage, improving water mobility and gas production. Recent flushing showed positive results, and it is planned to apply this in 20% of reactive wells.
 - b. Microbial techniques to enhance oil recovery by breaking down organic materials. This technique facilitates the microbial communities involved in the dynamic process of methanogenesis to further increase the methane production from the biodegradation of coal over timescales.
 - c. Well Uptime Improvement: Several measures have been taken to improve well uptime. 3½" tubing was introduced just above the pump to reduce HIT issues. Moulded rod guides and tubing rotators were added to reduce rod failure due to torque and/or well trajectory. Blast joints have been introduced, and tungsten and liner-coated tubing are being used to minimize wear and corrosion. Hydraulic Regulated Progressive Cavity Pumps (HRPCP) were also implemented to improve MTBF.
 - d. HVFR (High Viscosity Friction Reducer) has been used as it has very less residue as compared to Guar based linear gel to increase frac half-length as permeability is very low due to overburden stress.
 - e. Planned to drill Horizontal wells in deeper, tight CBM reservoirs to address permeability challenges which may have better productivity due to greater coal seam exposure.
- ▶ Extensive data acquisition programme underway in the deeper area to identify opportunities in deeper CBM to open-up newer areas of development & increase reserves base.
- ▶ The exploration of Shale gas is also underway in RG(East) CBM Block.



B. SP(West)-CBM-2001/1

The CBM Block SP(West)-CBM-2001/1, covering an area of approximately 500 Sq.km, was awarded to M/s Reliance Industries Ltd in 2002. The block is in Production since March 2017. Some of key highlights of the CBM project in SP(West) are:

- Drilling of more than 300 development wells.

- Current production rate is 0.88 MMSCMD.

- To ramp production from block, M/s RIL has initiated Multi-Lateral Horizontal Well Program, and the results are very encouraging. So far the plan is implemented in 40 wells which has resulted a production growth of about 40%. A schematic of Quad Lateral Horizontal Wells is presented below.

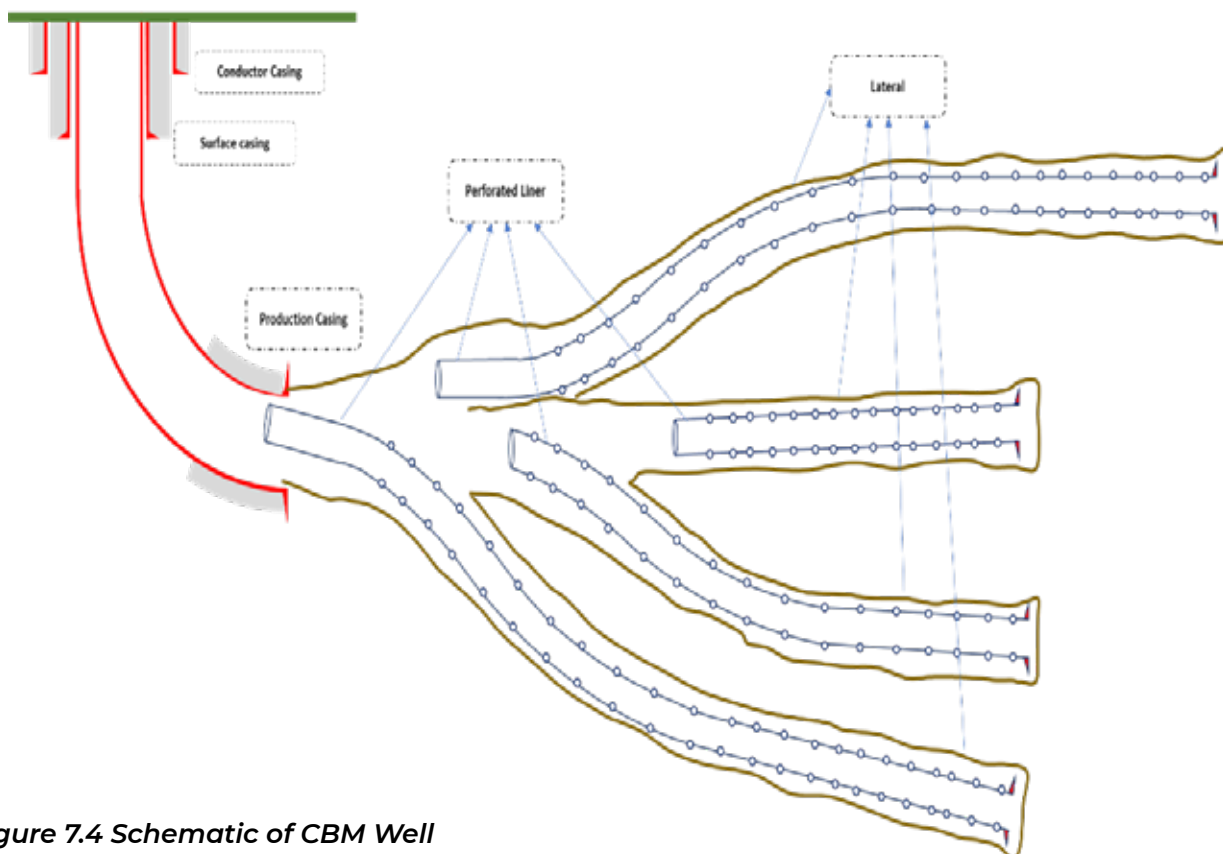


Figure 7.4 Schematic of CBM Well

• Future Prospects

1. Exploration Plans: GoI is in pursuit of establishing CBM resource in unexplored coal bearing areas by taking up exploration drive to further enhance the CBM acreages. Additionally, GoI is looking for alternatives as analysing the potentiality of CBM production in abandoned or de-coaled mines, which may not be suitable for coal mining.
2. Coal India and its subsidiaries are actively looking for coal bearing areas under their leasehold to award for CBM exploration and production. As per GoI policy 2018, wherein CIL and its subsidiaries were granted CBM rights in their leasehold areas, Bharat Coking Coal Limited (BCCL), a subsidiary of Coal India Limited has awarded one CBM blocks (area-24 Sq.km) in Jharia for CBM exploration and production.
3. The competitive gas price along with enhanced natural gas pipeline across the country has significantly enhanced the natural gas demand.
4. The projection of CBM up to FY 29-30 is shown in below table. This consists of production from existing 8 operational blocks. The recently awarded blocks (7) may also add production volume in coming years.



Table 7.6 CBM Production Projection

SI No	Block	Projection (MMSCM)				
		25-26	26-27	27-28	28-29	29-30
1	SOHAGPUR WEST	366.0	337.0	300.0	275.0	284.0
2	RANIGANJ EAST	406.0	507.5	487.1	493.0	564.4
3	RANIGANJ SOUTH	94.0	264.0	270.0	285.0	324.0
4	Bokaro	87.0	184.0	297.0	315.0	409.9
5	JHARIA	48.0	64.0	120.0	118.0	162.3
6	SOHAGPUR EAST	0.0	4.0	65.0	146.0	213.7
7	NK-CBM-2001/1	3.0	126.6	185.8	189.7	230.5
8	Raniganj North	0.0	37.0	125.0	120.0	177.0
Total		1004	1524	1850	1942	2366
MMSCMD		2.75	4.18	5.07	5.32	6.48

7.3.2 Shale gas

• Governing Policies:

1. In order to understand the prospectivity and untap the Shale Gas and Oil resource potential in India, GoI announced a Shale gas and Oil exploration policy on 14th October 2013 for the National Oil Companies (NOCs), ONGC, and OIL. The companies were required to carry out exploration in their PML areas.
2. In 2018, the Government of India notified a Policy framework for Exploration and Exploitation of Unconventional hydrocarbons in existing acreages under existing Production Sharing Contracts (PSC), Coal Bed Methane (CBM) Contracts and Nomination fields.
3. In Oct 2018, a policy framework to promote and incentivize Enhanced Recovery Methods for Oil and Gas was notified by the Ministry of Petroleum and Natural Gas, Government of India. Under this policy, fiscal incentives are provided from the first day of the entire production from future discoveries of unconventional hydrocarbons (Shale Gas/Oil and Gas Hydrate). Operators employing enhanced recovery techniques for shale gas extraction are offered reduced royalty rates and various tax benefits. These incentives are designed to make the deployment of such technologies more economically attractive.

• Shale gas exploration in Country

Under the Shale Gas Policy - 2013, 50 blocks in

4 basins; Assam, Krishna Godavari, Cauvery & Cambay were identified by ONGC, and 6 blocks in 2 basins; Jaisalmer and Assam were identified by OIL in the Phase-I of exploration which ended in April-2017.

Basin-wise work carried out by ONGC in its nomination areas is as below:

Table 7.7 Basin-wise ONGC work in Shale Gas

Basin	Phase-I Blocks	Blocks Taken up for Drilling	Actual Wells Drilled
Cambay	28	16	17
KG	10	6	8
Cauvery	9	1	3
Assam	3	2	2
Total	50	25	30

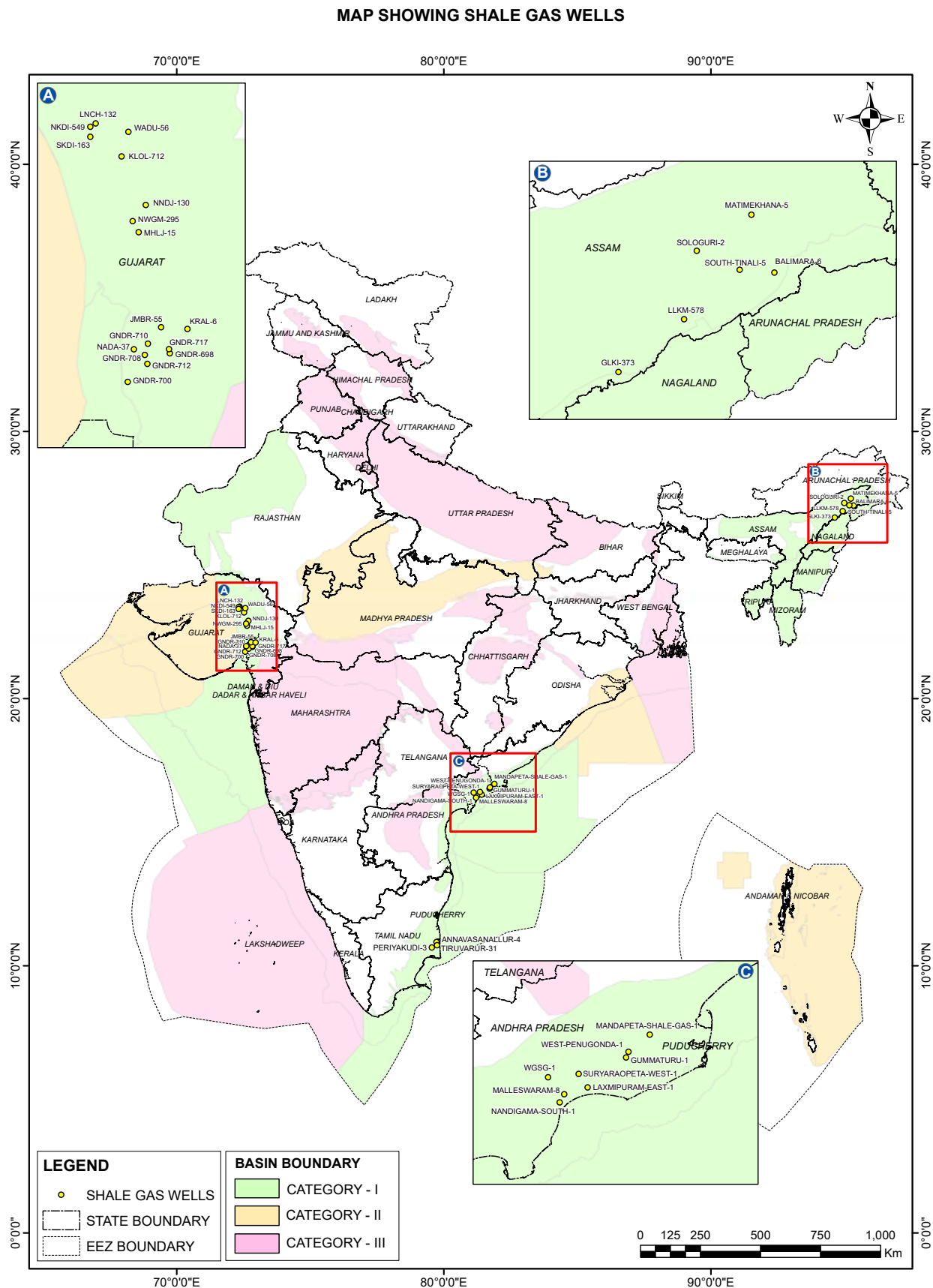
Basin-wise work carried out by OIL in its nomination areas is as below:

Table 7.8 Basin-wise OIL work in Shale Gas

Basin	Phase-I Blocks	Blocks Taken up for Drilling	Actual Wells Drilled
Jaisalmer	1	1	1
Assam	5	0	3
Total	6	1	4



Figure 7.2: The geographical distribution of shale gas/oil is shown in below map



• Future Prospects

1. In 2018 Govt. has allowed E&P operators in India to explore and develop all unconventional hydrocarbons (CBM, shale oil and gas, tight gas) under existing production sharing contracts, CBM blocks, and nominated licenses. This allows the exploration and development of shale gas/oil in existing PML areas.
2. The Barren Measure formation underlying the Raniganj formation in Raniganj Coalfields (West Bengal & Jharkhand) is rich in organic content consisting dominantly of shale interbedded with silt which is considered to have good potential for Shale gas. Preliminary studies carried out by existing CBM operators i.e. M/s GEECL & M/s EOGPL incorporating all the available datasets have shown presence of prolific thickness of Barren Measure shale across the basin.

7.3.3 Gas Hydrates

• Governing Policies:

The National Gas Hydrate Program (NGHP) is an initiative by the Government of India aimed at exploring and exploiting gas hydrates as a potential source of energy with an objective to conduct comprehensive surveys to assess the potential of gas hydrate resources in Indian offshore areas, particularly in the Krishna-Godavari (KG), Mahanadi, and Andaman basins.

• NGHP Expeditions

1. NGHP Expedition 01 (2006):

The first expedition was primarily an exploratory mission to identify gas hydrate

occurrences. Under NGHP-01, 39 holes were drilled to explore the presence of gas hydrate at 21 sites in Krishna-Godavari (KG), Mahanadi, Andaman, and Kerala-Konkan Basin. Except for Kerala-Konkan, the presence of gas hydrate was established in all three basins, but such gas was proved to be non-exploitable with existing technologies.

2. NGHP Expedition 02 (2015):

This phase focused on delineating gas hydrate deposits in the Krishna-Godavari Basin and assessing their potential for future production. Advanced drilling techniques and sophisticated logging tools were used to gather detailed data on the hydrate-bearing sediments. 42 holes were drilled at 25 sites in KG and Mahanadi Basin. Sand reservoirs with gas hydrate were located at two places in KG Basin and the considered to be prospective for future production testing.

3. NGHP Expedition 03 (Ongoing):

The current phase aims at testing production technologies and developing pilot projects for gas hydrate extraction. It involves the deployment of innovative drilling and production technologies to evaluate the feasibility of commercial gas hydrate production.

• Future Prospects

Given India's growing energy demands and the need for energy security, the NGHP plays a crucial role in identifying and developing these unconventional hydrocarbon resources. Last meeting of the NGHP committee was held in 2019. Committee suggested for acquiring of More data and drilling of more wells in Phase-2. Phase-3 i.e. Production phase will be carried out later after acquisition of more data.





8

Environmental and Regulatory Oversight in India's Upstream Oil & Gas Sector

8.1 Introduction

8.1.1 Energy and Ecology: Navigating Environmental Governance in India's Oil & Gas Sector

India's oil and gas sector is often hailed as the lifeblood of industrial progress and national security. With nearly 85% of its crude oil requirements imported, the country's energy security critically depends on tapping domestic reserves found in the deserts of Rajasthan, the forests of Assam, and the deep waters of the Krishna-Godavari basin.

Yet, the story of hydrocarbon development is not just one of exploration and engineering, it is equally a narrative of environmental responsibility. The challenge lies in ensuring that the "black gold" beneath our feet does not tarnish the green canopy above.



We do not inherit the earth from our ancestors; we borrow it from our children.

-Native American Proverb



Environmental governance plays a pivotal role in ensuring that petroleum operations ranging from seismic surveys and exploratory drilling to production and pipeline transport are conducted responsibly, with minimal adverse impact on ecosystems and communities. A robust environmental framework not only safeguards biodiversity and natural resources but also builds public trust, ensures compliance with



international commitments (e.g., Sustainable Development Goals, the Paris Agreement), and fosters long-term sustainability in the energy sector.

A well-structured regulatory regime, including Environmental Clearance (EC), Forest Clearance (FC), and other statutory approvals, is thus essential. It enables sustainable development while maintaining investor confidence.

In this context, environmental governance refers to the institutions, processes, and legal frameworks that regulate how humans interact with the environment, particularly in high-impact industries like oil and gas. It encompasses regulatory approvals, environmental impact assessments (EIA), public consultations, and continuous monitoring of ecological and social impacts throughout the project lifecycle.

8.1.2 Looking Ahead: Governance as a Growth Enabler

Environmental governance is often perceived as a compliance burden, a procedural necessity that slows down project execution. However, this view is rapidly changing. Increasingly, governance is being recognized as a strategic enabler of growth in the oil and gas sector.

Projects that proactively adhere to environmental norms early in their lifecycle tend to face fewer litigations, encounter smoother approval processes, and are better positioned to attract responsible investment. In an era of heightened ESG (Environmental, Social, and Governance) scrutiny, environmental compliance is no longer optional, but it is a marker of credibility and foresight.

The Government of India has acknowledged this shift and is actively pursuing regulatory reforms to make environmental governance more transparent, efficient, and investor friendly. Key initiatives include:

- ▶ **Ease of Doing Business Measures:** Enhancing procedural clarity and accelerating the clearance process.
- ▶ **Digitization of EC/FC Processes:** Introduction of platforms like PARIVESH 2.0, which provide end-to-end online submission, monitoring, and tracking of clearances.
- ▶ **Policy-Level Exemptions for Low-Impact Exploration:** Special provisions for activities such as seismic surveys or geophysical

studies that have negligible environmental impact.

These reforms not only reduce transaction costs and project delays but also align with India's broader goals of energy security, climate action, and sustainable development. As the sector matures, governance is emerging not just as a regulatory requirement, but as a driver of resilience, innovation, and long-term value creation.

8.1.3 Overview of Statutory Oversight: Roles and Responsibilities of Central and State Authorities

India's oil and gas sector operates within a multi-layered statutory framework, involving a constellation of institutions at both the central and state levels. These authorities collectively ensure environmental oversight, facilitate regulatory compliance, and uphold the principles of sustainable development in project planning and execution.

Central Authorities

- **Ministry of Environment, Forest and Climate Change (MoEF&CC):**

The MoEF&CC serves as an apex authority for evaluating the environmental implications of projects and activities. It is responsible for:

- ▶ Granting statutory environmental, forest & wildlife clearances after due assessment of potential impacts.
- ▶ Reviewing compliance with stipulated conditions under various environmental, forest & wildlife laws, rules, and regulations.
- ▶ Overseeing the Environmental Impact Assessment (EIA) processes and maintaining platforms such as PARIVESH 2.0 for digital clearance workflows.

- **Ministry of Petroleum and Natural Gas (MoP&NG):**

As the administrative ministry for the sector, MoP&NG plays a facilitative and policy-oriented role, including:

- ▶ Formulating and updating sectoral policies and guidelines.
- ▶ Coordinating with MoEF&CC and state agencies for smooth processing of statutory approvals.



- ▶ Enabling regulatory reforms to support responsible and timely hydrocarbon exploration & development.

- **Directorate General of Hydrocarbons (DGH):**

Operating under MoP&NG, DGH functions as the technical regulator for upstream oil and gas activities. Its key responsibilities include:

- ▶ Supporting project developers in navigating the clearance process.
- ▶ Identifying and addressing procedural inefficiencies to expedite licensing and approvals.
- ▶ Facilitating inter-agency communication and ensuring adherence to regulatory frameworks.

In recent years, DGH has spearheaded several initiatives to streamline and fast-track clearance processes. These reforms which are reflected through evolving MoEF&CC policies and procedural relaxations are documented in detail in later sections of this chapter.

- **State Authorities:**

State Forest Departments, State Pollution Control Boards (SPCBs), and District Administrations play a crucial ground-level role in the clearance and compliance ecosystem. Their responsibilities include:

- ▶ Conducting field verifications and site assessments.
- ▶ Forwarding and evaluating clearance proposals, including forest diversion cases.
- ▶ Managing public hearings and consultations under the EIA Notification.
- ▶ Issuing Consent to Establish (CTE) and Consent to Operate (CTO) under pollution control laws such as the Water and Air Acts.
- ▶ Monitoring local environmental compliance and addressing community grievances.

This interconnected governance framework ensures that project approvals in the oil and gas sector are subjected to rigorous scrutiny, balanced decision-making, and institutional accountability, thereby aligning economic growth with ecological preservation.

8.2 Statutory Clearances and Licenses for Oil & Gas Projects in India

8.2.1 Expediting Clearances for E&P Operations: The Oilfields (Regulation and Development) Act, 1948 (amended in 2025) and the Petroleum and Natural Gas Rules, 1959

Environmental protection is not merely a regulatory requirement rather it is often deeply rooted in the cultural and traditional practices of many communities across India. As such, aligning environmental priorities with stakeholder interests and the legislative framework is critical to achieving sustainable and inclusive development in the oil and gas sector.

The Oilfields (Regulation and Development) Act, 1948, as recently amended in 2025, along with the Petroleum and Natural Gas Rules, 1959, empowers the Central Government to regulate the development and management of oil and gas blocks across the country.

- **Traditionally:**

- ▶ Onshore Petroleum Exploration Licenses (PEL) and Petroleum Mining Leases (PML) were issued by State/UT Governments, based on allocations made by the Centre.
- ▶ Offshore blocks, in contrast, were managed and leased directly by the Central Government.

Upon award of a block and execution of the relevant production-sharing or revenue-sharing contract, operators are required to:

- ▶ Obtain a Petroleum Exploration License (PEL) for non-commercial, exploratory operations.
- ▶ Secure a Petroleum Mining Lease (PML) for commercial development and production.

Recognizing the need to simplify and accelerate the licensing process, the 2025 amendment to the Oilfields Act introduces a significant reform: the creation of a unified "Petroleum Lease". This new lease model replaces the bifurcated PEL and PML structure with a single comprehensive lease covering both exploration and production phases.





- **Objectives of the Unified Petroleum Lease:**
 - ▶ Streamline administrative procedures
 - ▶ Reduce delays associated with multiple approval stages
 - ▶ Improve operational efficiency
 - ▶ Facilitate faster initiation and scaling up of exploration and production (E&P) activities

Clearances Beyond Licensing

In addition to acquiring the requisite PEL, PML, or the new Petroleum Lease, project developers must secure a suite of statutory clearances to commence operations. These include:

- ▶ Environmental Clearance (EC) under the EIA Notification, 2006.
- ▶ Forest Clearance (FC) under Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980 (amended in 1988 and in 2023).
- ▶ Wildlife Clearance under the Wildlife (Protection) Act, 1972 (where applicable).
- ▶ Consent to Establish (CTE) and Consent to Operate (CTO) under the Water and Air Acts issued by State Pollution Control Boards.

While the lease confers preferential access rights to the block, no exploration or production activity can commence without securing these mandatory clearances. This layered process ensures that resource development proceeds

responsibly, with due regard for environmental conservation, biodiversity, and community interests.

8.2.2 Evolution of Statutory Clearances in India

The concept of environmental protection in India is deeply rooted in constitutional mandates, cultural values, and legislative instruments. Under the Environment (Protection) Act, 1986, the term environment is broadly defined as:

"Environment includes water, air and land and the inter-relationship which exists among and between water, air and land, and human beings, other living creatures, plants, micro-organisms and property."

This holistic view emphasizes the interdependence of natural and human systems, forming the basis for India's comprehensive statutory framework governing environmental clearances and sustainable development.

Constitutional Mandates for Environmental Protection

Environmental stewardship is enshrined in the Constitution of India, creating both a directive obligation on the state and a fundamental duty for citizens:

- ▶ **Article 47:** Directs the State to improve public health and the standard of living.
- ▶ **Article 48A:** Mandates the State to protect and improve the environment and safeguard forests and wildlife.



- ▶ **Article 51(A)(g):** Places a fundamental duty on every citizen to protect and improve the natural environment, including forests, lakes, rivers, and wildlife.

These provisions serve as the foundation for the legal and regulatory framework that governs environmental clearances in the country.

• Key Environmental Laws and Their Evolution

Over the decades, the Government of India has enacted and strengthened several environmental laws to ensure responsible industrial development, especially in sectors such as oil and gas. The key legislations include:

1. Wildlife Protection Act, 1972

- Ensures the protection and regulation of wildlife to maintain ecological integrity.
- Requires Wildlife Clearance for projects located within or near protected areas (PAs) or eco-sensitive zones (ESZs).

2. Water (Prevention and Control of Pollution) Act, 1974 (Amended in 1988)

- Aims to prevent and control water pollution and maintain or restore the “wholesomeness of water.”
- Established the Central and State Pollution Control Boards (CPCB/SPCBs) for enforcement.

3. Forest (Conservation) Act, 1980 (Van (Sanrakshan Evam Samvardhan) Adhiniyam, amended in 1988 & 2023)

- Regulates the diversion of forest land for non-forest use.
- Mandates compensatory afforestation and clearance from MoEF&CC.
- Amended in 1988 and 2023 to improve efficiency and ecological safeguards.

4. Air (Prevention and Control of Pollution) Act, 1981

- Provides for the prevention, control, and abatement of air pollution.
- Empowers pollution control boards to set emission standards and enforce compliance.

5. Environment (Protection) Act, 1986

- Grants **sweeping powers to the Central Government** to protect and improve environmental quality.
- Serves as an umbrella legislation under which rules like **EIA, hazardous waste, and CRZ** notifications have been issued.

6. Biological Diversity Act, 2002

- Aims to conserve biological diversity and regulate access to genetic resources.

7. Environmental Impact Assessment (EIA) Notification, 2006

- Introduced a **tiered clearance system**:
- **Category A projects:** Appraised at the national level by MoEF&CC.
- **Category B projects:** Appraised at the state level by SEIAA and SEAC.
- Mandates public consultation and expert scrutiny to assess environmental impacts before approval.

8. Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 (FRA)

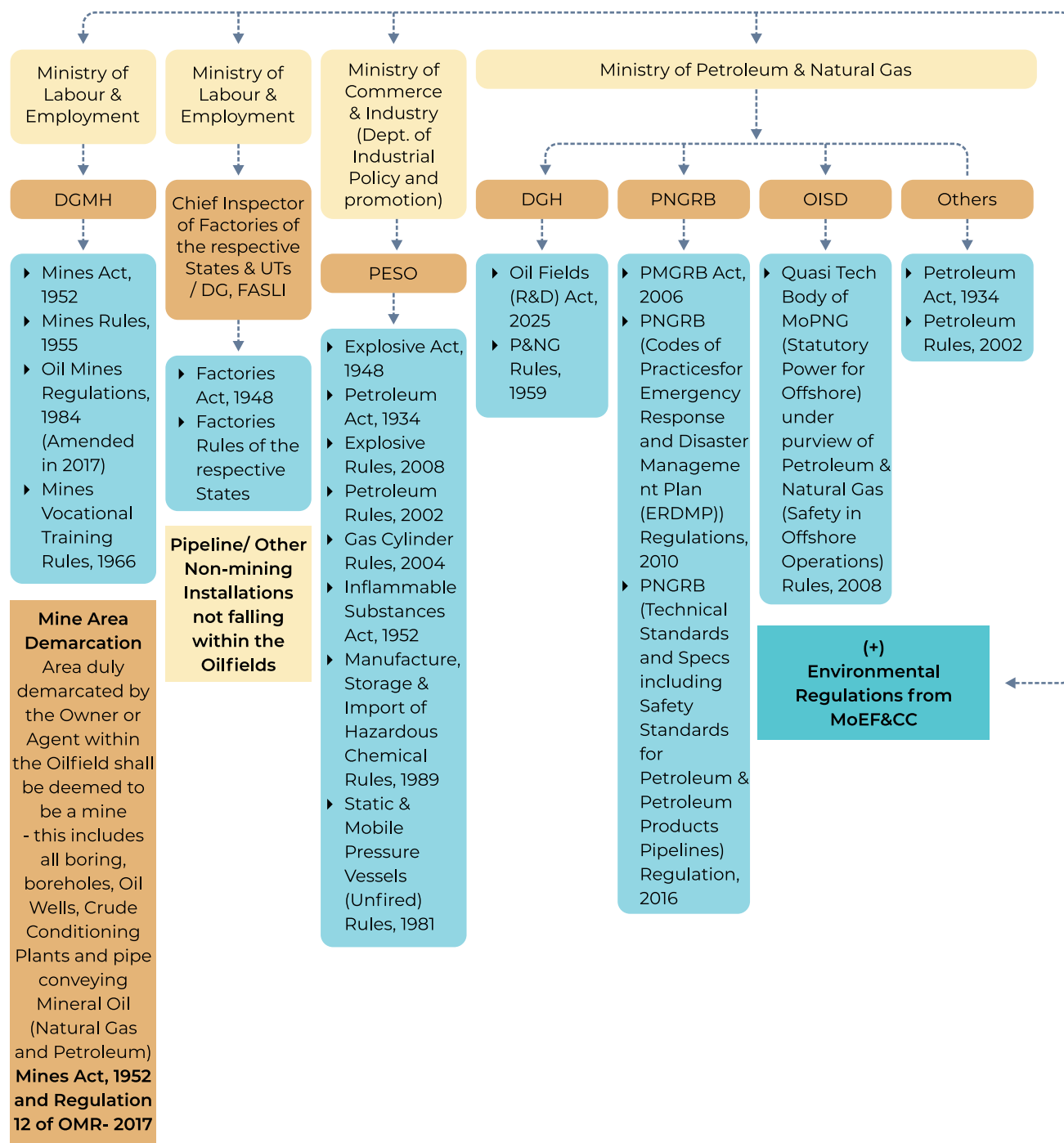
- Recognizes the traditional rights of forest-dwelling communities.
- Requires project proponents to obtain consent from Gram Sabhas prior to forest land diversion.

Together, these laws and rules constitute a comprehensive statutory architecture that guides the statutory clearance process for oil and gas in India. Their evolution reflects India's growing emphasis on balancing economic development with ecological sustainability, a principle now deeply embedded in national environmental policy.

8.2.3 Environmental Regulations for Oil & gas industry

Upstream O&G E&P operations are governed under multiple Rules/ Regulations/ Acts under different Ministries as detailed below:



Figure 8.1: Rules and Acts for governing O&G E&P operations by different Ministries.

Additionally, there are other applicable environmental regulations from MoEF&CC for the oil & gas industry, which are as follows:

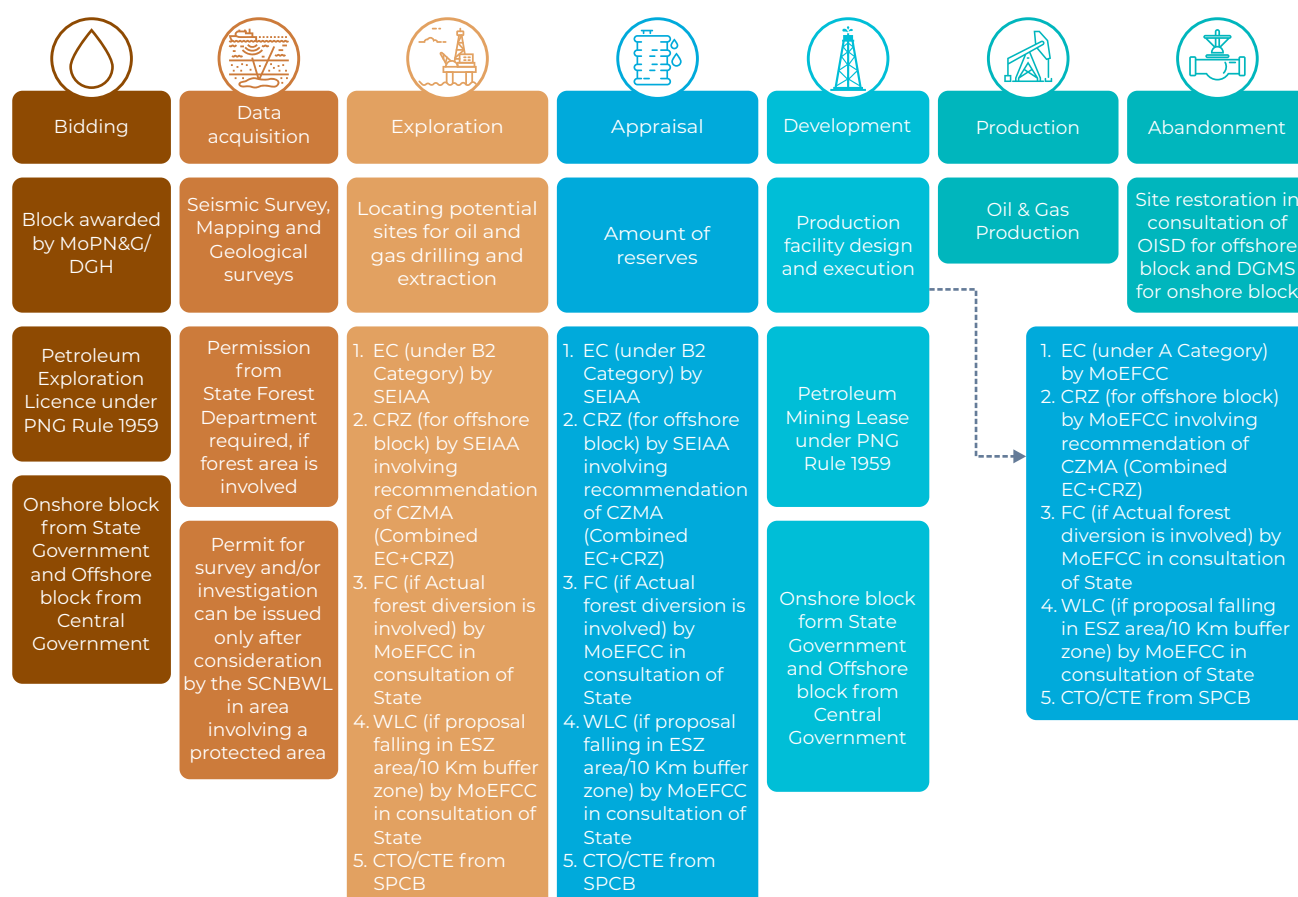
1. Indian Forest Act, 1927
2. Wildlife (Protection) Act, 1972.
3. The Water (Prevention and Control of Pollution) Act, 1974.
4. Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980 (amended in 1988 and 2023).
5. The Air (Prevention and Control of Pollution) Act, 1981.
6. The Environment (Protection) Act, 1986.
7. The Environment (Protection) Rules, 1986.



8. The Hazardous Waste (Management and Handling) Rule, 1989.
9. Manufacture, Storage and Import of Hazardous Chemicals Rules, 1989.
10. GSR 176(E) April 1996 - under the purview of EP Act, 1986 - Guidelines for Discharge of Solid Waste, Liquid Discharge and Gaseous Emissions by Oil Drilling and Gas Extraction Industry.
11. GSR. 546(E) dated 30th August 2005 under the purview of EP Act, 1986: Solid waste, drill cutting and drilling fluids disposal for onshore drilling operation.
12. Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 - For FCs.
13. EIA notification, 2006 vide S.O 1533 dated 14th September 2006: Not applicable beyond 12 Nautical Miles.
14. Hazardous Wastes (Management, Handling and Transboundary Movement) Rules, 2008.
15. GSR 186(E) 18th March 2008 under the purview of EP Act, 1986: Oil Refineries Standard and Notification.
16. Coastal Regulation Zone (CRZ) Notification, 2011.
17. Island Protection Zone (IPZ) Notification, 2011.

Figure 8.2: Basic Regulatory Requirements over Project Lifecycle of an Oil Block.

Environment Related Clearance: In the Lifecycle of Hydrocarbon Blocks

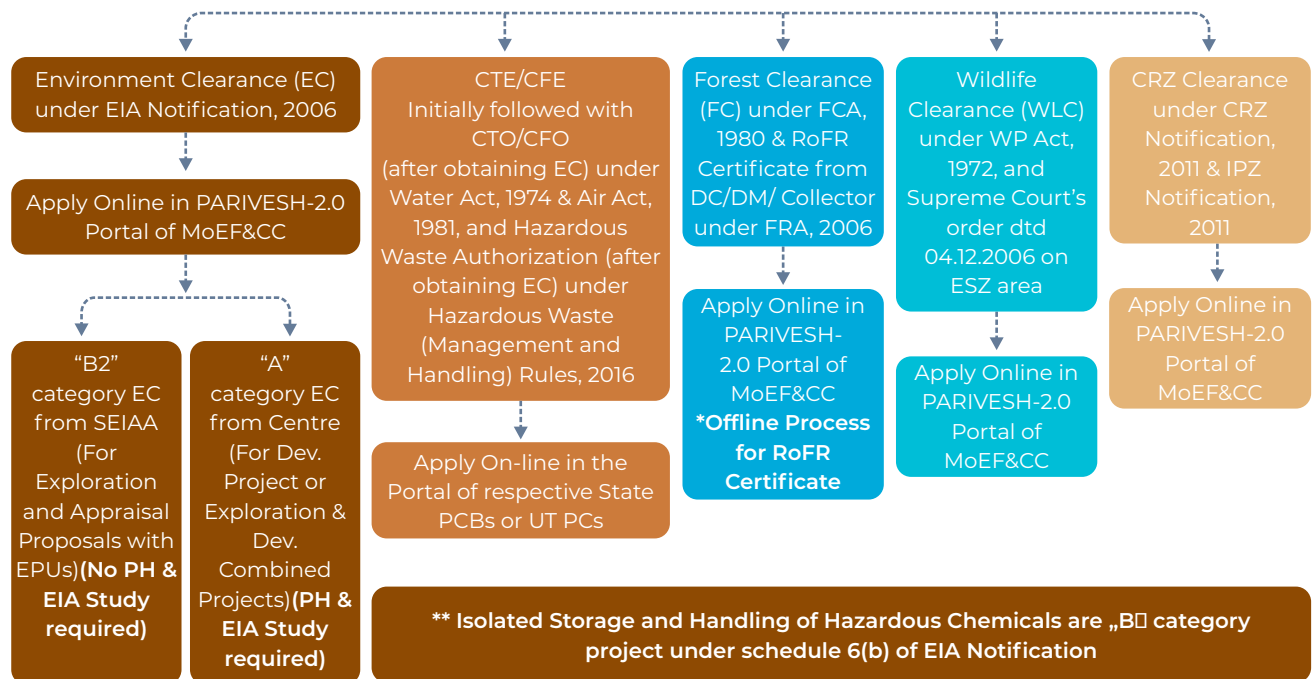


8.2.4 Statutory Environmental Approvals Required For “On shore offshore E & P activities

A chart showing the statutory approvals required

for oil & gas E&P activities under purview of various environmental regulations/Supreme Court's order and Forest Right Act is given as follows.

Fig 8.3: Statutory Environmental Approvals Required for O&G E&P Activities



8.3 Special Provisions, Relaxations & Enabling Reforms

In recent years, the Government of India has introduced a series of targeted reforms, procedural simplifications, and institutional innovations to streamline statutory clearances for the oil and gas sector particularly upstream E&P operations while maintaining robust environmental safeguards.

These reforms reflect a growing recognition that efficient governance can coexist with environmental responsibility, and that procedural clarity is essential to reduce delays, improve ease of doing business, and encourage investment in critical energy infrastructure.

8.3.1 PARIVESH 2.0 Portal – Overview and Benefits for Oil & Gas Operators

In 2023, the Ministry of Environment, Forest and Climate Change (MoEF&CC), Government of India, launched PARIVESH 2.0, a next-generation digital platform to facilitate statutory environmental clearances. Designed to improve efficiency, transparency, and predictability, the

portal manages proposals for Environmental Clearance (EC), Forest Clearance (FC), Wildlife Clearance (WL), and Coastal Regulation Zone (CRZ) clearance. For India's energy-intensive sectors like oil and gas, PARIVESH 2.0 represents a major step in aligning environmental governance with ease of doing business.

• Key Features of PARIVESH 2.0:

- ▶ **Unified Proposal Dashboard:** A single-window interface where project proponents can monitor the progress of all their proposals EC, FC, WL, and CRZ in one integrated view.
- ▶ **Automated Workflows and Alerts:** In-built automation and digital alert systems help reduce processing time, eliminate delays due to manual intervention, and ensure time-bound disposal of proposals.
- ▶ **GIS and Inter-Platform Integration:** Real-time access to spatial layers (forest boundaries, eco-sensitive zones, CRZs, wildlife corridors, etc.) by integrating with Bhuvan, Bhunaksha, and state forest databases.



- ▶ **Structured Communication Tools:** Enables seamless query management, digital submissions, and scheduling of site inspections through a centralized communication channel.
- **"Know Your Approval" (KYA): A Game-Changer for Hydrocarbon Projects**
 - ▶ Prevents cost overruns from delayed appraisals or rejected proposals.
 - ▶ Encourages proactive stakeholder engagement by clarifying approval requirements upfront.
 - ▶ Enhances project bankability and investor confidence through regulatory transparency.

One of the most transformative features of **PARIVESH 2.0** is the **"Know Your Approval (KYA)"** module, a pre-application advisory tool that allows project developers to assess the likely statutory requirements **before** submitting a formal proposal. For the oil and gas industry, which operates in diverse ecological and geographical zones, this feature is especially critical.

- **How KYA Supports Oil & Gas Operators:**
 - ▶ **Geo-Referenced Risk Screening:** By uploading project location coordinates or selecting the area on the GIS map, operators can instantly identify whether the proposed site falls within forest areas, eco-sensitive zones, CRZs, wildlife habitats, or tribal land governed under the Forest Rights Act.
 - ▶ **Automated Regulatory Checklist:** Based on the project's location and nature (e.g., onshore/offshore drilling, pipeline construction, seismic survey), KYA auto-generates a list of applicable statutory approvals including EC, FC, WL, and CRZ along with the relevant forms, templates, and timelines.
 - ▶ **Early Red Flag Identification:** Helps project proponents avoid ecologically sensitive or legally restricted zones in the planning phase itself, minimizing chances of proposal rejection or prolonged appraisals.
 - ▶ **Informed Decision-Making:** By enabling early-stage project design optimization, land parcel vetting, and risk mitigation, the KYA tool helps operators plan bankable, compliant, and low-risk proposals.

• **Sectoral Benefits for Oil & Gas Projects**

For hydrocarbon operators, especially in remote or high-sensitivity regions like the North-East, coastal belts, and tribal forest zones, PARIVESH 2.0 combined with KYA functionality offers a competitive edge:

- ▶ Reduces uncertainties and timelines in the clearance process.

In summary, PARIVESH 2.0 particularly the **"Know Your Approval"** module marks a transformative step in pre-emptive environmental governance. It empowers oil and gas operators with foresight, clarity, and control, enabling them to align energy development goals with ecological sustainability and compliance certainty.

8.3.2 Dedicated Committees for Proposal Appraisal and Recommendation

To ensure timely, transparent, and technically sound decision-making in the clearance process, the Ministry of Environment, Forest and Climate Change (MoEF&CC) has constituted a system of dedicated appraisal and recommendation committees. These bodies operate at both the central and state levels, with domain-specific expertise tailored to the nature and scale of the proposed projects.

• **Central-Level Committees**

- ▶ **Expert Appraisal Committees (EACs):** EACs are sector-specific bodies constituted under the EIA Notification, 2006. These expert panels evaluate proposals requiring **Environmental Clearance (EC)** under **Category A**, which includes projects with significant environmental impact. EACs assess the proposals based on Terms of Reference (ToR), public consultation outcomes, and Environmental Impact Assessment (EIA) reports. Their recommendations form the basis for final approvals by the MoEF&CC.

• **State-Level Committees**

- ▶ **State Environment Impact Assessment Authorities (SEIAAs):** For Category B2 projects which generally pose lower environmental risk, SEIAAs function as the nodal authority for appraisal and clearance at the state level. They are supported by State Expert Appraisal Committees (SEACs) for technical evaluation and ensure that decentralized decision-making aligns with national standards while reducing procedural burden on smaller-scale projects.



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- **Forest Clearance Appraisal Committees**

- ▶ **Project Screening Committees (PSCs):** For projects involving Forest Clearance (FC), the MoEF&CC has mandated the constitution of multi-stakeholder Project Screening Committees at the state and regional levels. These committees typically include:
 - Divisional Forest Officer (DFO).
 - Nodal Officer (Forest Department).
 - State Government representatives.
 - Project proponent.
 - Regional Office representative of MoEF&CC.
- ▶ PSCs serve as a platform for concurrent discussion, early-stage gap identification, and real-time resolution of inter-departmental queries. This integrated approach improves both appraisal quality and proposal throughput, especially for linear and exploration projects in forest areas.

- **Operational Efficiency and Prioritization**

All these committees follow pre-notified

schedules, operate under defined turnaround times, and adopt prioritization protocols for projects of strategic importance such as those in the energy, infrastructure, and defence sectors. The use of digital tools like PARIVESH 2.0 further enables centralized calendar management, documentation tracking, and decision monitoring.

This structured framework ensures that environmental governance is not only rigorous and consultative but also responsive to the developmental needs of sectors like oil and gas, where time-bound clearances are essential to project viability.

8.3.3 Delegation of Powers to States and Integrated Regional Offices (IROs)

In an effort to streamline environmental governance and enhance the efficiency of project appraisals, the Ministry of Environment, Forest and Climate Change (MoEF&CC) has delegated specific appraisal and approval powers to its Integrated Regional Offices (IROs) and, in certain cases, to State Governments. This decentralization initiative is a strategic reform aimed at reducing procedural bottlenecks and ensuring faster decision-making for eligible project categories.



• Operational Impact and Benefits

Shortened Clearance Timelines: Empowering regional and state-level entities enables faster scrutiny and reduces the administrative load on the central ministry, accelerating project initiation.

- ▶ **Improved Inter-Agency Coordination:** IROs function as localized hubs, facilitating better coordination between the MoEF&CC, State Forest Departments, Pollution Control Boards, and project proponents.
- ▶ **Context-Specific Appraisal:** Regional offices are better positioned to understand the ground realities, local biodiversity considerations, and socio-environmental sensitivities, resulting in more contextually informed decision-making.
- ▶ **Support for Strategic and Remote Regions:** This delegation is especially impactful in areas like the North-East, Western Offshore, and other ecologically sensitive zones where timely mobilization is critical for exploration and production (E&P) operations.

By decentralizing authority, MoEF&CC has not only enhanced the responsiveness of the clearance system but has also reaffirmed its commitment to balanced environmental stewardship in the facilitation of critical energy infrastructure.

8.3.4 Ease of Doing Business Reforms in Statutory Clearance Processes

In line with the Government of India's broader agenda to improve the ease of doing business, especially in strategic sectors like oil and gas, the Ministry of Environment, Forest and Climate Change (MoEF&CC) has implemented a series of procedural reforms to make the statutory clearance ecosystem more efficient, transparent, and investor-friendly. These reforms aim to minimize bureaucratic hurdles, accelerate timelines, and integrate technological advancements without compromising on environmental safeguards.

• Key Reforms Benefiting the Oil & Gas Sector

1. **Streamlining Documentation and Inspections:** Redundant documentation requirements have been systematically removed, and standardized templates for proposal submissions have been introduced. In place of traditional site inspections which

are often time-consuming, risk-based assessment protocols have been adopted. These are increasingly supported by geo-tagged photographic evidence, drone footage, and remote sensing data, reducing the need for repeated field visits and expediting appraisal timelines.

2. **Comprehensive Digitization and Process Automation:** Statutory clearance mechanisms have undergone end-to-end digital transformation, especially across EC and FC workflows. The integration of platforms such as PARIVESH 2.0, Gati Shakti, and the Forest Survey of India (FSI) database enables real-time tracking, document management, and inter-agency coordination. Automation of proposal routing, status updates, and alerts ensures minimal human intervention and reduced processing delays.
3. **Time-Bound Grievance Redressal and Escalation Protocols:** MoEF&CC and State Environmental Impact Assessment Authorities (SEIAAs) have institutionalized structured grievance redressal mechanisms. These are now governed by fixed timelines, with clear escalation matrices embedded into online systems. Project proponents can track responses, and seek timely resolution through dedicated helpdesks, ensuring greater accountability and reduced uncertainty in the clearance process.

Collectively, these reforms mark a paradigm shift from reactive compliance to proactive facilitation, reinforcing India's commitment to sustainable industrial development. For the oil and gas industry in particular, these measures translate into faster project initiation, lower administrative overheads, and a more predictable regulatory environment, all of which are critical for attracting long-term capital and ensuring energy security.

8.3.5 Policy-Level Relaxations

In recent years, a slew of initiatives has been undertaken by the Directorate General of Hydrocarbons (DGH), in collaboration with the Ministry of Environment, Forest and Climate Change (MoEF&CC), to streamline and expedite the grant of statutory environmental clearances. These efforts have resulted in several significant policy reforms and procedural relaxations tailored to the unique operational needs of the Exploration & Production (E&P) segment of the



oil and gas sector.

This section captures these developments in a chronological format for ease of reference and classifies them under the respective types of clearances:

8.3.6 Environmental Clearance (EC)

• 2020

▶ January 2020: Exemption of Exploratory Drilling from EIA Requirements

MoEF&CC, through its notification dated 16th January 2020, reclassified onshore and offshore oil and gas exploration activities as **Category B2** projects under the EIA Notification, 2006.

This was a landmark step as B2 projects are:

- Exempt from public hearing and EIA report preparation.
- Not being subject to the scoping stage (Terms of Reference).

This reclassification acknowledged that exploratory activities typically have minimal, temporary environmental impacts that can be mitigated through standard environmental safeguards.

• 2021

▶ March 2021: Streamlining of 'Essential Details Sought' (EDS)

MoEF&CC issued an Office Memorandum on 15th March 2021 placing a **30-day limit** on the pendency of proposals awaiting response on EDS. Beyond this period, proposals would no longer remain in the Ministry's pending list.

This intervention, long advocated by DGH, addressed persistent delays caused by back-and-forth communication between project proponents and the Ministry.

• 2022

▶ April 2022: Simplification of Coastal Regulation Zone (CRZ) Clearance for Exploratory Drilling

Earlier, separate approvals were required under both EIA and CRZ notifications, leading to protracted timelines. Based on DGH's representation, MoEF&CC issued clarificatory OM (dated 26.04.2022 and

29.11.2022) stating that Category B projects in CRZ areas could now be appraised solely by the State Environment Impact Assessment Authority (SEIAA), enabling a combined EC & CRZ clearance at the state level without central appraisal. This major procedural shift has helped fast-track drilling in CRZ zones.

▶ May 2022: Rationalization of EC Validity Linked with Forest Clearance (FC)

In line with Supreme Court directions (Lafarge case), MoEF&CC clarified that:

- The validity of ECs granted for projects involving forest land shall be reckoned from the date of Stage-II FC, or a maximum of two years from EC date, whichever is earlier. This was a long-pending concern since delays in obtaining Stage-II FC led to premature EC expiry. The rationalization effectively safeguards the EC validity window without compromising environmental safeguards.

▶ May 2022: Extension of EC Validity and Relaxation in Public Hearing Requirements

- **EC Validity for Hydrocarbon Projects** extended to **10 years**, with a **1-year extension** provision.
- **Notice period for rescheduling public hearings** reduced to **15 days** (from 30 days).
- Public hearings can now be presided over by the **Sub-Divisional Magistrate**, if confined to one subdivision, enabling quicker coordination.

▶ June 2022: Standardization Measures

- **Baseline Data Validity:** As per OM dated 08.06.2022, the validity of baseline data and public hearing reports for EC application has been fixed at **three (3) years**, reducing redundant data collection.
- **Certified Compliance Report (CCR) Timelines:** The OM also mandates issuance of CCRs within **3 months** from the date of application. This streamlines EC amendments and expansions, especially for drilling additional wells.



- **2023**

- ▶ **April 2023: Provision for Split Transfer of ECs** - As per EIA Notification amendment (S.O. 1832(E) dated 21.04.2023), a prior EC (except for mining projects) may now be:

- **Split among multiple legal entities.**
- **Transferred to another legal entity** via a simple application through the PARIVESH portal, subject to Regulatory Authority approval and EAC recommendation.
- This enhances business flexibility in asset transfer and joint development models.

- **2024**

- ▶ G.S.R. 703 (E) dated 12-11-2024 and G.S.R.702 (E) dated 12-11-2024 which exempts from obtaining CTE separately.

8.3.7 Forest Clearance (FC)

Forest Clearance (FC), governed by the Forest (Conservation) Act, 1980, is mandatory for diverting forest land for non-forest purposes. Several reforms in the past six years have greatly eased procedural complexities for the E&P sector.

- **2019**

- ▶ **September 2019: Guidelines for Seismic Surveys in Forest Areas**

MoEF&CC clarified via OM dated 30.09.2019 that:

- Up to 80 shot holes per sq km with max size 6.5 inches are permissible.
- Permission for seismic surveys may be granted at the State level, and approval under Section 2(iii) of the FCA is not required at the time of PEL grant.

- **2021**

- ▶ **August 2021: Non-applicability of FCA to Extended Reach Drilling (ERD)**

After technical consultations with DGH and industry stakeholders, MoEF&CC clarified that ERD wells, which drill laterally into forest areas from outside, do not require FC, provided certain conditions are met (OM dated 12.09.2023).

This technology-friendly move aligns Indian policy with global best practices.

- **2022**

- ▶ **November 2022: Separate Category for "Mining of Mineral Oil" on PARIVESH**

In response to DGH's proposal, MoEF&CC created a distinct classification for oil and gas projects under the PARIVESH portal. This distinction from conventional mining ensures that FC application formats and workflows are sector-relevant and user-friendly.

- **2023**

- ▶ **June 2023: Notification of Forest (Conservation) Rules, 2023**

Key reforms under the new rules include:

- **Project Screening Committees (PSCs)** at the State/UT level to pre-screen applications.
- **Regional Empowered Committees (RECs)** empowered to appraise and recommend mid-size projects (up to 40 ha) and linear infrastructure projects.
- **Delinking of PEL/PML from FC** where no physical intervention is involved.
- **Creation of Land Banks** for Compensatory Afforestation (CA), allowing CA in other States under consent-based flexibility.
- **Accredited Compensatory Afforestation (ACA):** Private landowners can now raise plantations and offer them to project developers, and these are also **eligible for Green Credits** under the Green Credit Policy Rules, 2023.

8.3.8 Advances vis-à-vis Protected Areas (PAs)

Protected Areas, including national parks and wildlife sanctuaries, are surrounded by Eco-Sensitive Zones (ESZs) where regulated development is allowed. Clearance for oil and gas projects near PAs often hinges on the final notification of ESZs, and approval from the National Board for Wildlife (NBWL).

- ▶ In 2019, DGH identified 52 Protected Areas near high-potential E&P zones for prioritization of ESZ finalization.



- ▶ As of now, 36 ESZs have been finalized, with 16 under active review by State and Central authorities. This focused intervention by DGH has significantly improved regulatory clarity for projects falling within 10 km of Protected Area boundaries.

8.3.9 North-East Region-Specific Provisions

Recognizing the ecological sensitivity, biodiversity richness, and logistical challenges unique to the North-Eastern region of India, several targeted initiatives and policy flexibilities have been introduced to facilitate responsible hydrocarbon exploration and production in this strategic area.

• Key provisions include:

- ▶ **Establishment of North-East Hydrocarbon Coordination Cells:** Dedicated coordination cells have been set up within the Directorate General of Hydrocarbons (DGH) to fast-track the appraisal and processing of statutory clearance proposals related to the region. These cells serve as nodal points for inter-agency coordination, helping reduce delays caused by regional complexities and administrative bottlenecks.
- ▶ **Forest Clearance (FC) Flexibilities:** Given that many North-Eastern states have high forest cover, special provisions have been made under the Forest (Conservation) Act to relax compensatory afforestation (CA) norms.
 - States may be allowed to undertake CA in non-forest areas or degraded forest lands where suitable forest land is not available.
 - The MoEF&CC has acknowledged the region's natural forest richness and limited availability of non-forest land, allowing for customized afforestation strategies that support both environmental and operational goals.
- ▶ **Extended Timelines for Exploration Phases:** In recognition of the terrain, climatic, and socio-political challenges in the region, the Ministry of Petroleum and Natural Gas (MoP&NG), in coordination with DGH, provides additional time for completing exploration phase commitments. These region-specific timeline relaxations ensure that Production Sharing Contracts (PSCs) and Revenue Sharing Contracts (RSCs) do

not penalize operators for delays that are beyond their control due to local constraints.

These tailored measures collectively aim to balance environmental stewardship with the strategic need for energy security in one of India's most underexplored yet resource-rich frontiers.

8.4 Institutional Mechanisms for Regulatory Facilitation at MoP&NG/ DGH

To ensure timely and coordinated regulatory approvals for hydrocarbon projects, particularly in ecologically and administratively sensitive regions, the Ministry of Petroleum and Natural Gas (MoP&NG) and the Directorate General of Hydrocarbons (DGH) have instituted several robust mechanisms. These initiatives aim to enhance inter-agency coordination, strengthen stakeholder engagement, and build institutional capacity to address complex challenges related to Environmental and Forest Clearances.

8.4.1 Empowered Coordination Committees (ECCs)

An Empowered Coordination Committee (ECC), chaired by the Cabinet Secretary and comprising representatives from key ministries including the Ministry of Environment, Forest and Climate Change (MoEF&CC), Ministry of Defence, Department of Space, and others was constituted in May 2019 through a Gazette Notification issued by MoP&NG. The ECC was established to streamline and expedite the grant of statutory approvals and clearances for hydrocarbon projects.

Since its inception, seven ECC meetings have been held, leading to the resolution of several long-pending issues and the formulation of key policy decisions. These efforts have significantly facilitated the development of the oil and gas sector.

8.4.2 North-East Coordination Committee (NECC) Cell at DGH

To fast-track regulatory clearances for Exploration and Production (E&P) projects in the North-East region of India, DGH established the North-East Coordination Committee (NECC) on July 2, 2021. Operating under the Hydrocarbon Clearance Cell (HCC) within DGH, the NECC collaborates with the E&P operators of NE region, North-Eastern state governments, and other stakeholders.

As part of this initiative, young professionals



designated as "Hydrocarbon Professionals (HPs)" have been appointed as centralized points of contact for all regulatory matters. Currently, 17 professionals are deployed across various departments in Assam, Arunachal Pradesh, Tripura, and other states. This initiative is aimed at resolving bottlenecks and ensuring timely issuance of statutory clearances.

8.4.3 Coordination with Central and State Governments

DGH has actively pursued delayed statutory clearances including Environmental, Forest, Wildlife, and Coastal Regulation Zone (CRZ) approvals as well as Petroleum Exploration License (PEL) and Petroleum Mining Lease (PML) cases with various State Governments. Since 2023, approximately 60 cases of green clearances and PEL/PML have been resolved.

States where coordination efforts have been particularly intensive include Arunachal Pradesh, Andhra Pradesh, Gujarat, Assam, Tripura, Madhya Pradesh, Tamil Nadu, Nagaland, and Jharkhand.

In addition to regular interactions with state governments, DGH participates in the

Fortnightly Regional Coordination Meetings (FRCMs) between Integrated Regional Offices (IROs), state authorities, and user agencies, as well as the monthly Inter-Ministerial Coordination Meetings (ICMCs) convened by MoEF&CC to expedite oil and gas sector proposals.

8.4.4 Stakeholder Workshops

DGH has played a pivotal role in organizing workshops involving Central Government officials and E&P industry representatives. These workshops aim to provide comprehensive support to operators, share DGH's available infrastructure, and promote a knowledge-sharing ecosystem through seminars, training sessions, and panel discussions.

In collaboration with MoEF&CC, the Environment Division of DGH has conducted four sensitization workshops focusing on recent policy reforms in Environmental and Forest Clearances. These workshops were held on June 10, 2022; September 30, 2022; June 20, 2023; 2024, and March 2025. They have fostered productive dialogue between the Government and E&P operators, helping to understand regulatory requirements and ensure compliance.





8.4.5 Technical Studies and Capacity Building

To promote long-term regulatory preparedness and fill knowledge gaps related to environmental compliance, MoP&NG and DGH have sponsored several technical studies and capacity-building projects:

- I. **Guidelines for EIA Studies Beyond 12 NM in Offshore Area:** While MoEF&CC guidelines currently apply only to offshore operations within 12 nautical miles (NM), DGH initiated a study through CSIR-National Institute of Oceanography (CSIR-NIO) to develop Environmental Impact Assessment (EIA) guidelines for operations beyond 12 NM. The resulting handbook, finalized in March 2023, offers comprehensive guidance across

the full lifecycle of E&P projects-including exploration, drilling, production, and site restoration.

- II. **Handbook on Environmentally Relevant Clearance Procedures:** To streamline statutory clearances and reduce project delays, DGH, in association with the Assam Science Technology & Environment Council (ASTEC) prepared a comprehensive handbook titled "Handbook for Environmentally Relevant Procedures and Guidelines for Upstream E&P Operations." The report identifies procedural bottlenecks and offers practical guidance for compliance. It was launched by the Hon'ble Minister of Petroleum and Natural Gas during the UrjaVarta 2024 conclave.

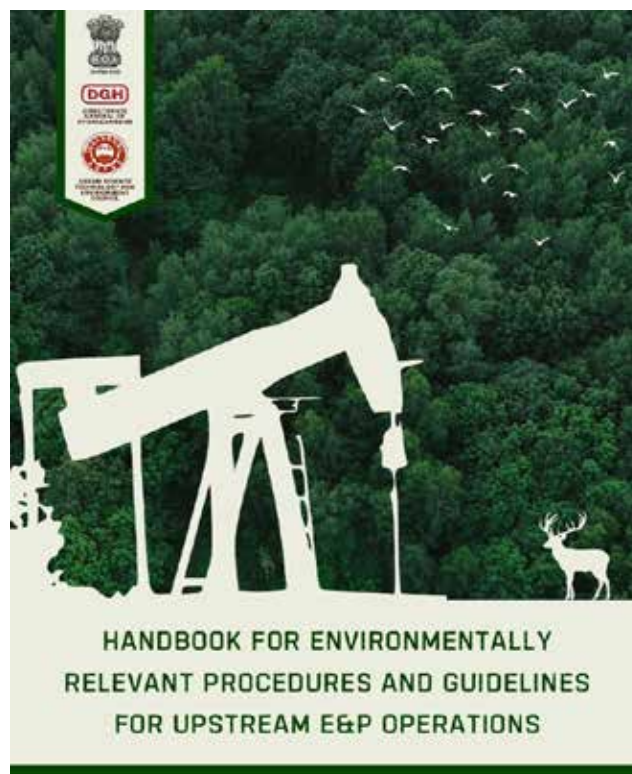


III. Documentary on Environmental Aspects of E&P Activities: To address public misconceptions comparing E&P activities with conventional mining, DGH produced a documentary highlighting the environmental, social, and economic aspects of responsible oil and gas operations. It emphasizes community health, skill development, conservation efforts, and adherence to best practices.

IV. Study on Climate Change Impact on Offshore E&P Activities: DGH, in collaboration with CSIR-NIO, Goa, has initiated a study to assess the vulnerability of offshore infrastructure to extreme weather events and climate change. The project includes the development of a Standard Operating Procedure (SOP) to provide early warnings and guide risk mitigation for offshore operations.

V. Wildlife Institute of India's (WII) Study on ERD/HDD Impact on Wildlife and Biodiversity: Following MoEF&CC's directive (dated 12.09.2023), which permits Extended Reach Drilling (ERD) in forest areas without prior Forest Clearance under specific conditions, DGH in association with Wildlife Institute of India (WII) has initiated a study which aims to assess the impact of ERD/Horizontal Directional Drilling (HDD) technologies and surface facilities on wildlife and above-ground biodiversity outside forest boundaries. Baseline data collection involving the study is underway in the North-East region. The findings will support the development of region-specific guidelines to ensure compliance and minimize ecological impact.

This suite of institutional mechanisms underscores MoP&NG and DGH's commitment to fostering an enabling environment for hydrocarbon exploration and production while ensuring robust



environmental safeguards and stakeholder engagement.

8.5 Key Developments During the Year

8.5.1 Amendment to the Oilfields (Regulation and Development) Act and Its Implications

The recent amendment to the Oilfields (Regulation and Development) Act, 1948 through the Oilfields (Regulation and Development) Amendment Act, 2025, marks a pivotal shift in legislative clarity and sectoral governance. The amendment explicitly removes the term "mining" from the definition of "mineral oil" and introduces the term "Petroleum Lease" in lieu of "Petroleum Mining Lease," thereby formally distinguishing petroleum operations from conventional mining activities.

This statutory refinement acknowledges the unique technical, environmental, and operational characteristics of petroleum exploration and production. Historically, oil and gas projects have been evaluated under regulatory frameworks and procedural norms intended for conventional mining, resulting in systemic bottlenecks. These include:

- ▶ Imposition of mining-specific environmental and forest clearance conditions that do not correspond to the nature of E&P operations.



- ▶ Extended delays in obtaining Forest and Wildlife Clearances due to procedural redundancies.
- ▶ Interruptions in the timely execution of hydrocarbon projects, adversely affecting India's energy security goals.

The amendment is a crucial step toward resolving these issues. By formally separating E&P activities from mining, it lays the foundation for more appropriate, efficient, and technically relevant regulatory treatment of oil and gas projects, ultimately expediting approvals and enhancing sectoral efficiency.

8.5.2 Summary of Green Clearances Facilitated

The Directorate General of Hydrocarbons (DGH) has continued its proactive role in securing statutory clearances and licenses for Exploration and Production (E&P) activities. During the year, DGH successfully facilitated approvals for:

- ▶ Approx. 40 Petroleum Exploration Licenses (PELs) and Petroleum Mining Leases (PMLs), expediting the regulatory processes and enabling project execution.
- ▶ Approx. 30 environmental clearances including Environment Clearances (EC), Forest Clearances (FC), Wildlife Clearances (WLC), Consent to Operate (CTO), and clearances in Eco-Sensitive Zones (ESZ), with a major focus on the North-East region of India.

These developments are expected to catalyze economic benefits by generating revenue for State Governments, creating employment, improving local infrastructure, and contributing significantly to India's energy landscape.

8.5.3 HIGHLIGHTS

Final grant of FC on December 24, 2024 of Duarmara block, which was pending since April, 2022

The site-specific Forest Clearance application for the Ningru PML of OIL in Arunachal Pradesh, submitted on November 1, 2018, received final approval on October 30, 2024.

Final FAC approval dated 27.01.2025 for conducting R & D study at an identified location of OIL in Assam in respect to ERD proposal

21 long-pending PEL/PML were resolved, including:

5 cases under the Discovered Small Field (DSF) regime in Andhra Pradesh;

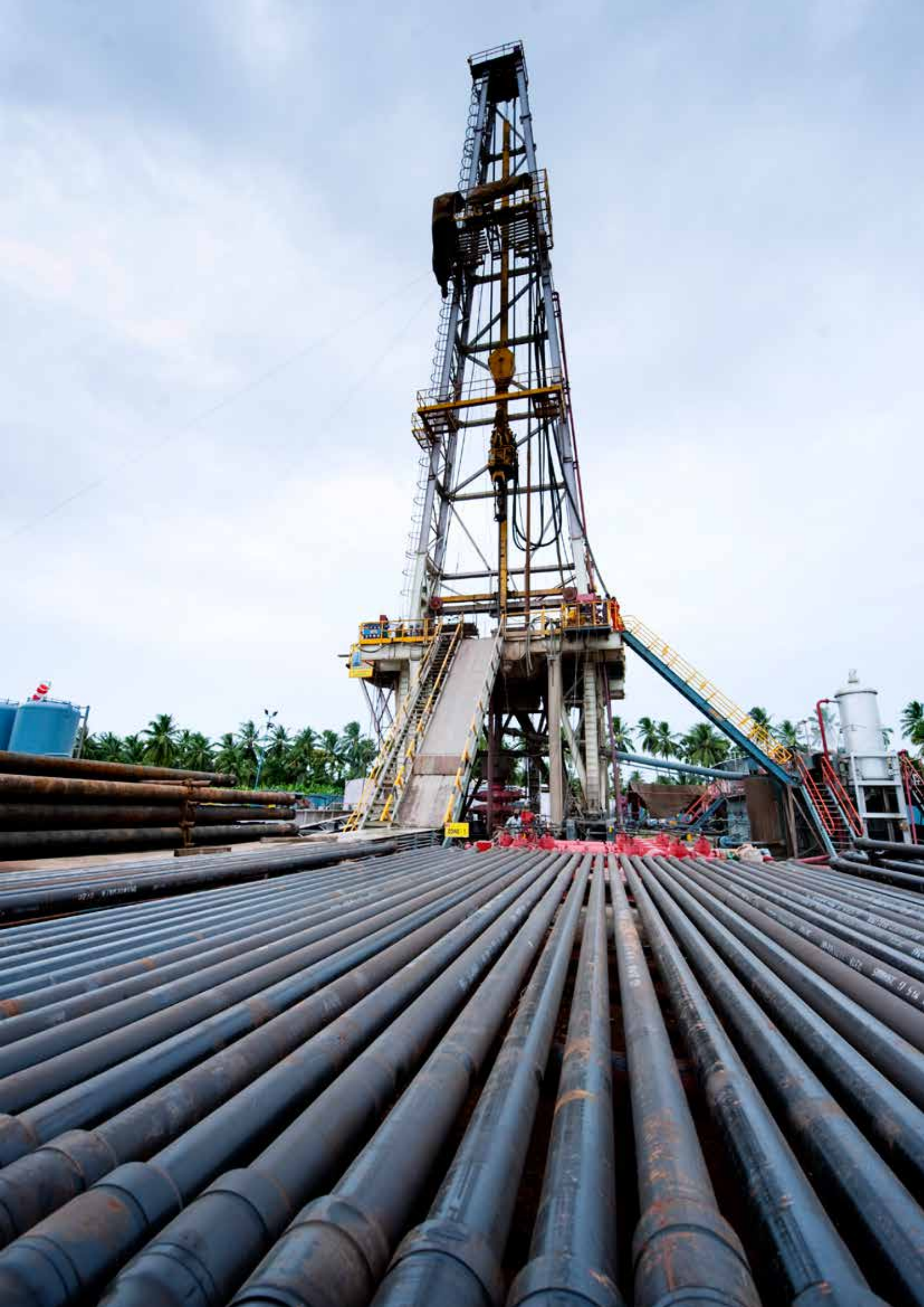
- ▶ **1 Pre-NELP case in Andhra Pradesh;**
- ▶ **14 cases in Tamil Nadu;**
- ▶ **1 Petroleum Mining Lease (PML) for Ashokenagar-1 in West Bengal.**

8.6 Reference Materials for Operators

▶ Key statutory documents and links:

- Ministry of Environment, Forest and Climate Change website (<https://moef.gov.in>)
- PARIVESH Portal (<https://parivesh.nic.in>)
- EIA Notification, 2006 and amendments
- Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980(amended in 1988 and 2023)
- Wildlife Protection Act, 1972 and its amendment
- Coastal Regulation Zone Notification 2011/2019
- Water (Prevention and Control of Pollution) Act, 1974 and its amendment
- Air (Prevention and Control of Pollution) Act, 1981 and its amendment
- DGH Operational Guidelines/ Handbook (https://dghindia.gov.in/assets/downloads/env_handbook.pdf)
- Relevant State policies and SOPs





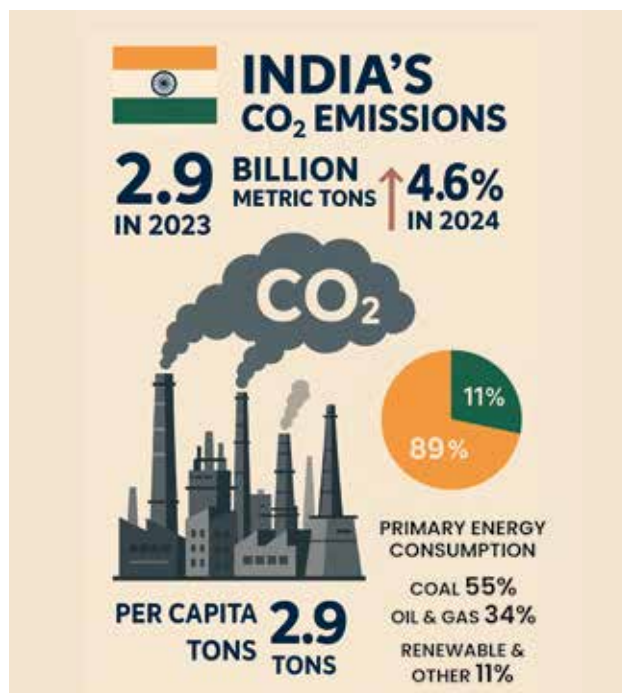


9

Hydrocarbon Efficiency & New Energy

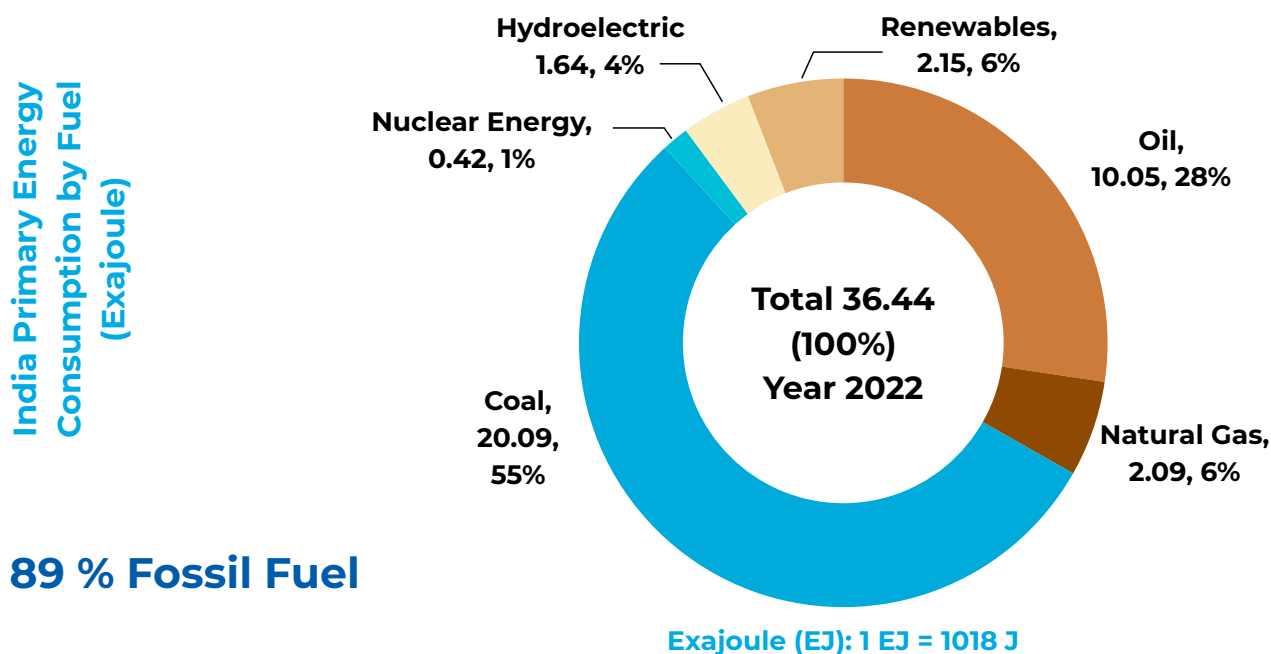
9.1 Introduction & Background

India is one of the fastest-growing major economies in the world, propelled by a large and dynamic market, alongside significant domestic reforms and sustained investments in energy and infrastructure. However, this rapid development has been accompanied by a notable increase in carbon emissions behind the USA and China. According to IEA report of 2023, India's CO₂ emissions stood around 2.9 billion metric tons in 2023 and it was projected to increase by 4.6% in 2024. However, India's per capita CO₂ emissions are estimated to be around 2.9 tons of CO₂ equivalent (tCO₂e), according to the Global Carbon Budget 2024 report. This is significantly lower than the global average of 6.6 tCO₂e.



According to the Energy Statistical Review of World Energy data 2023, 89% of India's primary energy consumption is from fossil fuels, with coal

contributing 55% and oil and gas contributing 34%. Renewable and other energy sources account for only 11% of energy usage (36.44 EJ).

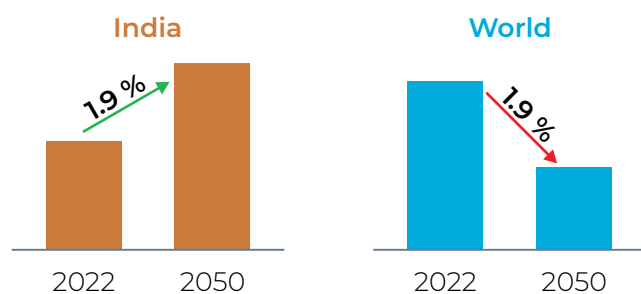


89 % Fossil Fuel

Source : Energy Institute Statical Review of World Energy, 2023

India's primary demand is expected to double by 2050, with a Compound Annual Growth Rate (CAGR) of 2.4% - 2.6%, while oil and gas consumption is projected to grow with a CAGR of 1.9%. In contrast,

The world's oil and gas consumption is expected to decrease with a CAGR of 2.6% until 2050.



Projected Oil and Gas consumption (MMTOE)

Source : For 2050, Accelerated Scenario of BP stats considered

As part of India's climate commitment, the Hon'ble Prime Minister during the 26th Conference of Parties (COP26) in Glasgow, announced PANCHAMRIT initiative (five key climate action points): 1) achieving 500GW non-fossil energy capacity by 2030 2) 50% energy from renewables by 2030, 3) cutting one billion

tonnes of carbon emissions by 2030 4) reducing carbon intensity by 45% by 2030 and 5) reaching net zero emissions by 2070.

9.2 Navigating Energy Transition through Advancing Hydrocarbon Efficiency and Integrating New Energy in E&P Operations

India's reliance on fossil fuels and its growing energy demands, coupled with the goal of achieving NetZero by 2070, require a holistic approach. This should include transitioning to enhancing hydrocarbon efficiency, renewable energy sources, , promoting the electrification of transportation, expanding forest cover, investing in carbon capture and storage, reducing dependency on coal, implementing supportive policies, fostering international collaboration, raising public awareness, encouraging green technology investments, and developing adaptation strategies. In line with India's Net Zero goal, major oil and gas companies have set ambitious net-zero targets as follows: ONGC aims for 2038, Oil India for 2040, Reliance for 2035, and Vedanta for 2050.

Oil and gas remain critical to our energy mix,



even as we transition toward renewables and cleaner fuels. The upstream oil and gas sector must evolve to stay relevant in a transitioning global energy landscape. Pragmatism in upstream development means integrating decarbonization efforts without compromising energy security, affordability, and economic growth. Setting clear targets and preparing roadmaps for net-zero is essential to driving a low-carbon energy future.

India's energy landscape is undergoing a transformative shift as the nation embraces a sustainable future. Central to this evolution is the imperative to enhance hydrocarbon efficiency while simultaneously integrating new energy sources. This dual approach is pivotal in ensuring energy security, reducing reliance on imported fossil fuels, and mitigating adverse environmental impacts, including greenhouse gas emissions and air pollution.

9.3 Establishment of Hydrocarbon Efficiency & New Energy (HENE) Department

To support oil and gas operators in achieving their net-zero targets, promote decarbonization, and integrate new energy sources into existing operations, DGH established the Hydrocarbon Efficiency and New Energy (HENE) Department, which was launched by the Hon'ble Minister of Petroleum & Natural Gas, Shri. Hardeep Singh Puri during the Inaugural edition of UrjaVarta on 11th July 2024.

The department focuses on driving decarbonization practices, improving hydrocarbon efficiency, and integrating new geological energies, including geothermal and natural hydrogen, into existing oil and gas operations.

9.4 Key Actionable Areas of HENE Department

The HENE department is tasked with advancing innovations and policies that drive energy transition in the upstream oil and gas sector with the emerging geological energy sources within the current E&P framework. The functions roles and responsibilities of HENE is given below.

9.4.1 Key Functions

- ▶ To advise the Ministry on hydrocarbon efficiency and decarbonization in the upstream oil and gas sector.
- ▶ To assess basin wise CO₂ storage potential for CCUS, Geothermal and Natural Hydrogen energy prospectivity.
- ▶ To advise the Government in the development of composite licenses/leases that synergizes oil and gas, geothermal, white hydrogen, and renewable (wind and solar) energy projects.
- ▶ To expand the National Data Repository (NDR) of DGH by incorporating data on new geological energy such as geothermal energy and natural hydrogen.

9.4.2 Roles and Responsibilities of HENE

The HENE shall have following Roles and Responsibilities of HENE

- ▶ Develop reporting standards for GHG emission measurements and verification mechanisms in upstream operations. It will collaborate with national and international agencies to establish emission reporting standards.



- ▶ Engage with operators on emission abatement actions, integration of new geological energy and renewable energy.
- ▶ Collaborate with agencies for field-wise assessments of reservoir characterization in different basins for geological carbon storage, across different geological storage options.
- ▶ Conduct assessment of the natural Hydrogen potential in different basins and propose suitable policy reforms. Propose hydrogen gas sampling in new and existing contracts.
- ▶ Extend support to Ministry of New and Renewable Energy on formulation of Geothermal policy .
- ▶ Formation of the Hydrocarbon Efficiency and New Energy (HENE) department will also partially fulfil the objectives outlined in the Oilfields (Regulation and Development) Amendment Act, 2025, Section 6, Subsection C-VII, new clauses (o), (p), and (q).

9.5 Advancing Hydrocarbon Efficiency and Integrating New Energy

In 2024-25, the Directorate General of Hydrocarbons (DGH) has taken significant strides in advancing India's upstream energy landscape toward decarbonization, sustainability, and energy security. Building upon global trends and national initiatives, DGH continues strengthening its role as a key facilitator in integrating hydrocarbon efficiency with emerging clean energy domains such as carbon capture, utilization, and storage (CCUS), geothermal energy, and natural hydrogen.

The global energy sector in 2025 is witnessing transformative shifts driven by climate commitments, rapid technological advancements, and economic imperatives for energy diversification. Natural hydrogen, geothermal energy, and CCUS have emerged as pivotal technologies for decarbonization, with increasing international collaboration, policy development, and pilot deployment. DGH has aligned its strategic vision with these trends by systematically assessing regulatory landscapes, technical frameworks, and innovations across global jurisdictions, positioning India at the forefront of these technologies. DGH simultaneously initiated the hydrogen exploration program recommending mandatory integration of hydrogen prospecting into existing upstream surveys.

In CCUS, DGH supports multiple stakeholders, evaluating technical proposals from various academic institutions and companies, facilitating inter-agency coordination and ensuring timely appraisals. Key recommendations were embedded into the amendment of PNG Rules under the ORDA Amendment Act 2025, supporting comprehensive energy block licensing, regulatory clarity, and carbon storage safety standards. To drive sustainable and efficient upstream operations while integrating emerging geological energies like natural hydrogen and advancing geothermal energy, DGH's action plan is built upon the following strategic pillars:

9.5.1 Initiative for CO₂ Sequestration

The HENE Department will facilitate the Exploration & Production (E&P) operators, academic institutions, and R&D institutions through data share as per policy in support of pilot projects focused on Carbon Capture, Utilization, and Storage (CCUS). These initiatives aim to assess the carbon sequestration potential across a variety of geological settings, including:

- ▶ Depleted oil and gas reservoirs
- ▶ Deep saline formations, with an emphasis on onshore basins
- ▶ Fractured basement (basaltic) reservoirs
- ▶ Existing coalbed methane (CBM) fields through enhanced recovery using CO₂ injection

• Technical Partnerships and R&D Collaboration for CO₂

Strategic collaborations with leading R&D institutions to advance research and development in the following areas:

- ▶ Enhanced hydrocarbon recovery and geological carbon dioxide (CO₂) storage
- ▶ Feasibility assessments of CO₂ injection and associated risk analysis
- ▶ Evaluation of CO₂ storage efficiency and comprehensive environmental impact assessments

9.5.2 Standardizing MRV Frameworks for GHG Management

Under the Monitoring, Reporting, and Verification (MRV) framework, it is proposed that standardized reporting protocols and verification mechanisms



be developed to track key emissions milestones in alignment with the Net Zero targets of all E&P operators. This includes collaboration with national and international agencies to establish unified greenhouse gas (GHG) reporting standards. Additionally, periodic reviews will be conducted to monitor emission and the effectiveness of abatement measures.

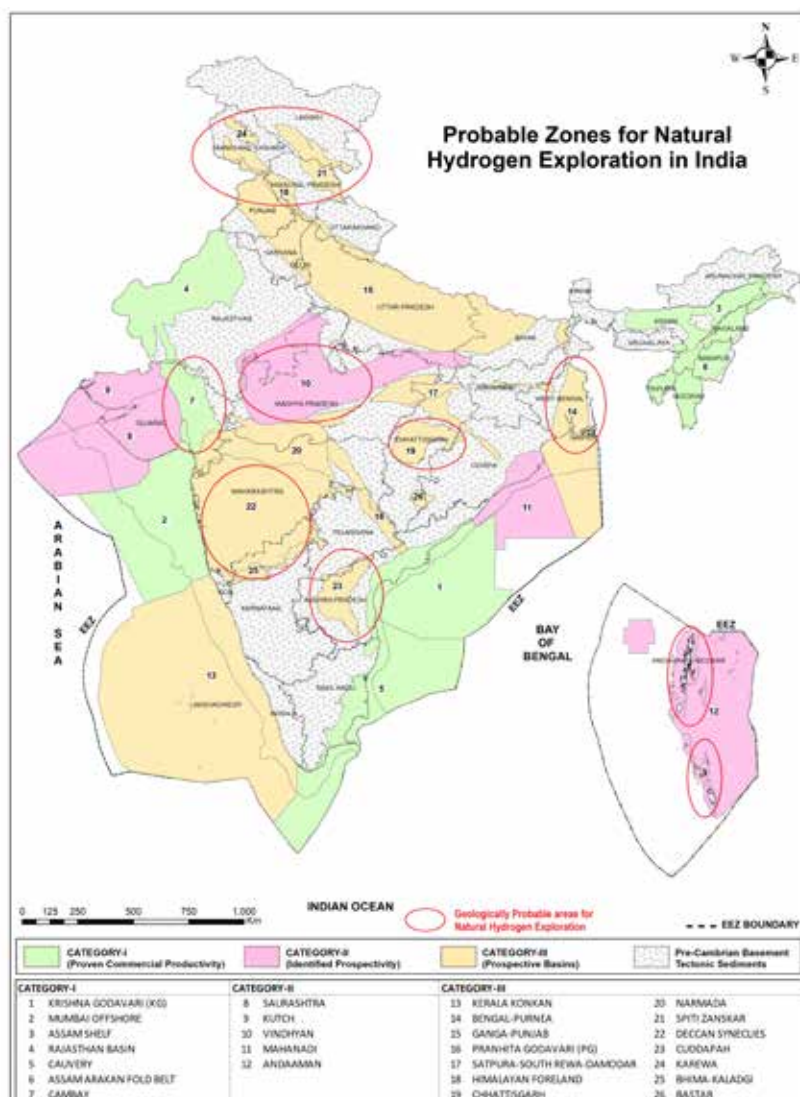
9.5.3 Natural Hydrogen Initiative

By early 2025, natural hydrogen has emerged as a key focus in global energy exploration, with over 150 exploratory licenses and more than 30 field investigations underway. Active geological mapping, gas flux monitoring, and pilot drilling across at least 12 countries reflect the growing international interest in its energy potential.

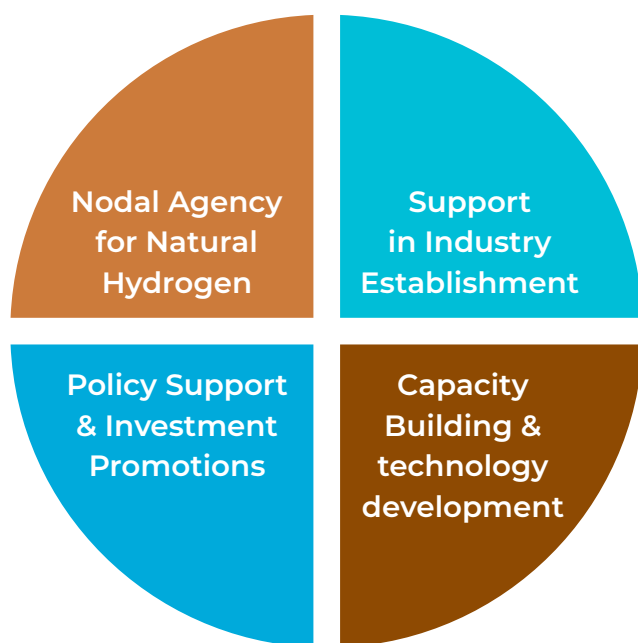
Preliminary analysis by the HENE team suggests that India's geological setting, marked by sedimentary basins and stable cratonic regions,

holds significant promise for natural hydrogen accumulation. Global analogues indicate that natural hydrogen can occur in diverse geological environments, including rifts, back-arc basins, intracratonic basins, Precambrian cratons, Banded Iron Formations, mineralized belts, mid-ocean ridges, and orogenic zones. Given these similarities, India's subsurface conditions are considered highly conducive to hydrogen generation, reinforcing the need for a focused national exploration strategy. Most probable zones to initiate natural hydrogen exploration are shown in the above map.

India's first breakthrough came in February 2025, when the Geological Survey of India (GSI) confirmed finding of natural hydrogen in the South Andaman district of the Andaman and Nicobar Islands. The gas was identified within serpentinized pillow basalt formations, marking a pivotal step from theoretical potential to proven resource.



To initiate this effort, DGH is initiating natural hydrogen exploration in India by introducing provisions for hydrogen gas sampling during exploratory drilling and integrating hydrogen detection into on-site gas chromatography systems, particularly in ultramafic, serpentized, and volcanic formations where hydrogen is likely present. To further support these activities, a joint data integration platform involving E&P operators and key stakeholders may be established to facilitate systematic data sharing and analysis of natural hydrogen systems. Additionally, on a voluntary basis, E&P operators are encouraged to incorporate natural hydrogen exploration into upcoming geoscientific studies, such as gradiometric, seismic, geospatial, and geochemical surveys, targeting promising geological settings like Precambrian cratons, ultramafic rocks, and ophiolitic belts to accelerate the identification of hydrogen-rich areas in India.



An International Conference on Exploring the Potential of Natural Hydrogen: A Sustainable Energy Frontier was organized by the Indian Institute of Petroleum and Energy (IIPE), Visakhapatnam. The discussions at the conference also reinforced the Directorate General of Hydrocarbons (DGH)'s initiative to advance natural hydrogen exploration in India by utilizing the country's vast and geologically diverse landscape, which offers significant potential for natural hydrogen generation and accelerated exploration efforts.

Experts highlighted the considerable potential for natural hydrogen within India's diverse geological framework. Key regions identified

included the South Andaman area, the Dharwar Craton, the Cuddapah and Bhima Basins, the Singhbhum Shear Zone, Northeast and Central India, Sohna, Ladakh, and the Nagaland-Manipur ophiolites and geothermal systems.

The Directorate General of Hydrocarbons (DGH) is driving this by facilitating data support and aligning natural hydrogen initiatives with existing oil and gas exploration frameworks. These efforts are expected to significantly accelerate the growth of a robust natural hydrogen sector, thereby strengthening India's clean energy infrastructure, and advancing its long-term sustainability goals.

9.5.4 Geothermal Data Integration

The Directorate General of Hydrocarbons (DGH), through its Hydrocarbon Efficiency & New Energy (HENE) Department, is advancing geothermal energy as a strategic decarbonization lever within upstream operations.

- **Contribution to National Policy and Regulatory Framework**

- ▶ HENE has provided critical inputs to the Ministry of New and Renewable Energy (MNRE) for the Draft Geothermal Policy. Contributions include requirement of basin-specific resource mapping, and strategies for the integration of geothermal development within existing oil and gas operations.

- **Support for Composite Energy Licensing**

- ▶ To promote integrated energy exploration, HENE will support the framework of the Composite Energy Licensing Framework. This framework enables joint exploration of oil, gas, geothermal, and natural hydrogen resources within a single block, accelerating the upstream sector's transition to integrated energy systems.

9.5.5 Achievements and Institutional Leadership (2024-2025)

The Hydrocarbon Efficiency and New Energy (HENE) department within DGH formalized its strategic role and delivered concrete outcomes:

- **Policy and Regulation**

Contributed to the draft amendments of the PNG Rules and the National Geothermal Policy, promoting holistic energy integration. aligned with the Oilfield Regulation and Development



Amendment Act, 2025. This includes guidance on licensing, environmental monitoring protocols, and long-term liability provisions.

• Technical Evaluation

Reviewed and appraised geological CO₂ sequestration and natural hydrogen exploration proposals from operators and academic institutions.

• Natural Hydrogen Framework

Preliminary geological analyses has been carried out to identify the promising zones for natural hydrogen exploration, based on global case studies. Additionally, a framework was drafted and initiated to enable hydrogen gas sampling as part of routine gas analysis during oil and gas drilling operations.

• Knowledge Partnerships

DGH has signed a Memorandum of Understanding (MoU) with IIT Bombay to advance decarbonization in the upstream petroleum sector. This collaboration focuses on research aimed at enhancing hydrocarbon recovery (EOR) and assessing the potential for CO₂ storage.



The key objectives include:

- ▶ Evaluating the geological characteristics and behavior of depleted reservoirs to determine their suitability for CO₂ storage, including assessments of injection feasibility and associated risks.
- ▶ Characterizing the geological properties and hydrogeological conditions of saline aquifers to assess their suitability for CO₂ storage, along with evaluating storage efficiency and conducting environmental impact assessments.
- ▶ Developing a comprehensive analytical framework to monitor CO₂ injection, migration, and containment within saline aquifers, ensuring compliance with

regulatory standards and environmental safeguards.

• Stakeholder Engagement

DGH organized a stakeholder interaction on “Upstream Decarbonization: Hydrocarbon Efficiency and New Energy Pathways” on November 26, 2024. The discussions identified strategic priorities for decarbonization and new energy integration within existing upstream operations.



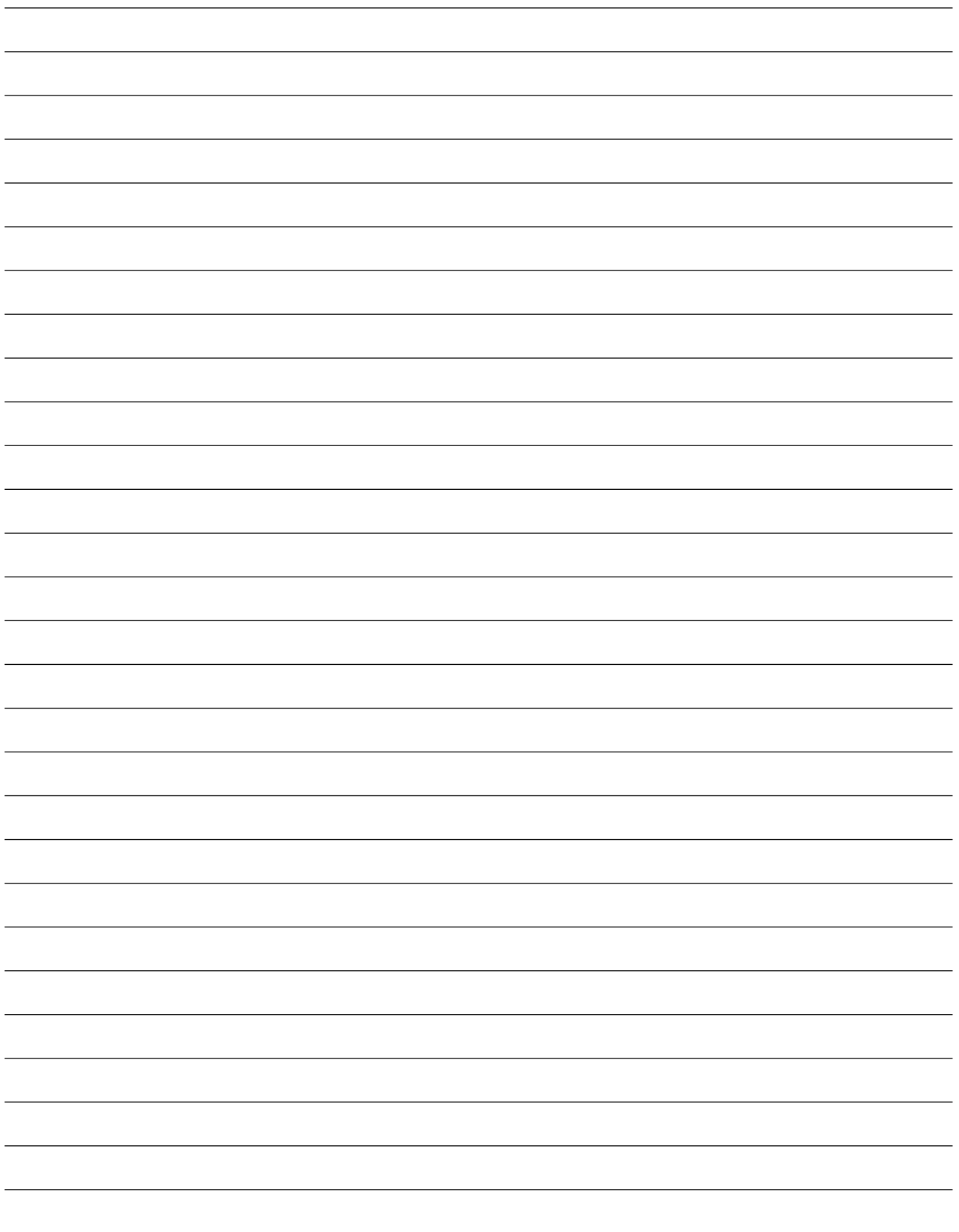
The discussions focused on key priorities such as minimizing methane emissions, advancing the adoption of Carbon Capture and Storage (CCS) technologies, and exploring clean energy alternatives like Advanced Geothermal Systems and Natural Hydrogen Exploration. Emphasis was also placed on the need for robust policy support to drive the integration of decarbonization strategies and the adoption of new geological energy sources. This collaborative dialogue marks a significant step toward aligning upstream activities with India's climate commitments and accelerating the transition to a low-carbon energy future.



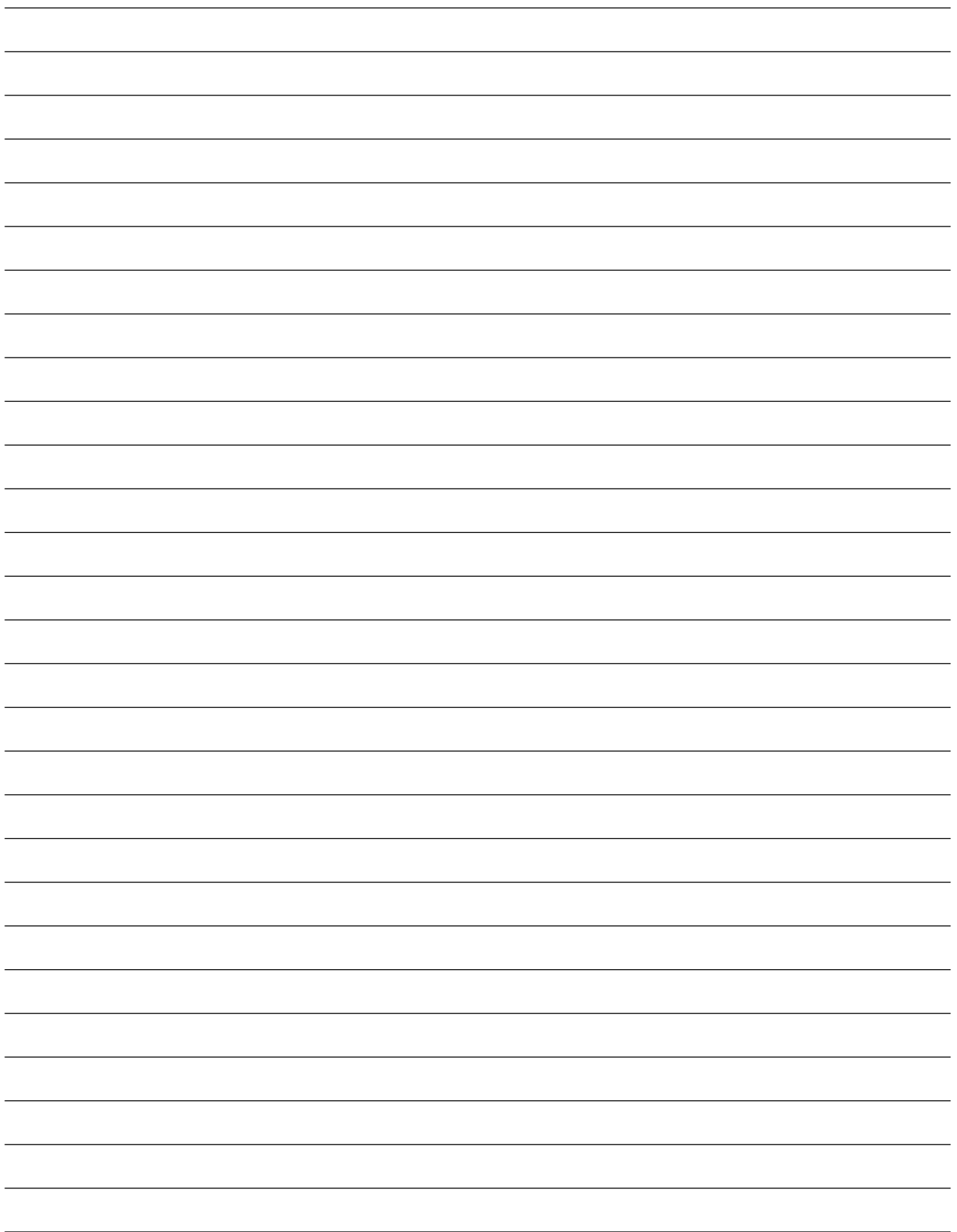


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